# A manifest of `docs/reports`

Two hundred and ninety independent mathematical research packages, classified by subject

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## 1 Scope and conventions

This document catalogues the two hundred and ninety research packages stored under `docs/reports`. The collection originally held one hundred and four reports; after two consolidations it contributes one hundred and two maintained reports. The other one hundred and eighty-eight came through ProveIt's incoming-reports procedure (`docs/incoming`) — six in batch 36, one each in batches 40, 41 and 44, two in batch 60, one each in batches 62 and 70, three in batch 72, sixteen in batch 73, two in batch 74, three in batch 75, two in batch 76, thirty-four in batch 77, three in batch 78, two in batch 79, two in batch 82, one in batch 83, five in batch 85, one each in batches 86, 87, 89 and 91, four in batch 100, one in batch 101, three in batch 102, four each in batches 103, 104 and 105, seven in batch 106, five in batch 107, ten in batch 108, eight each in batches 109 and 110, five in batch 111, seven in batch 112, eight in batch 113, five in batch 114, two outside the numbered batches (Report 274 and a two-way automata manuscript of 7 October), twelve in batch 138 and one in batch 143 — whose archives remain in the repository's history. Batches 100–113 are clusters of one session bundle of numbered Research Reports (arrival `60f54ea06`); batch 114 is the seven other archives of two arrivals of the same day. Later batches also extended existing reports; their entries name the batch. Batch 80 opened no report here: it extended one report and replaced the programs of three Parts by corrected code editions; batches 88, 90, 92, 93, 96, 97 and 98 likewise only extended reports. Batches 81, 84, 94, 95, 99, 115–137 and 139 brought nothing here; the manuscripts of batches 81 and 84, like part of batches 83, 86, 87, 89, 90, 92, 93, 95, 96 and 97, went to the surreal-number collection, one batch-85 manuscript, the five of batch 94 and two of batch 95 went to `Analysis/Transseries`, three manuscripts of batch 97 and the thirteen archives of batch 99 went to the Fabius drafts tree in `Analysis/FabiusFunction`, and the one hundred and six Gowers–Szemerédi manuscripts of batches 115–137 went, by Vladimir's direction, to one report beside the Lean development of Gowers's proof, `local-quantitative-refinements` in `Combinatorics/Ramsey/Research/GowersSzemeredi`. Of the manuscripts of 7 October that arrived with them but do not concern Gowers's argument, three came here (the two-way automata report and Parts of the A373271 and A328716 reports) and the rest went to new topic directories outside this collection, which `docs/incoming` lists. The archive of the batch-114 arrivals that continues the A082528 report was held for batch 110, which placed it as that report's Part III. Batch 138, the arrivals of 8 October, opened the categories `convex-geometry` and `probability`, placed two further reports among the generating functions and added Parts to three OEIS reports. Its twelve new reports here are single manuscripts placed without a write, by Vladimir's direction of 9 October 2026, so their entries describe the delivered packages. Its other manuscripts went to `local-quantitative-refinements` (a hundred and seventh source), to research directories in `Analysis/ErdosSimilarity`, `Combinatorics/BooleanFunctions` and `Combinatorics/Sidorenko`, to `Analysis/Polylogarithms` and to the Fabius drafts tree; batch 139 went to `Analysis/Polylogarithms` only. Batch 143 opens `set-theory` with one complementary closure-kernel report from three manuscripts; the seeded manuscript is the base, and a reconciliation README records their overlap and limits without rewriting the delivered articles. Batch 144 extends the same report with six manuscripts on pure subset closure, hereditary saturation, certificate coding and machine engineering; it opens no new report and

retains the collection total of 290. Batches 145 and 146 add six more manuscripts on equality erasure, empty-seed closure, exact grammar compression and a finite-sort obstruction to the reduced calculus. Batch 147 adds genuine ternary Bottom and a complete Hilbert reference; batch 148 adds a PA obstruction and complete ground-clause search. Batch 149 adds binding certificates, shared-template tableaux and a 268-state complete-logic compiler control. Batch 150 adds captured EPSC, two-parameter EPSC2, a protected closure rule and quantified-NAND tableaux, with separate initializer and compiler controls. The same report now reconciles fifty-six sources after batches 158–159's proof-interface, countdown and partial-salvage additions; the collection remains 290 reports. The next paragraphs describe the historical collection before those two consolidations. On 29 September 2026 the two Lipparini Problem 6.2 reports were unified, and the finite-height transfinite-word report was integrated with its guarded-block companion. Both proof routes and all distinct extensions are retained; old directories contain redirects, not extra catalogue entries. The seven surreal-number reports that this catalogue used to carry were moved out to a repository of their own, with their history, and are no longer listed here. They arrived as one hundred and twenty-five ZIP archives, in seven deliveries, which have been unpacked and grouped into the subject hierarchy described in Section 2. The archives themselves are no longer kept in the repository.

One hundred and twenty-five archives yield one hundred and three reports for two different reasons. Twenty archives duplicated another: sixteen pairs, three three-way clusters and one four-way cluster. Two more were folded into a neighbour not because they repeated it but because they shared a subject — the three reports on term counts in the derivatives of power towers are now one article with a common framework and three parts. That consolidation is editorial, and the distinction matters: those three were put through the duplicate sweep and cleared it, and their three theorems are three different theorems. The seventh delivery of eighteen contained no duplicate at all, which is worth recording: a sweep that finds nothing is evidence about the collection, not a sweep that failed. Each pair has been merged into one report that proves the shared theorem once and keeps everything both originals built on top of it. Where the two reached a result by genuinely different routes, both proofs are kept and marked as alternatives, with a sentence on what each one buys; the merged shuffle report keeps three independent finite certifications of one theorem, and the merged report on surreal option graphs kept two different five-vertex counterexamples for the same reason before it moved out with the other surreal-number reports. A merged entry below names both of its source archives and says what was doubled.

Not every pair aimed at one problem is a duplicate. The two Gonshor reports both prove his product-birthday bound, but on domains that are incomparable in both directions, so neither theorem contains the other and they are catalogued separately; the same holds for reports answering different numbered problems of one paper, noted where it occurs.

Most packages follow the same shape: a typeset article (`*.pdf`) with its complete LaTeX source, a `README.md` summarising the result, usually a `code/` directory with exact-arithmetic verification scripts, a `data/` or `results/` directory holding the recorded output of those scripts, and often `STATUS.md`, `sources.md` or `notes/` files auditing the sources consulted and the precise limits of the claim. The batch 143 closure-kernel reconciliation retains the base LaTeX source and selected supporting files; its README identifies the member articles and PDFs recoverable from the committed arrival archives.

Two cautions apply uniformly and are therefore not repeated in the individual entries. First, these are AI-assisted research drafts: none has been independently refereed or fully checked in a proof assistant. Standalone checked components are identified with their limited scope. Several continue a Lean or Rocq project elsewhere in ProveIt; that confers no formal status on them, and their entries say which project they continue. Second, the descriptions below record *what each report sets out to settle and what it claims to prove.* The catalogue does not itself verify those claims; individual reports retain their review and verification records. Where a report itself flags a limitation — a prior disproof, an imported theorem, a correction to the source, or an exception left open — that is noted, because it changes what the report is.

The naming of the directories is editorial. Several distinct reports attack the same published problem, and some arrived under colliding archive names, so each directory has been given a name describing its own angle on the problem; the merged directories usually name their combined scope. The September 29 consolidations retain the base directory names to preserve existing links. The original archive names are recorded with every entry.

## 2 The directory tree

| Category | Subcategory | Reports |
|---|---|---|
| enumerative-combinatorics | — | 27 |
| ordinals-and-order-types | ordinal-arithmetic | 3 |
| | wqo-powersets-and-statures | 4 |
| | friendly-order-types | 3 |
| | games-on-ordinals | 3 |
| | transfinite-words | 1 |
| | (top level) | 13 |
| generating-functions-and-asymptotics | oeis-sequence-asymptotics | 144 |
| | binary-product-coefficient-counts | 3 |
| | (top level) | 6 |
| log-concavity-and-unimodality | — | 14 |
| hankel-determinants | catalan-and-ballot | 4 |
| | somos-and-elliptic | 3 |
| | growth-and-runs | 2 |
| | (top level) | 1 |
| congruences-and-valuations | supercongruences | 4 |
| | iterated-series | 3 |
| | (top level) | 4 |
| automata-and-formal-languages | — | 9 |
| tetration-and-digit-stabilization | — | 6 |
| graph-theory | — | 3 |
| quaternionic-analysis | — | 1 |
| jacobian-conjecture | — | 4 |
| galois-theory-and-radicals | — | 2 |
| hilbert-tenth-problem | — | 12 |
| convex-geometry | — | 1 |
| probability | — | 9 |
| set-theory | — | 1 |
| **Total** | | **290** |

More than a quarter of the collection consists of counterexamples and corrections rather than proofs. Ordinal Chomp, open-query games, compositional tree series, the secant numbers, the decimal-length bound for tetration stabilization, power-tower stabilization, iterated-power digit stabilization, cluster variables, chromatic coefficients of cycles, products of $q$-integers, domination roots, Hardinian array constants, the mixed-base counterexample, recursive exponents, Gregory sign thresholds and binary substitution discrepancy all refute, or replace with a proved statement, an assertion published or catalogued elsewhere. Several reports whose main outcome is a proof still correct the statement they prove: the Somos-8 certificates, the shifted Catalan Hankel polynomials and the Rueppel determinants record a sign error, a missing one-unit shift and a numbering ambiguity in their sources, and the independence-system thresholds report refines its target after finding that the original conjecture had already been disproved by someone else.

## 3 Ordinals and order types

## 3.1 Friendly order types

Three reports on Isa Vialard's *friendly order type* $f(P)$, the rank of the tree of open-ended bad sequences in which every chosen point has an incomparable partner in the current residual. All three prove the same finite formula; they differ in what they build on top of it, and in how they prove it — one by a forest-rank monovariant that survives moves deleting a whole upset, one through a lemma bounding the cut maxima. A fourth report proved the formula by the first report's argument almost verbatim and has been merged into it.

**1. Friendly Order Types Through Incomparability Components**
`ordinals-and-order-types/friendly-order-types/incomparability-components-and-infinite-tails/friendly_order_types.pdf`
from `friendly_order_type_research (1).zip`

The same component formula, together with an exact friendly rank for connected infinite well-partial-orders: writing $o(C) = \lambda + n$ with $\lambda$ the limit part and $T(C)$ the finite set of points with finite principal upsets. Includes an exhaustive finite verification and a proof audit.

**2. Friendly Order Type Through Incomparability Components**
`ordinals-and-order-types/friendly-order-types/matroid-rank-and-ordinal-sums/article.pdf`
from `friendly_order_type_research (3).zip`

Identifies $|P| - c(P)$ with the graphic-matroid rank and the incidence-matrix rank of the incomparability graph. Adds a substitution formula, a finite Cartesian-product formula, and additivity over arbitrary well-ordered ordinal sums; evaluates every well-partial-order whose incomparability components are finite, and gives countable counterexamples delimiting stronger infinite extrapolations.

**3. Friendly Order Types of Finite Posets and Ordinal Products**
`ordinals-and-order-types/friendly-order-types/components-forests-and-ordinal-products/friendly_order_types.pdf`
from `friendly_order_types_research.zip + friendly_order_type_research.zip`

Proves $f(P) = |P| - c(\mathrm{Inc}(P))$ for a finite poset $P$, where $c$ counts connected components of the incomparability graph (isolated vertices included), and classifies the witnesses: every maximum friendly subset is $P$ minus one minimal root per component, counted by $\prod |\mathrm{Min}(C_i)|$. With Vialard's width-transfer theorem this gives ordinal width $\omega^{f(P)}$ for the Dershowitz–Manna multiset ordering. On the transfinite side it proves $f(\alpha_1 \times \cdots \times \alpha_r \times B) = \Theta$, the natural product, except that it is the predecessor of $\Theta$ when every $\alpha_i$ is a successor and $B$ has a greatest element, and classifies arbitrary finite ordinal boxes. Also: spanning-forest certificates, exact monotonicity under added comparisons, a bivariate generating function, and explicit four-point factors with equal $(o, h, w, f)$ whose products differ. *Merged; the Cartesian-product formula is proved twice, once by corner deletion and once by a chain-grid spanning-subgraph argument.*

## 3.2 Well-quasi-ordered powersets and statures

Four reports on the open questions in Section 6 of Abriola, Halfon, Lopez, Schmitz, Schnoebelen and Vialard, *Measuring well-quasi-ordered finitary powersets* (arXiv:2312.14587v2), and on the corresponding topological invariant.

**4. Cofinal strata and ordinal absorption**
`ordinals-and-order-types/wqo-powersets-and-statures/cofinal-strata-of-finitary-powersets/article.pdf`
from `cofinal_strata_ordinal_powersets.zip; batch 42: ProveIt_Ordinal_Heights_Matchings.zip; batch 36: ProveIt_Nonuniform_Powerset_Heights.zip, ProveIt_Persistent_Frontier_Heights.zip, persistent_frontier_heights.zip, ProveIt_Ordinal_Heights_Research.zip`

For $S_\alpha(Q)$, a finite poset $Q$ with each point replaced by a chain $\alpha = \omega^\rho$, proves $o(K(S_\alpha(Q))) = \sum_k \omega^{\rho \otimes k} c_k(Q)$ and $h(K(S_\alpha(Q))) = \alpha \cdot \mathrm{height}(M(Q))$, where $c_k$ counts size-$k$ maximal antichains

not dominated by a larger one. The maximal-order-type theorem allows nonuniform pure well-partial-order fibers and is proved by a finite cofinal-stratification argument. Parts II and III, merged in batch 36 from four manuscripts that each prove it, supply the nonuniform height formula the original left open. For positive pure fibers $\omega^{\rho_q}$, $h(K)$ is the largest ordinary ordinal sum, along a chain of maximal frontiers, of the largest fiber leaving the frontier at each step plus the largest fiber of the last frontier; after Cantor-normal-form refinement the same formula covers nonzero limit fibers. For arbitrary ordinal fibers, finite and successor ones included, the height is given instead by a finite dynamic program over downsets with fallback weight 1, and the frontier compression is not valid there. Adds exact point ranks, NP membership of the threshold problem for the explicit pure-block encoding, and a separate part for well-partial-order fibers with $h(K(P_q))$ a positive pure power. Part IV, added in batch 42, shows that for height-two skeletons with uniform fibers the height is $\alpha(\nu_{\mathrm{ur}}+1)$, with $\nu_{\mathrm{ur}}$ the maximum uniquely restricted matching number, so deciding the height is NP-complete already when every fiber is $\omega$.

**5. Intrinsic ordinal approximations for finitary powersets**
`ordinals-and-order-types/wqo-powersets-and-statures/intrinsic-ordinal-approximations/article.pdf`
from `intrinsic_ordinal_approximations.zip`

Studies the intrinsic invariant $a_o(A) = \sup\{o(B)+1 : \mathrm{Idl}(B) \hookrightarrow A\}$. Proves the universal bound $a_o(A) \le h(P_f(A))$ with equality exactly when every finitely generated downset of $A$ is an algebraic dcpo, gives the closed formula $h(P_f(\prod_i \alpha_i)) = \sup(\bigotimes_i (1+a_i)+1)$, and shows the universal identity fails on $\lambda + F$.

**6. Ordinal Order Maps and Finitary Powersets of Ordinal–Finite Grids**
`ordinals-and-order-types/wqo-powersets-and-statures/ordinal-order-maps-and-grids/ordinal_order_maps.pdf`
from `ordinal_order_maps_research.zip`

Exact maximal order types and heights for finite Hoare powersets of $\beta \times Q$ with $Q$ a finite poset: for $\beta = \omega^{\gamma_0} + \cdots + \omega^{\gamma_{m-1}}$, $o(\mathrm{Mon}(Q,\beta)) = \bigoplus_{s\ \mathrm{isotone}} \omega^{\bigoplus_q \gamma_{s(q)}}$.

**7. Sharp ordinal bounds for Hoare powerspaces**
`ordinals-and-order-types/wqo-powersets-and-statures/hoare-powerspace-statures/article.pdf`
from `hoare_statures_research.zip`

For all Noetherian spaces, $1 + \|X\| \le \|HX\| \le 2^{\|X\|}$, with witnesses attaining both endpoints at every ordinal and $T_1$ upper witnesses; the older $\omega^\alpha$ bound is attainable at infinite input stature $\alpha$ exactly when $\omega \cdot \alpha = \alpha$; the complete spectrum at $\omega + n$ is $\omega \cdot j(U) + (j(Q) - j(U))$. Extends the known well-quasi-order results of Abriola et al. to arbitrary Noetherian spaces.

### 3.3 Transfinite words

One unified report on Harry Altman's finite-alphabet sharpness question (arXiv:2409.03199v2, Example 3.15), combining three source manuscripts. The common height-two formula is stated once. The finite-height induction and the more detailed guarded-tail investigation are retained as distinct routes.

**8. Finite-alphabet transfinite words: exact order types and guarded periodic blocks**
`ordinals-and-order-types/transfinite-words/order-types-below-omega-squared/article.pdf`
from `omega2_order_types_proposed_solution.zip + ordinal_words_research.zip + finite_alphabet_transfinite_words.zip`

Claims $o(V_k(P)) = \omega^{\omega^{N_k(P)-1}}$ for $|P| \ge 2$ and each positive finite $k$, using canonical finite atom posets with $N_{k+1} = |P| + J(I_k(P)) - 1$. At height two this is $\omega^{\omega^{n+j(P)-2}}$; empty and singleton alphabets are explicit exceptions. Retains the selected-tail-family classification, its universal-only exception, chain/antichain extremality, exact-length strata, and canonical linear-time token comparison. The guarded proof has both natural-product and elementary amplification routes;

the finite-height proof uses protected separators. Two independent implementations and their separate audit records are preserved. Neither an all-height union formula nor a higher-cutoff selected-family classification is claimed.

### 3.4 Ordinal arithmetic

Three reports on ordinal and infinitary arithmetic, including a consolidation of two independent proposed solutions to Problem 6.2 of Paolo Lipparini, *Monotone infinitary operations on ordinals* (arXiv:2505.00424v2): define and study the least weakly monotone, $e$-special strictly increasing operation on countable sequences of ordinals.

**9. An Explicit Formula for the Minimal $e$-Monotone Infinitary Ordinal Sum**
`ordinals-and-order-types/ordinal-arithmetic/lipparini-minimal-infinitary-sum/article/explicit_ordinal_sum.pdf`
from `lipparini_problem_6_2_proposed_solution.zip + ordinal_operation_research.zip`

The same three-case formula for the least admissible operation is proved by two retained routes: profile ranks with arithmetic absorption and finite-product lower bounds, and corrected blocks with threshold interpolation and finite-part restoration. The latter avoids the product-rank lemma; both import Lipparini's characterization of the weaker $S$. Retains the multiset quotient, bounded-poset heights, finite-cap ranks with alternative proofs, equality on comparable inputs, cardinality preservation, and counterexamples to grouping and continuity. The correction is zero, a finite increment, or one *natural* summand $\omega$. Both implementations and their separate verification records are preserved.

**10. Jacobsthal Products All at Once**
`ordinals-and-order-types/ordinal-arithmetic/jacobsthal-products-all-at-once/article.pdf`
from `Jacobsthal_Products_All_at_Once.zip`

Question 2.16 of Altman, *Intermediate arithmetic operations on ordinal numbers* (MLQ 63 (2017) 228–242), asks for an all-at-once order-theoretic model of finite and infinitary Jacobsthal products, rather than one built from parenthesized binary multiplication or a limit. Gives a constructive affirmative answer: a Cantor-normal-form comparison rule on actual tuples or finite-support functions, invariant under order isomorphisms and extending the componentwise order, whose definition never evaluates a Jacobsthal product. Adds a restricted maximality characterisation and a finite-tail reduction. The associativity identity and the binary normal-form formula are classical and are not claimed here; the construction is not claimed to be preserved by arbitrary order embeddings.

**11. Pairwise Coherence Does Not Guarantee Completion**
`ordinals-and-order-types/ordinal-arithmetic/pairwise-coherence-of-infinitary-products/ordinal_product_coherence.pdf`
from `ordinal_product_coherence.zip`

Answers the pairwise-coherence sufficiency question of Lipparini, *Noncommutative infinitary semigroups* (arXiv:2605.28413v1, Remark 7.1(b)), negatively. On the two-element group $C_2$, retaining every finite product and declaring every infinite all-$e$ product to be $e$ gives a partial system satisfying full pairwise regrouping, even for arbitrary partitions, yet with no injective extension defining all finite and $\omega$-indexed products; two elements is minimal. Adds: for any monoid $M$ the canonical partial domain $P(M)$ is coherent, and for directly finite $M$ it completes exactly when the unit group is trivial; and for every $r$ a system on $3r + 4$ elements coherent through nesting depth $r$ but conflicting at depth $r + 1$, so no fixed finite depth of the coherence test characterises completability.

## 3.5 Games on ordinals

### 12. An ordinal Chomp transition at 2

ordinals-and-order-types/games-on-ordinals/
ordinal-chomp-transition-at-two/ordinal_chomp_counterexample.pdf
from ordinal_chomp_transition_counterexample.zip + ordinal_chomp_counterexample.zip

For a numerical semigroup $S$ ordered by $x \leq_S y$ iff $y - x \in S$, and the ordinal extension ordered by ordinary ordinal subtraction (so $S^2$ is the *lexicographic*, not coordinatewise, square), Chomp on $S$ is a second-player win while Chomp on $S^\sigma$ is a first-player win for every $\sigma \geq 2$, and $\omega \cdot 4 + 4$ is the *least* winning opening. Hence $\mathrm{ch}(\langle 4, 5, 6\rangle) = \mathrm{ch}(\langle 4, 6, 9\rangle) = 2$, a negative answer to Question 4.6 of Rivero Herrera (arXiv:2504.07317v3) and a refutation of Conjecture 4.1 of v1, already for three small natural-number generators. The proof reduces the infinite game to a finite spectral statement via a finite-window recurrence and an exact finite-state certificate, then transfers it by an explicit response procedure. *Merged; the shared $\langle 4, 6, 9\rangle$ certificates were verified to hold the same 253 rows entry for entry, and the two reports' poisoned and unpoisoned conventions are reconciled by an explicit dictionary.*

### 13. Every ordinal as a point-separating game value

ordinals-and-order-types/games-on-ordinals/
point-separating-game-values/ordinal_separation.pdf
from ordinal_separation_research.zip

For every ordinal $\lambda$, orders $\{0, 1\}^\lambda$ lexicographically and gives it the topology of all downward-closed sets, and the refinement generated by downward-closed sets minus finite sets. The point-separating game value is exactly $\lambda$ for both: the first space is compact $T_0$, the second compact $T_1$, and for infinite $\lambda$ neither is Hausdorff. Taking $\lambda = \omega^2 + 1$ supplies a compact $T_1$ witness of cardinality continuum for Problem 1.2 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3), with no CH, forcing or large-cardinal hypothesis. Hausdorff witnesses are not claimed. This is a different problem of that paper from the one the open-query games report answers.

### 14. Open-query games on ordinal spaces

ordinals-and-order-types/games-on-ordinals/open-query-membership-games/article.pdf
from ordinal_membership_games.zip; batch 52: ProveIt_Finite_Output_Open_Query_Games.zip; batch 56: ProveIt_Filter_Profiles_Open_Query_Complexity.zip; batch 65: ProveIt_Closure_Word_Duality.zip; batch 68: ProveIt_Universal_Word_Profiles.zip

A negative answer to Problem 1.3 of Chiozini, Csernák and Soukup (arXiv:2510.05754v3): two copies of $[0, \omega]$ each have set-membership number 1, but their topological sum has value 2. The general theorem computes the invariant for every nonempty Hausdorff space of finite Cantor–Bendixson height $h$ as $\lceil \log_2(h + 1) \rceil$ when the last nonempty derivative is a singleton and $\lceil \log_2(h + 2) \rceil$ otherwise. Part II, added in batch 52, replaces the membership bit by one of $q$ output labels: the optimal depth is $\lceil \log_2 L \rceil$ for the least number $L$ of ordered layers with open suffixes; on Hausdorff spaces of finite height $L \leq (q - 1)h + s$, sharply, with $s$ the number of labels on the last derivative; a topological sum of components of depth at most $d$ has depth at most $d + \lceil \log_2 q \rceil$, sharply; and computing the optimal depth is polynomial for two labels but NP-complete for three, even on disjoint chains. At $q = 2$ it reproduces Part I. Part III, added in batch 56, answers Part II's question of when that bound is attained, for spaces whose derived set is finite and nonempty: a coloring with $q_c$ colors needs exactly $q_c + \tau(G_c)$ layers, $\tau$ the directed feedback vertex number of a color digraph; the bound is attained exactly under a bin-partition condition on the partition capacities of the neighborhood filters; and at three labels a sum of three one-point free-ultrafilter spaces has value 2 while a sum of three convergent sequences has value 3. Part IV, added in batch 65, drops the separation and finiteness hypotheses for a fixed coloring: a layer word is admissible exactly when it contains as subsequences all reduced words whose nested color closure is nonempty. It classifies the finite-depth behaviors, shows that the longest such word has the length of the least Cantor–Bendixson height of a Hausdorff representative, gives $q_c + \tau(G_c)$ layers whenever the second derivative is empty, bounds finite-poset obstructions by $mq(q - 1)^{m-1}$

points, and proves the product law $\rho(X \times Y) = \rho(X) + \rho(Y) - 1$ for that longest length $\rho$. Part V, added in batch 68, computes the worst case over colorings at every finite height on locally compact or sequential (more generally rank-accessible) Hausdorff spaces: $\max_c L = h(q-1) + \min(q, t)$ with $t$ the size of the last nonempty derivative, also when $t$ is infinite, attained by one coloring constant off a countable set; it adds product and sum laws and an inverse theory of layer and query-depth spectra. Local data determining an arbitrary fixed coloring at height two, and a characterization of the spaces attaining that worst case (rank accessibility is sufficient, not necessary), remain open.

### 3.6 Order-theoretic invariants without a subcategory

Thirteen reports whose objects sit outside every subcategory above: the minor order on finite graphs, two on sites of filters on $\omega$, congruences of finite lattices, a translation-invariant ideal, the lexicographic order of the well-orderings of the reals, measurable prediction in Glazer's box games and hat guessing (batches 87, 90, 92 and 96), named elementary embeddings in urelement set theory (batches 89, 93 and 96), and five reports of batch 114: random bits, finite catalogues and arbitrary cuts of models of arithmetic, noisy parity on finite and infinite cubes, robust neutral choice rules, Freiling symmetry from finite blacklists, and bounded-width compression of power sets. They share a template but no main argument.

**15. Ordinal complexity of graph minors**
`ordinals-and-order-types/graph-minor-ordinals/graph_minor_ordinals.pdf`
from `graph_minor_ordinals.zip`

For finite simple graphs of maximum degree at most two under minor containment, computes maximal order type, ordinal antichain-tree width and descending-tree height: $o(D) = w(D) = \omega^{\omega \cdot 2}$ and $h(D) = \omega$, with exact values $\omega^{b+k}$ and $\omega^{b+k-1}$ for the subfamilies bounded by path size $b$ and cyclic-component count $k$. The general theorem is that a wqo $C$ of connected graphs with $o(C) = \alpha < \varepsilon_0$ has finite disjoint-union closure of type $\omega^\alpha$, proved by a component-splitting argument, since minor containment is not multiset embedding. A corollary is that continuity fails for increasing unions of minor-closed classes. The report is explicit that it does not compute the maximal order type of the minor order on all finite graphs and settles no named conjecture.

**16. A two-valued obstruction to gluing Fréchet powers**
`ordinals-and-order-types/frechet-two-valued-obstruction/article.pdf`
from `frechet_two_valued_obstruction.zip`

For the presheaf $F \mapsto A^\omega/{\sim_F}$ on proper filters on $\omega$ containing the cofinite filter, with the dense Grothendieck topology, proves in ZF that it is a sheaf if and only if $|A| \leq 1$. The counterexample is an explicit matching family of constant sections using two values, on a covering sieve with countably many generators each of countable base; the distinction between the number of sieve generators and the size of the full sieve is essential to it. The question is Massas, *A Semi-Constructive Approach to the Hyperreal Line* (2023), p. 517 after Fact 5.10, repeated in the 2024 dissertation *Duality and Infinity*, p. 249.

**17. Bounded-character Fréchet filter sites**
`ordinals-and-order-types/frechet-bounded-character-sites/article.pdf`
from `filter_complexity_research.zip, Frechet_Cardinal_Boundaries.zip`

Continues the previous entry's non-claim about other filter sites. For the site $P_\kappa$ of free filters on $\omega$ generated by fewer than $\kappa$ sets modulo the cofinite filter, represents the Booleanization $C_\kappa$ as a regular-open algebra on the free ultrafilters and proves that $P(\omega)/\mathrm{Fin}$ is order dense in it exactly for $\kappa \leq \mathfrak{p}$; above $\mathfrak{p}$ no complete Boolean homomorphism preserving the modulo-finite events exists in either direction. The atoms are the ultrafilters of character below $\kappa$, so $C_\kappa$ is atomless exactly for $\kappa \leq \mathfrak{u}$; $\mathfrak{h}$ is the first size of a family of events that cannot be decided jointly on a dense set of regions. Computes the associated sheaf by Boolean mixtures, with full Boolean-valued transfer, countable saturation of ultrafilter specializations, and sizes $\mathfrak{c}$, $2^{\mathfrak{c}}$, $2^{2^{\mathfrak{c}}}$ for the raw power and the two

sheafifications. A three-stage example uses Millán's forcing model as an imported input. Merged from two manuscripts whose indexings differ by a successor shift; the second one's closure-based results are stated at successor stages only.

**18. A sharp cover bound for congruences of finite lattices**
`ordinals-and-order-types/lattice-congruence-cover-bound/sharp_cover_bound.pdf`
from `lattice_cover_bound_research.zip`

If a finite $n$-element lattice has an element with at least $k \geq 3$ upper covers then $|\mathrm{Con}(L)| \leq 2^{n-k-1}$, with equality exactly for the glued sums $C_s \oplus M_k \oplus C_t$. This replaces the multicolour Ramsey threshold $R_{k-1}(k)$ of the source's Lemma 4.4 (arXiv:2603.11454) by $k$ itself. Adds a sharp nonextremal gap, a fan-label ideal refinement, and a bound on the size of Czédli's skeleton in terms of congruence density. All congruence machinery, including Freese's maximum, is proved from scratch. The target is an informal suggestion in the source, not a numbered conjecture.

**19. A non-Baire translation-invariant ideal**
`ordinals-and-order-types/non-baire-translation-invariant-ideal/article.pdf`
from `non_baire_ideal_research.zip`

An affirmative construction for Question 1 of Filipów and Tryba, *Densities for sets of natural numbers vanishing on a given family* (J. Number Theory 211 (2020) 371–382; arXiv:2309.00982, p. 7): a translation-invariant ideal that is not Baire and carries a perfect translation-almost-disjoint family.

**20. Lexicographic orders of the well-orderings of the reals**
`ordinals-and-order-types/lexicographic-well-orderings-of-reals/article.pdf`
from `batch 76: lexicographic_well_orderings_of_reals.zip + Lexicographic_Well_Orderings_of_Reals (1).zip + lexicographic_well_orderings_research.zip`

With $\kappa = 2^{\aleph_0}$ and $\mathcal{W}$ the order of all well-orderings of $\mathbb{R}$, as enumerations compared at their first unequal real entry, proves in ZFC that $|\mathcal{W}| = 2^\kappa$, that every linear order of size at most $\kappa$ embeds, even into each fixed-type slice, and that an ordinal embeds exactly when it is below $\kappa^+$; yet the binary lexicographic coding length and the real representation rank of $\mathcal{W}$ are both $\kappa^+$, so $\mathcal{W}$ embeds into no bounded union of strata, in particular not into the slice $\mathcal{W}_\kappa$, which is equimorphic with the cube $2^\kappa$ in lexicographic order and with the Kanovei–Shelah ultrafilter index. Also: each stratum $\mathcal{W}_{\lambda+n}$, $\lambda$ a limit of cardinality $\kappa$, is equimorphic with $2^\lambda \times n!$; $\mathcal{W}^\gamma_{\mathrm{lex}}$ is equimorphic with $\mathcal{W}$ exactly for $0 < \gamma < \kappa^+$; adjacency and the $n!$ blocks; the exact point-character and gap spectra (every gap has a countable side; exactly $2^\kappa$ gaps), a $(\mathrm{cf}\,\kappa, \mathrm{cf}\,\kappa)$ gap of $\mathcal{W}_\kappa$, a dense finite-condensation quotient equimorphic with $\mathcal{W}$, bounded surreal sign orders and an ultrafilter index built from well-orderings. The cube hierarchy re-proves a corollary of `point-separating-game-values`, cited first; real-power strictness is Kuhlmann's theorem; no priority is claimed, and no intrinsic embedding criterion or classification of the gaps of the longer strata. *Merged from three independent manuscripts proving the same core: each shared theorem printed once, with the other arguments as second routes. Each answers a question of another: the ZFC non-embedding into $\mathcal{W}_\kappa$, which one had only under $2^{<\kappa} < 2^\kappa$, and the cut and character spectra and the quotient another had asked about.*

**21. Countable information, uncountably many boxes: measurable prediction, hat-guessing surplus, adaptive extensions, query and randomness thresholds**
`ordinals-and-order-types/measurable-box-games/article.pdf`
from `batch 87: glazer_box_games_research.zip (Part I); batch 90: hat_guessing_research.zip (Part II); batch 92: adaptive_box_games_research.zip (Part III), hat_query_thresholds_package.zip (base) + hat_inspection_frontier.zip (Part IV); batch 96: hat_randomness_frontier.zip (Part V)`

Part I takes up the question that ends Section 5 of Glazer's *A choiceless box game paradox* (arXiv:2211.10474): does ZF prove that three players facing continuum many boxes of bits can guarantee that at most one of them guesses wrong? It separates a measurable choice of box and guess, a measurable success event and actual executability, and answers the question for

the first class only. For a standard Borel set of boxes, alphabet size $q$ and $m$ blind output maps measurable for the cylinder $\sigma$-algebra, the guaranteed number of correct guesses is exactly $\lfloor m/q \rfloor$, without assuming the success events measurable; a countable team has a fair simultaneous measure extension; measurable success forces a countable target range; a continuous selector has a success event of inner measure 0 and outer measure 1. Blind binary maps on $n$ boxes are labelled perfect matchings of the cube, executable maps exactly the recursively sliceable ones (the two classes first differ at four boxes); their number $F_n = 1, 2, 9, 232, 206065, \ldots$ satisfies an exact recurrence and the all-orders expansion $F_n = e^{\gamma 2^n} n^{-1}(1 - \frac{3}{2n} + \frac{17}{4n^2} - \frac{12}{n^3} + O(n^{-4}))$ with a growth constant $\gamma \approx 0.438158$, not Euler's; a doubly exponential rarity bound, information inequalities, an oracle extraction of finite losing prefixes, and a Busy Beaver comparison that is conditional on an encoding and uses a program-length running-time function, not Rado's score function. Neither counting sequence was in the OEIS on 3 October 2026. Part II (batch 90) answers both questions of Eldredge's Remark 6.8 (arXiv:2508.02828) on the surplus $D_n = S_n - n/2$ of correct guesses in the infinite binary hat game: Hamming-syndrome blocks give a computable strategy with $D_n = \log_2 n + O(1)$ almost surely, and for every positive $g = o(n)$ a continuous strategy with $D_n/g(n) \to \infty$ almost surely, so no sublinear envelope is universal; the coding mechanism is credited to Ebert–Merkle–Vollmer. Its finite corollary settles Part I's question Q9 for $N = 2a(2^r - 1)$ binary players and threshold $a2^r$: the optimum is exactly $1 - 2^{-r}$, over legal trees and blind maps alike. Part III (batch 92) classifies every extension of the product measure to the success events of adaptive players by measurable densities on a random set of feasible success patterns, which answers Part I's question Q5, with an atomic-mass law, a Hall polytope of joint laws and kernels for countable teams; the minimax $\lfloor m/q \rfloor$ holds for full Borel and universally measurable blind outputs, two of the three clauses of Q2, while for Baire-property outputs an invariant comeagre null set blocks the averaging proof and that clause stays open. Part IV (batch 92) merges two independent manuscripts that answer Part II's question on the cost of inspecting hats by the same construction: a common bound on adaptive queries makes divergence of the surplus null and meagre, and the exact expected-query threshold for deterministic trees is 2, attained by a computable strategy with fewer than two expected queries per player. The second source adds a lower bound for every nondecreasing cost, a central limit theorem and a classification at minimum depth that settles Q9 for $(m(2^r - 1), 2, m(2^r - 1))$ at threshold $m2^{r-1}$ for every $m \geq 1$. Two sentences calling the remaining cases of Q9 open are retained and refuted: the batch-90 one by Part I's own three-box team (Remark 15.1), and the batch-92 one on every $q \geq 3$ case above $\lfloor m/q \rfloor$ by zero-query constant guesses at a shared target, which attain $m/(qt)$ whenever $t$ divides $m$ (Remark 15.2). Part V (batch 96) answers the almost-sure clauses of Part IV's question 57.2, on that cost when random bits are free: with independent private coins, or with a public seed that is almost surely finitely valued, the admissible uniform expected inspection caps are $(1, \infty)$, the endpoint excluded (at cap 1 private coins succeed with probability at most $1 - 1/5184$), whereas one shared random integer of countably infinite support attains cap 1 with every individual expected cost below 1 and arbitrarily small entropy. A prescribed public seed with values in a standard Borel space permits cap 1 iff it is not almost surely finitely valued, and if the cap must hold conditionally on the public seed the caps are $(1, \infty)$ with private coins and $[2, \infty)$ without. The tools are a sign bound by projection onto the players who inspect nothing, owner-indexed Walsh accounting and a stopped supermartingale lemma; Part IV's thresholds and Bonami's lemma are re-proved with credit. The private positive-probability threshold at cap 1 stays open. The manuscript's hypothesis "finite positive-probability support" makes its finite-public theorem false as worded, since a seed with one atom and a diffuse part permits the endpoint (Remark 75.6); the write prints the hypothesis as probability-one concentration on finitely many values, with the delivered wording on record, corrects the abstract's "countably supported" seed to countably infinite support, and moves three unproved statements, among them the unsupported statement that its proofs were independently reviewed, to the questions. The publication review `47ab39154` then refuted two sentences of the write itself, retained in Remark 75.9: the guide's extension of the gap 1/5184 to finite public seeds (a two-valued public gate succeeds

with probability at least 9999/10000 below cap 1) and an equality in the proof of Corollary 75.7, now an inequality; neither theorem changes. Glazer's question itself is not answered; Part I's Q1 is not the "Question 1" of the surreal report `polish-models-of-omnific-arithmetic`. The Hilbert's-tenth programme reviewed Part I's computability claims (`f15962acd`, not its probability, enumeration or asymptotic claims), routed Part II's archive from its READMEs only (`bcc1a4438`) and reviewed the publication of Parts III–IV within a stated scope (`42b49fdbd`, the source of Remark 15.2); for Part V it read the manuscript in two bounded passes (`18825ee16`, `1eab36bb3`, both reading the seed hypothesis as atomic), authenticated the placement (`608e06daa`), confirmed the hypothesis correction independently (`c52f7d055`) and reviewed the publication (`47ab39154`). No priority; not formalized.

### 22. Naming elementary embeddings: atom components, Replacement, reflection, commuting actions and relations among injections

`ordinals-and-order-types/naming-elementary-embeddings/article.pdf`
from `batch 89: Naming_Elementary_Embeddings_Research.zip (Part I); batch 93: commuting_actions_replacement.zip (Part II); batch 96: atom_actions_research.zip (base) + commuting_injections_research.zip (Part III)`

In Yao's kernel-ideal models $M_\kappa(A)$ of urelement set theory with Replacement and choice (objects whose transitive closures involve fewer than $\kappa$ atoms), every injection $f$ of the atoms lifts to a canonical amenable elementary self-embedding $j_f$ fixing every pure set, and the initial elementary self-embeddings are exactly these lifts. When finitely many lifts are named, so that the schemes range over formulas containing them, Replacement holds exactly when the small type blocks of the atom graph have fewer than $\kappa$ atoms (for $\kappa = \omega$: finite components and a finite small part). Hence every one-name expansion keeps Replacement iff $\operatorname{cf}\kappa > \omega$, every expansion by two or more names iff $\operatorname{cf}\kappa > 2^{\aleph_0}$, and at $\aleph_2$ universal two-name preservation is equivalent, externally, to CH (a characterization, not a proof or refutation of CH). Collection fails at limit cutoffs already for $\Delta_0$ formulas of the base language and holds in full exactly at an uncountable successor $\kappa$ with fewer than $\kappa$ component types; full transitive reflection has the same criterion; two individually safe automorphisms can be jointly unsafe. The kernel-ideal construction is Yao's and the lifting idea is not claimed new. The same Collection spectrum is proved independently in Part VI of the surreal report `birthday-cutoffs-and-hereditary-sets`, whose Part VIII (batch 90) contains the case $\kappa = \omega$ of the criterion for finitely many named permutations, proved independently. Part II (batch 93) names commuting maps: every tuple of commuting permutations keeps expanded Replacement iff $\operatorname{cf}\kappa > \omega$, which answers Part I's question on commuting automorphisms in full; with commuting injections added the threshold is $\operatorname{cf}\kappa > \omega$ for at most one injection and $\operatorname{cf}\kappa > 2^{\aleph_0}$ for two or more, and two commuting proper injections, each harmless alone, can jointly destroy Replacement; for commuting permutations the atom kernel of definable closure is computed exactly (Part I's Question 10.1 in that case); for a finitely generated group named by generators every action keeps Replacement iff the group is finite ($\kappa = \omega$) or $\operatorname{cf}\kappa$ exceeds its number of conjugacy classes of subgroups; one uniform evaluator for an action of $\bigoplus_\omega C_2$ with orbits of at most two atoms destroys Replacement whenever $\operatorname{cf}\kappa \le 2^{\aleph_0}$ while separate names keep it; and set forcing that keeps $\kappa$ a cardinal preserves the verdict for a fixed action (the fixed-action part of Question 10.10). Part VIII of the surreal report is credited as prior at $\kappa = \omega$; one imported group-existence theorem (Ivanov–Ol'shanskii, read through Smith) is not proved, and its primary check is an open question. Part III (batch 96) merges two independent manuscripts on relations among named injections, which answer Part II's Question 23.3 by the same main theorem: for a finitely generated commutative monoid acting by injections, with $G$ the group completion of its cancellative image and $d$ the rank of $G$ modulo the units, the connected labelled types number $|\mathrm{Sub}(G)|$ if $G$ is finite, $\aleph_0$ if $G$ is infinite and $d \le 1$, and $2^{\aleph_0}$ (with a rigid Cantor family) if $d \ge 2$; $d$ is decidable from a finite commutative presentation; every such action keeps Replacement for all $\kappa$, iff $\operatorname{cf}\kappa > \aleph_0$, iff $\operatorname{cf}\kappa > 2^{\aleph_0}$ in the three cases; the relation $f^m = g^n$ is enumerated by necklace counts, with rational Catalan numbers in the coprime case; and forcing adds a connected type for $d \ge 2$ iff it adds a real. The base adds pure-real rooted codes of components, a centralizer-orbit count that generalizes a

lemma of Part VIII of the surreal report to injections and infinite components, and the exact atom kernel of the objects definable from given atoms and finitely many pure reals, for any finite tuple of injections; so Part II's Questions 23.1 and 23.2 and Part I's Question 10.1 are answered only with pure real parameters, and ordinary definability stays open. Results proved by both sources are printed once, with the second proof as a marked route. Re-scoped at the write: the second source's "commutative answer" to Question 23.5 decides only the commutative-monoid analogue, and the base's countability argument for the parameter-free definable kernel is metatheoretic, kept as prose and asked as Question 37.17. The Hilbert's-tenth research tree read the second manuscript in full and selected interfaces of the base before placement (`18825ee16`), authenticated the placement (`b16786caf`) and reviewed the write within a bounded scope without finding a mathematical defect (`6d78c238d`); `6d06f8952` corrected the README's sentence that no review of Part III existed, retaining it as a review remark. It continues the formal projects `SetTheory/ZF` and `SetTheory/BoundedConsistency` semantically and proposes a formalization route; their formula type has no atom predicate, and nothing here is formalized. Unrefereed; priority not established.

### 23. Random bits, finite catalogs and arbitrary cuts of arithmetic: exact noise thresholds, Boolean universality and the strength of induction

`ordinals-and-order-types/random-bits-arithmetic-cuts/article.pdf`
from `batch 114: Random_Bits_Arithmetic_Cuts.zip`

For a countable model $M$ of PA and an external random predicate $P \subseteq M$, the coding cut $\mathrm{Cut}(P)$ is the initial segment of those $a$ for which $P \cap [0, a)$ has an internal binary code. For pairwise-independent bits with biases $p_t$, modal bit $b$ and $\varepsilon_t = \min(p_t, 1 - p_t)$, almost surely $\mathrm{Cut}(P) = \mathrm{Cut}(b) \cap \{a : \sum_{t<a} \varepsilon_t < \infty\}$ (external sums). Fair bits give the standard cut, announced by Elliot Glazer (2023, for PA "or even Q"); the PA case is reconstructed with the report's own formula, and neither the Q case nor uncountable models are claimed. Every cut is the almost-sure coding cut of a full-support nonatomic product law; for independent arrays a family of conjunction cuts is realizable iff it is inclusion monotone, and every finitary Boolean observable has the intersection of the cuts of its minimal sensitive supports. A proper cut refutes $\mathsf{I}\Sigma_1(P)$; $\mathrm{Cut}(P) = M$ iff $(M, P)$ satisfies $\mathsf{I}\Delta_0(P, \mathrm{E})$; for zero-modal noise, $W = \sum_a p_a$ separates three regimes (full induction; $\mathsf{I}\Delta_0(P, \mathrm{E})$ without $\mathsf{I}\Sigma_1(P)$; neither). Also category and Baire-class results, the sharp pairwise atom bound $1/(N+1)$ or $1/(N+2)$ (Benjamini–Gurel-Gurevich–Peled, credited), and finite catalogs: bounded budgets iff $\sum \varepsilon_i < \infty$, an exact Hamming-layer formula, the normal transition (Kontoyiannis–Verdú, Chen–Effros–Kostina, credited) and a Poisson staircase for $p_N = \lambda/N$. The write proves Remark 13.1: $\mathrm{Cut}(P) = M$ makes $P$ a class, hence not strongly CP-generic (Abdul-Quader–Schmerl), partly settling Question 5; all fourteen questions stay open and priority is not established. Only the passing remark that PA has no finite models is formalized (`Logic/PeanoArithmetic/NoFiniteModel`; the independent check added same-named copies in an unimported optimization file); beta coding and a definable-cut induction failure of `Logic/PeanoArithmetic/NotFinitelyAxiomatizable` are ingredients, and the route of Section 11 is a proposal. Unrefereed; not formalized.

### 24. Noisy parity on finite and infinite cubes: exact coloring, inspection budgets, asymptotic laws and measurable choice

`ordinals-and-order-types/noisy-parity-cubes/article.pdf`
from `batch 114: Noisy_Parity_Finite_and_Infinite_Cubes.zip`

A uniform configuration $X$ of finitely or countably many bits is compared with a copy $Y$ in which one coordinate, drawn with ordered weights $p_1 \geq p_2 \geq \cdots$, is flipped and every bit is flipped independently with probability $\eta$; one measurable label $f$ succeeds when $f(X) \neq f(Y)$. For $0 < \eta < 1/2$ the optimum is $\max_k [1 + (1 - 2\eta)^k (2P_k - 1)]/2$, $P_k = p_1 + \cdots + p_k$, attained by a prefix parity on $K$ coordinates, $K$ finite also on the countable cube, with at most one adjacent tie and a first-sign-change certificate; under strictly decreasing weights the optima are exactly the signed prefix parities, with a quantitative stability gap. The exact expected-inspection frontier is linear between the points $(k, w_k)$, $k \leq K$, and constant beyond, attained by shared mixtures

of adjacent prefix parities; a spectral simulation replaces every measurable rule by a random parity $\chi_S$ with $\Pr(S = T) = \hat{f}(T)^2$, at mean cost equal to the total influence. At zero noise the measurable supremum 1 is not attained. A total parity label exists under AC, and over ZF exactly under a restricted choice of $E_{\mathrm{ev}}$-classes; it is measurable for a biased product measure iff $\sum \min(q_i, 1 - q_i) < \infty$; for summable heterogeneous noise the optimization over all subsets of $\mathbb{N}$ is compact and, in ZFC, a label with a measurable success event attains its value. Small-noise laws for uniform weights (saturation at $\eta n = 1$), geometric weights (a logarithmically periodic term), regularly varying and Zipf weights. The write adds, with proofs, Remark 5.2 (no deterministic decision tree attains the frontier at a non-integer budget $0 < c < K$) and Remark 6.4 (on the countable cube the stability gap tends to 0 as $\eta \to 0$). Two classical lemmas of Part IV of `measurable-box-games` (total influence at most the expected number of inspections; the Walsh degree of a depth-$d$ tree) are re-proved and printed as second routes. The machinery is classical (O'Donnell; Geschke–Lubarsky–Rahn and Serafin; Conley–Miller), priority is not established, and fifteen further questions stay open. The independent check found nothing to correct. Unrefereed; not formalized.

**25. Robust choices that never settle: an exact finite–infinite puzzle about resilience, choice and randomness**

`ordinals-and-order-types/robust-neutral-choice/article.pdf`
from `batch 114: robust_choices.zip`

A neutral rule $f : \{-1,1\}^n \to \{-1,1\}$ ($f(-x) = -f(x)$) is robust to $r$ edits at $x$ when no $r$ coordinate changes flip its output. The largest probability of robustness is $2^{1-d} \sum_{j \le (d-1)/2-r} \binom{d}{j}$, $d$ the largest odd integer $\le n$, attained for every $r$ at once by majority of $d$ voters, so an even committee does no better than one fewer member: a corollary of Kleitman's diameter theorem, credited to Kleitman and to Huang–Klurman–Pohoata, with the reduction and the constructions proved. A sharp prefix inequality bounds $\|\mathbb{E}[f \mid \mathcal{F}_m]\|_\infty$ by $2^m / V(m, r)$ times the vulnerability, $V$ the Hamming-ball volume, with the best dimension-free constant. If one-edit vulnerability tends to 0 the decisions are mixing and change sign infinitely often on a conull comeagre set; a Hamming-code construction separates one-edit from two-edit resilience; a subsequence makes the decision map a continuous open surjection with density within $e^{\pm\eta}$ of fair Bernoulli measure. In ZFC a neutral rule invariant under finite changes exists on all infinite streams, none is measurable and neither fibre has the Baire property; for a product of Bernoulli($p_i$) laws a completed-measurable one exists iff $\sum (p_i - 1/2)^2 = \infty$, a deduction from Kakutani's dichotomy. The write proves Proposition 11.1 (for several labels and a group with no global fixed label the fibres are not all measurable and not all Baire), the regularity half of Question 6; re-scopes Questions 1 and 8 (existence of the rule is equivalent over ZF to choosing one of each complementary pair of $E_0$-classes); and in Remark 10.1 derives the category half of the ZFC theorem from proved declarations of `SetTheory/Cardinals/Cardinals/Countable/GenericErgodicity.lean`, given complementation as a homeomorphism, which the library does not define. The finite statements were checked exhaustively in dimensions 1–5 (65,814 neutral rules); ten questions stay open; no priority is claimed. The independent check found nothing to correct. Unrefereed; not formalized.

**26. From finite blacklists to Freiling symmetry: exact avoidance thresholds, critical tournaments, choice-sensitive colorings and perfect safe sets**

`ordinals-and-order-types/freiling-symmetry-blacklists/article.pdf`
from `batch 114: From_Finite_Blacklists_to_Freiling_Symmetry.zip`

In a $k$-bounded blacklist system each point refuses at most $k$ others; a set is safe if no two members refuse each other. Every system on $n$ points has a safe set of size $\lceil n/(2k+1) \rceil$, which is sharp; at the critical size $n = (r-1)(2k+1)$ every system without a safe $r$-set has conflict graph $(r-1)K_{2k+1}$ with each block a regular tournament, and these systems are counted exactly. The least number of safe pairs is $\max\{0, \binom{n}{2} - kn\}$, $A_r(n,k) = \binom{n}{r} - kn\binom{n-2}{r-2} + O(n^{r-2})$, and $A_3(n,1)$ is exact. The partition of an arbitrary system into $2k+1$ safe sets is Theorem 3 of de Bruijn and Erdős (1951), credited; over ZF the report proves BPI $\Rightarrow$ SAP$_k$ $\Rightarrow C_2 \wedge \cdots \wedge C_{k+1}$ and SAP$_k$

on well-orderable sets. A Borel relation with countable sections on a perfect Polish space has a perfect safe set, a definable strengthening of the conclusion of Freiling's axiom of symmetry (the axiom itself is not proved), and almost every i.i.d. sample from an atomless measure is an infinite safe set. The write proves Propositions 12.16–12.18: the pseudoforest Conjecture 12.5 for $n \leq 9$ (an unshipped exact computation reaches $n \leq 18$) and Conjecture 12.6 for $n \in \{2k+1, 2(2k+1)\}$; $\mathrm{SAP}_k \Rightarrow C_3$, new for $k = 1$; and BPI $\Rightarrow \mathrm{SAP}^+_{\mathrm{fin}} \Rightarrow \mathrm{SAP}_{\mathrm{fin}} \Rightarrow C_{\mathrm{fin}}$, with $\mathrm{SAP}_{\mathrm{fin}}$ de Bruijn–Erdős's countable-colour theorem; it identifies two of the counts as OEIS A133686 and A007079. The independent check showed, with Jech's Theorem 7.16, that $C_2$ does not imply $C_3$ over ZF, so $\mathrm{SAP}_1$ is strictly stronger than choice for pairs; its exact strength stays open, and all seventeen further items are stated with what is missing. No priority is claimed for the finite refinements, and the free-set literature was not surveyed. Unrefereed; not formalized.

**27. Bounded-width power-set compression: exact finite thresholds, choiceless infinite obstructions and a defect extension of Glazer's chain-fiber theorem**
`ordinals-and-order-types/bounded-width-power-set-compression/article.pdf`
from `batch 114: Bounded_Width_Power_Set_Compression.zip`

Studies maps $f : \mathcal{P}(X) \to X$ whose fibers contain no antichain of $r+1$ sets, the property $\mathrm{SC}_r(X)$, extending Glazer's $r = 1$ theorem (his accepted MathOverflow answer, imported). With $W_n = \binom{n}{\lfloor n/2 \rfloor}$, a finite poset of width $w$ needs exactly $\lceil w/r \rceil$ labels (Dilworth), so $q_r(n) = \lceil W_n/r \rceil$, with middle-layer and grouped symmetric-chain certificates; $\mathcal{P}([n]) \to [n]$ exists iff $W_n \leq rn$, so the least admissible width is $\delta(n) = \lceil W_n/n \rceil = 1, 1, 1, 2, 2, 4, 5, 9, 14, 26, \ldots$, $\delta(n) = \sqrt{2/\pi}\, 2^n n^{-3/2}(1 + O(1/n))$, and the largest admissible $n$ is $N(r) = L + \frac{3}{2}\log_2 L + \frac{1}{2}\log_2(\pi/2) + O(1)$, $L = \log_2 r$. Over ZF an infinite set with $\mathrm{SC}_r$ is Dedekind-infinite, with an injection of $\mathbb{N}$ definable from $f$ and finitely many points, so no Dedekind-finite or amorphous set has it; through Forster's theorem no infinite $X$ with $X \times 2 \preccurlyeq X$ and no infinite well-orderable $X$ has it, so ZFC proves $\mathrm{SC}_r(X)$ iff $X$ is finite, nonempty and $W_{|X|} \leq r|X|$. The ZF case $r \geq 2$ is posed as open, with the profile of a counterexample. Wrong as printed, and kept with a numbered Remark 9.10 of the write: the definition of an antichain code does not require injectivity, and then the antichain-transfer lemma and theorem are false ($X = \{0, 1, 2\}$, $r = 1$, $\Theta$ constantly $\emptyset$, where $\mathrm{SC}_1(X)$ holds); requiring injectivity repairs them, and the only code the source constructs is injective, so the ZFC classification stands, as the independent check confirmed. That code is Pruss's for $r = 1$ in the same MathOverflow thread, credited by the write. The source's statement that ProveIt already has finite-poset and Boolean-lattice work is scoped: no Lean or Rocq file treats Dilworth, Sperner or poset width. Seventeen further items (F1–F17) are stated with what is missing; priority is not established. Unrefereed; not formalized.

## 4 Automata and formal languages

Nine reports on state complexity, synchronization, language hierarchies, crossover closures, additive complexity, the asymptotic enumeration of automata and size lower bounds for two-way automata. Three are merges: the shuffle bound arrived four times, the DFAO reversal bound twice, and the last report joins a manuscript on accessible automata with one on strongly connected automata that adapts its method.

**28. The six-state shuffle bound**
`automata-and-formal-languages/shuffle-six-state-bound/shuffle_six_state.pdf`
from `shuffle_six_state_research (1).zip + shuffle_six_state_research.zip + Shuffle_Six_State_Theorem.zip + six_state_shuffle_research.zip`

For complete DFAs with $m, n \geq 2$ and $\min(m, n) \leq 6$, the worst-case deterministic state complexity of the shuffle is $F(m, n) = 2^{mn-1} + 2^{(m-1)(n-1)}(2^{m-1} - 1)(2^{n-1} - 1)$, over a finite alphabet allowed to depend on $m$ and $n$; the unrestricted conjecture stays open. The route is a reachability theorem in the universal shuffle automaton — a subset is reachable exactly when it meets the initial row and the initial column — proved by induction on $(m+n, |S|)$ down to a finite certificate obligation.

*Merged from four reports. Two shipped the same 10,642 witnesses; a third supplies an independent database of 10,619 witnesses sharing no witness with them, so the finite step is certified twice over; a fourth adds a fifth analytic reduction that cuts the obligation to 453 certificates and replaces the solver search with a deterministic enumeration. All three routes are kept.*

### 29. A coloring obstruction to maximal binary DFAO reversal

automata-and-formal-languages/dfao-reversal-coloring-obstruction/article.pdf

from dfao_reversal_research.zip + binary_dfao_reversal_research.zip; batch 39: ProveIt_Three_Output_Reversal.zip; batches 40–41: ProveIt_DFAO_Eventual_Optimality.zip, ProveIt_Binary_DFAO_Reversal.zip, ProveIt_Binary_Reversal_Exactness.zip; batch 43: ProveIt_Four_Output_Reversal.zip, ProveIt_Four_Output_Reversal (1).zip; batch 67: ProveIt_Near_Linear_DFAO_Reversal.zip, ProveIt_DFAO_Spectral_Cancellation.zip

For transformations $a, b$ on an $n$-element set and an output map to $k$ labels, proves $|\tau\langle a, b\rangle| \leq k^n - k! + g(k) < k^n$ for $3 \leq k \leq n$, with $g(k)$ the largest order of a permutation of $k$ symbols — the binary-alphabet nonattainment question of Davies, arXiv:1705.07150v2, Section 5. The mechanism is a collision-orbit graph, a rightmost singular factorization, and a cyclic relabeling bound. Adds an abelian-permutation extension, a structural universal bound matching the published lower construction on all 372 pairs with $7 \leq n \leq 30$, and polynomial-time missing-coloring certificates. *Merged; the two originals ran the same three-case argument, and both a qualitative short proof and the counting proof are kept.* Part II, added in batch 39, settles three outputs exactly: $R_2(n, 3) = 3^n - 3(2^A + 2^B - 2) + B$ for every $n \geq 7$, with $(A, B)$ the closest coprime split of $n$, together with rigidity of the extremizers; the formula matches all twenty-four $k = 3$ rows of the finite table. Part III, merged in batches 40–41 from three manuscripts, settles every $k \geq 4$ past an explicit threshold: $R_2(n, k) = k^n - \mathrm{Bip}_k(a_n, b_n) + a_n b_n$ once $n \geq N(k) \leq 3k^4$, and for all $n \geq \max(7, k + 1)$ when $k = 4, \ldots, 7$ (computer-assisted below $N(k)$), with the lower construction imported from Davies; a non-effective eventual route for every $k \geq 3$, 919 finite certificates and a second proof of the $k = 3$ formula are kept. Explicit exactness for $k \geq 8$ below $\min(N(k), T(k))$, with $T(k)$ the threshold of Part V, stays open. Part IV, merged in batch 43 from two manuscripts on $k = 4$, re-derives that case with an analytic threshold $n \geq 26$ (against 212), and adds stability, the missing-orbit inventory, the extremal output maps and a minimal order-15 recurrence. Part V, added in batch 67, proves the formula for every $k \geq 4$ once $n \geq T(k) = 4k \log k + O(k)$ instead of $n \geq 3k^4$ (so for all $k$ up to $cn/\log n$, $c < 1/4$), with a theta-function limit of the optimal deficit when $n/k^2$ tends to a constant and a Gaussian law in between. Part VI, added in batch 67, determines the minimal recurrences of the explicit witness deficit for every fixed $k$: order 9 at three outputs (against Part II's 12), a product-12 cancellation on the odd residues but not in the full sequence at nineteen outputs, the only product-12 cancellation at any $k$, and explicit tail numerators.

### 30. Synchronizing Dżyga's automata by a cyclic retract

automata-and-formal-languages/dzyga-synchronizing-retract/article.pdf

from Dzyga_Automata_Cyclic_Retract_Research.zip

For the Figure-3-complete automaton $A_k$ of Conjecture 8 in Szykuła's 2026 survey, exhibits an explicit word $R_k = p(ec^{r-1})^{r-2}e$ with $r = k + 4$ that sends every state to $q_2$ for every $k \geq 1$, of expanded length $8k^2 + 54k + 92 = \frac{8}{9}n^2 + \frac{22}{3}n + 16$ for $n = 3k + 6$. The proof is a cyclic retraction with two projected reflections whose discrepancy supplies one adjacent merger, and does not depend on the finite computations; shortening the reflected tail by one state gives a nonsynchronizing automaton with a dihedral permutation quotient. This settles the synchronization clause of the stated model, not the numerical threshold conjecture, and the universal lower bound $\tau(q_0) \geq 4k + 8$ is not proved.

### 31. Gaps in bounded-shuffle hierarchies

automata-and-formal-languages/insertion-degree-spectra/article.pdf

from shuffle_spectra_research.zip; batch 41: ProveIt_Binary_Insertion_Spectra.zip; batch 66: ProveIt_Sharp_Insertion_Isolation.zip

A counterexample to the insertion-degree No-Gap Conjecture of Hughes, arXiv:2608.27755v1,

Section 11, followed by positive results: every finite set of positive integers containing 1 is the insertion-degree spectrum of two nonempty finite languages over a suitable alphabet, and any prescribed finite list of degrees above 1 is realizable with every other output of degree 1. A ternary pair has spectrum $\{1, 3\}$ with sole difficult output `abcba`, whose length 5 is the minimum possible for a gap witness; a binary pair has spectrum $\{1, 2, 4\}$ while all 18 singleton source-pair spectra are initial intervals, so source-pair ambiguity can erase a degree. The separate iteration-depth no-gap conjecture is proved for arbitrary languages. Part II, added in batch 41, fixes the alphabet: every finite set containing 1 is the spectrum of finite *binary* languages with the single inserted word $a^m$, so two letters always suffice; its $m^2$ length bound holds in its full-coverage class only. Part III, added in batch 66, determines that class exactly: an isolated degree-$m$ output with $g$ gaps between its $a$'s (the two end gaps included) and $T$ letters $a$ exists exactly when $g \geq m$ and $T \geq \max(g(m-1)+1, \tau_m)$, with $\tau_3 = 12$ and $\tau_m = 5m - 2$ for $m \geq 4$, so the minimum lengths for $m = 3, 4, 5$ are 14, 21 and 27. It characterizes the extremal centers and the mandatory targets, and at $m = 3$ classifies the shortest target languages: $3^9 = 19{,}683$ of them, of minimum size 37, attained by 512.

### 32. Constrained crossover closure

`automata-and-formal-languages/constrained-crossover-closure/article.pdf`
from `ProveIt_Constrained_Crossover.zip` (batch 44); batch 66: `ProveIt_Crossover_Stabilization.zip`, `ProveIt_Crossover_Classification.zip`; batch 71: `quadratic-finite-rank-package.zip`; batch 73: `rational-rank-slopes-package.zip`, `dfa-crossover-package.zip`, `common-path-rank-package.zip` (the earlier editions `dfa-crossover-package (1).zip` and `common-path-rank-package (1).zip` superseded, not printed)

Answers the question Hughes (arXiv:2608.27755v1, Section 4) leaves open on the complexity of deciding $L \otimes_c L = L$ for the constrained crossover of a context-free language: it is co-r.e.-complete. Adds an aligned-rank no-gap theorem for iterated crossover and effectivity results in which Presburger quantifier elimination is used only as an algorithmic step. Related to, but using nothing from, the insertion-degree report above. Part II, added in batch 66, shows finite stabilization of regular seeds PSPACE-complete for NFA input, already over two letters; the upper bound also covers DFA input, improving Part I's EXPSPACE bound (Part VI settles DFA input); in the stabilizing case the rank is at most quadratic in the number of states. Part III, added in batch 66, decides regularity, inclusion and equality of the full closures of context-free seeds, classifies the finitely stabilizing sparse one-marker seeds into regular, linear non-regular and non-context-free closures, and shows that thresholding the exact stabilization depth is coNP-complete for succinct ray input. Part IV, added in batch 71, shows that the quadratic order is attained for NFAs: the largest finite rank $F(s)$ of a binary language recognized by an $s$-state NFA satisfies $(s-3)^2 \leq F(s) \leq (2s+5)^2$ for $s \geq 4$, the lower bound replacing Part II's linear one. Parts V–VII, added in batch 73, answer the questions Parts II and IV left. Part V: on every live residue the largest rank of a length-$n$ hull word is $\sigma_r n + O(1)$ with a rational, computable slope $\sigma_r$, and $F(s) \leq 2(s-1)^2 + 3$. Part VI: finite stabilization of an explicit complete binary DFA is coNP-complete, and the largest finite rank of an $s$-state complete binary DFA satisfies $(s-7)^2 + 2 \leq F_{\mathrm{DFA}}(s) \leq s^2 + 3$ for $s \geq 8$. Part VII: $F(s) \leq s^2 + 8s$ for binary NFAs ($s^2 + \max(8, 2q-4)s$ over $q$ letters), so $F(s)/s^2 \to 1$ and $F_{\mathrm{DFA}}(s)/s^2 \to 1$, with an exact algorithm and $F(1), \ldots, F(4) = 1, 2, 5, 7$, the last three computer-assisted. The exact values beyond four states and the lower-order terms stay open. Lemmas the new Parts re-prove are cited, not reprinted, and two lemmas shared by Parts VI and VII are printed once.

### 33. A cut-sensitive attack on the first open MIX grammar case

`automata-and-formal-languages/mix-grammar-cut-sensitive/mix_grammar_report.pdf`
from `mix_grammar_research.zip`

The selected question — whether $L_3 = \mathrm{MIX}$ for the canonical tuple grammars of Yuyama, *On the Kanazawa–Salvati Conjecture* (arXiv:2609.13871v1) — is *not* settled here, and the report says so at the front. What it does contain, with complete written proofs, is a cut-sensitive membership recurrence, a broad three-component positive family, an infinite family in $L_3 \setminus L_2$,

and an exhaustive computation through length 21.

**34. Natural densities of the Tribonacci word's additive complexities**
automata-and-formal-languages/tribonacci-additive-complexity/article.pdf
from tribonacci_additive_density_research.zip

Answers Remark 19 of Popoli, Shallit and Stipulanti (arXiv:2410.02409). With $\beta$ the Tribonacci constant, the additive complexity of the Tribonacci word takes values in $\{3,4,5\}$ with natural densities $d_3 = (21\beta^2 - 38\beta + 5)/22$, $d_4 = (-55\beta^2 + 72\beta + 67)/22$ and $d_5 = (17\beta^2 - 17\beta - 25)/11$, all positive and summing to 1, with a power-saving error $O(N^\theta)$, $\theta = \log(7/5)/\log\beta$, at *every* integer cutoff rather than only along Tribonacci cutoffs. The 76-state output automaton is reconstructed independently rather than copied, and certified by evaluating its degree-76 characteristic polynomial at the adjacency matrix.

**35. Accessible and strongly connected deterministic automata, to every fixed order**
automata-and-formal-languages/accessible-and-strong-automata/article.pdf
from batch 77: accessible-automata-reproducibility.zip + strong-automata-reproducibility.zip

For a fixed alphabet of $k \geq 2$ letters, with $t = k(1 - e^{-t})$, $\rho = e^{-t}$, $v = t/k$ and $c = 1 - k\rho$, Part I proves that the number $a_n$ of rooted accessible complete deterministic transition structures on $n$ states (OEIS A006689, A006690) satisfies $a_n / S(kn+1, n) = \sum_{j \leq J} c_j n^{-j} + O(n^{-J-1})$ for every $J$, with $c_0 = c$, $c_1 = k\rho v/(2c)$ and the small-state counts entering first at $c_k$. The route is an exact first-failure renewal with two boundaries and a finite coefficient algorithm; $c_1, c_2, c_3$ are tabulated. Part II adapts the method, with a strict checkpoint, to strongly connected structures (A027834/A027835, A006691/A006692, A304312/A304313), transfers the expansion through the classical formal logarithm, confirms the identification that the A304312/A304313 records leave conjectural, and gives exact cyclic automorphism sectors with a uniform divisor-tail bound and a self-contained proof of Liskovets's existing unrooted formula. The leading equivalents (Korshunov, Bassino–Nicaud, Lebensztayn) are classical, and priority of the refinement is left unresolved: Korshunov's 1978 survey was not obtained. The Lambert-$W$ inverses and threshold brackets are instances of the transseries volume. *Merged from two manuscripts; the passages Part II repeats are printed once, in Part I.* Minimal DFAs of finite languages, a different object, are counted in the report `a082161-airy-amplitudes`.

**36. Binary two-way automata and corank budgets in Brauer monoids**
automata-and-formal-languages/binary-two-way-corank/article.pdf
from ProveIt_Binary_Two_Way_Corank.zip (arrival 964621dd2, 7 October 2026, outside the numbered batches)

For every $h \geq 2$ an explicit language $B_h$ over $\{0,1\}$, recognized by a $(6h+2)$-state 1NFA, such that every $s$-state 2DFA for it satisfies $8s + 2 \geq 2(5/2)^{\lfloor (h-2)/9 \rfloor}$; hence an $n$-state binary 1NFA ($n \geq 14$) can need at least $\frac{1}{4}(5/2)^{\lfloor (n-14)/54 \rfloor} - \frac{1}{4}$ 2DFA states. The ingredients are a repetition-independent corank budget for words in Brauer idempotents, used twice in a relation-addition amplifier ($p = q = 5$ optimal for it), a symbol-local sparse-slot matching representation of an $s$-state 2DFA in degree at most $2(a+2)s + 2$, and a two-register compiler with a binary decoder; also a sharp support bound for idempotents with deficit, an integer recurrence for the bound and eight research questions. Its binary nondeterministic complementation bound is conditional on an unreproved premise of a companion OpenAI manuscript; the relation-corner and tree-contour architecture is adapted, with attribution, from an external openai/math release, whose Apache-2.0 licence and third-party notices are staged as the report's nested licence. Written as a single-source report; its independent check confirmed the headline bound, rebuilt the sparse-slot representation (Theorem 3.1) from the text alone, and added two precisions on the trust boundary (Lemma 5.2 is the companion's lemma, only outlined here; the exact dependency set of Theorem 1.1). Unrefereed; not formalized.

## 5 Hankel determinants and continued fractions

## 5.1 Somos sequences and elliptic curves

### 37. A proof of Barry's Hankel–Somos-6 conjecture

hankel-determinants/somos-and-elliptic/barry-somos-6-hankel-recurrence/article.pdf
from barry_somos6_research.zip

Conjecture 7 of Paul Barry, arXiv:2211.12637: for the Hankel determinants $H_N$ of a four-parameter Jacobi-type generating function, $H_N H_{N-6} = \alpha H_{N-1} H_{N-5} + \gamma H_{N-3}^2$ with $\alpha = t^2(r+2)^2$ and $\gamma = t^3((s+t)(r+s+t+2)^2 + t(r+2)^3)$. Also proves a homogeneous four-parameter form and a seven-parameter polynomial Hankel theorem making exceptional specialisations rigorous; the verifier never divides by a determinant.

### 38. Finite Hankel Certificates for Three Somos-8 Conjectures

hankel-determinants/somos-and-elliptic/somos-8-hankel-certificates/article.pdf
from somos_hankel_proofs.zip

Conjectures 11, 12 and 13 of the same Barry paper, by importing Hone's genus-two Hankel theorem (arXiv:1907.05204v3, Thm. 5.4) and supplying a binomial translation, a polynomial-minor specialisation valid on the degenerate locus, and a rank-bound lemma that turns four symbolic rows into an all-index identity. Identifies a sign error in the source's Catalan expression for Conjecture 13.

### 39. Elliptic addition and Jacobi continued fractions

hankel-determinants/somos-and-elliptic/
elliptic-addition-and-jacobi-fractions/article.pdf
from elliptic_continued_fraction_research.zip

A consistently indexed, non-torsion version of Barry's 2023 Conjecture 4, with the coordinate formulas of Conjecture 3: every continued-fraction tail is given explicitly and removing one layer is shown to be elliptic translation. Hypotheses: characteristic zero, $Y^2 + aXY + bY = X^3 + cX^2 + dX$ nonsingular, $b \neq 0$, and $P = (0,0)$ of infinite order; order-three and order-four examples show the regular fraction genuinely needs the last one.

## 5.2 Catalan and ballot moments

### 40. Shifted Catalan Hankel Polynomials

hankel-determinants/catalan-and-ballot/shifted-catalan-hankel-polynomials/article.pdf
from catalan_hankel_conjecture_solution.zip; batch 39: ProveIt_Catalan_Root_Collisions.zip; batch 42: ProveIt_Cyclotomic_Catalan_Recurrences.zip; batch 53: Two_Endpoint_Catalan_Confluence.zip; batch 55: ProveIt_Interior_Collision_Research.zip; batch 57: ProveIt_Catalan_Growing_Shifts.zip

For $D_N^{(m)}(a,b) = \det(aC_{i+j+m} + bC_{i+j+m+1})$, proves the denominator-exponent assertion on page 4 of Barry, arXiv:2011.10827v1, and shows the separately printed Conjecture 2 needs an explicit one-unit shift correction. The minimal characteristic polynomial is $(X^2 - (a+2b)X + b^2)^{m(m-1)/2+1}$, so the minimal order is $m(m-1)+2$; all exceptional reduced denominators, the exact coefficient formula, numerator degree and reciprocal symmetry are supplied. Part II, added in batch 39, replaces $aC + bC'$ by an arbitrary polynomial multiplier and computes the root collisions: exact sector degrees with nonzero leading constants, the minimal recurrence order off resonance, a secondary-Hankel factorization and a certified root-free algorithm for the recurrence order. Part III, added in batch 42, gives every characteristic multiplicity for one repeated root at a cyclotomic resonance, and shows that the question it answers needed one more parameter: the sign of $z^L$, not only $t$ and $\mathrm{ord}(z^2)$. Part IV, added in batch 53, answers the endpoint half of Part II's question on uniform limits: for fixed-size clusters of roots approaching the endpoints 0 and 4 ($d$ roots in all) the normalized determinant is a convergent series in even powers of $1/(N + d/2)$, locally uniform through every collision inside a cluster, with a factorized leading profile, an explicit first interaction term, and ratio to the fixed-endpoint determinant tending to 1 exactly when the outward displacements are $o(N^{-2})$. Part V, added in batch 55, answers the other half: for fixed-size

clusters of roots colliding on the scale $1/N$ at fixed bases other than 0 and 4, the determinant is an exact finite sum of exponential sectors whose amplitudes are convergent series in $1/(N+d/2)$, uniform through every collision, with an explicit first correction (this expansion is not even), an explicit remainder bound, and factorized amplitudes for several separated clusters. For two equal-multiplicity roots inside the spectrum the limit is an oscillatory Jacobi Hankel determinant, the first correction oscillates as $\sin(2(N+t)\theta)$, and simple zeros move by a convergent phase-dependent expansion. Limits uniform up to the endpoints remain open. Part VI, added in batch 57, lets the shift grow with the determinant size: for real $a \geq 0$ the ratio $D_N^{(m)}(a,1)/D_N^{(m)}(0,1)$ tends to 1 exactly when $aN(N+m+1)/(4m+2) \to 0$, at any relative growth of $N$ and $m$; the size-biased inverse Jacobi zeros follow a shifted Marchenko–Pastur (Catalan-moment) law, and three critical scales, an algebraic free energy and a uniform saddle prefactor describe the crossover.

**41. Product formulas for ballot-polynomial Hankel determinants**
hankel-determinants/catalan-and-ballot/
ballot-polynomial-hankel-determinants/article.pdf
from Cigler_Hankel_Conjectures_13_15.zip

Conjectures 13, 14 and 15 in Section 5 of Johann Cigler, arXiv:2111.14492v3. An explicit coefficient product gives the sign and exact support of the normalised shift-$k$ Hankel determinant of the ballot moments and positivity of every residual coefficient for $n \geq 1$; the same products give the conjectured generating-function denominators, exact numerator degrees and reciprocal numerator identities. Adds strict log-concavity of the residual coefficients and their limiting binomial profile.

**42. Rectangular Schur polynomials and Cigler's Conjecture 8**
hankel-determinants/catalan-and-ballot/cigler-conjecture-8-schur/article.pdf
from cigler_conjecture8_research.zip

A complete proof of Conjecture 8 in Cigler, arXiv:2111.14492v3, Section 4 (printed pp. 16–17), for the Hankel determinants of $b_m(t) = \sum_r \binom{\lfloor m/2 \rfloor}{r}\binom{\lceil m/2 \rceil}{r} t^r$, including the numerator-polynomial degree and reciprocity assertions. A different conjecture of that paper from the ballot-polynomial entry (Conjectures 13–15) and from the parity entry (Conjecture 16).

**43. A parity factorization for Cigler's Conjecture 16**
hankel-determinants/catalan-and-ballot/cigler-conjecture-16-parity/article.pdf
from cigler_conjecture16_research.zip; batch 56: ProveIt_Cigler_Schur_Recurrences.zip; batch 66: ProveIt_Cyclotomic_Pole_Collapse.zip; batch 70: ProveIt_Cigler_Conjecture17_Resolution.zip

A complete proof of Conjecture 16 in Cigler, arXiv:2111.14492v3, Section 6 (pp. 21–22), on the Hankel polynomials, together with a sharp stabilization statement: the report also determines the first coefficient at which the conjectured stabilization fails. Part II, added in batch 56, proves the report's experimental rectangular-Schur formula: with $a = \lfloor k/2 \rfloor$ and $b = \lfloor (k-1)/2 \rfloor$ the normalized determinant is $s_{(n^a)}(1^a, t, (t^2)^b)$ for all $k \geq 1$ and $n \geq 0$. It determines the exact minimal recurrence at each shift, with reduced denominator $\prod_{q=0}^{k-1}(1-t^q z)^{1+\lfloor q(k-1-q)/2 \rfloor}$ over $\mathbb{Q}(t)$ and at every real $t \neq 0, \pm 1$, proves Cigler's denominators (84) and (86) minimal, and shows that the positive reciprocity sign printed in his (85), part of Conjecture 18, is false when $k = 2a$ with $a$ even. Part III, added in batch 66, determines the reduced denominator and the minimal recurrence at every complex parameter: at a primitive root of unity of order $\ell \geq 3$ at most one residue class loses pole order, decided by an explicit parity test, and it loses one order except for a loss of three at primitive fourth roots when $k \equiv 1 \pmod 4$. Part IV, added in batch 70, proves Cigler's Conjecture 17 in sign-corrected form: once his denominator $(1-t)^a(1-t^2)^{ab}$ is cleared, the determinant splits into $k$ blocks, polynomials in $t$ whose coefficients are integer-valued polynomials in $n$, of exact degrees $q(k-1-q)$ in $t$ and $\lfloor q(k-1-q)/2 \rfloor$ in $n$, with explicit leading terms in $n$; their reciprocity sign is $-1$ exactly when $4 \mid k$. The sign printed in Cigler's (83) fails for every $k \equiv 1 \pmod 4$ from 5 on (his displayed block for $k = 5$ is palindromic), so the conjecture as printed is false. Its own README explains why this is a different problem from the other Cigler entries in this collection.

## 5.3 Arithmetic Hankel determinants

The numerator formula recorded as a conjecture in OEIS A122251, with A122249 as its binary special case. Two reports attacked it and proved it the same way, sharing even their worked counterexamples; they are merged here into one. The merged report sits at the category's top level.

### 44. A Proof of an OEIS Hankel-Numerator Conjecture

hankel-determinants/a122251-numerators-and-denominators/report.pdf
from hankel_numerator_research.zip + hankel_oeis_conjecture_proof.zip

For the Hankel determinant

$$H_n(m,a) = \det(1/(a+m(i+j)))_{0\le i,j\le n},$$

the reduced numerator is $\prod_{p\mid m} p^{\,n(n+1)v_p(m)+2\sum_{k\le n} v_p(k!)}$: independent of the coprime offset $a$, a perfect square, and supported on the primes dividing $m$. Taking $m=p$ and $a=1$ settles the OEIS A122251 conjecture, with A122249 as the binary case. Also gives a closed form for the reduced denominator and all its local valuations, a sharp gcd theorem over the allowed offsets realised at just two explicitly constructed shifts, the extension to unequal row and column steps $\det(1/(a+m_1 i+m_2 j))$, recurrences, digit-sum asymptotics, and a Lambert-series generating function with a natural boundary. *Merged; the determinant is evaluated three independent ways, and a second proof of the numerator formula runs through an explicit inverse matrix and MacMahon's boxed plane partitions.*

## 5.4 Growth laws and run statistics

### 45. A proof of the Apéry Hankel growth conjecture

hankel-determinants/growth-and-runs/
apery-hankel-determinant-growth/apery_hankel_growth.pdf
from apery_hankel_research.zip; batch 73: apery_hankel_research.zip (a different manuscript under the same archive name); batch 77: apery-shifted-release.zip

For $A_k=\sum_j \binom{k}{j}^2\binom{k+j}{j}^2$ and $D_n=\det(A_{i+j})$, proves $\log D_n = n(n+1)\log\frac{17+12\sqrt{2}}{4}+O(n)$ — the limit conjectured in the September 2026 revision of OEIS A228143 — using Edgar's moment-density theorem. Also an explicit law for $\det(A_{m(i+j)+r_n})$ with $r_n/n\to\rho$, and non-D-finiteness of the formal generating series of $D_n$. Part II, added in batch 73, determines what Part I left open: by sharp Szegő asymptotics, $h_n/\Lambda^{2n}\to K=(\pi C/2)\exp\langle\log\varphi\rangle\approx 0.6334$, an arcsine average of Edgar's density (a quadrature value, not interval-certified), so $D_n/D_{n-1}\sim K\Lambda^{2n}$ and $\log D_n = n(n+1)\log\Lambda+(n+1)\log K+o(n)$, with $C=17+12\sqrt{2}$ and $\Lambda=C/4$; it refines Part I's stride theorem at $\rho=0$. Its re-proof of Part I's main theorem is printed as a corollary, not a new result. Part III, added in batch 77, proves for $H_N(r)=\det(A_{r+i+j})_{0\le i,j<N}$ and $s=r/N$ in compact subsets of $(s_*,\infty)$, $s_*=3\sqrt{2}/4-1$, the full equivalent $H_N(r)=M(s)N^{-1/24}e^{N^2A(s)+NB(s)}(1+c(s)/N+O(N^{-2}))$ with an exact Selberg normalization, by adapting Charlier–Gharakhloo's determinant-ratio analysis to a logarithmic potential; also that the floor-shifted sequence is eventually increasing, and rational-residue inverse thresholds, whose inversion cites the transseries volume. Its quadratic term $A(s)$ is Part I's stride theorem at $m=1$ and is not claimed as new; strides $m\ge 2$, $s\le s_*$ and the prefactor of $D_n$ stay open.

### 46. Binary Runs in Rueppel Hankel Determinants

hankel-determinants/growth-and-runs/rueppel-binary-run-determinants/rueppel_hankel.pdf
from rueppel_hankel_research.zip

For the Rueppel series $r(x)=\sum_{j\ge 0}x^{2^j-1}$, the polynomial identity $H_n(tx+1/r(x^2)) = (-1)^{1+\varepsilon+\lfloor R(n)/2\rfloor}(\varepsilon+(R(n)-\varepsilon)t^2)$, where $R(n)$ counts runs of equal bits in the binary expansion of $n$ and $\varepsilon$ is the low bit of the Gray code $n\oplus\lfloor n/2\rfloor$. Settles statements 8, 10, 13 and 23 of Barry's 2021 paper (not the identically numbered ones elsewhere), with a correction.

# 6 Congruences and $p$-adic valuations

## 6.1 Supercongruences

### 47. Congruences for OEIS A348410 at every prime

congruences-and-valuations/supercongruences/a348410-all-prime-congruences/article.pdf
from a348410_supercongruence.zip + A348410_Supercongruence_Research.zip

For $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$ proves the odd-prime cubic supercongruence $v_p(A(Np) - A(N)) \geq 3v_p(N) - \varepsilon_p$ with $\varepsilon_3 = 1$, $\varepsilon_p = 0$ for $p \geq 5$, and no coprimality condition; $(\alpha, \beta) = (2, 1)$ is Bala's A348410 conjecture. Separately proves an exact binary leading-defect formula, $A(m2^r) - A(m2^{r-1}) \equiv 2^{3r-3}\alpha\beta\binom{(\alpha+\beta+1)m-1}{m-1} \bmod 2^{3r-2}$, hence the all-prime bound $3r - v_p(12)$ and sharp denominators 12 and 24. An appendix audits arXiv:2104.10754v1, whose framing conclusion and weighted harmonic assertion fail on its own displayed examples. *Merged; the odd-prime and binary halves run on different mechanisms — a doubling automorphism, and reflection with formal integration by parts — and both are kept.*

**Not a duplicate of *signed-binomial-supercongruences*, and checked.** That report's $a_{r,s,t}(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$ specializes at $t = 1$, $(r, s) = (-\beta, -\alpha)$ to the coefficient studied here, and the two odd-prime statements then agree. Neither contains the other. That one is general in the extraction step $t$, which is fixed at 1 here; this one adds the prime 2 with an exact defect and its sharpness, every prime at once, minimal common denominators, the primitive-cyclic-word interpretation, the Lambert/Euler product, the algebraic generating functions, a spacing-three counterexample, and a sufficient criterion beyond the family. The proofs are different arguments: that one imports Jacobsthal's congruence, this one uses none and pairs a quadratic-tail estimate with a coefficient estimate.

### 48. Signed binomial supercongruences and OEIS A352373

congruences-and-valuations/supercongruences/
a352373-signed-binomial-supercongruences/signed_binomial_supercongruence.pdf
from OEIS_A352373_supercongruence.zip

For $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$ proves $a(mp^h) \equiv a(mp^{h-1}) \bmod p^{3h}$ for $p \geq 5$ and $a(m3^h) \equiv a(m3^{h-1}) \bmod 3^{3h-1}$, with $m$ not required coprime to $p$. Covers A352373, A348410, A351856 and A351857 as specialisations; the only non-elementary import is the classical Jacobsthal binomial congruence.

**Not a duplicate of *all-prime-congruences*, and checked.** Setting $t = 1$ and $(r, s) = (-\beta, -\alpha)$ turns the theorem below into that report's odd-prime tower, and the moduli agree after reindexing the multiplier. Neither contains the other: this one is general in the extraction step $t$, that one is fixed at $t = 1$ but proves the prime 2 case, the all-prime bound, the sharp denominators and much else. The two proofs are genuinely different — this one imports Jacobsthal's congruence, that one does not use it at all.

### 49. Apéry numbers under exponentiation and series reversion

congruences-and-valuations/supercongruences/
a267220-apery-transform-supercongruences/article.pdf
from Apery_OEIS_Supercongruences.zip, apery_oeis_conjectures(1).zip

Proves both supercongruence conjectures contributed by Peter Bala to OEIS A267220 on 17 October 2024, for every integer parameter including negative and zero. With $A(x) = \exp\sum_{n\geq 1} a_n x^n/n$ for the Apéry numbers $a_n$, $F(x) = x/\mathrm{Rev}(xA(x))$, $u_m(n) = [x^n]A(x)^{mn}$ and $v_m(n) = [x^n]F(x)^{mn}$: for odd $p$ and $R = \nu_p(N)$ both families satisfy $\nu_p(u_m(N) - u_m(N/p)) \geq 2R + \nu_p(m)$, the unsigned statement for $v_m$ holds at *every* prime including 2, and a *signed* refinement $(-1)^{mN}u_m(N) \equiv (-1)^{mN/p}u_m(N/p)$ recovers full precision at $p = 2$, where the unsigned bound genuinely loses one power. Merged from two reports that share a seed lemma and a Möbius/dilogarithm layer but diverge at the crux; both routes are kept, and the merged text is careful that the signed and unsigned 2-adic statements are different statements, not a sharper and a weaker version of one.

### 50. Four Motzkin congruences

congruences-and-valuations/supercongruences/a001006-motzkin-congruences/article.pdf
from motzkin_congruences.zip

Proves all four conjectures in Peter Bala's comment of 10 February 2022 on OEIS A001006, with sharp ranges. The engine is a prime-power block-transfer theorem: for odd $p$ and $q = p^{k-s+1}$, $h = mp^{s-1}$, the Motzkin numbers satisfy $M_{mp^k+d} \equiv T_h M_d$ for $0 \le d \le q-3$, with explicit corrections at $d = q-2$ and $d = q-1$; the boundary case gives $M_{np^r-2} \equiv \frac{\varepsilon_p^r - 1}{2} T_{n-1} - U_{n-1} \pmod p$. Adds the exact first failures beyond the two initial-block ranges, arbitrary-multiplier and all-prime-power boundary extensions, a prime-modulus digit algorithm, and examples showing why unweighted repetition modulo a prime cube is generally false. Discusses Burns, Kohen and Pan–Sun rather than claiming novelty.

## 6.2 Congruences for iterated and compositional series

### 51. Arithmetic of a Self-Iterating Generating Function

congruences-and-valuations/iterated-series/a396807-self-iterating-series/article.pdf
from A396807_congruences_and_prime_power_lifts.zip

Both conjectures in OEIS A396807 for $A(x) = x + A^{[5]}(x)A^{[6]}(x)$ (functional iteration): $A(x) \equiv x/(1-x)$ and $A^{[k]}(x) \equiv x/(1-kx)$ modulo 10 for every integer $k$, negative iterates included. Adds a sharp modulus theorem for $F = x + cF^{[r]}F^{[s]}$, rational lifts at every selected prime power, and minimal period 46860 for A396807 modulo 100 starting at index 1.

### 52. Prime-power congruences for compositional tree series

congruences-and-valuations/iterated-series/
compositional-tree-series-congruences/article.pdf
from compositional_tree_congruences.zip

For $A_l(x) = x\exp(A_l^{\circ l})$, shows the OEIS A396805 claim $a_5(n) \equiv n \pmod 5$ is *false*: first failure $n = 23$, with exception set $\{20j+3,\ 20j+4 : j \ge 1\}$ and residue 2 throughout. Proves eleven other conjectural assertions in A396803–A396806 via a general reduction of coefficients mod $p^\alpha$ to typed rooted trees on fewer than $p^\alpha$ labels, yielding eventual periods, rational residue generating functions, and an exact classification of $a_l(n) \equiv n \pmod p$.

### 53. Lacunary iterates modulo four

congruences-and-valuations/iterated-series/a168362-lacunary-iterates-mod4/article.pdf
from batch 73: ProveIt_A168362_Mod4_Theorem.zip; batch 114:
Digit_Sum_Divisibility_in_Lacunary_Iteration.zip

For $F(x) = \sum_{j\ge 0} x^{2^j}$ and its compositional iterates $G_m = F^{\circ m}$, OEIS A168362 is $a(n) = [x^n]G_n(x)$, and the entry records a sparse 2026 pattern for $a(n)$ mod 4. Proves $[x^N]G_m(x) \equiv 0 \pmod 4$ for all $m, N \ge 1$ whenever $N$ has at least three binary ones, gives exact bit formulas at powers of two and sums of two powers of two, and so classifies the diagonal completely: apart from $2, 3, 4, 6$, $a(n) \equiv 2 \pmod 4$ exactly when $n = 2^r + 2^s$, $r > s \ge 1$, $r \ge 3$, and $(r-s) \bmod 2^s$ lies in an explicitly computable residue set $R_s$. This confirms every exception printed in the entry; their number up to $X$ is $\kappa \log_2 X + O((\log\log X)^3)$ with $\kappa = \sum_s |R_s|/2^s \approx 2.32393$. The ratio conjecture $a(n+1)/a(n) \sim en$ stays open. Part II (batch 114) proves, for every prime $p$ and every integer iterate $m$ of $F_p(x) = \sum_j x^{p^j}$, inverse iterates included, that $[x^N]F_p^{\circ m}$ vanishes unless $N \equiv 1 \pmod{p-1}$ and has $p$-adic valuation at least $(s_p(N)-1)/(p-1)$, $s_p$ the base-$p$ digit sum; this answers Part I's question on the moduli $2^q$ for every $q$, and $[x^{2p-1}]F_p(F_p(x)) = p$ shows that Part I's suggested odd-prime cutoff "digit sum at most two" fails (a re-scoped research question, not a retracted theorem). It adds a composition-closed class containing every weighted seed, an explicit first residue of the second iterate, three sharp families, optimality through the Fuss–Catalan inverse of $x + x^p$, sparse and density-one bounds, $p$-automaticity of each fixed row by a Cartier construction, and a coefficientwise $p$-adic iteration group with sufficient periods; at $p = 2$ it re-proves Part I's Theorem 1.2 and Propositions 5.1 and 6.1, printed as second routes.

It records that the EXAMPLE lines of OEIS A168365 and A168366 describe an iteration other than the plain composition of their data and programs (nothing submitted). The independent check of the Part II write found an error of Part I: the one-bit formula (16) of Theorem 5.2 is false with its first binomial read as an integer ($m = 2$, $r = 1$) and right with it read mod 2, the reading the classification uses; the printed statement is kept with a dated correction. Unrefereed; not formalized.

## 6.3 Valuations, divisibility and periodicity

**54. Binary Carries and the Divisibility of Two-Stack-Sortable Permutation Numbers**
congruences-and-valuations/a000139-binary-carries-parity/research.pdf
from a000139_carries_research.zip

Proves Peter Bala's parity conjecture for OEIS A000139, $a(n) = 2(3n)!/((n+1)!(2n+1)!)$: $a(n)$ is odd exactly when $n$ is odd and its binary expansion contains no `11`. The full valuation is $v_2(a(n)) = s(n) + s(n+1) - s(3n)$ in binary digit sums. Two independent parity proofs, plus exact valuation distributions, recurrences, asymptotics, extremal formulas and limit laws.

**55. Polynomial divisibility of height-one factorial ratios**
congruences-and-valuations/factorial-ratio-polynomial-divisibility/article.pdf
from bala_factorial_divisibility_research.zip; batch 74: OEIS_Fractional_Factorial_Theorems.zip

A general rational-root criterion for uniform polynomial divisibility of balanced integral factorial ratios, applied to Peter Bala's uniform-product conjectures for OEIS A211417 and A295431: explicit multipliers for every product length, with the least multipliers certified in eight particular examples. Includes an atlas of the 52 sporadic height-one ratios with recurrences, generating functions and asymptotic expansions. Part II, added in batch 74, proves integrality and algebraic generating functions for A347854–A347858 through a rational-dilation theorem for height-one factorial ratios with composite denominators, settling the Gamma-interpolated values Part I left open for these five entries; it also proves $6(6n+1)(12n-1) \mid \mathrm{A295432}(n)$ and gives finite-monodromy certificates and moment laws. Its re-proof of Part I's A295431 theorem, with larger constants and no optimality, is printed only as a marked second route.

**56. Integrality of the Brent–Glasser–Guttmann asymptotic coefficients**
congruences-and-valuations/a321941-asymptotic-coefficient-integrality/article.pdf
from batch 73: A321941_Arithmetic_Conjectures.zip; batch 86: oeis_research_bundle.zip (its Part I; its Part II is Part V of a088714-bell-scale-growth)

Brent, Glasser and Guttmann introduced rational coefficients $d_k$ in the all-orders asymptotic expansion of a Hadamard product involving the exponential integral, conjectured that $r_k = 64^k d_k$ (OEIS A321941) is always an integer, and observed $r_k \equiv \binom{2k}{k} \pmod{32}$ for $k \geq 3$. Proves both, by keeping a quadratic invariant of the underlying second-order recurrence rather than only its symmetric-square recurrence, and extends them to polynomials $P_k(t) \in \mathbb{Z}[t]$ with $P_k(1) = r_k$, even for $k > 0$, with a coefficientwise congruence modulo 32. Consequences: the scale 64 is minimal, denominator bounds are sharp at infinitely many indices and exact when the index has at most four binary ones, and the inverse expansion has no denominator primes other than 3. The analytic expansion itself is the published theorem. Part II, added in batch 86 from part of a two-subject manuscript, proves the negativity conjecture $r_k < 0$ for every $k \geq 3$ through an exact radial-oscillator heat kernel, with a majorant bound for $k \geq 32$ and an exact certificate below; shows that Part I's constant edge $P_k(0)$ is the leading scale, $r_k \sim -16^k (k!)^2/(\pi^{3/2} k^{3/2})$, with a complete expansion in $1/k$; and finds a logarithmic singularity at 16 of the order-two Borel transform $\sum d_k s^k/(2k)!$, radially only, so its continuation stays open.

**57. When the Secant Numbers Become Periodic**
congruences-and-valuations/secant-number-periodicity/article.pdf
from secant_periodicity_research.zip

Refutes the OEIS A000364 conjecture that the secant numbers from $n = 1$ are purely periodic modulo every $m$: at $m = 27$, $a(1) \equiv 1$ but $a(19) \equiv 10$. The exact replacement is that pure

periodicity from index one holds if and only if no odd prime cube divides $m$; the least eventual period always divides $\varphi(m)$. Determines the least eventual period and exact onset for every modulus, with densities and a certified mean onset.

## 7 Tetration and digit stabilization

**58. A Counterexample to a Decimal-Length Bound for Tetration Stabilization**
tetration-and-digit-stabilization/congruence-speed-decimal-length-bound/article.pdf
from tetration_counterexample.zip

An explicit 2544-digit base $A = 1 + qu$ ($q = 5^{2547}$) for which $\nu_{10}(T_{n+1} - T_n) = \min(1 + 2548n, 2547(n + 1))$ at every height, so the congruence speed at height $\mathrm{len}(A) + 2$ is 2548 while the permanent value is 2547. This contradicts the decimal-length form of Conjecture 1 of Marco Ripà, *The congruence speed formula* (NNTDM 27(4), 2021). The report audits the later Ripà–Onnis confirmation: their weaker valuation bound is compatible with the counterexample, the decimal-length bound is not.

**59. Tetration, perfect powers, and frozen decimal digits**
tetration-and-digit-stabilization/perfect-power-minima-for-frozen-digits/article.pdf
from tetration_perfect_power_minima.zip

Determines $\rho(n)$, the smallest $n$th perfect power $x^n > 1$ whose eventual number of newly frozen terminal decimal digits is exactly $n$: $\rho(1) = 2$, $\rho(2) = 49$, and $\rho(n) = B(s)^n$ for $n \geq 3$ with $s = n - v_2(n)$ and $B(s) \in \{2^s \pm 1, 3 \cdot 2^s \pm 1\}$ according to $s$ mod 4. All other roots are ruled out by a deterministic quadratic size barrier plus small exact certificates.

**60. Phase-shift classification for integer tetration**
tetration-and-digit-stabilization/
phase-shift-classification/tetration_phase_classification.pdf
from tetration_phase_classification.zip

Proves the 21-word phase-shift conjecture in OEIS A376842, the 14-word even-base restriction from the appendix of Ripà's 2025 paper, and the 28-word prescribed-anchor conjecture in OEIS A376446. The mechanism compares the 2-adic and 5-adic valuations of $D_b = T_{b+1} - T_b$ at the first permanently constant-speed height: strict 2-adic dominance gives phase 5, strict 5-adic dominance a multiplicative orbit in $\mathbb{F}_5^\times$, and balanced valuations exactly four further anchored words.

**61. A Sharp Convergence Theorem for the $q$-Newton Series of Subcritical Tetration**
tetration-and-digit-stabilization/q-newton-series-convergence/article/
tetration_qnewton.pdf
from tetration_qnewton_research.zip

Proves convergence of the explicit $q$-binomial interpolation series of MathOverflow question 259278 for every real base $1 < a < e^{1/e}$, with $\log a = qe^{-q}$: the grouped series converges absolutely and locally uniformly for $\mathrm{Re}\, z > -2$, converges conditionally on $\mathrm{Re}\, z = -2$ except at $z = -2 + 2\pi ik/(-\log q)$, and diverges for $\mathrm{Re}\, z < -2$. Adds strict complete monotonicity and uniqueness under an eventual log-convexity condition.

**62. Stabilization of integer power towers**
tetration-and-digit-stabilization/power-tower-stabilization/article.pdf
from tetration_stabilization_research.zip + tetration_exact_digits.zip

Refutes the claim that every prime base ending in 9 reaches its permanent decimal congruence speed by height 2 (Ripà, Conjecture 2; arXiv:2208.02622v1, Conjecture 4.1): 2749 is the smallest prime counterexample, with $H(2749) = 3$. Proves exact formulas for $C_a(h)$, $V(a)$ and $H(a)$, an exact natural density for onsets, and infinitely many primes at each prescribed onset via CRT and Dirichlet. Also resolves the three conjectural comments in OEIS A324017 — two proved, the third disproved, with $3 \uparrow\uparrow 2 \equiv 27$ and $3 \uparrow\uparrow 3 \equiv 59 \pmod{64}$ — via $T_{h+r} - T_h \equiv -2q^h \pmod{q^{h+1}}$ for even $q \geq 4$, so exactly one radix-$q$ digit freezes per height. Adds a factorization-free

digit-lifting algorithm, modular saturation across Knuth arrow ranks, and a sign-sensitive local $p$-adic Lambert $W$ formula. *Merged; the shared one-digit-per-height theorem was proved identically in both originals and is proved once here.*

**63. Delayed digit stabilization: a counterexample and sharp bounds**
`tetration-and-digit-stabilization/iterated-power-digit-stabilization/delayed_digit_stabilization.pdf`
from `digit_stabilization_research.zip`

For $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$ and $D_a(b) = S_a(b) - S_a(b-1)$, answers negatively the question whether $D_a(b) = 1$ for all $b \geq 3$: the smallest counterexample is $a = 3 \cdot 2^{99} + 1$, with $S_a(2) = 100$, $S_a(3) = 102$, $D_a(3) = 2$. Proves minimality, an exact onset formula, the smallest counterexample at every stage, the natural density of exceptions, and arbitrary-radix extensions. The article is explicit that $a^{10^b}$ is not genuine tetration and derives the tower formulas separately.

# 8 Log-concavity, unimodality and positivity

**64. Single curves do not force log-concavity**
`log-concavity-and-unimodality/cluster-variable-log-concavity-counterexample/article.pdf`
from `annular_single_lamination_counterexample.zip`

An unpunctured annulus with one marked point on each boundary and a single unit-weight simple lamination curve gives, after mutations $2, 1, 2$ with the two mutable variables specialised to 1, the polynomial $1 + 3q^2 + 2q^3 + 4q^4 + 2q^5 + q^6$; the consecutive coefficients $3, 2, 4$ form a strict valley, refuting Conjecture 1 of arXiv:2508.04396v3 and conflicting with its general unpunctured-surface unimodality statement. The geometric realisation is proved, not merely the matrix calculation, and a fixed seed gives infinitely many such examples.

**65. Cycles refute infinite log-concavity for chromatic coefficients**
`log-concavity-and-unimodality/infinite-log-concavity-of-cycles/article.pdf`
from `chromatic_cycle_logconcavity.zip`

For the cycle $C_n$ with absolute chromatic coefficients under the iterated operator $L(a)_k = a_k^2 - a_{k-1}a_{k+1}$: infinitely log-concave for $3 \leq n \leq 11$; first failure at depth 5 for $n = 12$, depth 4 for $13 \leq n \leq 16$, and depth 3 for $n \geq 17$. So $C_{12}$ is the smallest cycle counterexample (no minimum over all graphs is claimed), with an explicit third-iterate witness polynomial valid for all $n \geq 3$.

**66. Oscillating Log-Concavity in Geometric Generating Functions**
`log-concavity-and-unimodality/oscillating-log-concavity-geometric-transform/article.pdf`
from `oscillating_log_concavity_research.zip`

Challenges 2 and 3 in Section 3 of Heim and Neuhauser, arXiv:2302.13327v1, for the *geometric* transform $1/(1 - uG(z))$. For divisor-sum weights, every real exponent and every $u > 0$ give infinitely many strict log-concavity and log-convexity indices with eventually bounded gaps; likewise for integer power weights with exponent $\geq 2$, the exponents 0 and 1 being explicitly exceptional. For square weights at $u = 1$, every four consecutive indices carry both signs (sharp), each with density $1/2$ in every arithmetic progression. A subdominant-pole theorem is the common mechanism.

**67. A counterexample and a sharp spacing-three criterion for products of $q$-integers**
`log-concavity-and-unimodality/q-integer-product-unimodality/report.pdf`
from `q_unimodality_attack.zip`

The necessity half of Conjecture 5.4 of Connelly, Ito, Martinez, Shevchenko and Yang (arXiv:2605.12822v1) is false: $(1+q)^6(1+q^3)$ is symmetric and weakly unimodal although no $a_i$ is divisible by 3 and $b = 2$ exceeds the asserted bound, and the example lies inside the source's claimed range. The report also proves the proposed sufficient condition for every $r \geq 2$, a necessary degree bound in the absence of an $r$-divisible ordinary factor, and a complete classification of spacings two and three.

### 68. Sharp size thresholds for failures of log-concavity in independence systems

log-concavity-and-unimodality/independence-system-thresholds/article.pdf
from sharp_log_concavity_thresholds.zip

For each rank $r \geq 4$, determines the smallest ambient ground-set size admitting failure of ordinary, ordered and ultra log-concavity, and shows the three thresholds coincide across all independence systems, those satisfying Xie–Xu's augmentation bound, and those with additive augmentation parameter 3 (values 7/6/5, 7/7/6, 8/7/7, and $r+1$ throughout for $r \geq 7$). The initial target, Xie–Xu Conjecture 4.8 (arXiv:2408.09152v1), had already been disproved by Alex Chengyu Li in September 2026; the report states this and refines the target to the rankwise minimum-size question.

### 69. A sharp order-five threshold for higher-order Stirling subset log-concavity

log-concavity-and-unimodality/higher-order-stirling-rows/article.pdf
from stirling_log_concavity.zip

For $S^{(r)}_{n,k}$, the number of partitions of $n + (r-1)k$ labels into $k$ blocks of size $\geq r$, proves that all rows are log-concave if and only if $1 \leq r \leq 5$, with strict inequalities in the interior — the log-concavity assertion of Deb–Sokal Conjecture 1.4(c) (arXiv:2507.18959v1). Also proves the cycle cases $r = 3, 4$; the fifth-order *cycle* conjecture is explicitly left unresolved, and the included operator counterexample is flagged as not being a counterexample to it.

### 70. A sum-of-squares proof of Ulas's real-zero conjecture

log-concavity-and-unimodality/stern-polynomial-positivity/stern_conjecture_note.pdf
from stern_conjecture_research.zip

Conjecture 4.4 of Maciej Ulas, arXiv:1909.10844v1, for the Stern polynomials indexed by $a_n = (4^n - 1)/3$ and $h_n = 2(4^n - 1)(2 \cdot 4^n + 1)/3 + 1$: for every $n \geq 0$ and every real $t$, $W_n(t) = (A_{n+1}(t) - tA_n(t)/2)^2 + 3t^2 A_n(t)^2/4 > 0$. The adjacent-polynomial quadratic identity behind it was already in Ulas's paper and is re-derived, not claimed as new.

### 71. Two-coin counterexamples to generalized Shepp–Olkin thresholds

log-concavity-and-unimodality/bernoulli-entropy-quasiconcavity/article.pdf
from bernoulli_entropy_research.zip; batch 43: proveit_product_entropy.zip; batch 57: ProveIt_Alphabet_Entropy.zip

For every finite real $q > 1$, both the Rényi-$q$ and the Tsallis-$q$ entropy of a sum of independent Bernoulli variables fail even *quasi*concavity as a function of the parameter vector, already on the open square $(0,1)^2$ with two coins. This disproves the proposed thresholds — Rényi 2 and Tsallis $3.65986\ldots$ — adjoined to Conjecture 4.2 of arXiv:1503.01570 and restated in arXiv:1510.05390, Section 8. Part II, added in batch 43, settles the subunit joint-concavity question for the joint bit vector only: product Rényi entropy is concave in every dimension for $0 < q \leq 2$, and product Tsallis entropy is concave for every $0 < q < 1$ exactly when $n \leq 9$. Part III, added in batch 57, answers Part II's question on larger alphabets for full categorical simplices: for $0 < q < 1$ product Tsallis entropy on $\Delta_{k_1} \times \cdots \times \Delta_{k_n}$ is concave exactly when $\sum_i \Theta_{k_i}(q) \leq 1$ for an explicit alphabet profile $\Theta_k$; at $q = 1/2$ three factors with a common alphabet of $k$ symbols are concave exactly when $k \leq 17$; concavity for every choice of $n$ alphabets holds exactly when $nq \leq 1$; and for $1 < q < 2$ concavity reduces to positive semidefiniteness of an $n \times n$ compression matrix. For the ordinary sum Part I's question stays open; an external claim about it is recorded as unverified.

### 72. Nonreal roots survive in polynomial iterative equations

log-concavity-and-unimodality/nonreal-roots-in-iterative-equations/article.pdf
from nonreal_roots_research.zip; batch 41: ProveIt_Increasing_Spectral_Core.zip; batch 42: ProveIt_Orientation_Reversing_Spectra.zip; batch 54: ProveIt_Exact_Algebraic_Degree.zip

Here $P[f] = \sum_j a_j f^{\circ j}$ means a combination of *iterates*, not a polynomial in pointwise values. Gives a negative answer to Problem 6.1 of Draga–Morawiec (arXiv:1503.00570), which proposed deleting nonreal maximal-modulus characteristic roots: an explicit odd, strictly increasing, globally bi-Lipschitz homeomorphism realises a cubic whose nonreal root cannot be discarded. Adds a classification of the circular root spectra that occur. Part II, added in batch 41, classifies the

minimal iterative polynomials of increasing homeomorphisms with roots on several circles, through a universal spectral core and an affine rigidity theorem. Part III, added in batch 42, does the same for decreasing homeomorphisms with $P(0) \neq 0$: two-cycle and one-sided spectra, a decreasing core with one or two witnesses, and a finite-smoothness classification. Part IV, added in batch 54, answers Part III's question on algebraic degree: for a polynomial coordinate $H$ and $f = H \circ (a\,\mathrm{id}) \circ H^{-1}$ the degree of $f$ over $\mathbb{R}(x)$ is $\deg H / \deg W$, where $W$ is the largest normalized right compositional factor of $H$ with $W(at) = a^{\deg W} W(t)$, and it is exactly $\deg H$ for every coordinate of Part III's finite-smoothness theorem. For every odd $n \geq 5$ it gives real-analytic involutions of algebraic degree $n$, with distortion arbitrarily close to 1, whose graph extension has Galois group $S_n$, so they are not expressible by radicals.

**73. Canonical binomial decomposition does not preserve real roots**
`log-concavity-and-unimodality/binomial-decomposition-real-roots/article.pdf`
from `mu_welker_recursive_decomposition.zip + recursive_binomial_counterexample.zip`

A negative answer to Question 3.10 of Mu and Welker (arXiv:2503.24076v1) for their canonical decomposition $F(t) = G(t) + tH(t)$ built from the integer binomial expansion of every coefficient: real-rootedness of $F$ does not pass to the parts. Two cubic witnesses are carried, one with a repeated root and one with simple roots, each with its own digit table, discriminant pair and exact nonreal zeros; the report also gives the minimal witness within cubics, two all-degree thresholds neither of which implies the other, and a two-sided limit. *Merged; the two originals were two write-ups of one proof program, each already containing both witnesses with the headline and the co-star swapped.*

**74. A first-triple criterion for log-concavity of MDS weight distributions**
`log-concavity-and-unimodality/mds-weight-log-concavity/article.pdf`
from `mds_log_concavity_research.zip`

For $d = n - k + 1 \geq 2$, $k \geq 3$ and $q > d$, the compressed formal MDS weight distribution $(1, A_d, \ldots, A_n)$ is log-concave if and only if the single inequality $A_{d+1}^2 \geq A_d A_{d+2}$ holds — the whole sequence is controlled by its first triple.

**75. A uniform Hankel obstruction for higher-order Stirling subset polynomials**
`log-concavity-and-unimodality/stirling-hankel-obstruction/article.pdf`
from `stirling_hankel_obstruction.zip`

Proves the higher-order negative assertion of Deb–Sokal Conjecture 1.4(d), with a stronger statement at every positive shift: for $r \geq 3$ and $a \geq 1$ the $3 \times 3$ Hankel determinant $\det(s_{r,a+i+j}(x))$ of the row-generating polynomials fails the required positivity. Clause (d) of that conjecture; clause (c) is the subject of the higher-order Stirling rows entry.

### 76. Normalization by the matching number for weighted matching-support polynomials

log-concavity-and-unimodality/matching-rank-normalization/article.pdf
from batch 72: ProveIt_Weighted_Rank_Five_and_Two_Shore_Covers.zip, strictness_equality_package.zip, ProveIt_Weighted_Rank_Four_Stability.zip, ProveIt_Sharp_Full_Hall_Classification.zip, ProveIt_Unit_Weight_Rank_Normalization_Counterexample.zip, ProveIt_One_Shore_Counterexamples.zip, ProveIt_Sharp_Rank_38_Hall_Endpoint.zip, ProveIt_Sharp_Rank_38_Full_Hall_ULC.zip, ProveIt_Last_Hall_Gap_Separation.zip, ProveIt_Rank_Two_Matrix_Blocks_and_Conditioning.zip, ProveIt_Three_Vertex_Marginals_and_Star_Cores.zip, ProveIt_Rooted_Hall_Sectors_and_Cubic_Obstruction.zip, ProveIt_Rank_Six_Finite_Last_Gap.zip, ProveIt_Rank_Six_Second_Newton_Inequality.zip, ProveIt_Rank_Six_Common_Exterior_Theorem.zip, ProveIt_Rank_Six_Nested_and_Conditional_Results.zip, ProveIt_Rank_Six_Conditioning_for_Every_Core.zip, ProveIt_Rank_Six_Complete_Core_Cubic.zip, ProveIt_Rank_Six_Matching_Hole_Cores.zip, ProveIt_Rank_Six_Two_Complete_Rows.zip, ProveIt_Rank_Six_Two_Complete_Columns.zip, ProveIt_Rank_Six_At_Least_Six_Core_Edges.zip, ProveIt_Eventual_One_Shore_Scaling_Through_Rank_Seven.zip (with a second manuscript nested inside it), ProveIt_Eventual_Five_Plus_Three_Covers.zip, ProveIt_Four_Plus_Four_Top_Gap.zip, ProveIt_Eventual_One_Shore_Scaling_Through_Rank_Eight.zip; batch 73: matching-rank-four-package.zip, all-two-by-three-cores-result.zip, two-element-matroid-lorentzian-result.zip, sharp-weighted-rank-boundary-result.zip; batch 100: ProveIt_Rank_Six_Third_Newton_Inequality.zip (bundle Report 81)

Continues Parts XI–XIII of the preorder root-polytope report on whether the bipartite matching-support polynomial $p_G$, with positive vertex activities, is ultra-log-concave of the order of the matching number $\nu(G)$; merged from twenty-seven manuscripts, one of them delivered only inside another's archive. Part I proves it for arbitrary positive activities on both shores whenever some minimum vertex cover meets one shore in at most two vertices, through an explicit transversal-matroid lift, so for every graph with $\nu(G) \le 5$; with that report's rank-six Hall counterexample the least matching number of a weighted failure is exactly six. Equality in a normalized inequality occurs only for balanced star forests, covers with two vertices on each shore give real-rooted support polynomials, and the full Hall family is ultra-log-concave for all activities exactly when one core has at most two vertices. Part II answers the unit-weight and one-shore versions negatively: a connected all-unit graph with 6916 vertices and $\nu = 3450$ fails, as does a 125-vertex graph with $\nu = 50$ weighted on one shore only, and for one common activity on the right core of a full Hall graph the least failing matching number is exactly 38. Part III, at $\nu = 6$ with all left activities one, proves the second and last of the five inequalities for every graph (the first holds in general), and all five for several core families. Part IV proves eventual ultra-log-concavity, after a common scaling of the right part of a minimum cover, for every graph with $\nu(G) \le 8$ and unit left activities, and locates the least matching number of a persistent failure in $[9, 38]$. Results that the preorder report already proves are cited there, not reprinted, and theorems proved by several manuscripts are printed once with the other proofs as second routes. Part V, added in batch 73 from four manuscripts written in parallel with batch 72, shows that at unit weights and matching number three a nonreal zero needs fifteen vertices, attained by a unique graph; proves a two-element Lorentzian deformation of every matroid basis polynomial and an all-rank first-layer positive-semidefiniteness theorem (which proves Part III's three-vertex marginal theorem without its 5,339-monomial certificate); gives a stable seed for two-neighbour star cores; and shows by three 13-vertex examples that the two-tail lifts do not extend to three tails. What those manuscripts re-prove of Parts I and III is cited or kept as second routes; six further archives of the batch (two earlier editions, four re-proofs or special cases) were not placed. Part VI (batch 100), from one further manuscript (thirty-two in all), proves that for a genuine $3+3$ minimum cover with unit left activities the quartic $F_4 = 15s^4 + 10p_1s^3 + 6p_2s^2 + 3p_3s + p_4$, with every right variable retained, is Lorentzian, so the third inequality $9p_3^2 \ge 16p_2p_4$ holds for every graph of matching number six with one weighted shore; with Parts I and III only the fourth, $8p_4^2 \ge 15p_3p_5$, stays open there. The proof is computer-assisted (exact class reductions, Rayleigh identities and a Bernstein certificate, all in exact arithmetic). Also, with arbitrary positive left activities and every matching number

$r$, $G \succeq dd^{\mathsf{T}}/r$ on signed vectors, generalizing Part V's first-layer theorem by the same route; a rank-three $4 \times 6$ regular-minor matrix on which the corresponding matrix bound, normalized by rank, fails on a signed vector (no positive-field failure is inferred); and nonnegativity of the cubic core-restoration coefficients of the fourth gap. Its first-gap equality proof is a further second route of the preorder report's theorem. Retained-variable Lorentzianity for other cover orientations is not claimed; the source's full regeneration of its two largest certificate arrays was not rerun here. The independent check (`f8accf37c`) found the write's Remark 46.1 (a minimum cover with exactly two right vertices also gives a Lorentzian $F_4$) valid. Unrefereed; not formalized.

**77. Rank ultra-log-concavity of preorder gamma and directed support polynomials at the actual degree**

`log-concavity-and-unimodality/preorder-gamma-rank-ulc/article.pdf`

from batch 73: `preorder-degree-four-part-1-of-4.zip, preorder-degree-four-part-2-of-4.zip, preorder-degree-four-part-3-of-4.zip, preorder-degree-four-part-4-of-4.zip, preorder-gamma-degree-three-package.zip, template26-structural-supplement.zip, role-cover-package.zip, three-core-rayleigh-result.zip, physical-cover-three-result.zip, bipartite-and-pendant-support-result.zip, independent-role-degree-three-package.zip, universal-sink-last-gap-package.zip, universal-sink-all-degrees-package.zip, universal-sink-five-core-package.zip, universal-sink-five-core-boundary-package-part-1.zip, universal-sink-five-core-boundary-package-part-2.zip`; superseded, not printed: `weighted-degree-three-package.zip, universal-sink-degree-four-package.zip`; batch 77: `universal-sink-five-core-cubic-package.zip`

Answers Research question 96 of the preorder root-polytope report: is the gamma polynomial $\Gamma_D = \sum_k \gamma_k t^k$ of a finite preorder, with $\gamma_k$ counting ordered pairs of disjoint $k$-sets joined along arcs, ultra-log-concave after normalization by its *actual* degree? Part I proves it for every preorder of actual degree at most four with unit activities: degree three by a Gallai–Edmonds reduction and exact integer-population certificates (a nonreal-rooted cubic needs fifteen vertices), degree four by bounded cores, 76 four-attachment templates and 50 global binomial-square identities, with a smaller second route for the largest certificate. Part II treats independent tail and head activities on arbitrary loopless directed relations: real stability of the signed monomer polynomial under every two-by-two role cover (generalizing the $2 + 2$ theorem of the matching-rank report), coefficientwise Rayleigh inequalities for a three-vertex shore, rank ultra-log-concavity under every physical vertex cover of size three and on bipartite physical graphs, a pendant-head stability criterion, and the full statement for independently weighted preorders of actual degree at most three. Part III adds a cloud of universal sinks to an arbitrary directed core: a sharp last-coefficient bound for every core size, and the statement at every actual degree for four-vertex cores and at actual degrees four and five for five-vertex cores. The degree-three and degree-four theorems are computer-assisted. Merged from eleven manuscripts and a six-page supplement, from eighteen archives (three of them handed over from the matching-support cluster); two earlier versions are superseded and theorems proved more than once are printed once with second routes. Section 14 of Part III, added in batch 77 from a further manuscript, proves $\gamma_2^2 \geq 3\gamma_1\gamma_3$ on the face $\gamma_4 = 0$ for every loopless directed five-vertex core with universal sinks and independent activities, with the constant 3 sharp even for preorder cores, by a degree-drop reduction to a computer-assisted two-sink lemma with 7,901 rational square certificates; so five-vertex cores are rank ultra-log-concave at every actual degree. For preorder cores it is a case of Part II's degree-three theorem, so the new content is the non-transitive core; Section 12's counterexample to the constant 3 has $\gamma_4 > 0$. Some certificate data over 2 MB, and Section 14's 95.1 MB of certificate shards, which cannot be regenerated, stay in the arrival commits, and the report says how to retrieve or rebuild them.

## 9 Quaternionic analysis

One report, first merged from ten manuscripts and since extended by two more. The subject is new to this collection, and the delivery was unusual: one expository survey and nine independent attempts at the same follow-up problem, all written on the same day. The nine agree, which is

the substance of the entry.

### 78. Slice regularity and the global inverse Fueter problem

quaternionic-analysis/slice-regularity-and-fueter-inversion/article.pdf
from quaternionic_analysis.zip and nine companion archives; batch 39: ProveIt_Fueter_Polynomial_Periods.zip; batch 64: ProveIt_Fueter_Global_Inversion.zip

One article, first assembled from ten manuscripts. Part I is the delivered expository survey: slice regularity and Cauchy–Fueter regularity side by side, with Cauchy formulas, Taylor and Laurent expansions, the regular product and the geometry of zeros, the argument principle, Rouché, open mappings, Schwarz–Pick and an intrinsic Riemann mapping theorem. Parts II–XI answer the follow-up question the other nine manuscripts all attacked independently: for an axially monogenic $g = P + IQ$, when is there a single-valued slice-regular $f$ with $\Delta_4 f = g$? The answer is a two-period obstruction. The closed one-forms $\omega_g = \frac{1}{2}(-P\,dx + Q\,dy)$ and $\theta_g = \frac{1}{2}((xP + yQ)dx + (yP - xQ)dy)$ have vanishing periods exactly when a global primitive exists, and then $f(x + Iy) = (x + Iy)C + V$ with $dC = \omega_g$, $dV = \theta_g$, unique modulo $q \mapsto qa + b$; the monodromy is affine, and the cokernel of the Fueter map has real dimension $8m$. The point the nine exist to make is that $\theta_g$ is computed from the target, with no first primitive chosen. **Nine independent derivations reaching the same two forms, the same $8m$ and the same logarithmic energy constant is the evidence the entry rests on.** Where they differed it was almost always notation — thirteen conflicts are recorded and resolved in `MERGE_NOTES.md` — but two were mathematical and would have produced a false statement if the merge had been mechanical: the size law at a critical singularity has genuinely different sharp constants in two norms that four manuscripts both called $M_g$, and the conserved flux recovering the slope residue needs the weight $3(\operatorname{Re} q - u) + \operatorname{Im} q$ rather than $q - u$, the coefficient 3 being exactly what makes it right-Fueter-regular. All ten source manuscripts and all ten verification programs are kept; the deduplication is in the article. Parts XII–XIII, added in batch 39 from an eleventh manuscript, answer the report's open problem on higher dimensions in large part: in ambient dimension $2h + 2$ one polynomial closed form of degree $2h - 1$ (at $h = 1$ the form $\theta_g + t\,\omega_g$) gives a complete obstruction, with a constructive inverse, a cokernel of dimension $2h\,d_V\,m$, an exact residue filtration and sharp size and energy constants. Part XIV, added in batch 64 from a twelfth manuscript, carries this across the real axis and to planar domains of arbitrary, also infinite, connectivity: every admissible polynomial period character is realized, a continuous linear right inverse of the period map exists exactly when the admissible period rank is finite, and a continuous nonlinear section exists always.

## 10 Graph theory

### 79. Mutual-visibility sets in Andrásfai graphs

graph-theory/andrasfai-mutual-visibility/article.pdf
from andrasfai_visibility_research.zip

Proves the generating function attributed to Andrew Howroyd in OEIS A391632, $\sum a(n)x^n = x(4 - 19x + 33x^2 - 40x^3 + 8x^4)/(1 - 10x + 29x^2 - 32x^3 + 8x^4)$, by an exact weight-preserving bijection between mutual-visibility sets of the $n$-Andrásfai graph and anchored closed walks of length $3n - 1$ in a four-state labelled graph. Adds the full bivariate generating function, an exact binomial formula for every cardinality, maximum-set classification, eventual polynomiality at each fixed deficit, exponential asymptotics, a central limit theorem, and linear-time recognition and exact uniform sampling.

### 80. A bicyclic 30-vertex graph with domination root $-4$

graph-theory/domination-root-minus-four-order-30/domination_root_30.pdf
from domination_root_30_research.zip; batch 67: ProveIt_Domination_Cycle_Obstructions.zip

An explicit connected planar graph on 30 vertices and 31 edges, treewidth 2 and domination number 10, with $D(G, x) = x^{10}(x + 4)Q_{19}(x)$ — a negative answer to question 2 of Section 4 in Alikhani and Griswold, arXiv:2608.00109v1, which asks whether 33 is the minimum order.

Their 33-vertex example had already disproved the original integer-root conjecture; no claim of global minimality at 30 is made. Three independent exact polynomial computations agree. Part II, added in batch 67, proves $1 \le |D(G,-1)| \le 3^{\beta}$ sharply for cycle rank $\beta$, a simultaneous divisibility budget for integer roots and their multiplicities, and the exact value sets of $|D(G,-1)|$: $\{1,3,7,9\}$ for $\beta \le 2$, so no root $-6$ at cycle rank at most two, and a ten-element set for $\beta \le 3$ (computer-assisted). Part I's zero-$C$ four-cycle search forces $|D(G,-1)| = 3$, so it can produce no integer root below $-2$ other than a simple $-4$; no root $-6$ or $-8$ is constructed.

**81. A marked-cycle proof of the trapezohedral domination conjecture**
graph-theory/trapezohedral-minimal-dominating-sets/article.pdf
from trapezohedral_domination_research.zip, A381190_proof_package.zip

Proves the generating function labelled conjectural in OEIS A381190: for the $n$-trapezohedral graph, the number of dominating sets that are inclusion-minimal among all dominating sets and induce a connected subgraph is $a_n = 2n(2p_{n-2} + p_{n-3})$ with $p_m = p_{m-2} + p_{m-3}$ (Padovan). The proof classifies such sets by a cyclic arrangement of gaps of length 2 and 3 with a restricted marked edge. Adds the size distribution, strict log-concavity, an unbiased sampler, a Binet formula and a central limit theorem.

## 11 Enumerative combinatorics

**82. A Twelve-Value Proof of the A181280 Formula**
enumerative-combinatorics/a181280-binary-matrix-formula/article.pdf
from A181280_research_package.zip

The binary-matrix formula recorded as Conjecture 20 by Kauers and Koutschan (2023) and still marked conjectural in OEIS A181280: an explicit spectral recurrence bound and a finite certificate resting on twelve decisive counts, together with further consequences. `STATUS.md` records that general C-finiteness was already proved in Kauers's December 2025 lecture slides.

**83. Esthetic Numbers, Folded Paths, and Algebraic Diagonals**
enumerative-combinatorics/esthetic-numbers-and-folded-paths/article.pdf
from esthetic_walks_research.zip

The diagonal conjecture $T(q, q-2) = A182555(q-2)$ in OEIS A377000, where $T(q,k)$ counts base-$q$ positive integers of length $k$ whose adjacent digits differ by one. Proves it exactly, proves the entry's other two conjectures as retrieved, and derives formulas for every fixed-offset diagonal $q = k + d$.

**84. Diagonal Hardinian Arrays: A Fixed-Parameter Asymptotic Formula**
enumerative-combinatorics/hardinian-array-diagonal-asymptotics/article.pdf
from hardinian_arrays_research.zip

For every fixed $r \ge 0$, $H_r(n,n) \sim C_r 4^{rn} n^{-r(r-1)/2}$ with an explicit $C_r$ in Gamma values. The formula confirms the constants tabulated by Dougherty-Bliss and Kauers (2024) but refutes their proposed restriction to powers of 2, 3 and $\pi$: $C_7 = 2^{18}5^2/(3^{45}\pi^3)$. Adds an exact finite determinant at fixed array size, an exact enumeration algorithm, and a limiting negative-binomial law for total deficits.

**85. An explicit formula for lucky parking spots**
enumerative-combinatorics/lucky-parking-spots/article.pdf
from lucky_parking_spots_research.zip

Proves Conjecture 13 of Butler, Hadaway, Lenius, Martens and Moats (JIS 29 (2026), Article 26.1.1): a closed form for $C(n,j)$, the number of parking functions in which *spot* $j$ is lucky. Adds an exact reflection law, all fixed right-edge columns, a closed form for OEIS A374533, Poisson edge limits, uniform bulk convergence, polynomial reciprocity and asymptotic expansions.

### 86. A hexagonal-lattice proof of the A195806 conjecture

enumerative-combinatorics/a195806-hexagonal-lattice/article.pdf
from A195806_hexagonal_proof.zip

An elementary derivation of the six residue-class formulas and the constant-coefficient recurrence recorded as conjectural in OEIS A195806, starting from the original triangular-array constraints rather than from interpolation. The steps are an integral six-parameter bijection, four independent translation intervals, a dihedral chamber reduction, and an exact lattice count.

### 87. Leaves of large type in the numerical-semigroup tree

enumerative-combinatorics/numerical-semigroup-leaf-types/article.pdf
from leaf_type_counterexample.zip

Disproves Conjecture 4.4 of Chappelon, Ramírez Alfonsín and Stamate, *Numerical semigroup tree: type-representation* (arXiv:2507.15006v2), with an explicit genus-34 counterexample, and establishes a sharp asymptotic order for the type of leaves in that tree.

### 88. Strict growth and the Catalan limit for adjacency-bounded 132-avoiding permutations

enumerative-combinatorics/adjacency-bounded-132-avoiders/article.pdf
from strict_growth_132.zip; batch 38: Closing_Catalan_Growth_Gap.zip; batch 43: ProveIt_Largest_Jump_Research.zip; batch 52: ProveIt_Macroscopic_Jump.zip, ProveIt_Macroscopic_Jump_Research.zip; batch 64: ProveIt_Critical_Jump_Moments.zip; batch 68: ProveIt_Polynomial_Rarity.zip; batch 71: 132_Avoider_Crossover_Package.zip; batch 72: General_132_Avoider_Crossovers_Package.zip, Uniform_132_Avoider_Spectral_Bridge_Package.zip, Relative_132_Avoider_Asymptotic_Package.zip, Theta_132_Avoider_Boundary_Package.zip, Integer_Count_132_Avoider_Asymptotic_Package.zip, Uniform_132_Avoider_Saddle_Sum_Package.zip, Growing_Macro_132_Avoider_Counts_Package.zip

For 132-avoiding permutations with every adjacent difference at most $m$ and growth constant $\alpha_m$: proves $\alpha_{m+1} > \alpha_m$ for all $m \geq 1$ — the remaining strict-inequality part of Nadler's Conjecture 2, which A. C. Li announced publicly on 8 September 2026, eleven days before Part I's date (a dated note in Part I records the announcement; Part I's proof does not use it, and it is not a retraction) — and the refinement $4 - \alpha_m = (2\log m + 4\log\log m + O(1))/m$, with $2\log\pi - 4\log 2 \leq \liminf E_m \leq \limsup E_m \leq 2\log\pi$. Part II, added in batch 38, closes that window: a Catalan-weighted supersolution and a growing cutoff give $E_m \to 2\log\pi$, with an all-orders inverse-log expansion. Its numerics at $m = 10^6$ (near 3.15) are far from the limit and are not evidence for it. The all-$m$ component ordering is still not claimed. Part III, added in batch 43, describes the joint limit near the boundary $m = n - d$: the distance $n - M_n$ of the largest jump converges in total variation at rate $n^{-1/2}$ to an explicit algebraic law, uniformly in $d = o(n)$. Part IV, merged in batch 52 from two manuscripts, treats the bulk regime $n - m \asymp n$: $\sqrt{n}\,P(n - M_n > un) \to (3/\sqrt{\pi})(1-2u)/\sqrt{u(1-u)}$ for $0 < u < 1/2$ (and $\to 0$ for $u \geq 1/2$), so for $1/2 < \theta < 1$ the fraction of 132-avoiders of length $n$ with every adjacent difference at most $\lfloor \theta n \rfloor$ is asymptotic to $3(2\theta-1)/\sqrt{\pi\theta(1-\theta)n}$, and the equivalent $3/\sqrt{\pi(1-\theta)n}$, which Part III declined to assert, is false. It also gives the exact constant of every moment $E(n - M_n)^p$, $p > 1/2$ (for instance $E(n - M_n) \sim 3\sqrt{n/\pi}$); for $\theta \leq 1/2$ it proves only $o(n^{-1/2})$ and polynomial lower bounds. Part V, added in batch 64, settles the critical moment, $E\sqrt{n - M_n} = \frac{3}{2\sqrt{\pi}}\log n + c_\star + o(1)$ with an explicit $c_\star \approx -0.9421$, and gives the first correction to every subcritical moment. Part VI, added in batch 68, gives the order of Part IV's fraction for every fixed $0 < \theta < 1$: it is $\Theta(n^{-\lfloor 1/\theta \rfloor/2})$, with an order-crossover scale $n^{1-1/(2(q-1))}$ near each $\theta = 1/q$ (so the fraction is $\Theta(1/n)$ at $\theta = 1/2$). Part VII, added in batch 71, finds the first amplitude and crossover function: $n$ times the fraction at $\lfloor n/2 + t\sqrt{n} \rfloor$ tends to $6 + 12t_+/\sqrt{\pi}$, locally uniformly in real $t$, so the fraction at $\lfloor n/2 \rfloor$ is asymptotic to $6/n$; and for $1/3 < \theta \leq 1/2$ the fraction at $\lfloor \theta n \rfloor$ is asymptotic to $I_3(\theta)/(2\pi n)$, with an explicit two-dimensional integral $I_3$ and $I_3(1/2) = 12\pi$. Part VIII, added in batch 72, finds every remaining fixed amplitude and reciprocal crossover: for $1/p < \theta < 1/(p-1)$ the fraction at $\lfloor \theta n \rfloor$ is asymptotic to $(p+1)/(2^p \pi^{(p-1)/2})\, I_p(\theta)\, n^{-(p-1)/2}$, with $I_p$ an explicit integral over a capped simplex, so every fixed $0 < \theta < 1$ now has an equivalent; and near every $\theta = 1/r$ the fraction, multiplied by $n^{r/2}$, tends to an explicit constant plus a multiple of $t_+^{r-1}$ in the scaled

window variable $t$ (at 1/3: $5/\sqrt{2\pi} + (729\sqrt{3}/(4\pi))\,t_+^2$). Part IX, merged in batch 72 from six manuscripts, lets the cap grow with $n/m \to \infty$: a logarithmic bridge to the fixed-cap growth rate uniformly for $n \geq c\,m\log m$, one saddle sum on that whole range, explicit Gamma-sum equivalents near $n \asymp m\log m$ and for $1 \ll n/m \ll \log m$, and a theta-function modulation for $n/(m(\log m)^{3/2}) \to \infty$ that tends to one only when $n \gg m(\log m)^2$; the equivalents pass to the actual class when $n(\log m)^2/m^2 \to 0$.

### 89. The first pattern failure in 123-avoiding permutations

enumerative-combinatorics/a273821-first-pattern-failure/article.pdf

from batch 74: A273821_First_Pattern_Failure.zip; batch 98: First_Pattern_Failure_Beyond_132.zip (Part II)

For a uniform 123-avoiding permutation of $[n]$, let $K_n$ be the largest number of its largest entries whose induced permutation still avoids 132. Proves the bivariate generating function of the joint counts $T(n,k)$ marked conjectural in OEIS A273821, by inserting successive minima and telescoping a Catalan convolution to $T(n,k) = 2^{k-1}\binom{2n-2k}{n-k-1} - \binom{2n-k-1}{n-k-1}$; every column $k \geq 2$ is hypergeometric with growth rate 4, answering the entry's growth question. Also an explicit discrete limit law for $K_n$ with exponentially weighted total-variation error $O(n^{-1})$, a crossover at $k \asymp \sqrt{n}$, two exactly separated exponential sectors for $k$ proportional to $n$, and, under the weight $y^{K_n}$, a transition at $y = 2$ with critical law $K_n/n \to \mathrm{Beta}(1, \frac{1}{2})$. Part II (batch 98) answers the report's first research question for the patterns $1(r+1)r\cdots 2$, $r \geq 2$: an exact generating function with a Chebyshev denominator and a positive coefficient sum, growth rate 4 in every fixed column, the critical weight $y_c = \sec^2(\pi/(r+2))$, the subcritical law as a sum of independent geometric variables, and an iterated long-pattern limit $K_r^\infty/r^2 \Rightarrow \sum_j E_j/(\pi^2 j^2)$ with a Wasserstein bound; for fixed $r \geq 3$ a uniform two-index estimate, a heavy-tailed critical deficit with sharp total-variation rate and moment threshold 1/2, the supercritical tilt, a displaced coexistence window and local complex zeros. Its $r = 2$ cases re-prove Part I as second routes; the write showed the uniform estimate false at $r = 2$, so $r \geq 3$ is necessary. The safe counts are Chow–West's and Krattenthaler's (their theorem numbers checked by the independent check of Part II, made on 7 October, which found no error; the growth constant is Chow–West's Corollary 4.1); constants are not uniform in $r$, no joint $r = r(n)$ limit is claimed, and the zero theorem is local. Unrefereed; not formalized.

### 90. Throwback Dynamics with Repeated Entries

enumerative-combinatorics/throwback-queue-dynamics/throwback_research.pdf

from throwback_research.zip

For the queue process that removes the head and reinserts it at the zero-based position equal to its weight: every token recurs if and only if at most $m$ input tokens have weight at most $m$, for every $m \geq 1$, with an explicit description of the recurrent tokens when that fails. Adds exact cycle periods and frequencies, a coprimality criterion for perfect mixing, Catalan and prime-parking counts for recurrent cores, and the largest labelled period of a core of size $r$.

### 91. Reciprocity and Matrix Duality for Preorder Polytopes

enumerative-combinatorics/preorder-polytope-reciprocity/article.pdf

from preorder_reciprocity.zip

Proves Conjectures 7.2, 4.2 and 4.4 of Athanasiadis and Chapoton, *Polytopes and posets associated to preorders* (arXiv:2605.26916v1), and the ordinary $q = 1$ case of their Conjecture 4.9 (not its full $q$-refinement). Two mechanisms: Ehrhart–Macdonald reciprocity after subtracting the all-ones vector, extending the double identity to independently weighted capacities; and duplication of both colour classes of the bipartite graph followed by Postnikov's lattice-point formula, giving the conjectured matrix transpose.

### 92. Support reciprocity and the $q$-zeta polynomial of preorder lattice-point posets

enumerative-combinatorics/preorder-q-zeta-reciprocity/article.pdf

from preorder_qzeta_reciprocity.zip + preorder_q_zeta_reciprocity.zip + preorder_q_reciprocity_research.zip

Proves the full $q$-refinement of Athanasiadis–Chapoton Conjecture 4.9, equation (21): for a

preorder $\tau$ on $[n]$, with $P_\tau$ the nonnegative integer vectors satisfying $a(I) \le |I|$ on every order ideal and $B_\tau$ its maximal elements, $Z_q(P_\tau, [-1]_q) = (-1)^n \sum_{b \in B_\tau} q^{-|\mathrm{supp}(b)|}$, together with the stronger identity at each exact support. The $q = 1$ case was already covered by the preorder-polytope reciprocity report and is cited, not reclaimed. *Merged from three reports that reached the same two theorems by three architecturally independent routes — an incidence inversion with Boolean localization, the same with a Hall-based privatization and a self-contained Euler identity, and a capacity-enumerator reciprocity with a transfer involution — and all three survive.*

### 93. An explicit shelling for preorder lattice-point posets

enumerative-combinatorics/preorder-shellability/article.pdf
from preorder_shellability_research.zip

An affirmative solution to Question 4.6 of Athanasiadis and Chapoton (arXiv:2605.26916v1, p. 13): an explicit pure shelling of the order complex of $P_\tau$ with its minimum removed, and a coning that shells the order complex of $P_\tau$ itself. Both complexes are therefore Cohen–Macaulay over the integers and over every field. A different question of that paper from the three other preorder entries.

### 94. Multiple-chain preorder polytopes

enumerative-combinatorics/multiple-chain-exponential-formula/article.pdf
from multiple_chain_exponential_formula_research.zip + multiple_chains_research.zip

Proves the exponential generating-function identity proposed in Section 9.5 of Athanasiadis and Chapoton (arXiv:2605.26916v1) for lattice points of multiple-chain preorder polytopes counted by number of nonzero coordinates, in a rectangular generalization with independent block width and height. Adds a duality corollary, an Ehrhart evaluation, a degree-$2^d$ root theorem with a quartic elimination certificate, two treatments of the $\gamma$-vector, a Stieltjes moment representation, the boundary constant in two forms, and a central limit theorem with exact variance. *Merged; the identity is proved twice, once through a subset encoding with a bounded symbol and once through slack meanders in a Laurent ring, and the eleven deliberate redundancies are listed in the text.*

### 95. Rationality and growth of stabilized permutation-weight series

enumerative-combinatorics/stabilized-permutation-weights/article.pdf
from stabilized_permutation_weights.zip

Studies the full stabilized generating functions of the permutation-weight $q$-Eulerian polynomials, beyond their initial connection with OEIS A256193: an all-$d$ finite-spectrum theorem, rationality, a universal integer denominator, the exact radius of convergence and residue-class asymptotics. The two-descent sequence gets an explicit formula in powers of 3, powers of 2 and Fibonacci numbers.

### 96. A Reflexive Root-Polytope Model for Preorder $h$-Polynomials

enumerative-combinatorics/preorder-root-polytopes/article.pdf
from preorder_root_polytopes.zip; batch 39: ProveIt_Height_Two_Preorder_Research.zip; batch 42: ProveIt_Cactus_Rigidity.zip; batch 44: ProveIt_Preorder_Gamma.zip, ProveIt_Preorder_Gamma_Research.zip; batch 55: ProveIt_Lorentzian_Support.zip, ProveIt_Support_Polynomials_Research.zip, ProveIt_ADE_Stability.zip, ProveIt_Matroid_Lifts.zip; batch 63: ProveIt_Marked_Tree_Stability.zip; batch 64: ProveIt_Two_Flow_Lattice_Sampling.zip; batch 66: ProveIt_Random_Tree_Stability.zip; batch 68: ProveIt_Hall_Bottlenecks.zip; batch 70: ProveIt_Matching_Rank_Research.zip; batch 71: ProveIt_Weighted_Rank_Three_Leaf_Compression.zip

Proves Conjecture 5.1(c) of Athanasiadis and Chapoton (arXiv:2605.26916v1): for a preorder $\tau$ on $[n]$ the directed root polytope $C_\tau = \mathrm{conv}(\{\pm e_i\} \cup \{e_i - e_j : j \le_\tau i\})$ is reflexive with $h^*(C_\tau, t) = h_\tau(t)$, and any pulling triangulation of its boundary is combinatorially the boundary of a simplicial polytope with that $h$-polynomial. Palindromicity and unimodality, Conjecture 5.1(a,b), follow, as do the full $g$-theorem conditions. The higher-dimensional augmented bipartite identity and preorder duality are credited to Dai, Hou, Liu, Thawinrak and Wang (arXiv:2608.16037v2). Part II, added in batch 39, treats height-two preorders through matching polynomials: $\gamma$-positivity in height two (Conjecture 5.2 there), a family showing that real-rootedness (Conjecture 5.3) fails — its first counterexample is credited to a MathDB posting by Shivam Patel — and crown

obstructions: an induced crown forces large minimal nonfaces in every lattice triangulation of $C_\tau$, so the boundary complex is not flag when the crown has at least six elements. Flag realizability in general (Conjecture 5.1(d)) remains open. Part III, added in batch 42, shows that exactly the cacti admit a faithful unit-phase minor model, a property strictly stronger than real stability. Part IV, merged in batch 44 from two manuscripts, proves Conjecture 5.2 for every finite preorder: $\gamma_k$ counts ordered pairs of disjoint $k$-sets joined by a $\tau$-increasing bijection, and a reflexive relation gives a palindromic polynomial exactly when it is transitive. Part V, merged in batch 55 from two manuscripts, shows that for a bipartite graph with shores of sizes $m$ and $n$ every support enumerator divided by $\binom{\min(m,n)}{k}$ is log-concave (the unweighted inequality is Theorem A of Röhrle and Ulirsch, arXiv:2402.15317, carried through the support-counting bridge), so every $h_\tau$ and every height-two $\gamma$-vector is ultra-log-concave; it classifies the palindromic support polynomials of all bipartite graphs, gives FPRASes for partition functions and for single coefficients, and shows exact counting #P-complete already in height two. Part VI, added in batch 55 with the stability sections of a further manuscript, shows that stability of the signed support polynomial is block-local, holds for the incidence–apex graph of a tree exactly when the tree is a finite or affine ADE tree, and is classified for books and bipartite multi-theta graphs; the books show it strictly stronger than real-rootedness. Part VII, added in batch 55, shows that the transversal matroid of a preorder determines the preorder up to isomorphism and that height-two preorder matroids generate exactly the gammoids under minors. Part VIII, added in batch 63, classifies stability for one apex adjacent to an arbitrary set of vertices of a tree (a corollary of Part VI, also proved directly), with exact $O(n^5)$ enumeration, optimization and repair of the stable apex neighbourhoods, a sharp diameter window, and real-rootedness of their size-counting polynomial exactly for paths. Part IX, added in batch 64, gives explicit polynomial-time two-flow bijections between the demand vectors of a bipartite graph with prescribed support and the supplier sets matchable to it (the correspondence itself is Oh's), and with them an almost-uniform sampler of actual lattice points of every finite preorder polytope. Part X, added in batch 66, answers Part VIII's question for a uniform random labelled tree with every vertex marked independently with probability $p$: the probability that the marking is stable is asymptotic to $\frac{9}{8}\sqrt{2\pi}\,p^3n^{5/2}e^{-pn}$ whenever $p\sqrt{n}\to\infty$, an explicit entire function governs the window $p\sqrt{n}\to\lambda$, and below it stability is asymptotically the event of at most two marks; for a fixed host the diameter determines the probability up to a polynomial factor, separating the typical rate $\log(1-p)$ from the annealed rate $-p$. Part XI, added in batch 68, answers Part V's question on normalization by the matching number negatively for positive vertex weights on both shores: a graph with 30 vertices, 104 edges, matching number eight and weights in $\{1,500\}$ has a weighted support polynomial that is not ultra-log-concave of order eight. A Hall-bottleneck family gives a rank-six phase diagram, the endpoint threshold for every core size and a lower bound asymptotic to $\frac{4}{3}\nu$ for any normalization depending on the matching number $\nu$ alone; the first rank-normalized inequality holds for every positively weighted graph, with a defect identity. Part XII, added in batch 70, answers the unit-weight version affirmatively for matching number at most three, by a sum-of-squares identity, and for a connected family of matching number four and disjoint unions, leaving unit weights from matching number four on and the one-shore version open. It also proves the first rank-normalized inequality for weighted directed relations, gives a $9\times 9$ rank-three bimatroid obstruction, exact neighbourhood-type kernels with an exact support sampler, and for every finite preorder the first gamma-vector inequality $\gamma_1^2\ge\frac{2d}{d-1}\gamma_2$, with $d$ the gamma degree, which also bounds the cost of computing the gamma vector. Part XIII, added in batch 71, merges leaves with a common single neighbour into one leaf of summed weight without changing the support polynomial, and so proves normalization by the matching number for arbitrary positive weights on both shores whenever a minimum vertex cover is split $1+(\nu-1)$ between the shores, hence for every weighted graph of matching number at most three. That left a positively weighted counterexample matching number four, five or six, with every minimum vertex cover meeting both shores in at least two vertices, and at matching number four, with or without weights, only covers split $2+2$. Batch 72 continued this line in a separate report, *matching-rank-normalization* (log-concavity section), to which dated

notes here now point: the unit-weight and one-shore versions fail (at matching numbers 3450 and 50), and the least matching number of a weighted failure is exactly six. Batch 73 opened a second separate report, *preorder-gamma-rank-ulc* (log-concavity section), which answers this report's Research question 96, rank ultra-log-concavity of the gamma polynomial at its actual degree, for unit activities through degree four and for independent activities through degree three. Conjectures 7.2, 4.2 and 4.4 of the same paper are proved in the preorder-polytope reciprocity report.

### 97. Largest Simplex Faces under Ordinal Sums

enumerative-combinatorics/
simplex-faces-of-order-and-chain-polytopes/Largest_Simplex_Faces.pdf
from Series_Parallel_Simplex_Faces_Research.zip

Answers the concluding open question of Freij-Hollanti and Lundström, *Simplex inequalities of order and chain polytopes of recursively defined posets* (Order 43 (2026), article 31). For $s_O(P)$ and $s_C(P)$, the largest dimensions of simplex *faces* of the order and chain polytopes: both are computable in $O(n)$ on any series-parallel decomposition, the order case carrying eight normalized endpoint states. For an ordinal sum of antichains, $s_O = r + \nu$ with $\nu$ a maximum $b$-matching of the layer path under capacities $\min(a_i - 1, 2)$, and $s_C = r + \max(0, k - 1)$. The extremal gap is sharp: $0 \le s_C - s_O \le \lfloor (n-2)/3 \rfloor$, attained for every $n \ge 2$.

### 98. Skew partitions, continued fractions and connected components

enumerative-combinatorics/skew-partition-continued-fractions/article.pdf
from A225114_proof_bundle.zip

Proves the repeated-power continued-fraction formula recorded as conjectural in OEIS A225114, which counts skew partition diagrams with no empty row or column. Unique concatenation gives $A(q) = 1/(1-P(q))$ for the edge-connected diagrams, and a boundary-path encoding followed by a weighted two-coloured Motzkin-path decomposition gives $P(q) = q/(1-2q-q^3/(1-2q^2-q^5/(1-2q^3-\cdots)))$, whence $A(q) = 1/(2-C(q))$ with $C = 1+P$. Adds a four-variable refinement, a formula by height, width and component count, a sharp truncation bound, exponential asymptotics and a central limit theorem for the number of components. **Explicitly not a novelty claim**: a prior-art check found Pasechnik's MathOverflow sketch of ten September 2024, and the entry credits Kurkov; the report says so first and credits both.

### 99. From mesh avoidance to Catalan inflation

enumerative-combinatorics/mesh-avoidance-catalan-inflation/article.pdf
from A289587_proof.zip, OEIS_A289587_proof.zip

Proves Scheuerle's generating-function conjecture in OEIS A289587 for permutations avoiding 321 and the mesh pattern $(12, 174)$, and the equal count for $(12, 234)$ by inversion: with $D = x^4 - 2x^3 - 5x^2 - 2x + 1$, $P = x^4 + 4x^3 + 11x^2 + 10x + 3$ and $Q = x^2 + 5x + 3$, the generating function is $(P - Q\sqrt{D})/(8x(1+x)^2)$. The headline structural result is the exact *occurrence count* $\mathrm{occ} = \binom{f}{2} + \sum_U \binom{|U|}{2}$, of which avoidance is the case $\mathrm{occ} = 0$. Merged from two reports that reached the identical generating function by the identical run-contraction substitution $u \mapsto u/(1+xu)$; the merge keeps *both* refinements, because they refine the same sequence by different statistics — excedances and records — and are genuinely different theorems, agreeing only in total. The two local predicates they are built on disagree on 31 of the 321-avoiders of length at most six, the smallest being 12, and the merged article proves where they coincide rather than assuming it.

### 100. Prescribed coefficient-lattice minima

enumerative-combinatorics/coefficient-lattice-minima/article.pdf
from coefficient_lattice_minima_research.zip

Settles the unnumbered realization conjecture after Lemma 1 of O'Bryant, *On Nathanson's Triangular Number Phenomenon* (arXiv:2506.20836): every nondecreasing list of even integers is the list of successive $L^1$-minima of the coefficient lattice of some finite integer set. The witness is the mixed-radix set $A = \{0, 1, h_1, h_1h_2, \ldots\}$, whose minima are exactly $(2h_1, \ldots, 2h_r)$ with $r+2$ distinct elements, and the radices may be given in any order.

### 101. Coefficientwise magic positivity through dimension ten

enumerative-combinatorics/magic-positivity-parking-polytopes/article.pdf
from magic_positivity_research.zip

A polytope is *magic positive* when its Ehrhart polynomial has nonnegative coefficients in the basis $t^j(t+1)^{n-j}$. Proves, in exact arithmetic, that the generalized parking-function polytope $X_n(b)$ is magic positive for every dimension $3 \le n \le 10$ and *every* positive integer parameter vector, uniformly in $b$ rather than on a sampled box — a partial solution of Conjecture 8.1 of Hill, Luo, Trinh and Vindas-Meléndez (arXiv:2607.15503v1). Dimension three is proved by printed algebraic formulas; the first-coefficient harmonic-number formula and the endpoint identities hold in every dimension. The conjecture in unrestricted dimension remains open.

### 102. Monotone blocks and reverse replies

enumerative-combinatorics/reverse-reply-avoidance-game/article.pdf
from reverse_reply_solution.zip

Proves Conjecture 2.9 of Ulfarsson (arXiv:2603.16004): for every $k \ge 3$ and every $n \ge R(k) = k^4 - 2k^3 - 3k^2 + 5k + 5$, the second player wins the normal-play permutation-avoidance game on $S_n$ with pattern length $k$ by always replying with the reversal, so the Sprague–Grundy value is 0. The proof is elementary, with no computer search: a minimal-support lemma isolates one pattern inside a family built from singletons plus one inflatable monotone interval, and inflation restores the original length so the reply is legal. $R(k)$ is asserted sufficient, not minimal.

### 103. Nonintegral tropical degree from two horizontal segments

enumerative-combinatorics/tropical-hilbert-nonintegral-degree/article.pdf
from tropical_hilbert_counterexample.zip

Computes the strict-witness tropical Hilbert function of Grigoriev (arXiv:2404.06440v2) exactly for a two-segment set: $\mathrm{TH}_{V_d}(k) = k + \min\{k, \lceil k/d \rceil\} + 1$. Taking $d = 3$ gives tropical degree $4/3$ — not an integer — and a quasipolynomial of least period 3, so eventual polynomiality fails. Every rational in $(1, 2]$ is realized, denominators are unbounded, and for $d = p/q > 1$ in lowest terms the least eventual period is exactly $p$. Both bounds are proved by hand.

### 104. Finite-prefix corrections for polyomino growth

enumerative-combinatorics/polyomino-growth-finite-prefix-corrections/article.pdf
from ProveIt_Polyomino_Research.zip; batch 40: ProveIt_Spectral_Saturation.zip

Continues the formal project `Combinatorics/Polyominoes/KlarnerConstant`, whose Lean-verified bound is $\lambda \le 9047/2000 = 4.5235$ for Klarner's constant. Proves a finite-prefix correction theorem for proper positive polynomial recurrences and applies it to Bui's seventeen-neighbourhood system with an exact marked-occurrence table through size 18, giving the computer-assisted, unformalized bound $\lambda \le 2249/500 = 4.498$ and $A_n \le (927884613/10^9)(2249/500)^n$. A second theorem gives balanced integer-monomial divergence certificates, complete for productive, strongly connected, nonlinear systems, and yields $4.52349 \le \mu_{\text{original}} \le 4.5235$ for the original recurrence — a statement about the recurrence, not a lower bound on $\lambda$. At intake, a fresh build of the delivered enumerator (the same program, not an independent one) regenerated the size-18 table and matched it exactly. Part II, added in batch 40, shows where the method stops: exact spectral saturation in the irreducible critical case, a square-root law, and an all-prefix barrier $\mu_N \ge 4.3149$ for Bui's map — a limit of this method, not a lower bound on $\lambda$ and not an improvement on 4.498.

### 105. Three consecutive square-pyramidal numbers: exact Frobenius formulas

enumerative-combinatorics/a069762-pyramidal-frobenius/article.pdf
from batch 75: OEIS_Pyramidal_Frobenius.zip

For $s_n = n(n+1)(2n+1)/6$, gives the Frobenius number $F_n$ of $\langle s_n, s_{n+1}, s_{n+2} \rangle$ (OEIS A069762, which had no formula) exactly: a quintic quasipolynomial of least eventual period six, $F_n = n^5/9 + 8n^4/9 + O(n^3)$, valid for every $n \ge 2$ apart from six exceptions, the last at $n = 23$. The proof is an explicit Apéry normal form after two gcd reductions. Also the exact genus, quasipolynomials for every gap-power sum, a triangular limit law $2(1 - x)$ for a uniform gap divided by $F_n$, the minimal generating-function denominator with an order-18 eventual recurrence,

and a convergent Puiseux inverse. Eventual quasipolynomiality itself is Roune–Woods' and Shen's theorem; the claim is the explicit formula. The inversion instantiates the transseries volume's Lagrange–Bürmann and staircase theorems, which the report cites.

### 106. A pole hierarchy for valley-monotone compositions

enumerative-combinatorics/valley-monotone-bargraphs/article.pdf
from batch 75: OEIS_Valley_Monotone_Bargraphs.zip; batch 98: OEIS_Bargraph_Critical_Windows_and_Extremes.zip (Part II)

Proves Conjecture 2.3 of Flórez, Ramírez and Villamizar (JCTA 2024): the number $d(n)$ of compositions of $n$ with weakly increasing interior valley heights (non-decreasing bargraphs by area) satisfies $d(n) = C\lambda^n(1 + O(\theta^n))$ with $\theta < 1$, $\lambda = 1.97642067794583\ldots$ and $C = 0.60390151906189\ldots$ certified by rational interval arithmetic; the digits printed in 2024 are the finite ratio at $n = 100$. The input is the stacks generating function of OEIS A001523; $d(n)$ itself has no verified OEIS number. Also infinitely many simple real poles increasing to 1, so the generating function is not D-finite, a finite-pole expansion to every radius below 1, a central limit theorem for bottom-level valleys, a terminal-component law and the asymptotics of the pole accumulation. Whether the real poles exhaust the spectrum is left open. Part II (batch 98) answers the extreme-height half of Part I's question 10.4: the tallest column $H_n$ satisfies $P(H_n \le h) = \exp(-\alpha n\rho^h)(1+O(\log n/n))$ uniformly in the central window, $\alpha = 0.45020217246\ldots$, with lattice limits along phase subsequences and no single continuous Gumbel limit, asymptotically independent of the Gaussian bottom-valley count. Weighting bottom-level valleys by $u$ makes their count geometric below, uniform after scaling at and Gaussian above the exact critical weight $u_c = \rho_2(1-\rho_2)(1-\rho_2^2)/(1+\rho_2^4) = 0.12218100900\ldots$ ($\rho_2$ the second real pole), with a uniform $1/n$ critical window to all orders, Dirichlet area fractions at several simultaneous critical levels, an exact $q$-product identity for the height tail, a fixed-order expansion of the height-cutoff pole, a Dirichlet mixture of Gumbel-type maxima at criticality, and Lambert-$W$ inverses that instantiate the transseries volume. Five statements the manuscript only sketched are proved in the write; the independent check found them valid and widened the range on which one lemma's proof gives uniqueness. The joint normal law for the additive statistics of question 10.4 stays open, and Part II's decimals are uncertified diagnostics. Unrefereed; not formalized.

### 107. Bridgeless toroidal maps

enumerative-combinatorics/a343093-bridgeless-toroidal-maps/article.pdf
from batch 86: oeis_advances.zip (its Section 2)

Proves Peter Bala's posted conjecture for OEIS A343093, the rooted bridgeless maps on the torus with $n$ edges: $\sum_{n\ge2} b_n t^n = \left(\frac{1}{4}\sum_{n\ge1}\binom{4n}{n}t^n\right)^2$, equivalently $U^2/(1-3U)^2$ with $U = t(1+U)^4$, from a bridge decomposition with the root accounted for (a re-proof of Courtiel–Yeats–Zeilberger's) and Bender–Canfield–Robinson's genus-zero and genus-one series. Also two positive finite formulas (the numbers count ordered pairs of binary words; no bijection with maps is claimed), an algebraic equation of degree four, and $b_n = (256/27)^n[1/24 - 7\sqrt{6}/(216\sqrt{\pi n}) + O(n^{-3/2})]$ with the $n^{-3/2}$ term. The manuscript's other two subjects are Part IV of `a088714-bell-scale-growth` and Part II of `a301981-unitary-divisor-partitions`; its statement that the Bell normalization of A088714 remains open is contradicted by another manuscript of the same batch. A bounded search is not a priority claim, and nothing was submitted to OEIS.

### 108. A finite positive amplitude for stressed Kunz words

enumerative-combinatorics/numerical-semigroup-stressed-amplitude/article.pdf
from Report274_Stressed_Kunz_Finite_Amplitude.zip (Report 274, arrival 764740f07, outside the numbered batches)

Numerical semigroups by genus in Kunz coordinates. Claims that the stressed words on $\{1,2,3\}$ satisfy $s_g = C_s\rho^g + O(\rho_2^g)$ with $0 < C_s < \infty$, $1/\rho$ the root of $(x^2+x^3)(x+x^2+x^3) = 1$, so that Zhu's polynomial-prefactor formulation (his Conjecture 7.3) holds for $s_g$ with exponent 0, against his numerical expectation of 1.6–1.7; also a two-pole expansion for depth at most three, a shrinking inverse bracket and the vanishing of every fixed inverse-power correction. The proof is self-contained: an exact boundary renewal, an activity transformation, a loop-aware container lemma,

dense-window standardization, rational certificates and early-one suppression. *Not independently verified:* the write and the independent check each read every proof and found no error, but the constants are existential and give no onset, Zhu's counts to genus 95 and the boundary series are still pre-asymptotic, and the source itself asks for an independent check, recorded as a further question; the check split Zhu's counts exactly into the separated and the early-one classes. It sits beside `numerical-semigroup-leaf-types` (the same objects, no shared theorem); Reports 270 and 271, which it names, were not delivered. Unrefereed; not formalized.

# 12 Generating functions and asymptotics

## 12.1 Coefficient counts of binary products

Three reports on counting coefficients of $\prod_i (1 + x^{G_i})$. The first two answer, independently, the universal-rationality part of Richard Stanley's MathOverflow question 431075 ("A conjectured rational generating function", 2022), which had no posted answer when checked on 19 September 2026. A third answer used the same mixed-base construction and the same enumerative argument as the first, and has been merged into it.

### 109. A Mixed-Base Counterexample to Stanley's Rationality Question

`generating-functions-and-asymptotics/binary-product-coefficient-counts/mixed-base-counterexample/article.pdf`
from `stanley_mixed_base_counterexample.zip + Stanley_Rationality_Counterexample.zip`

With $G_{2j+1} = 2^j$ and $G_{2j+2} = 3^j$ the exponent generating function is rational, while $\nu(2n) = 2^{n-\lfloor (n+1)\alpha \rfloor}$ for $\alpha = \log_3 2$ counts the coefficients equal to one in $\prod_{i \le N}(1 + x^{G_i})$ and its generating function is not rational, not D-finite, not P-recursive, and has a natural boundary. Classifies rationality in the family interleaving $a^j$ and $q^j$ with digit polynomial $1 + y + \cdots + y^{a-1}$ for all $2 \le a < q$ — Stanley's general $r = a - 1$ — as holding exactly when $q^u = a^v$. Adds a finite-quotient rigidity theorem, an abstract natural-boundary theorem for $\sum q^{\lfloor n\theta \rfloor} t^n$ at every irrational $\theta$, the exact limit set of $a_n/\lambda^n$, and a streaming recurrence making the point that non-P-recursive does not mean hard to compute. *Merged; the enumerative half was the same argument in both originals, but the two routes to non-rationality differ and both are kept — an elementary finite-state reduction mod p, and the rigidity route that reaches the stronger transcendence conclusions.*

### 110. Rational exponents, nonrational coefficient counts

`generating-functions-and-asymptotics/binary-product-coefficient-counts/recursive-exponent-construction/research.pdf`
from `stanley_counterexample.zip`

A different negative answer, from the recursion $a_0 = 2$, $a_1 = 1$, $a_n = a_{n-1} - 2a_{n-2}$, with $b_n = T_{n-1} + 2^n - 1 + a_n$, $T_n = T_{n-1} + b_n$, $G_{2n-1} = 2^{n-1}$ and $G_{2n} = b_n$: the counting series has a natural boundary on the unit circle, hence is neither rational, algebraic nor D-finite.

### 111. Coefficient residues in generalized Fibonacci products

`generating-functions-and-asymptotics/binary-product-coefficient-counts/kbonacci-coefficient-residues/article.pdf`
from `stanley_kbonacci_research.zip + quasifibonacci_parity_research.zip`

Settles Conjectures 6.2 and 6.3 of Stanley, *Theorems and Conjectures on Some Rational Generating Functions* (arXiv:2101.02131v3, pp. 23–24). For $k$-bonacci seeds each exceeding the sum of their predecessors, the number of odd coefficients of $\prod_{i \le n}(1 + x^{w_i})$ has generating function $(1 + 2z^k)/(1 - 2z + 2z^k - 2z^{k+1})$, independently of the seeds; the proof builds periodic signs of block length $k + 1$ and block product $-1$ and shows every signed finite product is flat. *Merged; the Conjecture 6.3 halves proved the same statement on the same domain, and the surviving report adds Conjecture 6.2 and the asymptotic analysis.*

## 12.2 Single-sequence OEIS studies

One hundred and forty-four reports on the growth or the closed form of a single OEIS sequence. One article counts the terms in the derivatives of three power towers — $x^{x^x}$, $x^{x^2}$ and $x^x$ — whose three parts reach three different answers by three different recurrences, and two reports added in batch 72 sit beside it: one proves the leading constant $\frac{1}{2}$ for $x^{x^a}$ at every integer $a$, hence for $x^x$ and $x^{x^2}$, and the other studies the $x^{x^2}$ coefficients at unbounded logarithmic deficit. Two more reports share a Bell-scale saddle-point method on unrelated sequences. Shared technique is not shared theorem, and none of these duplicates another. Thirteen reports came in batch 73, one in batch 74, one in batch 75, thirty-three in batch 77, five in batch 85, one in batch 91, seventy-eight in batches 100–113 and one in batch 138; they are the last one hundred and thirty-three entries below, in order of placement, the batch-77 ones grouped by the cluster that placed them (ascent sequences; iterated Bell numbers, cone walks and divisor weights; trees and networks; words, permutations, partitions and placements; moments, renewals and moving parameters). Four batch-73 reports merge two manuscripts that proved the same theorems independently on the same day (A139217, A022629, A215561, A279619): the shared theorems are printed once and the other proofs kept as marked second routes. Batch 75 brought a third and a fourth independent proof of the A022629 theorems, and a manuscript about two thirds of which re-proves results of the A215561 report; those duplicates are tabulated in the reports, not reprinted, and only their new results were added. Batch 77 did the same with a fifth independent proof of the A022629 theorems, a fourth of the A215561 ones, a third of the A279619 constant and a second of the A126764 corrections. Several of its new reports join a manuscript with addenda that nest it byte for byte; each paper is printed once. The four partition reports of batches 73–74 (A022629, A033552, A097356, A238016) study different products in different saddle regimes and share no proposition, nor do the batch-77 partition reports (A174065, A291698, A292507, A301746, A301981), which share at most a method with them; the A189281 and A330266 reports share a method but no theorem. The two batch-85 manuscripts on matrix compositions proved the same core theorems independently and are one report, printed in one notation with a dictionary; batches 85 and 86 also added Parts to the A189281, A000571, A088714 and A301981 reports. Batches 100–113 are clusters of one session bundle of numbered Research Reports (arrival `60f54ea06`): four new reports in batch 100, one in 101, three in 102, four each in 103, 104 and 105, seven in 106, five in 107, ten in 108, eight each in 109 and 110, five in 111, seven in 112 and eight in 113; many bundle Reports instead became Parts of reports above (batches 100–103 extended fifteen of them, batch 108 the A261781 report and batch 110 the A082528 report, after batches 97 and 98 had extended ten), and their re-proofs are printed as marked second routes. Two manuscripts of 7 October from outside the bundle were placed as Parts of the A373271 and A328716 reports. Batch 138 added Parts to the A308338, A094149 and A290354 reports and placed the A275672 report without a write. Where several bundle Reports on one sequence were merged, the entry names the base and every archive. Many of these manuscripts invert their asymptotics by Lambert $W$ and rounding; those steps are instances of the inversion theorems of the transseries volume in `Analysis/Transseries`, which the reports cite rather than claim.

### 112. Bell-scale growth of OEIS A088714

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a088714-bell-scale-growth/article.pdf`
from `A088714_asymptotic_growth.zip; batch 73: golden_moment_laws.zip; batch 76: A088714_Bell_Local_Estimates.zip; batch 86: oeis_advances.zip (its Sections 3-6), oeis_research_bundle.zip (its Part II)`

For the unique formal series with $A(x) = 1 + xA(x)^2 A(xA(x))$, proves $\log a_n = n\log n - n\log\log n - n + O(n\log\log n/\log n)$, so $(e\log n/n)\, a_n^{1/n} \to 1$ and the ordinary generating function has radius of convergence zero while the exponential one is entire. The lower bound comes from a single explicit path through an exact triangular recurrence, the upper from a Stirling-positivity comparison with derivatives of Touchard polynomials. The multiplicative equivalent is *not* claimed, and the

finer Bell-number form is labelled a conjecture with a sufficient ratio condition isolated. Part II, added in batch 73, treats A088714 and its companion A088713 as moment sequences: determinate moment measures, strict Hankel positivity, golden-ratio endpoint laws with exponents $(3-\sqrt{5})/2$ and $(\sqrt{5}-1)/2$, the ratio law $a_{n+1}/a_n \sim n/\log n$ and the far tail. Its re-proof of Part I's theorem is replaced by a pointer; the finer Bell conjecture stayed open there. Part III, added in batch 76 from a continuation manuscript: Part I's formal endpoint coefficient expansion, proved uniformly for $m \le C\log n$; the one-sided correction $\liminf(r_n - n/W(n)) \ge 3/2$ for $r_n = a_n/a_{n-1}$ (the conjecture needs 2); and a reduction of the finer Bell conjecture to one scalar defect estimate $E_n = O(W(n)^q/n^2)$, sufficient and in some form necessary, with a tilted-variance criterion, the complementary-endpoint bound $5/(r_n r_{n-1})$ and exact numerics to $n = 1600$. Its re-derivation of Part I's summability step is a pointer; $W(n)$ enters only as the Bell saddle of the transseries volume, which it cites. Parts IV and V, added in batch 86 from two manuscripts whose other subjects are the new report `a343093-bridgeless-toroidal-maps` and Parts II of the A301981 and A321941 reports. Part IV: both moment measures have strictly positive real-analytic densities on $(0,\infty)$ with full support and exact density and derivative equivalents at zero; the correction to $F(s) \sim s^{-\alpha}$ is a nonzero real-analytic antiperiodic profile in $\log_\varphi \log s$, certified nonzero by an interval computation, with a convergent expansion in $1/\log s$; and every fixed-shift expansion $c_n = \sum_{j\le K} \gamma_j a_{n-1-j} + O(a_{n-K-2})$ of the companion A088713. Part V: the complementary endpoint to every fixed order, which proves Part III's missing estimate; bounded factors and $r_n = n/W(n) + O(1)$; and the finer Bell conjecture (Conjecture 8.1), $a_n = C_* \mathcal{N}_n(1 + O(W(n)^5/n))$ with $0 < C_* < \infty$ unidentified, so $a_n \sim C_* B_n e^{W(n)^2+3W(n)}$ — proved there, not yet independently reviewed (the other batch-86 manuscript still calls it open). Its proof of the companion expansion is a second route to Part IV's, and its normalization of $c_n$ by the same $C_*$ is new. Part I's pointwise ratio expansion (8.7) stays open.

### 113. The exponential growth constant of OEIS A158415

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a158415-growth-constant/article.pdf`
from `A158415_asymptotic_growth.zip`

For the number of distinct values expressible with $n$ symbols from $E ::= 1 \mid \sqrt{E} \mid (E+E)$, proves a computable $\gamma > 1$ exists with $a(n)^{1/n} \to \gamma$ and $1.86204 < \gamma < 1.9208$, gives an effective characterisation as a supremum of algebraic finite-data bounds, and proves the boundary identity $\sum a(n)\gamma^{-n} = \gamma^2$, which rules out a pure $C\gamma^n$ prefactor. The rigorous lower bounds rest on a uniform injectivity theorem: any finite alphabet of representable values can serve as nested-radical digits, via an odd valuation at a prime above 2. No closed form for $\gamma$ is claimed, and terms 22–28 get certified intervals rather than exact values.

### 114. Dyadic scaling rigidity of 3AP-free permutations

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a003407-dyadic-scaling-rigidity/article.pdf`
from `ProveIt_Dyadic_Scaling_Rigidity.zip` (batch 40); batch 43: `ProveIt_3AP_Nonholonomicity.zip`

For $\theta(n)$, the number of permutations of $[n]$ with no three-term arithmetic progression (OEIS A003407), studies the dyadic scaling of $\theta(n)$: a continuous log-periodic growth profile with its exact fundamental period, rigidity of the profile under real rescaling, explicit losses and gains of the 2:1 gluing, and the empirical and sampling laws; the adjacent ratios $\theta(n+1)/\theta(n)$ have no limit. The parity recurrences and Sharma's Theorem 2.8, which the manuscript imports, are already proved in Lean in `Combinatorics/Ramsey` (the table $\theta(1..20)$ there rests on a `native_decide` certificate); the report's own theorems are not formalized. Part II, added in batch 43, proves $\theta$ is not P-recursive and its generating functions are not D-finite (the ordinary one is transcendental), using the published counts $\theta(64)$, $\theta(75)$, which nothing in ProveIt certifies; whether the generating function has a natural boundary stays open.

### 115. An exact coefficient formula for Ramanujan's theta function

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a260306-formula-proof/a260306_proof.pdf
from A260306_formula_proof.zip

Proves both finite formulas recorded in OEIS A260306, for every coefficient index, for the expansion $\theta(N) \sim \sum c_n N^{-n}$ defined by $\sum_{k<N} N^k/k! + \theta(N)N^N/N! = e^N/2$. The route is an exact Laplace-integral difference, an analytic change of variable with $F(\psi(w)) = w^2/2$, and Lagrange inversion reducing each $c_n$ to one coefficient extraction, then a finite change of basis into the entry's double and triple sums. Includes the analytic error estimate the identification needs, rather than only checking the two sums agree, and identifies A065973 as the reduced denominators.

### 116. The maximal positive coefficient of a Hermite polynomial

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a277280-hermite-maximum/article.pdf
from A277280_asymptotics.zip

An exact closed form for OEIS A277280, the largest coefficient of the physicists' Hermite polynomial $H_n$ with signs retained: with $k_n = \lfloor (2n + 6 - \lfloor \sqrt{8n + 21} \rfloor)/8 \rfloor$ and $d_n = n - 4k_n$, the exponent $d_n$ is the unique maximizer and $a(n) = n!\, 2^{d_n}/((2k_n)!\, d_n!)$. The distinction from the largest coefficient in absolute value is kept explicit throughout.

### 117. Asymptotics of OEIS A277364

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a277364-bell-asymptotics/A277364_asymptotics.pdf
from A277364_asymptotics.zip

For $a(n) = \sum_{k \le n/2} S(n,k)$, partitions of an $n$-set into at most $\lfloor n/2 \rfloor$ blocks, proves $a(n) \sim B_n$ with an explicit first correction: writing $r = W_0(n)$, $a(n) = \exp\{n(r - 1 + 1/r) - 1\}(1+r)^{-1/2}\,(1 - P(r)/n + O(r^2/n^2))$ for an explicit rational $P$. Also gives a parity-split equivalent for the omitted tail $B_n - a(n)$.

### 118. Term counts in the derivatives of power towers

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
power-tower-derivative-term-counts/article.pdf
from A293239_research_report.zip, A290268_partial_results.zip,
A281434_exact_algorithm_and_cubic_growth.zip; batch 42: ProveIt_A290268_Finite_Certificates.zip;
batch 70: ProveIt_A290268_Depth3_Article.zip; batch 72:
ProveIt_Lehmer_Comtet_Prime_Multiple_Nonvanishing.zip,
ProveIt_Lehmer_Comtet_Stirling_Residues_and_Candidate_Lists.zip,
ProveIt_Lehmer_Comtet_Uniform_Padic_Nonvanishing.zip, ProveIt_A290268_Depths_Five_Six.zip,
ProveIt_A290268_Count_Bounds.zip; batch 73: ProveIt_A290268_Eight_Depths.zip,
ProveIt_A290268_Depth7.zip

One article on three OEIS sequences that ask the same question of three functions: how many distinct monomials are in the fully collected $n$th derivative of $x^x$ (A293239), $x^{x^2}$ (A290268) and $x^{x^x}$ (A281434)? Part I proves once the apparatus all three had established separately — the canonical differential-polynomial form $f^{(n)} = fP_n$ or $fx^{-n}P_n$ with its first-order integer recurrence, the independence of logarithmic monomials that makes the count an invariant of $f$ rather than of a differentiation algorithm, the identity $a(n) = |\operatorname{supp} P_n|$, and the envelope bound. Two normalizations are needed, because neither clause covers all three in the frames the parts use. **None of the three headline questions is fully settled, and the article says so in its abstract**: for $x^x$ the count reduces exactly to zeros of the first-kind Lehmer–Comtet triangle A008296 and $a(n) = \Theta(n^2)$, but the closed formula needs an isolated nonvanishing step — and separately the recurrence printed in the entry for $n > 6$ is *disproved* at $n = 8$; for $x^{x^2}$ the conjectured expression is proved to be an upper bound and $a(n) = \Theta(n^2)$, but the article does not prove the constant $\frac{1}{2}$ (since 1 October 2026 the sibling report *power-tower-exponent-supports* does, for $x^x$ as well); for $x^{x^x}$ explicit bounds give $a(n) = \Theta(n^3)$ but not $a(n) \sim \frac{2}{3}n^3$, and there the obstruction is not a nonvanishing statement but a density one, since further zeros provably exist. A closing part sets the three side by side and records what they share beyond the framework:

every cancellation any of them can prove is a parity zero of the same falling factorial. This is a thematic consolidation, not a deduplication: the three were checked and are not duplicates, and all fifty numbered results of the three originals are preserved unchanged. Batch 42 added to the $x^{(x^2)}$ part depth certificates that close the nonvanishing problem at logarithmic deficits 3 and 4 for every $n$ and improve its lower bound on the count; the open region was then deficit at least 5. Where that manuscript re-derived results already in the report, it is printed as a second route. Batch 70 added a second proof of the deficit-three case (no zeros beyond the forced holes), by a harmonic sign law and positivity propagation in $k$ with its own finite certificate; that sign law proves a sign change the batch-42 data had only observed. Its manuscript's claim that deficit three was still open is corrected in the text. Batch 72 added three manuscripts to the $x^x$ part: Lehmer–Comtet nonvanishing on the prime-multiple offsets $\mu(p-1)$, $p > \mu$, under explicit conditions (in full for $\mu = 2, 4$, and for $\mu = 3, 5$ at odd $p$), Stirling residues that leave one candidate zero on $(p-1)^2$, and exact $p$-adic valuations on two families; the closed formula stays open. It added two to the $x^{x^2}$ part: deficits five and six are classified by exact moment seeds, so the open region was then deficit at least 7, every fixed deficit reduces to a singly exponential rectangle, and $a(n) \geq \frac{3}{8}n^2 + \frac{n}{2}\log n - \frac{n}{2}\log\log n - \frac{5}{4}n$ for $n \geq e^{10}$, still the best explicit bound. Re-derivations of earlier results are printed once and credited. The two sibling reports of the same batch, *power-tower-exponent-supports* and *a290268-unbounded-deficits*, are not part of this article; dated notes in it point to them. Batch 73 added two independent manuscripts to the $x^{x^2}$ part: deficit seven is classified twice, by three-anchor certificates of 51,699,091 and 48,113,025 cells, and deficit eight once (552,465,284 cells); at both only the reflection holes vanish, so the open region is now deficit at least 9. Their re-derivations of the deficit-five and -six classification, the largest-root minorant and weaker cutoffs and count bounds are printed as second routes or remarks.

### 119. Derivative supports of $x^{x^a}$

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/power-tower-exponent-supports/article.pdf`

from `batch 72: ProveIt_All_Integer_Exponent_Derivative_Supports.zip, ProveIt_A293239_Leading_Asymptotic.zip, ProveIt_A290268_Leading_Asymptotic.zip, ProveIt_Half_Exponent_Derivative_Supports.zip, ProveIt_Algebraic_Exponents_and_Cancellation_Frequencies.zip (with a second manuscript nested inside it), ProveIt_Irrational_Exponents_Effective_Cancellation_Bounds.zip; batch 100: ProveIt_Half_Exponent_Hermite_and_Finiteness.zip (bundle Report 59)`

Let $A_a(N)$ count the nonzero monomials $x^{a\ell-N}(\log x)^k$ in the $N$th derivative of $x^{x^a}$ divided by $x^{x^a}$. For every fixed positive integer $a$, proves $A_a(N) = N^2/2 + o(N^2)$, by saddle contours in the coordinate $\tanh(t/2)$, a convex-envelope coverage argument and an integer-triangle estimate; for $a = 1$ this is $a(N) \sim N^2/2$ for OEIS A293239 (density zero of the zeros of the Lehmer–Comtet triangle), and for $a = 2$ it is $\lim a(N)/N^2 = \frac{1}{2}$ for OEIS A290268, the constant the power-tower term-count report leaves open. The same leading asymptotic holds for $a = 1/2$, with two infinite families of cancellations off the reflection line. On a fixed factor defect $r$ it proves finiteness of the cancellations off an explicit reflection line for rational $a \neq 1$ ($r \geq 4$) and for algebraic $a$ of degree $D$ ($r \geq 4D + 2$), at most $(2\lfloor r/2 \rfloor)^2 - 1$ cancellation cells, effectively, for irrational $a$ and none for transcendental $a$, and exact square-root counting laws for rational $1/2 < a < 1$ on defects two and three. Every asymptotic statement is ineffective. Parts I–III are merged from seven manuscripts: the integer theorem was proved three times, and the separate proofs at $a = 1$ and $a = 2$, more explicit at those exponents, are printed as second routes. Part IV (batch 100), an eighth manuscript, shows for $x^{\sqrt{x}}$ that on each fixed defect $r$ the scaled residual tends to the Hermite polynomial $\mathrm{He}_r$ with an explicit first correction of relative order $N^{-1/2}$; for large $N$ it has $r$ simple real roots with explicit shifts, so at most $\lfloor r/2 \rfloor$ cells vanish per large order and $O_r(\sqrt{X})$ through order $X$. Also a Weierstrass model $V^2 = U^3 - 122400U + 9415000$ of the defect-four quartic and the count $X + (1 + 1/\sqrt{3})\sqrt{X} + O_R(1)$ of cancellations through order $X$ on the defects $0 \leq r \leq R$. Its residual, odd factor, Laguerre edges, defect-two and -three counts and both finiteness theorems re-derive Parts I–III at $a = 1/2$ by the same route and are printed as second routes; its contribution sentence, fair at its pin, is kept with a dated

correction, and its README's "integrality preserved" for the elliptic model is false read literally. No effective threshold or finite zero list on any defect $r \geq 4$. The write's outlook that the leading Hermite term holds for every $0 < a < 1$ was qualified by the independent check (`0c81c51e2`): its window $k = (2 - 1/a)N + O(\sqrt{N})$ meets the physical cells only for $1/2 \leq a < 1$. Unrefereed; not formalized.

### 120. OEIS A290268 at unbounded logarithmic deficit

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a290268-unbounded-deficits/article.pdf

from batch 72: ProveIt_A290268_Explicit_Finite_Ratio.zip, ProveIt_A290268_Airy_Fold.zip, ProveIt_A290268_Arithmetic_Phase.zip, ProveIt_A290268_Sparse_Cancellations.zip, ProveIt_A290268_Pointwise_Three_Halves.zip, ProveIt_A290268_Strict_Quadratic_Gain.zip, ProveIt_A290268_Direct_Contour.zip, ProveIt_A290268_Global_049_Bound.zip, ProveIt_A290268_Growing_Logarithmic_Degrees.zip (superseded, not printed)

Studies the integer coefficients $H_{d,k,q}$ behind the term count of the $N$th derivative of $x^{x^2}$ when the logarithmic deficit $d$ grows with $N$. A regular saddle branch ending at a fold $\kappa_c \approx 0.0281461$ carries a uniform phase law that proves eventual nonvanishing at every algebraic limiting ratio $k/d \to \kappa < \kappa_c$, with possible zeros only near explicit transcendental rays; an additive Airy law holds in the window $|k - \kappa_c d| = O(d^{1/3})$. Two-dimensional cosine laws with explicit phase curvatures give density one with $O(N^{5/3})$ exceptions on tail patches and positivity on the triangle $40k + 420q \leq 91d$; block congruences modulo primes, prime-unit families with exact valuations, all-depth central strips and an example showing that root geometry alone cannot prove the support conjecture complete it. The manuscripts' headline lower bounds ($(3/8 + c)N^2$, $3N^2/8 + CN^{3/2}$, $\liminf a(N)/N^2 \geq 28176/57247$) are kept in a table as superseded by $a(N) = N^2/2 + o(N^2)$ from the companion report on derivative supports; the phase laws behind them are not implied by it. Merged from eight manuscripts with shared material printed once; a ninth is wholly implied by the base and is described, not printed. The support conjecture stays open.

### 121. Iterated sums and products from one

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a352969-correctness-and-growth/article.pdf

from A352969_correctness_and_growth.zip

For $S_0 = \{1\}$, $S_{n+1} = (S_n + S_n) \cup (S_n \cdot S_n)$ and $a_n = |S_n|$ (OEIS A352969), proves correctness of the listed terms and the doubly exponential law $2^{2^{n-r_n-4}} \leq a_n \leq 2^{2^{n-1}}$, hence $\log\log a_n = n + O(\sqrt{\log n})$. The step is simultaneous, not repeated closure — a distinction the report is careful about.

### 122. Distinct third derivatives of a power tower

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a199085-exponentiation-derivative-values/article.pdf

from A199085_generating_function.zip

For OEIS A199085, the number of distinct values taken by the third derivative at $x = 1$ over all fully parenthesized exponentiations of $n$ copies of $x$, proves $a(1) = a(2) = 1$ and $a(n) = \lfloor (n^2-2)/3 \rfloor$ for $n \geq 3$, with the rational generating function $z/(1-z) + z^3/(1-z)^3 - z^7/((1-z)^2(1-z^3))$. The proof is for all $n$, not an extrapolation: every attainable pair of second and third derivatives is classified and a witness constructed for each admissible pair. Despite the shared vocabulary this is a different question from *power-tower-derivative-term-counts*, which fixes the function and varies the derivative order; here the order is fixed at three and the parenthesization varies.

### 123. Finite-fringe rigidity in congruence-restricted greedy dissociated sequences

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a139217-greedy-dissociated-fringes/article.pdf

from batch 73: finite_fringe_oeis_A139217_A139218.zip + ProveIt_OEIS_A139217_A139218.zip

OEIS A139217 and A139218 choose greedily the least positive integer congruent to 1, respectively 2, modulo 3 that keeps the set dissociated (all subset sums distinct). After three terms the signed subset-sum difference set is an interval with a fixed finite fringe of holes, {3} for A139217

and $\{1, 3, 6, 11\}$ for A139218, and a general propagation theorem keeps it so forever. Hence exact closed forms, for instance $a(n) = \frac{3}{4}\,2^n + 1$ for $n \equiv 0, 1$ and $\frac{3}{4}\,2^n - 2$ for $n \equiv 2 \pmod 3$ for A139217 ($n \geq 3$), rational generating functions and inverse expansions; both satisfy $a(n) = a(n-1) + a(n-2) + 2a(n-3)$, for A139217 only from $n = 6$, so the entry's range "$n > 4$" fails at $n = 5$. *Merged from two manuscripts that proved the same theorems independently on the same day; each shared theorem is printed once and credited to both, the second proof kept as a second route, and the cross-identities, remainder bound and tables that only one has are added.*

### 124. Five OEIS asymptotic conjectures for distinct-partition norms

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a022629-distinct-partition-norms/article.pdf`

from `batch 73: OEIS_Distinct_Partition_Norms.zip + A022629_Research_Package.zip; batch 75: A022629_research_package.zip, OEIS_Norm_Weighted_Partitions.zip; batch 77: oeis-fixed-power-asymptotics-result.zip; batch 100: ProveIt_Power_Weighted_Distinct_Partitions_Relative_Expansions.zip (bundle Report 89, base of Section 21) + ProveIt_OEIS_A022629_Asymptotics_and_Inversion.zip (bundle Report 82), ProveIt_Critical_Theta_Power_Weighted_Distinct_Partitions.zip (bundle Report 93)`

For $a_\alpha(n) = \sum N(\lambda)^\alpha$ over partitions $\lambda$ of $n$ into distinct parts, $N(\lambda)$ the product of the parts, proves the logarithmic growth formulas marked conjectural in A022629, A092484 and A265840–A265842, and with $K = \sqrt{2n}$, $r = \log K$, $c = \pi^2/(6\alpha^2)$ the expansion $\log a_\alpha(n) = \alpha K(r - 1 + c/r + c/r^2 + \cdots)$ to every order; a separate local-limit argument gives multiplicatively accurate saddle asymptotics with all Edgeworth orders, since exponentiating a truncated logarithm does not. Also a Lambert-$W$ inverse, a Gumbel law for the largest part, asymptotic to $(1+1/\alpha)\sqrt{2n}$, and the modulus of the first Fourier resonances. *Merged from two independent manuscripts with identical coefficients (one treats A022629, the other's $\alpha = 1$): shared results printed once, with the second proofs, an explicit differential-operator derivation and the results only the A022629 manuscript has.* Two batch-75 manuscripts, the third and fourth independent proofs of these theorems with identical coefficients, are recorded in one table rather than reprinted; their new results form Sections 15–19: sharp signed cumulant scales, a saddle remainder of every fixed order with first correction $\sim -3/(8\alpha K r)$, localization of the first index attaining a large value to within $1 + o(1)$ of an explicit inverse, eventual strict log-concavity and fixed-degree Jensen hyperbolicity; and the convolution powers $\prod(1 + k^\alpha q^k)^\mu$ (A297321, A022630, A022631) with $\mu$-independent coefficients, two further inverse coefficients and an exact check of the eighth forward coefficient. A batch-77 manuscript is a fifth independent proof of the core theorems (all logarithmic orders for $\alpha = 1$), again tabulated, not reprinted. Its new results form Section 20: the slot family $\prod(1 + k^\alpha q^k)^{k^\beta}$, $\alpha > 0$, $\beta \geq 0$ an integer, with two logarithmic orders, an exact-saddle expansion to every fixed order, a threshold inverse and strict monotonicity, which for $\alpha = \beta = 1$ proves Kotěšovec's conjectured leading term for OEIS A266891; and resummed "thermal" forms of the logarithm and the inverse, no sharper than the two-term truncations. Sections 21–22 (batch 100) come from three manuscripts of one author series. Two of them, for every real $\alpha > 0$ (the base) and for A022629, are the sixth and seventh proofs of the core theorems, again tabulated; Section 21 adds the forward and inverse coefficients $P_9$, $P_{10}$, $\Delta_9$, $\Delta_{10}$, the smooth extension to $t = 0$ of the lattice-minus-integral difference with coefficients $\xi_r(\alpha)$ holomorphic on $\Re\alpha > 0$ (for instance $\xi_0(1) = \frac{1}{2}\log 2\pi - 1$, $\xi_1(1) = 7/12 - \gamma$), an all-orders relative expansion keeping the exact integral, and an exact-range inverse that recovers the index by rounding at sequence values. The third, Section 22, lets $\alpha$ grow so that $\Lambda = \alpha(s-1)/M^{1/3}$ stays in a compact subset of $(0, \infty)$ ($t$ the exact saddle, $M = e^s$ its transition site; $\alpha \asymp (2n)^{1/6}/\log n$) and proves $a_\alpha(n) = e^{F(t)+nt}(2\pi B)^{-1/2}\{\Theta(\mu_R, Q_\perp) + o(1)\}$ with a theta multiplier, $\mu_R$ the mean number of parts at least two and $Q_\perp \to \pi^2/(3\Lambda^3)$ its variance after regression on the size, so that single-saddle dominance fails there; also a theta-adjusted inverse with error $o(M^{2/3})$. It answers in part the report's projects on multi-saddle expansions and growing powers; it is the Fourier-dual form of the lattice-peak transfer theorem of `a357825-theta-ballot-power-sums`, not a corollary (two of its hypotheses are unproved for this row), and it does not explain the fixed-$\alpha$ finite-size errors, which an uncertified cell decomposition places in the Fourier cells near $\pm 2\pi/M$. No effective onset,

convergence, optimal truncation or uniformity as $\Lambda \to 0$ or $\infty$; the two re-proofs' claims to settle or strengthen the posted conjectures are read as further proofs. The write completed two proofs; the independent check (`91ef1ef6f`) found its five additions valid and corrected one order claim: the theta shift of the threshold is of order at most $1/t$, and of exact order $1/t$ only away from two phases per period. Unrefereed; not formalized.

### 125. Three exceptions before every square

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a097356-sqrt-restricted-partitions/article.pdf

from batch 73: OEIS_SquareRoot_Partition_Research.zip; batch 74: A097356_Sharp_Envelopes_Transseries_Inversion.zip

OEIS A097356 counts partitions of $N$ into parts at most $\lfloor\sqrt{N}\rfloor$. Proves that the posted lower constant 0.0881548837… is exactly the limit inferior, while the literal eventual lower bound fails at precisely three indices in every square block, the three integers before the next square. The engine is a uniform all-orders expansion of $p_m(\alpha m^2 + \sigma m + \tau)$ with a finite Gaussian coefficient operator; for A097356 it carries a phase factor $\rho^{\{\sqrt{N}\}}$, giving the cluster set, limiting distribution, summatory law and an inverse with $O(1)$ index error once a piecewise-linear phase map is added to the Lambert-$W_{-1}$ core. A206227 and A206240 are covered too; the leading asymptotic is Szekeres's. A second derivation from batch 74, written independently with equal constants, is merged: its proofs are second routes, and its own results (exact shell-polynomial degrees, second-order envelope constants, the exact phase loss, a Fourier identity and an $O(N^{-1/2})$ rate for the limiting law) form an added section.

### 126. Partitions into Catalan numbers

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a033552-catalan-partitions/article.pdf

from batch 73: Catalan_Partitions_Research.zip; batch 98: catalan_partition_laws.zip (base) + Catalan_Partition_Statistics_Research.zip (Part II)

For OEIS A033552, partitions whose parts are distinct Catalan numbers, a complete expansion in inverse powers of a logarithmic size parameter with a nonconstant periodic amplitude, an explicit first correction and an inverse expansion for the threshold at which $p(n)$ reaches a given value, from a contour-controlled saddle estimate, a Barnes $G$ product identity and a Fourier series in $\Gamma(s)\zeta(1+s)$; also a phase-dependent limit law for the largest part index. The entry poses no asymptotic conjecture, and the methods are classical. Part II (batch 98) merges two independent manuscripts answering Part I's questions 10.4 and 10.5, with the more general one as base: the exact total-variation threshold for independence of the upper multiplicities from geometric variables; a corrected edge law $P(J_n - M \le h) = G_h(1 + C_h/m) + O(m^{-2})$ for the largest index, proved in both and printed once, with growing offsets, two-sided tails, all moments and periodic first corrections; for the number of parts, $t_n N_n$ tends to an explicit series $W$ of scaled exponential variables with no $1/m$ term in the mean ($S_1 = 1 + 4\pi/(9\sqrt{3})$); joint limits with the upper process; the law of $W$, whose tail series has the unit circle as natural boundary, with an inverse tail; a Brownian-bridge mass profile and a lacunary universality class. The write proved one manuscript's unproved covariance candidate, $m\,\mathrm{Cov}(t_n N_n, J_n) \to -S_1\Xi(\theta)$; the independent check found it valid and sketched a sharper remainder. No claim of either manuscript was found wrong; the conditioning framework is classical (Arratia–Tavaré, Fristedt), and no exponentially small sectors or interval certificates are claimed. Unrefereed; not formalized.

### 127. Central Poupard numbers

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a125054-central-poupard-numbers/article.pdf

from batch 73: Central_Poupard_OEIS.zip

For $A_n(c) = \sum_k \binom{n}{k} c^{n-k} T_k$ with $T_k$ the tangent numbers, $A_n(1)$ is OEIS A125054. Proves the continued fraction recorded there as a conjecture in December 2025 and $A_n(1) \equiv 3 \pmod 9$ for $n \ge 1$, by formal continued fractions. Analytically: an exact convergent decomposition into odd-pole exponential sectors, a uniform all-order endpoint–saddle expansion for $c = \tau n^2$ with a

first-order transition at $\tau_* \approx 0.2624665$ and a displaced finite-size window, mixture and Gaussian laws, located polynomial zeros, the same transition for moments of any density with tail $Cy^\beta e^{-\alpha y}$, and a Lambert-$W$ inverse.

### 128. Theta laws for high-power ballot sums

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a357825-theta-ballot-power-sums/article.pdf

from batch 73: OEIS_Theta_Ballot_Sums.zip; batch 98: ProveIt_OEIS_Lattice_Peak_Transfer_2026-10-05.zip (Part II)

For $a_n = \sum_h B(n,h)^n$, with $B(n,h)$ the ballot paths of length $n$ ending at height $h$ (OEIS A357825), the normalized ratio whose non-convergence the entry records is a Jacobi theta function at a drifting parity-sensitive phase times $e^{-5/6}$. Proves a uniform all-orders expansion in theta derivatives, the exact lower and upper limiting constants, the cluster interval and phase law, ratio and Turán asymptotics and eventual strict log-convexity; the multiset counterpart differs by an explicit exponential sector of relative order $n^3 2^{-n}$, and for $\sum_h B(n,h)^k$ transitions occur at $k \asymp n$ and $k \asymp n^{3/2}$. Bala's supercongruence stays open. Part II (batch 98, a late arrival) answers the report's question Q6 in a sufficient-hypothesis form: a general lattice-peak transfer theorem gives, under a local quadratic law, a Gaussian majorant and interior completeness, $S_n(r_n)/\Omega_n^{r_n} = \vartheta(q_n, \delta_n) + o(1)$ for power sums of a positive array on a moving lattice, with cluster sets, every fixed order, a multidimensional version and a first-order inverse. The write weakened the majorant to an exponential one that log-concavity supplies, completed two proofs, and proved that the ballot rows and the Mahonian rows of `a380274-mahonian-growing-powers` are instances, the latter by reorganizing that report's proof. The manuscript's identity for $\sum u^3 e^{-qu^2/2}$ drops a term and is corrected on record, and its claim that Part I's all-orders coefficients are "precisely" its own holds at first order only; the independent check fixed two wordings. The hypotheses are sufficient, not necessary, and nothing arithmetic is claimed. Unrefereed; not formalized; Part II does not say whether an AI assistant was used.

### 129. Fixed-composition excursions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a215561-fixed-composition-excursions/article.pdf

from batch 73: oeis_balanced_excursions.zip + late_growing_excursions.zip; batch 75: OEIS_Fixed_Content_Ballot_Words.zip; batch 77: oeis-ballot-asymptotics-20261001.zip; batch 101: A215561_Fixed_Alphabet_Asymptotics.zip (bundle Report 90, base) + A215570_Balanced_Ballot_Asymptotics.zip (bundle Report 86) (Part VI)

For words with $n$ copies of each of $r$ letters whose every prefix sum is at most its average (OEIS A215561), proves the fixed-row conjecture $A_r(n) \sim \kappa_r (rn)!/(rn\,(n!)^r)$ with $\kappa_r$ a positive algebraic number given by a finite kernel-root product, a complete expansion in inverse powers of $n$, D-finiteness of every row with a transcendental ordinary generating function for $r \geq 3$ (the specific A215570 recurrence is proved in Part IV, batch 75), a contact law and a Lambert-$W$ inversion. Part II holds the second manuscript's own results: the leading term for every finite labelled step set, a second all-orders route, second-order terms for $r = 2, 3$, a giant primitive component, eventual log-convexity and a rounding bound. *Merged from two manuscripts that proved the main theorems independently, with constants agreeing to the last digit; shared results printed once and credited to both, the different routes kept as second proofs.* Part IV, added in batch 75 from a third manuscript, proves Kauers–Koutschan's Conjecture 15, the order-three recurrence for A215570, by a positive Legendre-coefficient formula and a rational telescoping certificate printed in full, and computes the coefficients $\alpha_{5,2}, \ldots, \alpha_{5,8}$ of the five-letter row exactly. About two thirds of that manuscript re-proves results of Parts I–II, its row theorem only with relative error $O(n^{-1/5})$; those results are listed in one comparison table, not reprinted. Part V, added in batch 77 from a fourth manuscript written without knowledge of this report, re-proves the main theorems with the same constants, sextic, Hessian and $\alpha_{5,1}$; these are tabulated, not reprinted. It adds the first correction $\alpha_{4,1} = -\frac{37}{200}\varrho^3 + \frac{11}{40}\varrho^2 + \frac{13}{40}\varrho - \frac{259}{400} \approx -0.5097846849$ of the four-letter row A215562, with $\varrho = \varphi - \sqrt{\varphi}$ and $\varphi$ the golden ratio, the Hessian $H_4$, an operator-norm second route to the

global-arc bound and a nested-logarithm inverse for every row; $\alpha_{4,2}$, $\alpha_{4,3}$ and the rows $r = 6, 7, 8$ stay open. Part VI, added in batch 101 from two bundle reports also written without knowledge of this report, proves Theorems 3.1–3.3 a fifth and a sixth time (every fixed alphabet with the two extreme steps unmarked; five letters with the zero steps deleted); these are tabulated as second routes. New in them: $\alpha_{5,2} = \frac{36}{25} - \frac{63\sqrt{5}}{125}$ computed without the recurrence by the Gaussian-moment rule on the zero-free walk, agreeing with Part IV; the leading constant of the zero-free count $A_5^\circ(n)$; $A_5(n+1) \geq 35A_5(n)$, which the write extends to $A_r(m+n) \geq A_r(m)A_r(n)$, so every row with $r \geq 3$ increases strictly at every index; for odd $r$ a zero-step identity $A_r(n) = \binom{rn}{n} A_r^\circ(n)$; and a closed Hessian determinant. Report 86's "lattice factor 4" is shown not to be an instance of the lattice-peak transfer theorem of `a357825-theta-ballot-power-sums` (no growing exponent; it depends on the coordinates), and the two sources' "no proof found" and "still unproved" are corrected by dated notes; nothing in them was found wrong. The independent check of the write found every item valid and refined three wordings. Unrefereed; not formalized.

### 130. An exact gamma constant for an Apéry-like sequence

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a279619-level-seven-gamma-constant/article.pdf`

from `batch 73: A279619_Gamma_Asymptotics.zip + A279619_exact_asymptotics.zip; batch 77: level7-asymptotics-replay.zip; batch 97: A279619_Hankel_Positivity.zip (Section 17)`

Proves the asymptotic displayed in OEIS A279619 with the constant identified: $a_n \sim \sqrt{3\pi}\, 27^n/(\Gamma(\frac{1}{7})\Gamma(\frac{2}{7})\Gamma(\frac{4}{7})\, n^{3/2})$, with $\sum a_n/27^n = 3\Gamma(\frac{1}{7})\Gamma(\frac{2}{7})\Gamma(\frac{4}{7})/(8\pi^2)$, a second exact sum at $z = 1/125$, rational corrections to every fixed order with error bounds, ratio, root and critical-tail expansions, and a continued-fraction companion with a recessive alternating solution. The Lucas-congruence conjecture of the entry is not established. *Merged from two manuscripts that proved the constant independently on the same day by different routes — a hypergeometric pullback with Ramanujan's level-seven series, and an eta product at $i/\sqrt{7}$ evaluated by Chowla–Selberg — both kept.* A third manuscript, added in batch 77, proves the constant once more; its route is printed as Section 16: the Fricke involution of the weight-one theta series gives the singular coefficient directly at $\tau = i/\sqrt{7}$, the level-seven case of Guillera–Zudilin's modular limit formula, and an Epstein zeta function gives the eta value. It adds no theorem, but an explicit generator for the corrections of every convolution power, with coefficients in a polynomial ring over $\mathbb{Q}$, and rounding brackets for every fixed weight, whose inversion cites the transseries volume; its claim to answer O'Brien's Question 10.2 is not carried. A fourth manuscript, written with the report in hand and added in batch 97 as Section 17, answers its research question on log-convexity and Hankel positivity: $a_n$ is strictly log-convex for every $n \geq 1$, by elementary ratio barriers; the Hankel matrix $(a_{i+j})$ is strictly totally positive of order 5 but not totally nonnegative of order 6, with $D_6(1) = -2687577700940853734656 2624$, the positive part certified by 21 rational polynomials with positive coefficients and Fekete's finite-order criterion; $a_0, \ldots, a_{10}$ are the moments of a positive measure on $[0, \infty)$, the least $a_{11}$ such a measure admits is attained by a unique six-point measure, and the actual $a_{11}$ falls below it; every fixed-size shifted Hankel determinant has a full asymptotic expansion with an explicit generator, and a large enough shift restores any fixed order of total positivity. Its recurrence-only proof of the expansion is printed as a second route, and its Lambert-$W$ inverse for the determinants is an instance of the transseries volume's theorem. The write added Subsection 17.10: the generating function has a square-root singularity at $z = -1$ with coefficient $B_- \in (-0.2075, -0.1754)$, so no tail of $(a_n)$ is a Stieltjes moment sequence, the order of first failure of total positivity is finite for every shift and tends to infinity with the shift, and a positive measure on $\mathbb{R}$ with moments $a_n$, if one exists, is unique, supported in $[-1, 27]$ and charges every neighbourhood of $-1$; an independent check of these statements found no gap and added one sentence on an atom at 0. Not established: such a Hamburger measure, the optimal shifts for orders $\geq 6$ (finite data only), and the closed forms $B_- = -2\pi/(\Gamma(\frac{1}{7})\Gamma(\frac{2}{7})\Gamma(\frac{4}{7}))$ and its companion, observed to 200 digits. No claim of the manuscript was found wrong; the Hilbert's-tenth research tree triaged its archive within a bounded scope (`37af2f0d6`). Unrefereed; not formalized.

### 131. Spectral collisions and boundary asymptotics for zig-zag Eulerian numbers

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a205497-zigzag-eulerian-spectra/**article.pdf**

from batch 73: zigzag_oeis_research.zip

For the descent counts $z(n,k)$ of linear extensions of a fence (OEIS A205497), determines the exact reduced denominator of every fixed-column generating function; its degree, the minimal recurrence order, is $r_k = \frac{40}{27\pi^2}(k+1)^3 + O(k^2 \log k)$, about $80/(9\pi^2)$ of the usual unreduced order. Also a uniform relative-error bound, a Gumbel crossover $\exp(-e^{-s})$ when $n/(k+\frac{3}{2}) - \log n \to s$, and strict log-concavity near both edges for $k \le cn/\log n$; full-row log-concavity and real-rootedness are not proved. Two OEIS presentation corrections.

### 132. Winding sectors and the four-displacement restricted-permutation problem

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a000382-winding-correction/**article.pdf**

from batch 73: ProveIt_A000382_winding_correction.zip

For the permutations of $\mathbb{Z}/n$ with every displacement in $\{0,1,2,3\}$, a bijective proof by winding number of $\Phi_n = 2(L_n+1)$, $L_n$ the Tribonacci–Lucas numbers; the formula itself is already printed in OEIS A004306. OEIS identifies $\Phi_n$ with A000496 and with four times A000382, which first disagree at $n=8$: the recurrence marked conjectural in A000382 is proved, but for the sequence produced by Mendelsohn's 1961 formula, not for the permutation count, and the exact difference has generating function $z^4/(1-2z+z^4)$. Also an inverse transseries.

### 133. Fixed-multiplicity shuffles and balanced Smirnov words

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a330266-balanced-smirnov-poisson/**article.pdf**

from batch 73: ProveIt_Balanced_Smirnov_OEIS.zip; batch 77: balanced-smirnov-repair-reproducibility.zip; batch 101: A330266_Uniform_Tail_Repair.zip (bundle Report 98, Part III)

Proves Kirgizov's conjecture in OEIS A330266 for every fixed $k \ge 2$: the fraction of shuffles of $n$ ranks of $k$ cards with no two equal neighbours tends to $e^{-(k-1)}$, and the number of equal adjacencies is asymptotically Poisson$(k-1)$, with an exact Laguerre-kernel generating integral and explicit corrections, for $k=4$ $e^{-3}(1 - \frac{9}{8n} - \frac{3}{128n^2} + \frac{189}{1024n^3} + O(n^{-4}))$. Part I marks one tail estimate, on which the uniform expansion and the corrections rest, as a sketch; the limit law does not depend on it. Part II, added in batch 77 from a second manuscript, proves it for every fixed $k$: the factorial-moment domination $E(X)_M \le (k-1)^M$, an explicit uniform remainder with a Touchard-polynomial constant, an all-orders expansion in integer powers, a degree bound giving factorial cumulants $O(n^{1-r})$, and a sharper inverse along the count sequence, which cites the transseries volume. Every coefficient Part I displays is re-derived unchanged and cited, not reprinted; the second manuscript's contraction, ratio and local derivations are kept as second proofs. Part III, added in batch 101 from a third manuscript written from the same pin as Part II without knowledge of it, repairs the same step a second time; the steps that agree with Part II are mapped and cited, and two proofs are kept as further routes. New in it: the fourth-order terms for every $k$, for A330266 the term $+\frac{7395}{32768n^4}$; a remainder constant sharper by $2^{-(J+1)}$; and for unequal rank sizes an exact factorial-moment identity with the bound $E(X)_M \le (k_{\max}-1)^M$. The write adds a Poisson limit for bounded multiplicities and the local probabilities $P(X=j)$ to every fixed order. The fourth-order coefficients are an exact computer-algebra identity, not a hand derivation; growing $k$, divergence and optimal truncation, and a certified integer inverse stay open. The independent check found every item valid and corrected the wording of one proposition on unequal ranks (its degenerate cases). Unrefereed; not formalized.

### 134. Unconditional all-orders asymptotics for two OEIS permutation sequences

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a189281-path-forest-expansions/article.pdf

from batch 73: oeis_path_forests.zip; batch 74: ProveIt_Fixed_Gap_Asymptotics.zip; batch 85: OEIS_Rational_Collapse_and_Integrality.zip; batch 97: ProveIt_Moving_Gap_Permutations.zip (Part IV), ProveIt_A189281_Borel_Completion.zip (Part V); batch 101: Fixed_Displacement_Permutation_Asymptotics_Source.zip (bundle Report 83, Part VI), Report232.zip (Part VII)

OEIS A189281 and A110128 count permutations avoiding a prescribed signed or absolute difference between entries two positions apart. Proves the displayed higher expansions without the conjectured recurrences, through a stabilization theorem for tilings of long path forests: for any fixed offsets the violation count has an all-orders expansion $e^{\theta u}$ times polynomials, with a finite exact formula for every coefficient, extended through order sixteen. The conjectured recurrences are not claimed; Part I proposes a rational-function conjecture, which Part III proves. Part II, added in batch 74 from a manuscript that re-derives Part I's theorems independently (its coefficients through order twelve agree and are printed only as a check), adds shape-uniform remainders, comparison bounds with an $n = 12$ example, a closed $c_3(r, s)$ and integrality for oriented one-path families. Part III, added in batch 85, proves that conjecture for every order as the case $(2, 2)$ of a closed form, valid for every pair of directed offsets, of the stable moments (via a single-sum moment formula), and deduces that every oriented correction polynomial has integer coefficients, with an exact degree law; it adds an inverse-factorial ${}_2F_0$ kernel, an entire Borel transform of the coefficients (no summability claim), A189282–A189284 through $n^{-6}$ and certificates at $n = 64, 128, 256$. Its special cases of Part II's one-path results are credited to Part II. Part III does not cover the absolute case, and no Part proves the guessed order-8 recurrence of A189281. Part IV (batch 97) proves the expansion uniformly over every pair of path forests, in normalized subpath counts and with no hypothesis on the path lengths: the order-$n^{-J}$ term depends only on subpaths of at most $J + 1$ edges, sharply; one short path of $L$ vertices shifts the absolute-model avoidance probability by $-2e^{-2}n^{-L-1}$; and for moving offsets $r/n \to \alpha$, $s/n \to \beta$ the limit is Poisson$(\theta(1 - \alpha)(1 - \beta))$, with correction boundaries at $r/n = 1/q$. Part V (batch 97), for A189281, gives an explicit Dawson-function Borel kernel and the positive-ray sum $C_*(n)$, summable in no open sector, the sharp growth of the coefficients (the series has radius zero and they are not P-recursive), a truncation rule for $C_*$, the OEIS order-13 operator annihilating the analytic completion (not the counts), and a flat remainder $e\,a_n/n! - C_*(n)$ that is nonzero at least once in every four consecutive indices; two numerical statements of its source (a truncation index, and one sign change that is in fact fourteen) are corrected on record. Part VI (batch 101, bundle Report 83), a same-day third derivation of Part I's theorem that predates Parts I–II, prints only its additions: exact Bonferroni enclosures at $n = 100, 200, 1000$, the threshold ceiling bracket and an exact-digamma inverse (instances of the transseries volume, proofs completed by the write) and four literature items. Part VII (batch 101, bundle Report 232) treats two reciprocal absolute gaps at once, answering the several-gaps question of the earlier Parts: a joint two-mark expansion uniform on compact sets of $\mathbb{C}^2$, in which cycles enter first at order $n^{-2}$; A189358 (camel) $= e^{-4}n!(1 + 8/n + 48/n^2 + O(n^{-3}))$ and A137774 (knight) $= e^{-4}n!(1 + 4/n + 28/n^2 + O(n^{-3}))$, where multiplying the marginal laws would predict 36 and 8; and threshold inverses. Its single-gap marginals are Part I's, and one misstated Hölder step is corrected in place. The independent check of the batch-101 write found everything valid and narrowed one sentence (the $n = 1000$ enclosure checks Part I's $c_{11}$, not $c_{12}$). Unrefereed; not formalized.

### 135. Two Fourier peaks and a moving lattice maximum

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a039831-two-fourier-peaks/article.pdf

from batch 73: A039831_Two_Fourier_Peaks.zip

Proves Kotešovec's 2023 conjecture $M_n \sim 3(n/e)^n$ in OEIS A039831, for the largest coefficient of $\prod_{j \le n}(1 + q + q^3 + \cdots + q^{2j-1})$, with explicit corrections through relative order $n^{-4}$: rounding produces a log-periodic term at order $n^{-3}$, and a second Fourier neighbourhood at $q = -1$ a parity

term at order $n^{-4}$, which moves the maximizing exponent infinitely often. Also a zero-multiplicity law under Bernoulli smoothing and a Lambert-$W$ inverse. A039824 is not resolved.

### 136. The cubic boundary for restricted partitions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a238016-restricted-partitions-cubic-boundary/article.pdf

from batch 74: OEIS_Restricted_Partitions_Cubic_Boundary.zip; batch 98: OEIS_Restricted_Partitions_Crossover.zip (Part II)

For partitions of $N$ into parts at most $m$, proves $p_m(N) \sim N^{m-1}/(m!(m-1)!)$ if and only if $N/m^3 \to \infty$, a condition still called conjectural in OEIS A238016 and A258670 (sufficiency is Erdős–Lehner's; necessity is new here), with ratio $e^{1/(4c)}$ at $N \sim cm^3$, a uniform centred expansion for $N \geq c_0 m^3$, five corrections for A238608, exponential sectors at $N = q^m$, inverse charts and a Poisson law for equal adjacent parts. The cubic scale is the $\alpha \to \infty$ end of the A097356 report's saddle question, not an answer to it. Part II (batch 98) answers Part I's question on a uniform bridge below the cubic scale, and with it that end of the A097356 report's question: with $X = N + m(m+1)/4$, $\varepsilon = m^2/X$ and $V_m(X) = X^{m-1}/(m!(m-1)!)$, $\log(p_m(N)/V_m(X))$ has an expansion $m\mathcal{D}(\varepsilon) + E_0(\varepsilon) + \sum_j E_j(\varepsilon) m^{-j}$ to every fixed order, uniform for all $N \geq am^2$; the centred volume law holds iff $X/m^{5/2} \to \infty$, with ratio $\exp(-1/(72\lambda^2))$ at $X \sim \lambda m^{5/2}$; transitions at the scales $m^{2+1/(2r)}$ have rational constants $d_r$, infinitely many nonzero; and an analytic inverse chart. Several Part I results and Szekeres's constants are re-proved as second routes. Part II is not uniform as $N/m^2 \to 0$, and negativity of every $d_r$ is not claimed ($d_1, \ldots, d_{16} < 0$ is evidence only); the expansions are fixed-order. The independent check corrected one tabulated decimal and the order of a step in the write's all-orders inverse, and recorded floating evidence that the second-transition sign changes between $m = 1373$ and 1374. Unrefereed; not formalized.

### 137. L-convex polyominoes and a colored-partition comparison

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a126764-lconvex-polyominoes/article.pdf

from batch 75: A126764_Lconvex_Research.zip; batch 77: lconvex-asymptotics-reproducibility.zip; batch 101: L_Convex_Polyomino_Area_Asymptotics_Source.zip (bundle Report 95, Part III)

Proves the Guttmann–Kotěšovec asymptotic recorded as a conjecture in OEIS A126764 for fixed L-convex polyominoes of area $n$, $a_n \sim \frac{13\sqrt{2}}{768}\, n^{-3/2} e^{\pi\sqrt{13n/6}}$, with three correction terms in closed form and relative error $O(n^{-7/4})$. The leading term and first correction already follow from a squeeze $0 \leq p_n - 4a_n \leq 3(p_n - 2p_{n-1} + p_{n-2})$ ($n \geq 3$) against the coefficients $p_n$ of an eta quotient counting coloured partitions; a finite-cut identity and a full-circle estimate give the other two. Also an all-orders radial expansion of the generating function at $q = e^{-t}$ (not a coefficient expansion) and a $W_{-1}$ inverse, an instance of the transseries volume's Lambert and staircase theorems. The area generating function is a cited input; Part I's manuscript metadata says it was prepared with ChatGPT. Part II, added in batch 77 from an independent second manuscript, proves the all-orders coefficient expansion that Part I left open: from an exact decomposition $A = PB^2 + R$, with $P$ Part I's eta quotient, $B$ a partial theta function and $R$ exponentially smaller, $a_n = \frac{13\sqrt{2}}{768} N^{-3/2} e^{\pi\sqrt{13N/6}} (\sum_{r<M} c_r N^{-r/2} + O(N^{-M/2}))$ with $N = n - 1/6$, for every $M$, re-proving Part I's three corrections. Also eventual strict log-concavity, monotonicity by an injection, and a shifted $W_{-1}$ inverse with integer brackets of every order, again an instance of the transseries volume; Part I's boundary series equals $B$. Part II states that Guttmann and Kotěšovec print the reciprocal of the amplitude; checked in batch 101 against their arXiv version 3, the display has the reciprocal while the quoted decimal is correct. Part III, added in batch 101 from a third manuscript that knew Part I but not Part II, proves the all-orders coefficient theorem a third time, in the unshifted index, by a route with every estimate written out (positivity, the partition factor $(q;q)_N$ common to all later terms, and Chebyshev interpolation into the saddle window), where Part II sketches its circle estimates; it gives a duality proof of the decomposition, with a positive series and explicit bounds for the remainder, strict increase $a_{n+1} \geq a_n + 1$ for every $n \geq 1$, the limit $n - n_0(a_n) \to \eta = 1/6 + 36/(13\pi^2)$ for the leading Lambert root, so that rounding that

model is eventually right (wrong for $2 \le n \le 555$ and right for $556 \le n \le 8000$ by computation; no onset is proved), and exact terms to $n = 8000$, agreeing with all 2001 b-file terms. The write completes its saddle extraction and identifies its auxiliary polynomials as Al-Salam–Chihara polynomials; its novelty sentence, stale against Part II, is corrected. The independent check found every item valid and sharpened one constant. Unrefereed; not formalized.

**138. The logarithmic deficit of 120-avoiding ascent sequences**

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202061-ascent-120-deficit/article.pdf

from batch 77: oeis-a202061-report.zip + oeis-a202061-sharp-deficit.zip + oeis-a202061-second-order-report.zip + oeis-a202061-third-order-report.zip + oeis-a202061-allorders-report.zip

For ascent sequences avoiding 120 (OEIS A202061), with $\mu \approx 7.2959$ the largest root of $\mu^3 - 8\mu^2 + 5\mu + 1$, $D_n = n \log \mu - \log a_n$, $F = n^{1/3}(\log n)^{2/3}$ and $L = \log n$, five Parts prove, each refining the last: $D_n = \Theta(F)$ from an exact positive height-walk representation, which excludes every equivalent $C_0 \mu^n e^{-dn^\gamma} n^g$ with $\gamma \ge 1/3$, among them the stretch $n^{3/8}$ that Conway, Conway, Elvey Price and Guttmann estimated numerically, and that the generating function is not D-finite (Part I); $D_n/F \to C_\star \approx 2.2326$, explicit (II); the term $(7/3)C_\star F \log L/L$ (III); the next constant $\kappa \approx -1.9401$, which proves Part III's conjecture (IV); and, for every fixed $J$, an expansion of $D_n/(C_\star F)$ in powers of $1/L$ with universal polynomials $P_j(\kappa + (7/3)\log L)$ (V); each with its inverse. No multiplicative equivalent, amplitude or effective threshold is claimed, priority is not established, and the inverses are instances of the transseries volume. The A202062 report gives a second route to the growth rate $\mu$. *Merged from five manuscripts whose archives nest their predecessors byte for byte; each paper is printed once, in chain order, and one manuscript's restatements of Part I are cited.*

**139. Ascent sequences avoiding 000**

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202058-ascent-000-growth/article.pdf

from batch 77: oeis-a202058-report.zip + oeis-a202058-fine-report.zip + a202058-lognormal-report.zip + a202058-ratio-report.zip; batch 102: A202058_Ascent_Sequence_Growth_Source.zip (bundle Report 97)

For ascent sequences avoiding 000 (OEIS A202058), $T = 3\pi^2/8$ and $c_n = a_n T^n/n!$, Part I proves the factorial growth constant $(a_n/n!)^{1/n} \to 8/(3\pi^2)$ conjectured by Conway, Conway, Elvey Price and Guttmann, through the exact radius $T$ of the exponential generating function, with $\log c_n = O(n^{9/13}(\log n)^3)$. Part II proves $-O(n^{2/3}(\log n)^{5/3}) \le \log c_n \le O((\log n)^2)$, which excludes a factor $e^{cn^\sigma}$ with $c, \sigma > 0$, gives exact counts through 400, shows that $a_n/n!$ is not $\mathrm{PF}_3$, and conjectures $\log c_n \sim (2/3)(\log n)^2$. Part III proves that law in limsup and cumulative form, with $\log A(t) \sim (2/3)\log^2(1/(T-t))$; Part IV proves the ratio limit $a_{n+1}/((n+1)a_n) \to 8/(3\pi^2)$ with error $O(n^{-1/4}(\log n)^{3/4})$ and the lower bound $\log c_n \ge -O(\sqrt{n}(\log n)^{3/2})$. The pointwise law, log-concavity and a multiplicative equivalent stay open; the Lambert-$W$ inverses are instances of the transseries volume. *Merged from four manuscripts in chain order; Part III's re-proof of Part II's bounds is kept as a second route, Part IV's uncredited restatement of a Part II identity is credited, and statements a later Part supersedes carry dated notes.* Part V (batch 102), bundle Report 97, written without knowledge of Parts I–IV, gives a third proof of the root limit (error $o(n)$ only), by a finite Perron bound with buffered Eulerian blocks, and a second proof of the ratio limit, without a rate. New: the two-catalytic functional equation $H(x; s + x(t-1), t) = tH(x; s, t - xs(t-1)) - xst(t-1)$ with its coefficient recurrence, a weighted upper bound, and endpoint laws: the proportions of values used once, of unused available values, of distinct values and of ascents tend to $1 - 8/(3\pi)$, $8(\pi+1)/(3\pi^2) - 1$, $1 - 4/(3\pi)$ and $4/(3\pi) + 8/(3\pi^2)$, with exponentially small deviations; strict monotonicity $a_{n+1} > a_n$; exact counts through 395, equal to Part II's. Its inverse is weaker than those of Parts II and IV. Part V is the cap-two case of Part III of `a294220-ascent-multiplicity-caps`, whose Corollary 18.3 proves Report 97's operation count. Report 97's "unresolved" stretched factor is excluded by Parts II–III (dated notes); its pointwise law and its formal eikonal and Borel reading are further questions.

### 140. Exact enumeration of 201-avoiding ascent sequences

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a202062-ascent-201-enumeration/article.pdf

from batch 77: oeis-a202062-report.zip

For ascent sequences avoiding 201 (OEIS A202062), proves the cubic generating function conjectured by Guttmann and Kotěšovec, by an exact two-label generating tree whose two-catalytic kernel equation is solved by two rational invariants, one from an order-seven kernel orbit; the report finds their conjectured cubic equivalent to its own and a sign error in their printed final combination. For every $M$, $a_n = C\mu^n n^{-9/2}(\sum_{m\le M} c_m n^{-m} + O(n^{-M-1}))$, with $\mu \approx 7.2959$ the largest root of $\mu^3 - 8\mu^2 + 5\mu + 1$, $C \approx 13.4300$ in closed form and $c_m \in \mathbb{Q}(\mu)$, the first three exact; also a $W_{-1}$ inverse with ceiling brackets, an instance of the transseries volume's Lambert theorems. With Conway, Conway, Elvey Price and Guttmann's equal-growth theorem it gives $\lim b_n^{1/n} = \mu$ for A202061, a second route to Part I of the A202061 report. Novelty rests on a scoped literature check, and the envelope constants are not explicit.

### 141. Asymptotic enumeration and inversion for A122399

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a122399-surjection-diagonal/article.pdf

from batch 77: oeis-a122399-research.zip; batch 98: A122399_Boundary_Transitions_and_Inversion.zip (Part II)

For $a_n = \sum_k k^n k!\, S(n,k)$ (OEIS A122399), the diagonal of $1/(1-ue^y(e^x-1))$: an exact, absolutely convergent vertical-contour formula for the two-size numbers $A_{m,n}$ with a nonasymptotic tail bound; an all-orders saddle expansion uniform for $m/n$ in compact subsets of $(0,\infty)$; and $a_n = CD^n(n!)^2 n^{-1/2}(1 - 0.0600744\ldots/n + \cdots)$ through $n^{-4}$, with $D \approx 3.1610887$ (A317855) and the first correction in closed form. Also a controlled inverse with integer-threshold envelopes, whose initializer is an instance of the transseries volume's inverse-Gamma core; a central limit theorem for the number of occupied blocks; and $a_{n+\varphi(p^r)} \equiv a_n \pmod{p^r}$ for $n \ge r$, which proves Bala's conjecture that $p-1$ is a period modulo $p$; no least period is claimed. The leading equivalent is Kotěšovec's (OEIS, 2018), the methods are standard saddle methods, and no global novelty is claimed. Part II (batch 98) studies the defect $D = m - K$ of the number $K$ of blocks of a uniform structure counted by $A_{m,n}$: $D$ is exactly a sum of $m-1$ independent Bernoulli variables (real zeros of the block polynomials, by Harper's method; the case $n = 0$ is Frobenius's), with explicit Poisson bounds for all $(m,n)$; a degenerate, Poisson or Gaussian regime at $n = m(\log m + c)$ as $c \to +\infty$, fixed or $-\infty$, with $D \to \mathrm{Poisson}(e^{-c}/2)$; uniform lattice Edgeworth expansions to every fixed order; a complete expansion in the critical window with a rational coefficient algorithm; doubleton defects; an all-orders inverse for the all-singleton probability with the exact certificate $N_{1/e}(1000) = 6207$; and a diagonal local limit theorem with Part I's constants. It answers Part I's Questions 2 and 4 in part. Its "transseries" are depth-one power–log expansions, as a dated note records; the write proved a sketched degree bound and the sharpness of the count half of the growing Poisson window, and the independent check corrected that proof's closing sentence, which named an impossible regime. Large deviations, count asymptotics for $m/n \to \infty$ and a bridge to the saddle range stay open. Unrefereed; not formalized.

### 142. Asymptotics of iterated Bell diagonals (OEIS A139383, A261280)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a139383-iterated-bell-diagonals/article.pdf

from batch 77: oeis-iterated-bell-report-attribution-revised.zip + iterated-bell-proportional-addendum.zip (the first edition oeis-iterated-bell-report.zip superseded, not printed)

For the iterated Bell numbers $H(n,m) = n![z^n](e^z - 1)^{\circ m}$, with OEIS A139383 $= H(n,n)$ and A261280 $= H(n,n+1)$, Part I proves the leading equivalents those entries label conjectural, and for each fixed integer $k$ an expansion $H(n,n+k) = e^k C\, n^{2n-5/6} 2^{1-n} e^{-n}(1 + \sum_j P_{j,k}(\log n) n^{-j})$ to every fixed order, with an outward-rounded interval certificate that $2 < C < 4$ (the digits $C \approx 2.86540798$ are not certified), an exact coefficient generator, and smooth inverses with a

safe integer bracket. Part II, from an addendum conditional on Part I, extends the expansion to proportional depth $m = \lambda n + \delta$, uniformly for $\lambda$ in compact subsets of $(0, \infty)$, with an amplitude $D(\lambda)$, $D(1) = 2C$, proved strictly positive for every $\lambda > 0$, a floor-depth phase $e^{-\{\lambda n\}/\lambda}$, eventual monotonicity and inverses; the endpoint regimes stay open. Prellberg's 2002 bivariate formula and parabolic-coordinate method already give the leading equivalent: no priority is claimed for it, for the method or for the coefficients. The inverses are instances of the transseries volume. The superseded first edition differs only in that attribution; the check that Prellberg's broad printed theorem fails for $f(z) = z/(1-z)$ is shared with the Takeuchi report. A reciprocal note (batch 138) records that Part IV of `a290354-iterated-euler-diagonals` proves $I(\beta) = \beta 2^{\beta/3} h(\beta)$ for every $\beta > 0$ under four unrefereed inputs, two of them from this report, and that the forced value $C = 2.8654079726\ldots$ lies just below the three uncertified runs here.

### 143. Corner polyhedra and Schnyder labelings: polynomial exponents and limiting amplitudes

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a377922-corner-polyhedra-schnyder/article.pdf

from batch 77: oeis-bimodal-cone-research.zip + oeis-bimodal-full-asymptotics.zip; batch 103: Geometric_OEIS_Sequences_Cone_Exponents_and_Inverses_Source.zip (bundle Report 112; Part III)

Fusy, Narmanli and Schaeffer encode corner polyhedra (OEIS A377922) and 3-connected Schnyder labelings (A377920) as quadrant walks with a two-state parity and conjecture equivalents $\kappa\mu^n n^{-\alpha}$ with irrational exponents (their Conjecture 25). Part I proves $p_n = \Theta((9/2)^n n^{-\alpha_P})$ and $s_n = \Theta((16/3)^n n^{-\alpha_S})$ with the conjectured $\alpha_P = 1 + \pi/\arccos(9/16)$ and $\alpha_S = 1 + \pi/\arccos(22/27)$, by a one-unit cycle buffer, a random-clock transfer and a Fourier local limit; since both exponents are irrational, the generating functions of A377922, A377920 and A377921 are not D-finite; and first-threshold inverses with $O(1)$ error. Part II, from an addendum, proves the full equivalents with finite positive amplitudes $\kappa_P$, $\kappa_S$ (completing Conjecture 25), the coefficientwise equivalent $\tilde{s}_n \sim (16/19)^3 \kappa_S (16/3)^n n^{-\alpha_S}$ for A377921, and a Lambert-$W_{-1}$ inverse with a vanishing-width ceiling bracket, an instance of the transseries volume. No digits of $\kappa_P$, $\kappa_S$, no rates and no correction terms are claimed; A377923 is outside both theorems; the iid cone theorems are Denisov–Wachtel's, and neither manuscript claims priority. Part III (batch 103), from a manuscript written without knowledge of Parts I–II, proves a cone theorem at the logarithmic scale for Markov-additive walks on $\mathbb{Z}^2 \times E$ with a finite phase set $E$ and unbounded steps with a uniform exponential moment, killed on leaving the quadrant: under hypotheses (H1)–(H5) the exact-time kernel is $n^{-1-\pi/\theta+o(1)}$, with a survival bound under (H1)–(H4). Also new: a six-face sleeve injection giving $s_{n-6} \le \tilde{s}_n \le s_n$, hence (at the write, with Part I) $\tilde{s}_n = \Theta(\mu_S^n n^{-\alpha_S})$; non-D-finiteness from the logarithmic law alone (stated in general at the write); the exponential-moment certificate 189/13; $4 < \alpha_P < 5$, $6 < \alpha_S < 7$; three further references. What it says about the three sequences is a weaker second route, and its open amplitudes, ratio limit and $O(1)$ inverse are settled by Parts I–II (dated notes). `a348351-one-sided-rectangulations` proves its exponent by the same method; its walk satisfies (H1)–(H5). The independent check found the write's remarks valid and corrected a scale label and a coefficient in one proof. An amplitude or $\Theta$ form of the cone theorem stays open. Unrefereed; not formalized.

### 144. Precise asymptotics of the powered Catalan numbers (OEIS A113227)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a113227-powered-catalan/article.pdf

from batch 77: oeis-powered-catalan-report.zip + oeis-powered-catalan-fourier-addendum.zip

The powered Catalan numbers $a_n$ (OEIS A113227) count permutations avoiding the vincular pattern 1-23-4. From the known continued fraction Part I derives a positive discrete moment representation $a_n = \sum_\lambda \omega_\lambda \lambda^n$ whose transform is a Bessel quotient, locates its large spectral nodes, and proves $a_n \sim e^{e-2}(2\pi)^{-1/2} r^{-3/2} B_n(e)$, with $B_n$ the Touchard polynomial and $r = n/W(n/e)$, answering the question of precise asymptotics that Elizalde left open in 2006; also an expansion to every fixed order with explicit $c_1$, $c_2$, a canonical continuous inverse and an exponential-generating-function corollary. Part II, from an addendum, proves a zero-free complex wedge for the Bessel

weight, replaces the spectral sum by a lattice sum with relative error $O(e^{-dn})$ for every $d < \log 2$, and splits $a_n$, for each fixed $J$, into $J+1$ exact Fourier sectors, each with an all-orders expansion. No convergence, growing number of sectors, Borel sum or transseries is claimed, and the inverse is an instance of the transseries volume. The recurrence, models and continued fraction are prior art (Callan, Beaton et al., Gladkovskii); the sibling pattern 12-34 (A113226) has its own report.

### 145. Growing powers of Mahonian coefficients

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a380274-mahonian-growing-powers/article.pdf
from batch 77: oeis-mahonian-crossover-result.zip

For $S_n(r) = \sum_k M_n(k)^r$, with $M_n(k)$ the number of permutations of $n$ with $k$ inversions, proves that for every $r_n \to \infty$ the ratio $S_n(r_n)/C_n^{r_n}$, $C_n$ the central Mahonian number, is asymptotic to a theta function $\Theta_\delta(r_n/V_n)$, where $V_n = n(n-1)(2n+5)/72$ is the inversion variance and $\delta \in \{0, \frac{1}{2}\}$ depends on $n \bmod 4$: a Gaussian integral below the critical scale $r \asymp n^3$, a discrete Gaussian partition function on it and $1+2\delta$ above it, with the matching limit laws for the common inversion count of colliding random permutations. Also an explicit fourth-order correction, uniform expansions to every fixed order on the transition scale, an unnormalized multiplicative equivalent, and a Lambert-$W_0$ index inverse on a fixed critical path, an instance of the transseries volume. The fixed-power equivalent with three corrections, including OEIS A380274 ($r = 3$) and A380275 ($r = 4$), is in Wang's 2026 manuscript and is not claimed; priority for the growing-power extension is not established. A dated note relates the crossover to the critical theorem of the theta-ballot report; that report's general lattice-peak question is answered in a sufficient-hypothesis form by its Part II (batch 98), whose Proposition 21.1 makes this report's critical window an instance by reorganizing its proof.

### 146. Divisor-weighted partition products and assemblies (OEIS A301746 and A294363)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a301746-divisor-weighted-asymptotics/article.pdf
from batch 77: oeis-divisor-square-asymptotics-result.zip + oeis-divisor-assembly-result.zip

Two manuscripts on generating functions weighted by the divisor function $d(k)$. Part I treats OEIS A301746, the coefficients of $\prod_k (1+q^k)^{d(k)^2}$, and proves the conjecture recorded there. From the fourth-order Mellin pole at $s = 1$ it obtains $\log a_n \sim \sqrt{n}(\log n)^{3/2}/(2\sqrt{6})$, Kotěšovec's 2018 conjecture, with three further inverse-logarithmic corrections and the matching inverse. Part I also gives a relative expansion to every order in exact saddle cumulants, strict monotonicity and shrinking brackets for the integer threshold; the contributions of zeta zeros are left open. Part II treats A294363, $n![z^n]\exp(\sum_k d(k) z^k)$, whose Mellin kernel has a double pole: an explicit multiplicative equivalent through a shifted Lambert-$W$ saddle, an all-orders expansion, the OEIS logarithmic statement, and a factorial-core inverse. Its coefficients are rational in the Lambert parameter, and the inverse's absolute error tends to zero. The two Parts share a method template, also used by the batch-77 source of the A022629 report, but no theorem. No priority is claimed for the methods, and the Lambert, reversion and bracket steps are instances of the transseries volume.

### 147. Ascent sequences with bounded multiplicity (OEIS A294220)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a294220-ascent-multiplicity-caps/article.pdf
from batch 77: oeis-bounded-multiplicity-family-report.zip + oeis-fixed-sector-addendum.zip;
batch 102: Ascent_Sequences_with_Bounded_Multiplicities_Source.zip (bundle Report 99)

For ascent sequences of length $n$ in which every value occurs at most $b$ times (the array OEIS A294220; cap three is A317784), Part I proves for every fixed $b \geq 3$ that $(a_n^{(b)}/n!)^{1/n} \to \mu_b = 1/T_b$, with $T_b = \int_0^\infty \log E_b(v)/(E_b(v)-1)\, dv$ and $E_b$ the truncated exponential, with error $O_b(n/\log n)$ in $\log a_n^{(b)}$; also a Lambert-$W$ inverse in the length, the large-cap law $T_b = \pi^2/6 + (b+1)(\zeta(b+2) - 1) + O(b^{5/2}(4/9)^b)$ with explicit constants, and a $W_{-1}$ inverse in the cap with counterexamples

to naive rounding. The cap-two value $T_2 = 3\pi^2/8$, the constant $8/(3\pi^2)$ conjectured by Conway, Conway, Elvey Price and Guttmann, is taken from the A202058 report, not re-proved. Part II, from an addendum, writes $T_b - \pi^2/6$ as a convergent sum of positive Taylor sectors, proves that the remainder after any fixed number of sectors is asymptotic to the next sector, and expands each fixed sector to all orders in $1/b$; the remainder above is $\sim \frac{8}{9\sqrt{\pi}}\sqrt{b}\,(4/9)^b$. No prefactor or all-orders expansion in $n$, no joint regime $b = b(n)$ and no priority is claimed; the inverses are instances of the transseries volume. Part III (batch 102), bundle Report 99, which writes the cap as $r$ and knew neither earlier Part, proves for every fixed $r \geq 2$ the root and ratio limits $\mu_r$: the ratio limit is new for $r \geq 3$ and answers Part I's Question 2, while the root half and the case $r = 2$ are second routes. New also: endpoint laws for the proportions of values used $k$ times, of distinct values and of ascents, with exponential tails; the weighted growth formula $\lim N^{-1}\log(H_N(u)/N!) = -\log\chi(u)$ and a full large-deviation principle; and a histogram-preserving sorted recurrence counting in $O_r(N^{r+2})$ operations. Its Section 24 re-proves the large-cap results in $R = b + 1$ (its $-31/8$ is Part II's $-27/8$ in $b$) and adds a continuous cap, exact-sector envelopes and a two-candidate cap selection; its length inverse is weaker than Part I's. The write proves Report 99's remark that for $K = 1$ the derivative bracket of its envelopes improves to $1 + O((2/3)^R)$. Part V of `a202058-ascent-000-growth` is the case $r = 2$. No prefactor, central or local limit theorem, or classification of the finite-rate endpoint region is claimed.

### 148. Galled trees by gall count: the cube-root crossover, positive density and the near-maximal endpoint

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a399421-galled-trees/article.pdf`

from `batch 77: galled-density-reproducibility.zip + galled-crossover-reproducibility.zip + galled-endpoint-report.zip (the earlier edition galled-crossover-reproducibility (1).zip superseded, not printed)`

For $g(n, k)$, the rooted binary unlabeled simplex time-consistent galled trees with $n$ leaves and $k$ galls (OEIS A399421, row sums A397952), proves expansions to every fixed order in three disjoint regimes. Part I: for $k \asymp n^{1/3}$ the fixed-$k$ asymptotic acquires the factor $\exp(-2\gamma_0 k^{3/2}/\sqrt{n})$, $\gamma_0 = 1.1300337163\ldots$, with a series in $n^{-1/3}$ and an explicit first correction. Part II: for $k/n$ in compact subsets of $(0, 1/2)$, a saddle expansion in $1/n$, a central limit theorem with mean $0.19741431\,n$ and variance $0.05548588\,n$, local and large-deviation laws and integer inverse brackets; it records that the inspected versions of Agranat-Tamir et al. print $\gamma \approx 0.3846$ where their defining equation gives $0.3835758866\ldots$, the value credited on OEIS to Kotěšovec. Part III: for $n - 2k - 1 \asymp \sqrt{n}$, an expansion in $n^{-1/2}$ whose factor two comes from the parity of two moving square-root singularities. The generating function and the fixed-$k$ and scalar leading terms are prior work, the cube-root mechanism is Fuchs and Hao Yu's for leaf-labeled classes, matching the regimes stays open, and the inverses are instances of the transseries volume. *Merged from three manuscripts of one workspace, one per regime, with no theorem proved twice; the common model is printed once.* Five heavy triangle files are not shipped; the README regenerates them.

### 149. Airy amplitudes for relaxed trees, compacted trees and finite-language automata

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a082161-airy-amplitudes/article.pdf`

from `batch 77: relaxed-binary-trees-reproducibility.zip + compacted-binary-trees-reproducibility.zip + fixed-arity-airy-package.zip + ternary-airy-amplitudes-result.zip + larger-alphabet-automata-report.zip + oeis-dfa-finite-language-report.zip; batch 102: A082161_Positive_Airy_Amplitude_and_All_Orders_Source.zip (bundle Report 100) + A254789_Positive_Airy_Amplitude_and_All_Orders_Source.zip (bundle Report 101) + A331120_Minimal_Binary_Automata_All_Orders_Source.zip (bundle Report 103)`

Relaxed trees, compacted trees and minimal automata of finite languages satisfy one triangular recurrence family indexed by the arity $k$; their growth orders are due to Elvey Price, Fang and Wallner and to Ghosh Dastidar and Wallner, and the existence of a limiting amplitude was open. Proves positive amplitudes and Poincaré expansions to every fixed order in $n^{-1/3}$. Part I

(A082161): $a_n = \gamma\, n!\, 4^n e^{3zn^{1/3}} n\,(1 + \sum_k c_k n^{-k/3})$, $z$ the largest Airy zero, with three explicit logarithmic corrections. Part II (A254789): the same with $n^{3/4}$, exact positive renewal models, and cancellation of the $n^{-1/3}$ term in the logarithm of the compacted-to-relaxed ratio. Part III: every fixed $k \geq 3$ and all three models at scale $(n!)^{k-1}(k^k/(k-1)^{k-1})^n e^{\gamma n^{1/3}} n^{(2k-1)/3}$, with common first two corrections, first model-dependent defects of order $n^{-(k-2)}$, eventual decrease of the normalized ratios and a cancellation identity; its ternary section, the base proof, adds the third coefficient, and its comparison section bounds the automaton-to-relaxed ratio below by $\prod_{j=2}^{n}(1 - 1/(2j^{k-1})) \geq (2\sqrt{2}/\pi)\sin(\pi/\sqrt{2})$, uniformly in $k$ and $n$. Part IV (A331120): binary automata, with $8^n$ and $n^{7/8}$. No amplitude is evaluated, the expansions are not uniform in $k$, the Lambert-$W$ inverses are instances of the transseries volume, and the sources' claim that Ghosh Dastidar–Wallner use a false bound $U \leq q$ was not checked against the printed paper. *Merged from six manuscripts: the ternary one is the $k = 3$ case and base proof of the arity one, the comparison one the earliest form of its comparison step, and the automaton one repeats the compacted one's method and Jacobi appendix, printed once.* Part V (batch 102), from bundle Reports 100, 101 and 103 (relaxed, compacted, automata), which knew none of Parts I–IV, proves their binary theorems a second time in even time, by a positive semidefinite two-step Jacobi product, lazy-walk smoothing and three-lag signed memory. New: Report 100's Lemma 72.1 proves the relaxed amplitude positive from the exact recurrence alone, without the published lower bound every earlier Part uses (the write's Remark 66.1 applies it to Part I); the endpoint error $O(n^{-2/3})$; the fourth to sixth relaxed and compacted coefficients, with $r_n/(4nr_{n-1})$ and, by the write's Corollary 66.3, $c_n/(4nc_{n-1})$ through $n^{-3}$; for automata $d_4$, $c_4$, $\sigma_7$, the ratio through $n^{-7/3}$ and an explicit $X^{-1/3}$ inverse term. The independent check stated two dependencies of Corollary 66.3: Report 100's exact certificate suite and, through Part II, the published compacted lower bound. The amplitude values 166.952…, 173.127… and 76.438 are uncertified, Report 103's remark on a normalization in the AofA 2020 paper was not checked, and Reports 100 and 101 carry the author line "Research prepared for private review". Unrefereed; not formalized.

### 150. Airy asymptotics for tree-child networks

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a213863-tree-child-networks/article.pdf`

from `batch 77: d-combining-asymptotics-reproducibility.zip + tree-child-asymptotics-reproducibility.zip; batch 102: Fixed_Degree_Tree_Child_Networks_All_Orders_Source.zip (bundle Report 104) + A213863_Tableaux_and_Tree_Child_Networks_Source.zip (bundle Report 102)`

For maximally reticulated $d$-combining tree-child networks and their word diagonal $a_n$ (OEIS A213863 at $d = 2$), proves for every fixed $d \geq 2$ that $a_n = \gamma_d (n!)^{d-1} \Gamma^n e^{Bn^{1/3}} n^{\rho}(1 + \sum_r C_{d,r} n^{-r/3})$ to every finite order, with $\gamma_d > 0$ defined by convergent limits, $\Gamma = 4(d+1)^{d-1}/(d-1)!$, $B = 3z((d-1)/(d+1))^{2/3}$, $z$ the largest Airy zero, and $\rho$ explicit, by a contractive Jacobi reduction and polynomial–Airy profiles; ternary coefficients are given. Part II, for $d = 2$: explicit coefficients; every order for the total number of binary tree-child networks, $TC_n \sim (\sqrt{e}\,\gamma/12)(n!)^2 12^n e^{Bn^{1/3}} n^{-5/3}$, through the Pons–Batle identity proved by Lin, Liu, Liu, Liu and Xin; weighted all-orders expansions of the reticulation deficit around Poisson(1/2); an exact eventual total-variation identity with sharp rate $|B| n^{-2/3}/(48\sqrt{e})$; explicit inverses with integer brackets, instances of the transseries volume. In Parts I–II the total count for $d \geq 3$ has only its leading term, and no amplitude is evaluated or certified. *Merged from two manuscripts with one proof skeleton: the binary manuscript's own proof of the case $d = 2$ is kept as a second route, and four identical displays are printed once.* Parts III and IV (batch 102), from bundle Reports 104 (every fixed $d$) and 102 ($d = 2$), closed before the sources of Parts I–II arrived, re-prove the main theorem independently, in an exact gamma-product gauge with a new convergent amplitude formula and in the factorial gauge $3^p p!$; they are credited as third and fourth routes, and their priority sentences are qualified. New for every $d$: $C_{d,1} = B^2(8(d+1)^2 - 27(d-1)^2)/(810(d-1)^2)$, negative exactly for $d \geq 4$, and $C_{d,2}$ in closed form; at $d = 3$ the coefficient $C_{3,3}$ and, resting on Report 104's reading of a published proof (Chang et al., Lemma 3.24 and Theorem 1.9(ii)) that is not shipped and

was not checked, the total-to-maximal ratio $1 + O(n^{-1/2})$ and the first total-count correction; the $d = 4$ diagonal $1, 1, 91, 51821$ and the trap that A213864 is not the $d = 3$ sequence. New for $d = 2$: the fifth and sixth logarithmic coefficients $L_5 = B(108 - B^3)/8748$ and $L_6 = 2B^3/2187 - 1/162$, confirmed by Part II's generator at order nine and, at the independent check, by exact counts to $n = 7000$; the ratio $a_n/(12na_{n-1})$ through $n^{-3}$; and an $O(n^{-2/3})$ ratio and total-variation rate from published lemmas alone, without the Pons–Batle identity, weaker than Part II and kept as a second route. Fitted amplitudes $\gamma_2, \ldots, \gamma_5$ are uncertified diagnostics.

### 151. Historic tree asymptotics and a high-order obstruction

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a333497-historic-trees/article.pdf

from batch 77: historic-tree-report.zip; batch 103: A333497_B_Tree_Histories_All_Orders_and_Inversion_Source.zip (bundle Report 113; Part II) + A336009_B_Tree_Histories_Transseries_and_Non_D_Finiteness_Source.zip (bundle Report 115; Part III) + B_Tree_Order_59_Counterexample_to_the_Universal_Asymptotic_Conjecture_Source.zip (bundle Report 117; Part IV)

For reduced 5-historic trees (OEIS A333497), the exponential coefficients $h_n$ of the solution of $H''' = H^2$, $H(0) = H'(0) = H''(0) = 1$, proves a unique positive blowup point $\rho$, $180^{1/4} \le \rho \le 60^{1/3}$, and an expansion of $h_n/(30(n+2)!\rho^{-n-3})$ to every fixed grade with exact Gamma ratios, oscillating in $\log n$ through the exponent $\lambda = (13 + i\sqrt{71})/2$: the relative error is $2|K|n^{-13/2}\cos(\arg K - \frac{1}{2}\sqrt{71}\log n) + O(n^{-15/2})$ with $K \neq 0$, so it changes sign infinitely often. The same for A336009 ($G'''' = G^2$), with the exact real-mode coefficient $-1/18052070400$ from a conserved energy. Main result: the factorial-pole asymptotic of Burghart and Wagner's Conjecture 1 fails for every ODE order $r \ge 30$ (conjecture parameter $m = r - 1 \ge 29$), through an unstable conjugate pair (an exact rational bridge for $30 \le r \le 37$), a sign-variation cone, strict sign-regularity and a generalized stable manifold. The orders $4 \le m \le 28$ and Conjecture 2 stay open, no replacement asymptotic is given, and the fitted constants are not certified; the manuscript also reads the journal's Table 1 as listing the radius under its reciprocal's label. The Lambert-$W$ start and two-ceiling window of the inverse are instances of the transseries volume. Parts II–IV (batch 103), three manuscripts written without sight of Part I, prove its order-five ($m = 2$), order-seven ($m = 3$) and order-59 ($m = 29$, $r = 30$) results again by other routes (an explicit convergent invariant surface; a scalar majorant reaching $z = 0$; a cyclic sign-change cone invariant under Euler steps and the complete spectrum at $r = 30$ certified by exact inequalities), printed as marked second routes, their abstracts' novelty claims with dated notes. New: Part III proves that the exponential generating functions of A333497 and A336009 are not D-finite, so neither sequence is P-recursive (windings about the dominant singularity give infinitely many independent branches); the bounds $16800^{1/6} \le \rho_4 \le 840^{1/4}$; a blowup enclosure for every order; explicit polydisk radii 1183/960 and 121/6720 of the local series (a radius in $t$ needs a bound on $|C|$, which none gives). The write credits Mallet-Paret–Smith (1990) for the cyclic sign-change count and proves that the cyclic cone equals Part I's, so Part IV's lemma replaces one of Part I's two uses of the strong-$k$-positivity theorem; the independent check narrowed both statements (the other use, and signed couplings). Nothing was found wrong. Unrefereed; not formalized.

### 152. Asymptotic expansions of the Takeuchi numbers

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a000651-takeuchi-numbers/article.pdf

from batch 77: takeuchi-asymptotics-research.zip

For the Takeuchi numbers $T_n$ (OEIS A000651) and $w = W(n)$, proves $\log T_n = e^w(w^2 - w + 1) + w^2/2 - \frac{1}{2}\log(1 + w) + \log C + p_1(w)/n + p_2(w)/n^2 + O((1 + w)^8/n^3)$ with explicit rational $p_1, p_2$ and $0 < C < \infty$, without assuming the leading equivalent, which is Prellberg's (announced 2002); then every fixed order, each $p_j$ by a finite exact algorithm and one convergent quadrature, and an explicit $p_3 \sim w^4/72$, so $p_j = O(w^j)$ fails at $j = 3$. The transfer is a quantitative stability lemma for descending Markov chains with jumps close to $1 + \mathrm{Pois}(W(r))$. Also the Takeuchi-minus-Bell coefficients, and real and integer inverses that are instances of the transseries volume. The

manuscript states that the general theorem printed in Prellberg's 2002 slides omits a restriction ($f = z/(1-z)$, $a = z$, $b = 1$ gives $B_{n-1}$, not a multiple of $B_n$), without effect on the Takeuchi case; `a139383-iterated-bell-diagonals` makes the same check, printed only here. $C$ is not enclosed, and effective constants and optimal truncation are open.

### 153. Rooted identity trees with fixed maximum outdegree

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a116379-bounded-identity-trees/article.pdf

from batch 77: rooted-identity-trees-source.zip; batch 103: A116379_A116380_All_Orders_Identity_Trees_and_Inversion_Source.zip (bundle Report 130; Part II)

For rooted unlabeled identity trees with at most $d$ children at each vertex (OEIS A116379 for $d = 3$, A116380 for $d = 4$, whose leading asymptotic OEIS records as unknown), proves for every fixed $d \geq 2$ the square-root law $a_{d,n} = C\rho^{-n}n^{-3/2}(\sum_{j\leq R} D_j n^{-j} + O(n^{-R-1}))$ to every fixed order, with $C > 0$ and explicit $D_1, D_2$, from the signed root equation: $1/4 \leq \rho \leq 9/20$, a transversality bound with the constant $1-810/(319\sqrt{19})$ independent of $d$, a characteristic point and a unique dominant singularity. Tabulates $\rho, C, D_1, D_2$ for $d = 3, 4$ to 30 decimals, uncertified as delivered and certified in Part II ($\rho_3 = 0.40077450047\ldots$, $C_3 = 0.41431687066\ldots$) and inverts a specified finite model by $W_{-1}$, an instance of the transseries volume. Nothing is uniform in $d$; the derivative cancellation is Bell, Burris and Yeats's, and no priority is claimed, since Meir–Moon–Mycielski (1983) and Haigh–Kennedy–Quintas (1984) were not read in full. Part II (batch 103), from a manuscript written without knowledge of Part I, encloses $\rho$, $\tau$, $C$, $D_1$, $D_2$, $\lambda = \log(1/\rho)$, $b = -\frac{3}{2}\log\lambda - \log C$ and the Puiseux coefficients $\beta_1, \ldots, \beta_5$ for $d = 3, 4$ by a rational-interval computation (outward rounding at denominator $10^{90}$; radius brackets of width $10^{-45}$, the others at most $1.4 \cdot 10^{-41}$), so every entry of Part I's table is the correctly rounded value; this answers half of Part I's first further question ($D_j$ for $j \geq 3$, other caps and effective thresholds stay open). The certificate is computer-assisted, specialized to $d = 3, 4$, and uses the qualitative theorem to identify the radius. Also a second proof of Part I's theorem, an explicit binary family giving $\rho_d < 9/20$ strictly, a rational recurrence for the Gamma-ratio coefficients (proved equal to Part I's at the write) and an exact threshold algorithm. The manuscript credits Haigh–Kennedy–Quintas 1984 to "G. W. Furnas" (corrected at the write); the independent check withdrew a criticism the write had made of one of its proofs and corrected one digit of the write. Unrefereed; not formalized.

### 154. Tournament score sequences and weighted component asymptotics

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a000571-tournament-score-sequences/article.pdf

from batch 77: tournament-score-sequences-source.zip; batch 85: tournament_coexistence_research.zip (Part II); batch 98: tournament_supercritical.zip (Part III)

Refines Kolesnik's leading asymptotics for the score sequences of tournaments (OEIS A000571) and of strong tournaments (A351822), from the generating function $\exp A(z)$ of Claesson–Dukes–Franklín–Stefánsson: to every fixed order, $T_n^{\pm} = (e^{\pm\lambda}/(2\sqrt{\pi}))\, 4^n n^{-5/2}(1 + c_1^{\pm}/n + \cdots)$ with $\lambda = A(1/4) = 0.33023754398\ldots$, $c_1^{\pm} = -1/8 \pm 5\mu/2$, $\mu = A'(1/4)/4 = 0.67395260185\ldots$ and explicit $c_2^{\pm}$. For the count weighted by $u^m$, $m$ the number of strong blocks: fixed subcritical weights; at the critical weight an expansion in $n^{-1/2}$ with no $n^{-1}$ term; uniformly on a logarithmic window above it a half-power expansion with Kummer-function coefficients; and at $\tau = \frac{1}{2}\log n + 2\log\log n + \log(\mu\sqrt{\pi}/2) + s$ a coexistence profile proportional to $1+e^{-s}$. Inverses of specified models are instances of the transseries volume. Part I claims nothing for $\tau < 0$ or for growing orders, and the renewal mechanisms are credited to Blanchet and Glynn. Part II (batch 85) answers the report's first open question, uniformly for $s$ in compact sets: at that weight the score sequence is asymptotically a logistic mixture, with probability $1/(1+e^{-s})$ of one block of size $n - O_P(n/\log n)$ (residual strings within total variation $O(1/\log n)$ of two geometric renewal strings, Gamma$(2,1)$ count and deficit, stable residual fluctuations, a mixed Fréchet second maximum, every fixed inverse-logarithmic correction) and otherwise $n/a_c + O_P(n^{2/3})$ blocks with a 3/2-stable count and a Fréchet largest block; the giant probability converges at rate $O(1/\log n)$ around a Lambert-$W_{-1}$ centre, an instance of the transseries volume. Its coefficient input re-proves Part I's coexistence

theorem (printed as a note); the Gibbs-partition and big-jump mechanisms are credited to Stufler and others. Part III (batch 98) answers Part I's second question, the supercritical crossover, on a fixed small interval of weights $u \geq u_c$ with excess growth rate $0 \leq t \leq t_0$ ($t_0$ not effective): an exact change of measure to a tilted renewal, a pole–branch expansion of the weighted count to every fixed order uniformly on that interval, through the coalescence $nt = O(1)$ of the moving pole with the branch point, with a new $n^{-3/2}$ coefficient at the critical weight; tempered 3/2-stable block counts on the scale $tn^{2/3}$ and Gaussian counts beyond it, tempered Poisson large blocks, a Gumbel maximum with a Lambert-$W_0$ centre (an instance of the transseries volume) and a convergent Puiseux inverse. The write proved the manuscript's unproved claim that no continuous Gumbel limit exists at fixed $t > 0$. The $t = 0$ stable count law is the critical-composition (map-Airy) law of Banderier, Flajolet, Schaeffer and Soria and is not claimed new; expansions are fixed-order, the count and maximum theorems marginal, two sketched lemmas are further questions, and the joint theory across $\tau = 0$ stays open. The independent check of Part III, missed at the intake and made on 7 October, found no error, qualified the write's "the rest is new" and read Section IX.11 of Flajolet–Sedgewick, whose Theorem IX.16(ii) contains the $t = 0$ count law in local form. Unrefereed; not formalized.

### 155. Fixed forbidden distances in involutions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a239144-forbidden-distance-involutions/**article.pdf**
from batch 77: fixed-forbidden-distance-involutions.zip +
involution-refinements-attribution-revision.zip (its earlier edition
involution-refinements-reproducibility.zip superseded, not printed)

For involutions of $[n]$ avoiding the transpositions of a forbidden graph of fixed maximum degree, proves a uniform complex probability-generating function expansion to every fixed order in $n^{-1/2}$, with coefficients evaluated at the graph's own matching-root moments, and an exact split of the avoidance count into two Gaussian sectors of relative weights $e^{\pm\sqrt{n}}$, each with its own expansion. For a fixed finite set of forbidden distances the local statistics are eventually exactly affine in $1/n$, so the columns of OEIS A239144 have constant coefficients: $A_{n,r}/I_n = e^{-r}(1 + r/\sqrt{n} - r/(2n) - r(8r^2 - 12r + 3)/(24n^{3/2}) + O(n^{-2}))$. Section 9 holds the second manuscript's case $r = 1$: the all-orders law of the number of adjacent transpositions (A217876; zero column A170941), its total-variation distance from Poisson(1), a second proof by growing moments and a Touchard transform, a Chebyshev sector decomposition with sector ratio $(-1)^n e^{-2\sqrt{n}}(1 - 31/(12\sqrt{n}) + O(1/n))$, and coefficients through order eight. *Merged: the second manuscript proves the case $r = 1$ with the same coefficients* $31/24$, $-55/48$, $1717/1920$*; its inverse statements are given by pointer.* The Gaussian matching identity and the root-moment method are prior work (Godsil, Godsil–McKay, McLeod), the zero-count limit is Ganjtabesh–Steyaert's, the Poisson law Li's and $31/24$ Kotěšovec's. The inverses are instances of the transseries volumes.

### 156. All fixed order asymptotics for distinct parabolic double cosets

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a260700-parabolic-double-cosets/**article.pdf**
from batch 77: double-coset-report.zip

Let $p_n$ count the distinct subsets of $S_n$ that are double cosets of standard parabolic subgroups (OEIS A260700). Starting from Browning's exact enumeration, proves $p_n = Kn!\,\rho^{-2n}(\sum_{m \leq M} B_m(\rho) n^{-m} + O(n^{-M-1}))$ for every fixed $M$, with $\rho = \log 2$, $K = e^{-\rho^2/2}/(4\rho^2)$ and $B_1(r) = r^2/2 - r^3 - 11r^4/12 + r^5/4$; $B_2$, $B_3$, $B_4$ are explicit. The first correction proves Browning's higher-order conjecture with $c = -KB_1(\rho) = 0.10819705289\ldots > 0$, the sign proved without a numerical test. The tools are uniform generalized-Fubini pole extraction, summable bounds on the signed inner defects and a joint bound for the omitted long-cycle configurations of the outer Stirling transform, with a finite exact coefficient algorithm. The leading equivalent is Browning's theorem, novelty rests on a bounded search, and five questions stay open. Smooth Lambert-$W$ inverses and an eventual two-ceiling bracket for the integer threshold are instances of the transseries volume; the report keeps the counterexample $y = p_{25} + 1$ to the naive rounding

$N(y) = \lceil x_1(y) \rceil$. A companion of the A260952 report below; neither uses the other. Reciprocal notes of batch 113 point to `a173217-ordered-tuple-relations`, which propagates every Fubini pole through one signed-Stirling transform (a precedent for one step of its Question 3, not an answer), and to `a104779-total-kostka-sums`, whose asymptotics of A178718 and A321652 count every presentation separately and do not bear on $p_n$.

**157. Full support signed permutations and their asymptotic coefficients**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a260952-full-support-signed-permutations/article.pdf
from batch 77: Coxeter_Full_Support_Asymptotics.zip

Proves Kotěšovec's 2015 conjecture recorded in OEIS A260952, $a_m \sim -2^{m+1} m!/(9(\log 3)^{m+1})$, with the corrections $1 - 2\tau/m - 9\tau^2/(4(m)_2) - 27\tau^3/(2(m)_3) - 1755\tau^4/(16(m)_4) + O(m^{-5})$, $\tau = \log 3$, every further fixed order by a finite formula, and $a_m < 0$ for all $m \geq 1$. The $a_m$ are the asymptotic coefficients of the full-support elements of the Weyl groups of types $B$ and $D$ (A109253, A112225); they are written as a positive Stirling transform whose late expansion holds to every fixed order on the scale $\log(1+q)$, uniformly in $q$. The two exact endpoint sums of the signed-permutation model have separate all-order expansions for $1/4 < |q| < 4$ (not extended to $q \geq 4$), which give an all-order sparse-sign crossover: the number of negative entries tends to a Poisson law conditioned to be positive. The models are Bergeron–Hohlweg–Zabrocki's and the Stirling-transform construction Martin–Kearney's. Lambert-$W$ growth-index inverses with a stated rounding width are instances of the transseries volume. A companion of the A260700 report above; neither uses the other.

**158. Ordered long-jump rook paths**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a227578-ordered-rook-paths/article.pdf
from batch 77: ordered-rook-reproducibility.zip

Let $A(n,k)$ count the paths from $(n,\dots,n)$ to the origin that lower one coordinate by any positive amount at each step and stay weakly ordered (OEIS A227578; columns A059231, A227580, A227583, A227596). For every $k \geq 2$ the report proves an exact coefficient identity with a squared Vandermonde numerator, so each column is D-finite (no posted recurrence is certified), and, by the Raichev–Wilson smooth-point theorem, $A(n,k) = C_k (k+1)^{kn} n^{-(k^2-1)/2} (\sum_{j<M} d_j(k) n^{-j} + O(n^{-M}))$ for every $M$, with $C_k$ in closed form through a Mehta-type Gaussian integral, $d_1(k) = -(k-1)(k+1)(2k^4+8k^3+9k^2+6k+12)/(12k(k+2)^2)$ and $d_2(k)$ in closed form; every $d_j(k)$ comes from a finite generator and is a rational function of $k$ with poles only at $0, -1, -2$. This proves the general fixed-$k$ growth that OEIS records as a conjecture; the leading constants for $k \leq 5$ were already recorded there by Kotěšovec, and the methods are classical. The $W_{-1}$ inverse and the two-ceiling threshold enclosure are instances of the transseries volume. "Rook" means a move here, not a rook polynomial.

**159. Regular cyclic word covers**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a108242-regular-cyclic-word-covers/article.pdf
from batch 77: regular-cyclic-word-covers-reproducibility.zip

Counts sets and multisets of rotation classes of words of fixed length $\ell \geq 3$ over an $n$-letter alphabet in which every letter occurs exactly $r$ times. An exact slot-partition identity gives, for fixed degree, the ratio to the carrier $(rn)!/(\ell^{rn/\ell}(rn/\ell)!\,(r!)^n)$ to every order in $1/n$, with rational coefficients and absolute remainder control. For $\ell = 3$ it identifies OEIS A108242, A110105, A110104 and A110106 (following the examples and Mishna's source where OEIS titles disagree with the parameters), with corrections through $n^{-5}$, such as $1 + 8/(9n) + 104/(81n^2) + \cdots$, and the corrections each posted carrier induces. At the critical degree $r = \lambda n^{(\ell-2)/2}$ it proves an expansion to every fixed order in $n^{-1/2}$, uniform on compact positive $\lambda$-intervals, with multiplier $\exp(s\lambda^2/(2\ell) + \mathbf{1}_{2|\ell}\,\lambda/\ell)$, $s = \pm 1$; the even-length term comes from half-turn symmetry. The leading equivalents were already posted by Kotěšovec, and the literature comparison is an applicability screen, not a priority claim. A $W_0$ inverse with a lattice threshold enclosure is an instance of the transseries volume. The open cyclic-words question of the A330266 report concerns other objects and is not answered.

**160. Unitary divisor sum colored partitions and their asymptotics**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a301981-unitary-divisor-partitions/article.pdf
from batch 77: unitary-partition-report.zip; batch 86: oeis_advances.zip (its Section 7)

For the products $\prod(1-z^m)^{-\sigma^*(m)}$ (OEIS A301981) and $\prod(1+z^m)^{\sigma^*(m)}$ (A301982), $\sigma^*$ the unitary divisor sum, proves that the stretched-exponential equivalents conjectured in both OEIS entries are false: the ratio of coefficient to model does not eventually stay in any compact subinterval of $(0,\infty)$, by a least-height zeta-zero noncancellation (the Mellin kernel has denominator $\zeta(2s-1)$) and an Abelian transfer. Unconditionally $\log a_n = c\,n^{2/3} + o(n^{1/3})$ with $c = 3(A/4)^{1/3}$, $A = \pi^2/6$ for A301981 and $\pi^2/8$ for A301982, and the error is not $O(n^{1/4-\varepsilon})$; for each sequence separately, an error $O_\varepsilon(n^{1/4+\varepsilon})$ for every $\varepsilon > 0$ is equivalent to the Riemann hypothesis, which is neither proved nor disproved. Also a corrected equivalent at the explicit point $(2A/n)^{1/3}$ (without a rate), all-fixed-order exact-saddle and off-center Hermite expansions, and two-ceiling threshold brackets whose integer step is an instance of the transseries volume. Part I left open which one-sided divergence occurs; Part II (batch 86, from one section of a three-subject manuscript) proves that both do: by Landau's positivity principle with the surviving pole's order $m$ and leading Laurent coefficient, the ratio to the refuted model has $\liminf a_n/m_n = 0$ and $\limsup a_n/m_n = \infty$ for both products, with $\log(a_n/m_n)$ beyond $\pm K_\pm n^{\sigma_*/3}(\log n)^{m-1}$ infinitely often, hence $\Omega_\pm(n^{1/4})$ (exact under the Riemann hypothesis, with Part I), without the Riemann hypothesis or simple zeros; the inverse of the pure model errs by $\Omega_\pm((\log y)^{7/8})$. The refutation concerns OEIS text, not a repository claim, and no OEIS submission is implied.

**161. Log-periodic factors in radix-layer partition asymptotics**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a174065-radix-layer-partitions/article.pdf
from batch 77: radix-partition-reproducibility.zip

For $b \geq 2$ let $a_b(n) = [z^n]\prod_{j\geq 0,\, i\geq 1}(1+z^{ib^j})$; for $b = 2^m$ it counts partitions into parts $k$ with $v_2(k) \equiv 0 \pmod m$: ordinary partitions at $b = 2$, OEIS A174065 at $b = 4$, A393565 at $b = 8$. The report proves an expansion of $a_b(n)$ to every fixed order against $KN^{-\kappa}e^{2\sqrt{AN}}$, with explicit differential operators acting on a positive analytic 1-periodic function $H$ of $\log_b \sqrt{N/A}$, and shows by the zero-free line of $\zeta$ that $H$ is nonconstant for every $b > 2$. Hence the equivalents recorded in A174065 and A393565 have correct smooth constants but omit a persistent log-periodic factor; a separate Abelian argument excludes every constant-only equivalent, and the subsequential ratios fill exactly $[\min H, \max H]$, with $(\min H)(\max H) = 1$ from an exact radial reciprocity. Also eventual strict increase and log-concavity, and a two-ceiling threshold enclosure whose integer step is an instance of the transseries volume. The methods are classical Mellin analysis (Flajolet–Gourdon–Dumas, Madritsch–Wagner). The correction concerns OEIS text, not a repository claim, and no OEIS edit was made. The mechanism, but no theorem, is shared with the A033552 report.

**162. Full asymptotic expansions for partition sums of acyclic orientations**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a372395-acyclic-orientation-partitions/article.pdf
from batch 77: orientation-partition-asymptotics.zip

OEIS A372395 and A370613 sum the numbers of acyclic orientations of the complete multipartite graphs $K_\lambda$ over the partitions $\lambda$ of $n$, unrestricted or into distinct parts. The report proves $B_\pm(n) = A_\pm n!\, n^{-p_\pm} e^{C_\pm\sqrt{n}}(\sum_{j\leq J} a_{\pm,j} n^{-j/2} + O(n^{-(J+1)/2}))$ for every $J$, with $p_+ = 1$, $p_- = 3/4$, explicit amplitudes including a tilted Gamma-integral factor, and $a_{+,1} = -0.30756533\ldots$, $a_{-,1} = 0.23095021\ldots$; this answers the question of Sun (arXiv:2605.04006), to whom the exact Gamma representation and the constants $C_\pm$ are due. A uniform quadratic majorant controls the signed representation; a finite Gaussian-expectation algorithm gives every coefficient, and a second short route gives $a_{\pm,1}$. A corollary extends the expansion to the sums of $(-1)^{|V|}\chi_{K_\lambda}(-v)$ for fixed $v > 0$, with $\Gamma(n+v)/\Gamma(v)$ in place of $n!$. The expansions are Poincaré expansions, not transseries, and four questions stay open. The Lambert-$W$ inverse with its integer-threshold envelope is an instance of the transseries volumes.

### 163. Sparse nonattacking chess placements

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a201513-sparse-chess-placements/article.pdf

from batch 77: Sparse_Chess_Asymptotics.zip

For $n$ nonattacking kings (OEIS A201513) or knights (A201540) on an $n\times n$ board, measured against Kotěšovec's common leading equivalent $B_n = e^{-9/2}n^{2n}/n!$, proves rational expansions to every fixed order, four corrections explicit: $K_n/B_n = 1+7/(3n)+119/(9n^2)+11327/(810n^3)+90475/(1944n^4)+O(n^{-5})$ and $N_n/B_n = 1 + 37/(3n) + 914/(9n^2) + 330917/(810n^3) + 966583/(1944n^4) + O(n^{-5})$. For any fixed finite symmetric move set $S$ the count is $(n^{2n}/n!)\,e^{-(|S|+1)/2}$ times an all-orders series, also uniformly for $k$ pieces with $k/n$ in a compact positive interval, and the first correction is $c_1 = B + 1/3 - \tau$ (boundary loss $B$, triangle density $\tau$), which is why kings and knights share the leading term but not the next; wazir (A201511) and fers (A201861) coefficients through order four follow. The cluster-expansion method is credited to Davies–Jenssen–Perkins and Pulvirenti–Tsagkarogiannis. The coefficients depend on the carrier: against the Stirling-expanded carrier the first corrections are $9/4$ and $49/4$. The Lambert-$W$ growth-index inverse and its rounding qualification are instances of the transseries volume.

### 164. Asymptotic enumeration of tied football seasons

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a380592-tied-football-seasons/article.pdf

from batch 77: tied-football-report.zip

In a double round-robin league of $n$ labelled teams with 3/1/0 scoring, $a(n)$ (OEIS A380592) counts the seasons in which every team finishes on the same score. The report proves $a(n) \sim B_n\Theta(c_n)$ with $B_n = 3^{n(n-1)}n^{-1/2}(9/(56\pi n))^{(n-1)/2}e^{-9829/65856}$, $c_n = 8(n-1)/3 - 13/84$ and the positive 1-periodic theta function $\Theta(c) = \sum_j e^{-4\pi^2 j^2/9}e^{2\pi ijc}$, so the multiplier depends on $n$ mod 3 and cannot be replaced by 1. The proof uses mean-matching exponential tilts and a sharp global Fourier localization: a negative quartic gives an exact Gaussian majorant and an outlier bound handles the complement. On each residue class the conditioned common score tends to a discrete Gaussian centred at $c_n$, which is not its conditional mean. After checking the hypotheses of Isaev's complex cumulant theorem (arXiv:2508.16952), the external input, it gives every fixed order in $1/n$ with a finite exact algorithm and the first correction. The residue-aware inverse threshold enclosures need no Lambert function, the phase being quadratic, and are instances of the transseries volume's reversion and staircase theorems. The model and its exact counts are Jehn–Habermann–Lavrov's.

### 165. All-orders asymptotics and asymptotic inversion for A292507

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a292507-binomial-partition-transform/article.pdf

from batch 77: A292507_report_and_code.zip

For $a_n = \sum_k \binom{n}{k} b(k)$, where $b(k) = p(k) - p(k-1)$ counts the partitions of $k$ without parts 1 (OEIS A292507), proves an all-orders Poincaré expansion $a_n = C\,2^n n^{-3/2} e^{\beta\sqrt{n}}(\sum_{j\le M} c_j n^{-j/2} + O(n^{-(M+1)/2}))$ with $\beta = \pi/\sqrt{3}$ and $C = (\pi/6)e^{\pi^2/24}$, an additive form of Kotěšovec's logarithmic conjecture in the entry; $c_1, c_2, c_3$ are in closed form, and a finite Gaussian-moment algorithm gives every $c_j$ ($c_1, \ldots, c_6$ computed). Also an all-orders reversion of the real model in $X^{-1/2}$ and $\log X$ with a residual bound, and shrinking integer-threshold brackets with existence constants only, not certified thresholds; these are instances of the transseries volume's Lambert, perturbed-inversion and staircase theorems and are cited, not claimed. The only analytic input is Nemes' explicit partition-remainder estimate. The leading binomial-smoothing mechanism is prior art (A218481), the coefficient formula is Gutkovskiy's, and no publication priority is claimed; nor is convergence of the correction series or a complete transseries.

### 166. Permutations avoiding the vincular pattern 12–34 (A113226)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a113226-vincular-avoiders/article.pdf

from batch 77: a113226-asymptotics.zip + a113226-refined-addendum.zip; batch 101: Exact_Vincular_Avoidance_Asymptotics_Source.zip (bundle Report 87, Part III), Canonical_Vincular_Exponential_Sectors_Source.zip (bundle Report 94, Part IV)

For the permutations avoiding the vincular pattern 12–34 (OEIS A113226), equivalently Bevan–Cheon–Kitaev's $\{3, 2+2\}$-free naturally labelled posets, Part I derives the closed EGF $H(z) = \exp\{z/2 + q(3\arcsin q - \pi/2)/\sqrt{1-q^2}\}$, $q = e^{z/2}/2$, identifies the cumulants as A136127 shifted, and proves the stretched-exponential form of Bevan–Cheon–Kitaev's Conjectures 13 and 14 with exact constants and to all orders: $A_n \sim D\, n!\, \rho^{-n} e^{3cn^{1/3}} n^{-5/6}(1 + \sum_j a_j n^{-j/3})$ with $\rho = \log 4$, $c = (\pi^2/(4\rho))^{1/3}$ and $D = 2e^{-3}\sqrt{c/(3\pi)} \approx 0.0357$, where their fit gave about 0.032; $a_1, \dots, a_4$ are exact. Its Lambert-$W$ inverse is an instance of the transseries volume. Part II, an addendum, marks the distinct strict downsets and the isolated elements: a record-to-cycle bijection explaining the exponential transform, a complex-uniform saddle expansion, all-orders local large deviations for interior densities, Gaussian and local limit laws, and a conditional Poisson law of isolated elements with total-variation error $O(n^{-2/3})$. Endpoint densities and certified constants are not claimed; Part II answers Part I's questions 1–2 in part, question 3 stays open, and question 4 is answered for every fixed sector by Part IV. The two heavy exact tables are not shipped; the README regenerates them. Part III (batch 101), a same-day independent answer to Part I's assignment written against an earlier pin, re-derives the EGF, the recurrences, the A136127 identification and $a_1, a_2, a_3$ from the permutations themselves by a diagonal-state route (printed as second routes), gives integral forms of Elizalde's block functions, and proves that the EGF is differentially algebraic but not D-finite, so A113226 is not P-recursive. Part IV (batch 101), which continues Part III, gives a single-valued branch of period $4\pi i$ cut horizontally from every $\zeta_k = \log 4 + 2\pi i k$, the exact decomposition $A_n/n! = \sum_k H_k(n)$ for $n \geq 1$ into absolutely convergent full-cut sectors, and a complete expansion $H_k(n) = C_k \zeta_k^{-n} e^{3\kappa_k n^{1/3}} n^{-5/6}(\sum_j c_{k,j} n^{-j/3})$ of each fixed sector, whose $k = 0$ case is Part I's theorem; nothing is uniform in a growing $k$, and no transseries, resummation or certified sector value is claimed. The write proves four steps the sources assert, among them the displayed two-term reversion of Part III's inverse; the independent check found one of its sentences false (successive reversion terms, unlike their bounds, need not gain $X^{-2/3}/\log X$: the third term is of order $X^{-1}(\log X)^{-4}$ and the fourth of order $X^{-5/3}(\log X)^{-4}$), corrected with that counterexample in a dated note. Unrefereed; not formalized.

### 167. Asymptotic expansions and inversion for A124380

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a124380-signed-moment-asymptotics/article.pdf

from batch 77: a124380-reproducibility.zip

For $\sum_n a_n z^n = \sum_k z^k \prod_{j \leq k}(1 + jz)$ (OEIS A124380), proves an exact decomposition $a_n = I_n + (-1)^n J_n$ into a positive and a signed moment integral, the multiplicative equivalent $a_n \sim \frac{1}{2} e^{1/2} x \exp\{x^2(2L-1) + x(L+1) + L^2/8\}$ with $x = \sqrt{n/2}$ and $L = \log x$, which is more than Kotěšovec's logarithmic conjecture in the entry, and every fixed-order relative expansion with coefficients polynomial in $L$. The signed sector is exponentially smaller and has its own all-orders oscillatory expansion, with the error measured against its amplitude so that it stays valid near zeros of the sine, and coefficients $q_j^-(L) = (-1)^j q_j(L - i\pi)$. Also eventual strict log-convexity, an all-orders inverse in $X = \sqrt{y/W(y/e)}$, an instance of the transseries volume's inverse-Gamma core with no novelty claimed, and a conditional certification rule for the integer threshold; $T(y) = \lceil \nu(y) \rceil$ is not asserted. The combinatorial models are Chen–Fan–Zhao's and Cerbai–Claesson–Sagan's, no priority is claimed, and four research questions stay open.

### 168. Complete increasing subsequences in multiset permutations (A047909 and A268485)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a047909-beta-renewal-subsequences/article.pdf

from batch 77: beta-renewal-asymptotics-reproducibility.zip + beta-renewal-uniform-reproducibility.zip

Studies the probability $p_m(k) = H(m,k)(m!)^k/(mk)!$ that a word with $m$ copies of each of $1,\dots,k$ contains $12\cdots k$ (OEIS A047909, diagonal A268485). Its representation as the probability that $k$ independent Beta$(1,m)$ increments sum to at most 1, and the Gaussian limit, are El Maazouz–Pitman's and are credited, not claimed. Part I proves an expansion to every fixed order in the window $k = m + x\sqrt{m}$ with $P_1,\dots,P_4$ explicit ($P_1 = (x^2+8)/6$); on the diagonal only odd half-integer powers occur beyond the leading $1/2$, with four corrections computed. It adds smooth quantile and Lambert-$W$ count inverses with qualified integer brackets and all-orders rare-tail expansions for $k/m$ in compact subsets of $(0,1)$ or $(1,\infty)$. Part II, an addendum, joins the regimes: relative error $O(m^{-(R+1)/2})$ for both tails, uniformly on compact ranges of $k/m$ and through $k = m+1$, recovering Part I's coefficients by a second route, and both real Lambert-$W$ branches for tail thresholds. The inverses are instances of the transseries volume and the tilting method is classical (Esscher, Cramér, Daniels). Effective constants, growing orders, ratios tending to 0 or $\infty$, and certified rounding remain open.

### 169. Uniform asymptotics and moving zeros for the level-15 family of Cooper

generating-functions-and-asymptotics/oeis-sequence-asymptotics/cooper-level-15-moving-zeros/article.pdf

from batch 77: modular-crossover-replay.zip

For Cooper's level-15 polynomials $T_\varepsilon(n)$, of degree $n$ in the parameter $\varepsilon$, proves a two-singularity expansion with rates $\varepsilon+11$ and $\varepsilon-1$, uniform to every fixed order on the disc $|\varepsilon+5| \le 1/10$, with an additive error that stays valid where the two terms cancel; in the window $\varepsilon = -5 + u/n$ the profile is $e^{u/6} + 10(-1)^n e^{-u/6}$. Hence for every large odd $n$ there is exactly one zero with $|u - 3\log 10| < 1$, real and simple, at $\varepsilon_n = -5 + 3\log(10)/n + (162/125 - \frac{9}{2}\log 10)/n^2 + O(n^{-3})$, and no real zero in a bounded window for large even $n$. Explicit path, sheet and connection proofs give all ten constants of Cooper's Table 8 (levels 11, 14, 15 and 24), labelled conjectured and determined numerically there, including two negative-axis connections with exact eta multipliers. The formulas are Cooper's, the level-11 equivalent is Kotěšovec's (A284756) and the positive modular radial principle Guillera–Zudilin's. Also a finite generator for every correction order and a qualified, non-effective threshold inverse, an instance of the transseries volume. No historical priority and no global location of the zeros are claimed; the OEIS numbers of five of the sequences were not verified.

### 170. Moving fugacity asymptotics for distinct partitions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a291698-moving-fugacity-partitions/article.pdf

from batch 77: moving-fugacity-reproducibility.zip

For $a_n(\alpha) = [q^n]\prod_{k\ge1}(1+n^\alpha q^k)$, distinct partitions with one size-dependent weight $u = n^\alpha$ per part, proves, uniformly for $\alpha$ in compact subsets of $(0,\infty)$, an all-orders principal Bessel expansion with a finite generator for its coefficients and, through an exact Jacobi decomposition, finitely many conjugate resonance sectors with separated local integrals and sectorwise remainders. This yields the equivalents conjectured by Kotěšovec for A291698 ($\alpha=1$) and A292304 ($\alpha=2$), and shows that the elementary Sommerfeld form is relatively valid exactly when $\alpha \ge 1/2$. Also a specified monotone real continuation with a Lambert-$W$ inverse and its first correction, a formal reversion generator and an integer-threshold statement, instances of the transseries volume with no novelty claimed. The onset is slow: at $n = 20{,}000$ the principal term for $\alpha = 2$ is still off by about 5.6%. The sector count and depth are fixed; no growing-sector theorem, convergence or explicit constants are claimed, nor priority, and four questions stay open.

**171. Gamma constants and scaling limits for rounding extinction**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a082528-rounding-extinction/article.pdf
from batch 85: Rounding_Extinction_OEIS.zip; batch 110: Sequential_Power_Rounding_Gamma_Constants_and_Inverses_Source.zip (bundle Report 188, Part II) + Periodic_Rounding_Extinction.zip (Part III)

Starting from $x_1 = n$ and rounding down to a multiple of $k^m$ at stages $k = 2, 3, \ldots$, the first zero satisfies $\tau_m(n) \sim (c_m n)^{1/(m+1)}$ with $c_m = m\Gamma(m/(m+1))^{m+1}$ for every real $m > 0$: Cloitre's conjecture in OEIS A082528, with the constants $2\Gamma(2/3)^3$ and $3\Gamma(3/4)^4$ conjectured numerically for A082527 and A082528. The case $m = 1$ ($c_1 = \pi$, the Tchoukaillon sieve A002491) is Erdős and Jabotinsky's and is credited, not claimed. Also the uniform limit of the whole extinction trajectory, the same law for weights with $k(w_k/w_{k-1} - 1) \to m$, rational product certificates for $c_m$, the limiting distribution of rounding losses, the dependence on $m$, and a block construction showing that regular variation alone does not fix the constant. No error term (the $O(1)$ recorded in A073047 is not proved), no rate, nothing uniform in a moving $m$; Appendix B is an OEIS update draft that was not posted, and the OEIS terms the package uses are CC BY-SA 4.0 data. Part II (batch 110, bundle Report 188, written without knowledge of Part I) proves $B_p(k) = K_p k^{p+1} + O(k^{(p+1)^2/(p+2)})$ uniformly on compact $p$-sets, through a monotone integral coordinate and dyadic telescoping, with an effective all-index enclosure, a global minimal-path profile with rate and a positive-terminal law: it answers Part I's question on rates, its leading law is a second route to Part I's Theorem 1.1, and the write corrects the last digits of its decimals $c_2$, $c_3$ and its stale novelty sentences. Part III (batch 110, from a non-bundle archive held from batch 114) answers the question on schedules for nondecreasing weights with bounded scaled increments and a convergent augmented law — finite Gamma products, the activation jump $1/(2\pi) \to \pi/8$, a schedule with no limit, the exact spectrum $(0, 1/(m+1))$, Möbius recovery — and the bounded form of the sparse-exception question; sparse large jumps and the other questions stay open. The independent check confirmed every number and the write's four remarks, and corrected archive sizes, an intake remainder band and one perturbation sentence. Unrefereed; not formalized.

**172. Matrix compositions: exact spectra, Hankel products and uniform asymptotics**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a261781-matrix-compositions/article.pdf
from batch 85: Matrix_Compositions_Uniform_Asymptotics.zip (base) + OEIS_Matrix_Compositions_Research.zip; batch 98: a261781_hankel_zero_laws.zip (Part III); batch 108: Packed_Matrices_Asymptotics_Inverses_and_Joint_Limits_Source.zip (bundle Report 169, Part IV)

For matrix compositions with $k$ rows (OEIS A261780, A261781) and the diagonal A261784, Part I proves, for every column weight $u > 0$, that the reduced denominator of column $k$ is $\prod_{j \le k}((1+u)(1-z)^j - u)$, so the minimal eventual constant-coefficient recurrence has order exactly $k(k+1)/2$; the recurrence itself, with that denominator, is Munarini, Poneti and Rinaldi's (2009), and only minimality is new, which narrows one manuscript's claim to prove both posted conjectures. Also the maximal Hankel determinants as explicit products in $\mathbb{Z}[u]$ with their other zeros on $\Re u = -1/2$; a pole bound uniform in $n$ and $k$; the coverage window $n = k(\log k + s)$ with the sharp total-variation rate and a joint Gaussian–Poisson law of column and zero-row counts; uniform saddle expansions to every order on compact rays $n/k \in K \Subset (1, \infty)$, with the closed amplitude and two corrections for A261784; and a prescribed-column threshold. Part II, at $u = 1$, adds strict monotonicity of the growth-constant increments, every fixed order of the coverage probability, the entire Stirling kernel, the Poisson law of a Stirling defect, a third diagonal correction and a Lambert-$W$ index inversion with integer brackets, an instance of the transseries volume. The two manuscripts proved the same core theorems independently and permute their Greek letters, so the second is printed in the first's notation with a dictionary, its re-proofs tabulated. The Fubini pole and the positive Stirling transform are those of `a260700-parabolic-double-cosets`; the only formalized ingredient is the Fubini generating function in the Lean `Analysis/FabiusFunction` tree. No uniform statement for $T(n, k)$, no $n/k \to 1$, no exponentially small sectors; OEIS fixtures

are CC BY-SA 4.0 and the OEIS draft is not submitted. Part III (batch 98) answers Part I's question on the limiting distribution of the determinant zeros: the zero-counting measure of the reduced Hankel determinant, normalized by its degree, tends to the law of $-1/2 + iY$ with $Y$ Cauchy of scale $1/2$, at Kolmogorov distance $\sim (9/\pi^2)(\log k)/k$ (a sharp constant); cyclotomic multiplicities $E_m(k) = k^2 \log k/(\zeta(2)\psi(m)) + O(k^2)$ uniformly in $m$ ($\psi$ Dedekind's function); a first correction for smooth statistics, a divisor-sum edge law with the exact extreme zeros, a transition formula for every height, the moment transition at $p = 1$ and the full-degree limit in $s/k^2$. Its re-proof of the determinant product is printed as a pointer. The multiplicity estimate has an absolute error only, the sharp constant comes with no quantitative error, and the edge law concerns zeros of order $k^2$, not the largest few. The independent check of Part III, missed at the intake and made on 7 October, found no error and corrected the wording of its batch-98 reciprocal note on `a122399-surjection-diagonal`. Part IV (batch 108, bundle Report 169, written without knowledge of Parts I–III, Sections 34–44) reaches the diagonal A261784 through the Stirling transform and a single saddle: its amplitude (its $\beta_1$ is Corollary 7.4's $\beta$ as the same rational function), corrections $\beta_2, \beta_3$, column mean and variance, first inverse correction and integer brackets re-prove results of Parts I and II and are printed in full as marked second routes, a table renaming about forty of its letters, while the inverse terms in $x_0^{-2}$ and $x_0^{-3}$ (shown to hold for Part II's inverse too), an explicit pole-tail bound, a complex-uniform expansion in a column marker and a marker for cells with entry at least 2, and an independent joint limit of the column count (Gaussian) and the number of such cells (Poisson with mean $Lr = 2v = 1.1046\ldots$), transferred to the total excess, are new. It answers no question of the host; it bears on Part II's latent-defect question and answers an analogue, at the single ray $n/k = 2$, of its joint-laws question (the write's "settles one instance" was corrected by the independent check: the repeated-cell count, Poisson$(2v)$, is not a realization of the defect, Poisson$(v)$). Its questions 1–5 become Research questions 17–21 (the second re-scoped) and its sixth is printed once in Part II; fixed orders only, no local limit for the column count, and its $N = 800$ diagnostics stay uncertified. Unrefereed; not formalized.

### 173. The source boundary of extensional acyclic digraphs

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a182220-source-boundary/article.pdf`

from `batch 85: OEIS_A182220_Source_Boundary_Research.zip; batch 98: OEIS_Source_Boundaries.zip (base) + OEIS_EAD_Natural_Boundaries.zip (Part II)`

An extensional acyclic digraph on $n$ vertices has at most $n - \lceil \log_2 n \rceil$ sources, and every smaller positive count occurs: the formula still labelled conjectural in OEIS A182220, whose upper half is Tomescu's (2011, credited). The extremal classes are classified, so the boundary diagonals $b_n^{(d)} = u(n, n - \lceil \log_2 n \rceil - d)$ of A182162 have $b_n^{(0)} = u_q \binom{2^q - q}{n-q}$ ($u_m = \text{A001192}(m)$) and exact $(d+1)$-term formulas; a uniform covering bound gives a dyadic entropy law, the subsequential growth rates fill $[e^{\lambda_d}, e^{\lambda_{d+1}}]$ ($[1, 4]$ for $d = 0$), and the jumps at powers of two make every $b^{(d)}$ and $n!\, b^{(d)}$ non-P-recursive, a counterpart of a lemma of `a003407-dyadic-scaling-rigidity`. Also an all-orders expansion with $u_m$ kept exact and a total-variation binomial law for arc deficits. The source sieve and the collapse to transitive sets are known; no priority, no natural-boundary claim for the diagonal series in Part I, no formal proof; the OEIS-facing appendix is an unsubmitted draft, and the code's OEIS fixtures are CC BY-SA 4.0 data. Part II (batch 98) merges two independent answers to Part I's first question, with the one stronger on every shared theorem as base: for every fixed $d$ the circle $|z| = R_d = e^{-\lambda_{d+1}}$ is a natural boundary of the diagonal series $B_d$; at every root of unity $\zeta$ of power-of-two order the radial ratio is $\zeta(1 - aR_d)/(1 - aR_d\zeta) + O(t)$ at radius $R_d e^{-t}$, $a = 2^{d+1} - 1$; $\log B_d(R_d e^{-t}) = L^2/(2\log 2) + L \log L/\log 2 + O(L)$, $L = \log(1/t)$; two-sided $L^p$ bounds on every arc; and no bidisk. Also an abstract transfer lemma for coherent dyadic blocks, natural boundaries for every residue class of power-of-two step, a bijective proof of Tomescu's deletion recurrence and first block corrections. The write added, with proofs, that every arithmetic section has a natural boundary, and bounds for the $O(L)$ term; the independent check found them valid, derived when the radial ratio of a section has a limit, and corrected two

values in a numerical note. One manuscript's mention of "independent internal reviews" could not be verified and is not evidence of review. Nothing is claimed at non-dyadic angles or for growing defects. Unrefereed; not formalized.

### 174. Finite Fredholm truncation for long increasing subsequences

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a047874-long-increasing-subsequences/article.pdf

from batch 85: LIS_fixed_sector_research_package.zip

For the permutations of $N$ letters with longest increasing subsequence of length exactly $k$ (OEIS A047874) or at least $k$ (A214152), truncating a Bonferroni expansion over shifted rows of the Robinson–Schensted shape at $R$ terms gives globally holonomic arrays that agree with the true ones for $N < (R+1)(k+R)$, a cutoff shown sharp, so both arrays are D-finite in every fixed sector $N \le rk$: the sector question that Kauers and Wang (arXiv:2609.02220) raise informally, whose case $r = 2$ is their theorem. Hence P-recursiveness along every rational ray; single sums when $N \le 2k+1$; a uniform all-orders expansion of the tail on compact ranges of $(N-k)/k$, with polynomial coefficients; and five correction terms for A269021 (whose leading term is Kotěšovec's) and three for A267433. The holonomic-closure step is flagged for independent review; the guessed operators of the one-third sector, minimal orders and global holonomicity are not claimed, and the discrete Bessel kernel and holonomic summation are imported. Not formalized; OEIS fixtures are CC BY-SA 4.0 data.

### 175. Fixed-height shifted rectangles

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a181199-shifted-rectangles/article.pdf

from batch 85: Shifted_Rectangle_Asymptotics.zip (Part I); batch 98: Shifted_Rectangle_Boundary_Research.zip (Part II); batch 101: Report231.zip (Part III), Report233.zip (Part IV), ProveIt_Fixed_Height_Shifted_Strips_Asymptotics.zip (bundle Report 85) and ProveIt_Shifted_Strips_Rational_Diagonals_and_Transcendence.zip (bundle Report 91) (Part V)

For $T_m(n)$, the $m \times n$ arrays of $1, \dots, mn$ increasing along rows, columns, diagonals and downward antidiagonals (OEIS A181196; $m = 4, 5$ are A181198, A181199), five Parts. Part I proves an all-orders expansion at every fixed height $m$ with rational coefficients from a finite moment formula, via a random-threshold cut and a Gaussian–Vandermonde ensemble; for $m = 5$ this is Kotěšovec's conjectured $T_5(n) \sim 9 \cdot 5^{5n+1/2}/(2^{17}\pi^2 n^{12})$, proved without the guessed recurrence, with five corrections. The generating function is algebraic exactly for $m = 1, 2$. Also the ratio to ordinary rectangular standard Young tableaux, $2^{-m(m-1)}(1 + m(m^2-1)/(4n) + \cdots)$, whose normalizer is the transseries volume's rectangle-tableau asymptotic; GUE-type limit laws for the intermediate row lengths; eventual monotonicity and log-convexity; and a $W_{-1}$ inversion, an instance of the transseries volume. Part II (batch 98) proves that the generating function is the diagonal of a rational series, hence D-finite, at every fixed height (event-word binomial sums and Bostan–Lairez–Salvy), and gives the sharp all-order expansion of the rare count $R_m(n)$ of arrays whose first row ends before the last row starts, with $R_m(n)/T_m(n) \asymp n^{2m-5/2}4^{-n}$. The write adds the leading term of Part I's boundary error and narrows the source's framing: $R_m$ is a much smaller sector of that error, so the source does not answer Part I's question on it (its all-order expansion stays open), and of Part I's question 7 only D-finiteness is, not the growth of the minimal order. Parts III and IV (batch 101, Research Reports 231 and 233) prove, from the array definition and for every $n \ge 1$, the order-two, degree-nine A181198 recurrence and the order-three, degree-24 A181199 recurrence that OEIS displays as conjectural, and Kauers–Koutschan's Conjectures 18 and 19, the printed finite sums (the first for $n \ge 2$), through Pfaffian formulas for a Volterra transfer, one or two period integrals and explicit rational certificates; each adds a sixth correction, and Part IV evaluates effective error constants. The write shows that the A181198 operator has minimal order and that A181199 obeys no recurrence of order one; the independent check found and proved recurrences of lower degree at higher order (degrees 8 and 7 at orders 3 and 4 for A181198, 18 and 16 at orders 4 and 5 for A181199). Part V prints Reports 85 and 91 (1–2 October 2026, before Parts I–II and without knowledge of them), earlier independent routes to

the all-orders expansion, the algebraicity dichotomy and the rational-diagonal theorem; the report credits that chronology. Open: whether A181199 has a recurrence of order two, everything at $m \geq 6$ and any growing-height statement. The shifted hook formula and Sun's order-polytope reading are credited; the two OEIS drafts, which say not to mark the recurrences proved, are unchanged and not submitted. Unrefereed; not formalized.

### 176. Zero auxiliary clipping tables and A196460

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a196460-clipping-tables/article.pdf

from batch 91: A196460_asymptotics_and_inversion_sources.zip, A196460_sharp_uniform_truncations_sources.zip

For each fixed Boolean table $S \subseteq \{1, \ldots, K\}^2$ read on positive integers by clipping its two coordinates, one integer polynomial equation needs no auxiliary exactly when the table is closed under coordinate-flat replacement, and one strictly positive auxiliary otherwise. The closed tables of side $n+1$ number $C_n = 1 + a_n$, where $a_n$ is A196460. Part I gives exact counts and fixed-order forward, logarithmic and inverse expansions. Part II bounds all truncations of the exact finite sector expansion for $n \geq 32$, keeping the distinct terminal-sector convention for $C_n$. This is not a uniform growing-order logarithmic or inverse theorem: an exact integer threshold still needs an exact comparison. Coefficients and degree may depend on the whole finite table; no uniform paid integer compiler or universal operation bound follows. Part I's classification theorem is that of Research Report 69, source 38 of `signal-machine-collision-certificates`, restated with proof (dated notes in both reports); its inversion sections redo the companion transseries volume's connected-graph method, and its constants $b_k$, $c_k$ are A047863 and A002027, credited at the write. Two manuscripts, integrated and reviewed within the publication review's stated scope at `9f924ac47`; that review did not certify every analytic proof. Unrefereed; not formalized.

### 177. Strict twice partitions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a271619-strict-twice-partitions/article.pdf

from batch 100: Strict_Twice_Partitions_Phase_Asymptotics_Source.zip (bundle Report 182, base) + Nested_Partitions_Asymptotics_and_Inverses_Source.zip (bundle Report 181)

Twice partitions with a strict outer level. Part I treats OEIS A271619, $a(n) = [q^n] \prod_{k \geq 1}(1+p(k)q^k)$, blocks of distinct sizes: an expansion to every fixed relative order in $M^{-1/2}$, $M = \lfloor(\sqrt{8n+1}-1)/2\rfloor$, uniform through triangular boundaries, through two effective charges $M$, $M-1$ that exchange dominance near the triangular phase 9/16 over a persistent low-hole cloud ($C_0 = \prod(1+1/p(k)) = 7.6015\ldots$, constants enclosed in integers), with explicit first and second corrections, integer inverse enclosures, $\log a(n) = (2b/3)(2n)^{3/4} + O(\sqrt{n}\log n)$ for $b = \pi\sqrt{2/3}$, the failure of a single central minor-arc bound, a tame edge expansion through Bloch–Okounkov brackets, the crossover centre and a total-variation law for the outer count. It is an eventual theorem with no effective onset: at $n = 5000$ the leading formula is about 562 times too small. Part II treats A358836, $\prod(1 + q^k/(q;q)_k)$, blocks of distinct lengths: an exact Jacobi, eta and MacMahon reduction, every fixed order of the free energy and of the exact-saddle Edgeworth expansion, $\log a_n = \frac{4\pi}{3\cdot 24^{1/4}} n^{3/4} - \frac{\sqrt{6}}{24}\sqrt{n}\log n + O(\sqrt{n})$ with an elementary relative equivalent, controlled inversion, a Gaussian number of blocks and a Gumbel largest length independent of it. Two manuscripts, printed whole, sharing no theorem. Part II's expansion of the moving Jacobi product is identity (11) of `a291698-moving-fugacity-partitions` at a complex fugacity, derived independently; neither Part answers that report's Section 10.3 or question 7 of `a022629-distinct-partition-norms`, and the weight $p(k)$ is not regularly varying. The write proved two steps Part II gives in outline (the sharpened mean shift and the marker-derivative bounds); the independent check (`342a4583e`) found both valid. Further questions: the $\alpha = 1/3$ stretched-amplitude modulus law (an observation), a certified constant $C_P$, Part II's full marked expansion. Part I calls itself "rigorous"; no priority claim. Unrefereed; not formalized.

### 178. Asymptotics and inverses for a stable Hilbert series

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a126348-stable-hilbert-series/article.pdf

from batch 100: ProveIt_A126348_Hilbert_Asymptotics_and_Inverses.zip (bundle Report 84)

For $F(q) = \prod_{k\geq 1}(1+q^k/(1-q)) = \sum a_n q^n$, OEIS A126348, a stable Hilbert series of top-degree Grothendieck and Lascoux polynomials, proves a relative all-orders expansion at the quadratic logarithmic saddle $n = e^{2L}(L^2/2+L+\pi^2/6)$: the resummed leading term $\exp\{e^L B-1\}/\sqrt{2\pi e^{3L}V}$, with $B = L^2+L+\pi^2/3$ and $V = L^2+3L+\pi^2/3+1$, times rational corrections $C_j(L) = O((1+L)^j)$, with $C_1$, $C_2$ explicit and a finite rule for every order. Also an exact Jacobi and eta factorization of $F(e^{-t})$ with a convergent sector expansion of its theta quotient, a uniform power-log expansion of $\log F$, a minor-arc bound with no resonance sectors (the fugacity $1/(1-q)$ is large only near $q = 1$), an explicit range inverse with $\widehat{N}(a_n) = n + O(e^{-L}(1+L)^3)$, so that rounding recovers $n$, strict increase for $n \geq 2$ and a two-candidate ceiling bracket for integer thresholds. The factorization is not new: it is identity (11) of `a291698-moving-fugacity-partitions` at the $q$-dependent fugacity $u = 1/(1-q)$, which the manuscript, pinned before that report existed, does not cite; the credit is the write's. The Lambert start, the reversion and the threshold bracket are instances of the transseries volume, with no novelty claimed for the method. Everything is eventual: no effective onset or explicit constants, no exponentially small coefficient sectors or Stokes data, no exact interpolation, no priority claim. The manuscript's unproved statements (deep truncations of the formal inverse, the next correction $B_1$) are further questions; the write proved that the carrier root $L = L_0 + \sum d_k L_0^{-k}$, $L_0 = 2W(\sqrt{y}/2)$, is a convergent series ($d_1 = -1$, $d_2 = 5/2 - \pi^2/3$) and completed a sketched step of the complex-neighbourhood lemma. The independent check (`f8accf37c`) corrected one figure in that proof: the five-term sum leaves an error of $4.3 \cdot 10^{-6}$ at $y = 10^{12}$, not $3 \cdot 10^{-7}$. One manuscript; no claim found wrong. Unrefereed; not formalized.

### 179. Unlabeled Bipartite Multigraphs: Asymptotic Expansions, Rare Symmetries and Connectivity

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a007716-bipartite-multigraphs/article.pdf

from batch 100: ProveIt_A007716_Asymptotics_and_Inverses.zip (bundle Report 88, base) + ProveIt_A007716_Symmetry_and_A007718_Connectivity.zip (bundle Report 92)

OEIS A007716 counts bipartite multigraphs with $n$ edges, no isolated vertices and two distinguished shores up to shore-preserving isomorphism. With $w = W(n)$ and $B_n$ the Bell numbers, Part I proves the relative equivalent $a_n \sim B_n^2 e^{w^2/2}/n!$, whose divergent factor comes from parallel edges and which strengthens Pavlichin, Jiao and Weissman's logarithmic estimate, through an exact invariant-partition Burnside formula and a positive weighted count $Z_n$ with $a_n/Z_n = 1 + O(\log^3 n/n)$; a complete fixed-order expansion with rational coefficients $Q_j(w)$ ($Q_1$, $Q_2$ explicit) and a finite coefficient algorithm; uniform Bell shifts; smooth inverse models with integer-threshold brackets; non-P-recursiveness; and, for $r \geq 3$ shores, the identity term as a relative equivalent. Part II, a sequel, shows for the uniform measure on classes that a nontrivial vertex automorphism has probability $2w^3/n+O((1+w)^7/n^2)$, almost always a single leaf transposition, which answers Part I's question on rare symmetries at leading order; that disconnection has probability $\sim w^2/n$, usually through one isolated edge, with an exact fragment law; and, for the connected counts A007718, every fixed order, inverse brackets and non-P-recursiveness. Merged from two manuscripts, base Report 88; Part II's restated inputs are printed once, in Part I, and its letters are renamed to avoid clashes. The order-one Bell expansion is that of `a277364-bell-asymptotics`, credited; A120733 of `a260700-parabolic-double-cosets` is the labelled analogue. No claim was found wrong; further questions record the one-paragraph all-orders Bell-saddle argument, existential exponents, an asserted order-one remainder, $Q_2$ printed without derivation (the write checked its unmarked part numerically only) and Part II's reliance on Part I's sketches. No convergence, effective constants, Stokes data or priority is claimed. The independent check, missed at the intake and made on 7 October, needed no correction. Unrefereed; not formalized.

### 180. Leafless loopless multigraphs by number of edges

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a307316-leafless-multigraphs/article.pdf

from batch 100: A307316_A307317_Refined_Asymptotics_Connectedness_and_Inversion_Source.zip (bundle Report 131)

$A_m$ (OEIS A307316) and $C_m$ (A307317) count isomorphism classes of loopless multigraphs with $m$ edges and minimum degree at least two ($C_m$: connected), vertices and edges unlabelled. With $w = W(2m)$, $F(m) = m(w - \log w - 1 + 2/w)$ and $H_m = \exp\{F(m) + w^2/4 - 3w/2 - 1\}/\sqrt{2\pi m(w+1)}$, proves the polynomial sandwich $(1 - o(1))H_m \le C_m \le A_m \le (1 + o(1))e^w H_m$, hence $\log a_m = F(m) + w^2/4 + O(\log m)$ for either sequence with every fixed inverse-logarithmic order, $C_m/A_m = 1 - O(w^3/m)$ by a component convolution on isomorphism classes, and a first-threshold inverse $m_0 + O(1)$ that assumes no monotonicity; also an exact star-conditioning sandwich and a relative minimum-degree probability in the vertex-labelled composition model. *Trust boundary:* the upper half rests on Wright's 1972 theorem on simple graphs counted by edges, which the source checked only in an OCR'd reproduction and a contemporary abstract and the intake not at all; the lower bound, the labelled-model results and the Euler-product identities need no Wright input. Not claimed: a relative equivalent $A_m \sim H_m$ (the sandwich leaves the factor $e^w$), historical priority (Wright's and Lupanov's work must be credited), any transfer from labelled counts; the identification with symmetric kinematic-polynomial counts and A226919 is conjectural. These, the leading disconnected correction, minimum degree above two and full expansions are further questions; the exact data for $m \le 50$ are evidence only. The write proves $A_{m-2} \le A_m - C_m$ (adding a doubled edge is injective on classes) and that $(A_m - C_m)/A_m \sim w^4/(4m^2)$ would follow from any $A_m \sim \kappa H_m$; the independent check (`f8accf37c`) found this valid. The neighbour `a007716-bipartite-multigraphs` works on the scale $W(n)$, not $W(2m)$. One manuscript; no claim found wrong. Unrefereed; not formalized.

### 181. Exact asymptotics and inversion for column convex permutominoes

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a196275-column-convex-permutominoes/article.pdf

from batch 101: A196275_Exact_Asymptotics_and_Inversion_Source.zip (bundle Report 96)

For $a_n$, the column-convex permutominoes with $n$ rows and $n$ columns (OEIS A196275), a Bernstein-basis embedding turns Tomás's triangular recurrence (an external input, checked here only against the 200 b-file terms) into $b_n = a_n/(n+1)! = \frac{1}{2}\langle 1, T^n 1\rangle$ for a positive self-adjoint operator $T$ on $L^2[0,1]$ with an elementary resolvent. Hence the closed form $B(z) = -(2-z)\log(1-z)/(2z(2-z+\log(1-z)))$, an exact spectral measure (one atom at $h = 1/(1 - W_0(1/e)) \approx 1.38593$ off $[0,1]$ plus a positive density on $[0,1]$), and $a_n \sim k\,(n+1)!\,h^n$ with $k = (2h-1)(h-1)/2 \approx 0.34191$; the source presents this as a proof of the equivalent conjectured numerically by Beaton, Disanto, Guttmann and Rinaldi (2011), a literature status the intake did not check. The secondary term $r_n = b_n - kh^n \sim 1/(2n\log^2 n)$ has an inverse-logarithmic expansion to every order and a mixed inverse-power/inverse-logarithmic expansion with explicit coefficient functions and sector-level remainders; for a specified spectral interpolation the inverse is a convergent series of exponentially small sectors about the exact Gamma-envelope inverse; with $\mathcal{E}(x) = k\,\Gamma(x+2)h^x$, $\mathcal{E}(n) < a_n < \mathcal{E}(n+\frac{1}{2})$ for every $n \ge 1$, so rounding $\mathcal{E}^{-1}(a_n)$ recovers every index; and $B$ is not D-finite, so $(a_n)$ is not P-recursive. The write proves the disk estimate behind the convergent inverse series (Proposition 8.2) and an $L^2$ statement on the Bernstein-smoothed profile: its distance from the projection on the eigenvector at $h$ is exactly $\sqrt{2r_{2n}} \to 0$ (the write's bound $\le 1$, sharpened by the independent check). Not claimed: local asymptotics of the profile coefficients, optimal truncation, convergence of the logarithmic series, interval-certified inversion or global priority; the tail estimates of Theorems 5.1 and 6.1 are written in condensed form. The inversion steps are instances of the transseries volume's factorial-core and staircase theorems. Single source; the intake found no false claim. Unrefereed; not formalized.

### 182. Ascent Sequences Avoiding 100 or 110

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a202059-ascent-100-110-growth/article.pdf

from batch 102: Report243.zip (bundle Report 243, base) + Report241.zip (bundle Report 241)

Ordinary ascent sequences avoiding the pattern 100 ($a_n$, OEIS A202059), 110 ($d_n$, A202060), both 000 and 100 ($e_n$), or all three ($c_n$, Callan and Mansour's Class 36). Part I (bundle Report 243, the base) proves $\log u_n = n(\log n - 2\log\log n + \log 2 - 1) + O(n(\log\log n)^2/\log n)$ for $u = a, e$, so $(\log n)^2(u_n/n!)^{1/n} \to 2$, and brackets the linear constant of $d_n$ and $c_n$ in $[-\log 2 - 1, \log 2 - 1]$, through the triangular comparison count $b_n = \text{A098569}(n-1)$, a run-indexed marked-occurrence encoding, positive-row words with a stars-and-bars rank bijection and a repair of empty rows; also two-term first-crossing inverses. Part II (Report 241, written first) proves $\log u_n = n\log n - 2n\log\log n + O(n)$ for $c, a, d$ with an exactly counted fresh-label family, and keeps priority for the encoding, for Corollary 20.1, which refutes the scale $\Gamma(3n/4+1)\mu^n n^g$ conjectured by Conway, Conway, Elvey Price and Guttmann (EJC 2022) for $a$ and $d$, and every fractional factorial $\Gamma(\alpha n + 1)$, $0 < \alpha < 1$, even with a factor $e^{o(n\log n)}$, and for Corollary 21.2: $c$, $a$, $d$ are not P-recursive. Every main estimate of Part II follows from Part I; the implied results are kept as second routes and mapped in a table. The write proves Part I's aside on the leading order of its error (sharpened at the independent check to $L^2/4 - L + 1 + o(1)$), shows that $e_n$ is not P-recursive, bounds $d_n/a_n$ and $c_n/a_n$ on the exponential scale, and asks whether $d_n \le b_n$ always (true for $n \le 26$). The independent check corrected the write's claim that Theorem 6.1 of `a098569-self-modified-ascents` implies Part I's triangular lemma: it gives the sum estimate, not the single-term estimate. No relative equivalent, no constant for $d$ or $c$, no rounding rule for the inverses; nothing in either manuscript was found wrong. Unrefereed; not formalized.

### 183. Self Modified Ascent Sequences and Positive Diagonal Tables

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a098569-self-modified-ascents/article.pdf

from batch 102: Report242.zip (bundle Report 242)

Let $b_N$ count nonnegative upper-triangular integer tables with positive diagonal and entry sum $N$, equivalently self-modified ascent sequences of length $N$ ($b_N = \text{A098569}(N-1)$), and $c_N$ their binary members ($c_N = \text{A121690}(N-1)$); OEIS A098568 is the table by dimension. Proves an expansion to every fixed order at the exact gamma saddle, uniformly in bounded real dimension marks; a full-span local limit theorem for the dimension, with corrected mean $\sim 2N/\log N$ and variance $\sim 2N/(\log N)^2$, which answers the A098568 question on how the row terms are distributed; the elementary carrier $b_N = \sqrt{2/P(u)}\exp(Nu(u+1)/(u+2) + u/2 + u^2/4)(1 + O((\log N)^4/N))$ with $u(u+2)e^u = 2N$ and $P(u) = u^2 + 4u + 2$; the binary probability $c_N/b_N = e^{-u^2/2-u}(1 + \mathcal{R}(u)/N + O((\log N)^8/N^2))$ with an explicit rational $\mathcal{R}$ and an all-fixed-order quotient algorithm; and two-ceiling brackets for the inverse thresholds, with a counterexample at $b_{1000}$ to an exact-ceiling rule. The write proves the moderate deviations of the dimension that the source asserts, and records that the first saddle correction $C_1 \sim 1/(24N)$ stays negative until its sign change between $N = 15{,}864$ and $15{,}865$ (located by the independent check; the write had said between $10^4$ and $10^6$; uncertified numerics). The exact enumeration, the word/table bridge, the binary subclass and the $N\log N$ scale are prior (Bayoumi, El-Zahar and Khamis 1989 and later papers; Kotešovec's OEIS formulas), credited as the source credits them and not checked against the papers. Constants are non-effective; difference parameters $d \ge 2$, complex marks, higher moments, growing order and the binary collision structure are further questions. The same triangular sum is the comparison count of `a202059-ascent-100-110-growth`; its single-term estimate there follows from Section 3 here, not from Theorem 6.1 (an overclaim corrected in both reports at the independent checks). Single manuscript; unrefereed; not formalized.

### 184. Weak and difference ascent sequences (OEIS A336070)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a336070-weak-ascents/article.pdf

from batch 102: A336070_Weak_Ascent_Logarithmic_Correction_Source.zip (bundle Report 105, base) + A336070_Weighted_Levels_and_Large_Deviations_Source.zip (bundle Report 107) + Ascent_Sequences_Sharp_Remainders_and_Bounded_Inverses_Source.zip (bundle Report 108)

Weak ascent sequences (Bényi, Claesson and Dukes; OEIS A336070) and difference ascent sequences with a fixed parameter $d$ (Dukes and Sagan; $d = 0$ gives the Fishburn numbers, $d = 1$ A336070); $\mu = 6/\pi^2$. Part I (bundle Report 105, the base) proves for every fixed $d \geq 0$ that $(a_n^{(d)}/n!)^{1/n} \to \mu$ and $a_{n+1}^{(d)}/((n+1)a_n^{(d)}) \to \mu$, with exponential concentration of the ascent count, and for $d = 1$ that $\log(a_n/(n!\mu^n)) = \frac{1}{2}(\log n)^2 + O((\log n)^2/\log\log n)$, with a signed Lambert-$W$ inverse; the Fishburn asymptotic of Hwang and Jin is its external lower bound. Part II (Report 107) weights each level by $w > 0$: $\log H_n(w) = \log n! + n\log\mu + \frac{w}{2}(\log n)^2 + O((\log n)^2/\log\log n)$ compact-uniformly in $w$, a full large-deviation principle for the number of levels at speed $(\log n)^2$ with rate $x\log(2x/w) - x + w/2$, and an inverse. Part III (Report 108) reduces both remainders to $O(\log n)$, proves the correction $\frac{d}{2}(\log n)^2$ with remainder $O_d(\log n)$ for every fixed integer $d \geq 1$, threshold inverses with bounded additive error, and the moment bounds $\mathbb{E}E_n = \frac{w}{2}(\log n)^2 + O((\log n)^{3/2})$, $\operatorname{Var} E_n = O((\log n)^3)$. The write adds the zero-level atom with error $O(\log n)$ and the case $d = 0$ (coefficient 0) from the Fishburn asymptotic; the independent check restricted a doubling inequality in that proof to $i \geq 1$ and confirmed that a real $d > 0$ acts as $\lceil d \rceil$ because the $d$-ascent inequality is strict, so the coefficient is $\lceil d \rceil/2$. Report 105's binomial transform is Jelínek's Fact 5.1, credited at the write; the three manuscripts' citation of Mansour (2026) was not verified. No relative-error equivalent, amplitude, coefficient of a $\log n$ term, central limit theorem or uniformity in $d = d(n)$. *Merged from three manuscripts in chain order, superseded statements with dated notes; Report 108's Sections 3–4 repeat Report 107's and are printed once.* Unrefereed; not formalized.

### 185. Closed lambda terms (OEIS A135501 and A220894)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a135501-closed-lambda-terms/article.pdf

from batch 103: Closed_Lambda_Terms_All_Logarithmic_Orders_Source.zip (bundle Report 110, base) + Closed_Lambda_Terms_Growth_and_Binding_Geometry_Source.zip (bundle Report 109)

Counts closed lambda terms of size $n$ up to alpha-equivalence, with abstractions and applications of size one and variables of size $s \in \{0, 1\}$ (OEIS A135501 for $s = 1$, A220894 for $s = 0$); $r = s + 1$, $N = n + 1$. Part I proves $M_r(N) - O(\log N) \leq \log A_s(n) \leq M_r(N) + O(R_r(N))$ with $M_r(N) = (N/r)(t - 2 + 1/t) + N/2$, $t = W(4eNe^{-r/2})$ and $R_r(N) = N^{1-1/(3r)}(\log N)^{-2/3+1/(3r)}$, hence every fixed inverse-logarithmic order (polynomials $P_1, \ldots, P_6$ and a recurrence; the $n/\log n$ term is $(n/r)(\log\log n - \log 4 + r/2)/\log n$), and a controlled threshold inverse with an asymmetric bracket. Part II, written first, proves the weaker four-scale form and inverse (printed as a superseded second route; its undetermined $n/\log n$ coefficient is determined by Part I, and its inverse omits a nonzero constant), the skeleton sandwich, and a joint large-deviation principle at speed $n$ for the scaled numbers of abstractions and the maximum unary height of a uniform closed term, rate $(t - 1 - \log x)/r$ on $0 < x \leq t$. The coarse majorant, proved by both manuscripts the same way, is printed once in Part I. Proved at the write: the open ratio $R(b, u)$ of Part II satisfies $\log R(b, u)/u \to \frac{1}{2}$ at the moving saddle, and, after the independent check (which proposed the exponential tilt and corrected three sentences), Proposition 18.2 extends this to Part I's whole saddle window for large $N$ (a second check fixed that range). No equivalent, amplitude or fluctuation law; the exponent of $R_r$ is an upper-bound exponent; the literature statements were not checked against the papers. The Lean type `Term 0` of `Computability/CombinatoryLogic` is exactly the set counted; nothing is formalized. Merged from two manuscripts. Unrefereed; not formalized.

**186. One-sided rectangulations: logarithmic power and non-D-finiteness**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a348351-one-sided-rectangulations/article.pdf
from batch 103: A348351_Rectangulations_Exponent_and_Non_D_Finiteness_Source.zip (bundle Report 111)

One-sided, equivalently area-universal, rectangulations with $n$ rectangles (OEIS A348351) are counted, through the Asinowski–Cardinal–Felsner–Fusy bijection (used, not re-proved), by four-colour quadrant walks with bounded steps. With $\Gamma = (7+\sqrt{17})/2$, $\rho = (29-7\sqrt{17})/4$ and $\alpha = 1+\pi/\arccos(-\rho) = 2.9569294595\ldots$, the report proves $\log a(n) = n\log\Gamma - \alpha\log n + o(\log n)$, the exponent half of the conjecture $a(n) \sim c\,\Gamma^n n^{-\alpha}$ stated after their Proposition 4.23; the generating function is not D-finite ($\alpha$ is irrational, $2<\alpha<3$, by a conjugate of $2\cos\theta$ below $-2$); and $N(y) = \log y/\log\Gamma + (\alpha/\log\Gamma)\log\log y + o(\log\log y)$. The method: a Perron tilt with a bounded Poisson corrector, Whitt's martingale functional limit theorem, a lattice local limit theorem, exact-time interior bridges, wedge harmonic measure and annular transfer. Proved at the write: the walk satisfies hypotheses (H1)–(H5) of the cone theorem of `a377922-corner-polyhedra-schnyder`, Part III, so the exponent is also an instance of it (this report is the earlier, model-specific proof; neither source cites the other); the general criterion that nonnegative integers $\mu^n n^{-\alpha+o(1)}$ with irrational $\alpha$ have a non-D-finite generating function; $a(n+1) \geq 2a(n)$ for $n \geq 1$; and two of four condensed steps written out. The other two (the bridge limit, the uniformity of the annular transfer) are listed as an open question, as are the amplitude, an error term, corrections and a bounded-term inverse; non-D-finiteness does not exclude the exact multistate recurrence. Priority is not established. The independent check found every item of the write valid (cosmetic fixes only). Unrefereed; not formalized.

**187. The second logarithmic term for regular Tesler matrices**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a008608-tesler-matrices/article.pdf
from batch 103: A008608_Tesler_Second_Logarithmic_Term_Source.zip (bundle Report 106)

For the number $a_n$ of $n\times n$ regular Tesler matrices (OEIS A008608), the leading term $\log a_n \sim Cn^{3/2}$, $C = \pi\sqrt{8/27}$, is Balashov–Bulavenko–Molybog's (EJC 2026, Corollary 37). The report proves the second term: $Cn^{3/2} - \frac{3}{4}n\log n - 2n\log\log n - O(n) \leq \log a_n \leq Cn^{3/2} - \frac{3}{4}n\log n + B_U n + O(\sqrt{n})$ with $B_U = 0.2043366936\ldots$, so $\log a_n = Cn^{3/2} - \frac{3}{4}n\log n + O(n\log\log n)$. The upper bound is elementary (an eta identity and singleton elimination); the lower bound builds a feasible geometric cut profile and integer throughput vectors and applies the capacity inequality of Brändén–Leake–Pak (an external input whose proof ProveIt has not checked), with the assembly $-\frac{1}{4} - 1 + \frac{1}{2} = -\frac{3}{4}$. Also the threshold inverse $N(y) = x + (2C)^{-1}\sqrt{x}\log x + O(\sqrt{x}\log\log x)$, $x = (\log y/C)^{2/3}$, with a non-effective integer bracket. No constant is effective; no $O(n)$ remainder, linear term, equivalent or amplitude is claimed, and the novelty of $-3/4$ is uncertified. Added at the write: $B_U$ is only an upper-bound constant (for $9 \leq n \leq 26$ the linear quotient lies in $[-1.443, -1.421]$, uncertified), and a proof that the $2n\log\log n$ margin penalty is sharp for this family of margin vectors. The manuscript omits O'Neill (2018), whose bounds are $e^{O(n^2)}$; the write cites it with Pak's question whether $a_n = e^{\Theta(n^2)}$, which Corollary 37 and, alone, the report's elementary upper bound answer negatively. The independent check completed the sharpness proof (two upper bounds it presupposed), gave O'Neill's journal and arXiv numberings and credited the $n^{1.6}$ observation to Pantone. Unrefereed; not formalized.

**188. Self-complementary tournament score sequences (OEIS A345470 and A351869)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a345470-self-complementary-scores/article.pdf
from batch 103: Self_Complementary_Tournament_Scores_Leading_Constants_and_Inverses_Source.zip (bundle Report 116, base) + Self_Complementary_Tournament_Scores_Growth_and_Inverse_Source.zip (bundle Report 114)

Counts the self-complementary score sequences $C_n$ (OEIS A345470) and the strong ones $D_n$ (A351869), as sorted sequences, by exit times of integrated random walks. Part I proves $C_n \sim$

$A\,2^n n^{-3/4}$ and $D_n \sim e^{-\lambda} A\,2^n n^{-3/4}$ with one amplitude $A > 0$ for both parities, where $\lambda = 0.33023754\ldots$ is the constant of `a000571-tournament-score-sequences`, so the strong fraction is $e^{-\lambda} = 0.71875297\ldots$ (certified enclosure); a harmonic identity, a Lambert-$W_{-1}$ inverse with vanishing-width brackets and an eventual gap $N_C \le N_D \le N_C + 1$. Part II, written first, gives the first proof of $\Theta(2^n n^{-3/4})$ (Stockmeyer's conjectures), an $O(1)$ inverse and the monotonicity of $C_n$ (all $n$) and $D_n$ ($n \ge 2$); its other results are second routes. Wrong claim on record: both manuscripts copy the limit density printed in Denisov–Wachtel's Theorem 1 (AIHP 2015, journal version only), which has mass $0.1154552960\ldots$, not 1. The write proves that the density of Groeneboom–Jongbloed–Wellner's (2.24), cited in the same paper, has mass 1 and velocity marginal $f_\star(0) = \sqrt{2/3}\,\Gamma(3/4)^2/\pi = 0.3902762188\ldots$, about ten times Part I's printed $0.0377\ldots$; so $A = \sqrt{6}\,\Gamma(3/4)\pi^{-3/2} V_{DW}(1/\sqrt{2}, 1/\sqrt{2})$, and the lemma that used the printed shape is repaired. The independent check confirmed the corrected density by a dynamic programme of the conditioned walk to $n = 600$ (floating point, extrapolated: evidence, not a certificate) and corrected the write's supporting text (the printed density predicts $A \approx 0.0374$; Monte Carlo values replaced). $V_{DW}$ is not evaluated: counts suggest $A \approx 0.387$, uncertified. The $\lambda$ enclosure rests on Kolesnik's tail theorem; no rates or corrections. Merged from two manuscripts; re-proofs printed as marked second routes. Unrefereed; not formalized.

### 189. Inversion sequences solved by kernel orbits (OEIS A279544, A279567, A279569, A279558)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a279544-inversion-kernel-classes/article.pdf

from batch 104: Inversion_Sequences_A279544_A279567_Asymptotics_and_Inverses_Source.zip (bundle Report 118) + A279569_Two_Kernel_Asymptotics_and_Inverses_Source.zip (bundle Report 119) + A279558_Asymptotic_Expansion_and_Inverse_Thresholds_Source.zip (bundle Report 122)

Britt and Beaton (arXiv:2512.21943v3) gave generating trees and catalytic equations for the inversion sequences avoiding a triple of relations and stated, explicitly as non-rigorous, laws $a_n \sim C\mu^n n^{-3/2}$ for four classes. Three manuscripts prove them by one method: a kernel substitution reduces the catalytic equation to a scalar affine functional equation, solved along contracting Möbius-type orbits seeded at an algebraic kernel root; a global continuation argument excludes competing singularities, and an exact rational certificate shows that the square-root amplitude does not vanish. Part I treats class 214 (OEIS A279544, $\mu = 4$, $C$ to 52 decimals) and class 1509 (A279567, $\mu = 3 + 2\sqrt{2}$); Part II class 1953A (A279569, $\mu = 27/4$) and, through Martinez and Savage's Wilf-equivalence (their Theorem 60), the counts of class 1953B; Part III class 830 (A279558, $\mu = 27/4$). For each class: coefficient asymptotics to every fixed order with $C$, $c_1$, $c_2$ in certified intervals, strict monotonicity (for class 830 completed by a remark of the write), and a threshold inverse inside two ceilings, centred at a Lambert-$W_{-1}$ root in Parts I–II and expanded in powers of $1/\log M$ with coefficients polynomial in $\log\log M$ in Part III; the inverses are instances of the transseries volume. Kotěšovec's OEIS amplitudes agree with every certified interval, so no digits are new. Nonalgebraicity and $D$-finiteness (which Britt and Beaton believe fail), the corrections beyond $c_2$, effective constants (the inverse constants are existential) and earlier rigorous literature (not searched) stay open. Nothing in the manuscripts was found wrong; the independent check corrected the write's statement that every Part proves a Lambert-$W_{-1}$ inverse. Siblings from the same paper, by unrelated methods: `a279571-inversion-cone-walk` and `a279551-inversion-log-deficit`. *Merged from three manuscripts that share no passage; every Part is printed in full.* Unrefereed; not formalized.

### 190. Inversion sequences avoiding 100, 101, 201 (OEIS A279571)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a279571-inversion-cone-walk/article.pdf

from batch 104: A279571_Leading_Equivalent_and_Harmonic_Constant_Source.zip (bundle Report 125) + A279571_Exact_Asymptotic_Exponent_and_Non_D_Finiteness_Source.zip (bundle Report 123)

A positive change of weights turns Britt and Beaton's generating tree for class 2106 (inversion sequences avoiding 100, 101 and 201, OEIS A279571) into a centred two-colour Markov-additive

walk with unbounded steps in the quadrant, with wedge angle $\theta = \arccos(2/\sqrt{7})$ and $p = \pi/\theta \in (4, 5)$. Part I proves $a_n \sim C_A 9^n n^{-\kappa}$, $\kappa = 1 + \pi/\arccos(2/\sqrt{7}) \approx 5.4017$, with $0 < C_A < \infty$ characterized by forward and reversed killed harmonic functions and a positive endpoint series; a killed local limit theorem at every fixed pair of states; a threshold inverse with explicit centre and a ceiling bracket of vanishing width; that 1/9 is the only singularity on the circle of convergence; and, $\kappa$ being irrational, that the generating function is not $D$-finite. Part II, the earlier manuscript, proves the weaker $a_n = 9^n n^{-\kappa+o(1)}$ by geometric annuli and exact-time gluing, and alone derives the functional equations and kernel. The write checks Part II's walk against the log-scale cone theorem of `a377922-corner-polyhedra-schnyder` (its Part III): (H1)–(H4) hold, but (H5) fails at the origin endpoints the theorem states, so the exponent is an instance of that theorem's proof, stated as a proposition at fixed endpoint states, not of its statement; the independent check confirmed this and needed no correction. No digits of $C_A$ (an uncertified $a_n 9^{-n} n^{\kappa} = 87.65$ at $n = 1000$), no rate or correction terms, no local $\Delta$-domain expansion, no unconditional ceiling rule for the inverse and no general amplitude cone theorem are claimed; prior numerical estimates (Kotěšovec, Britt–Beaton) are credited. Nothing in the manuscripts was found wrong. *Merged from two manuscripts; the six blocks they share word for word are printed once, in Part I.* Unrefereed; not formalized.

### 191. The logarithmic deficit of two inversion sequence classes (OEIS A279551 and A279556)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a279551-inversion-log-deficit/article.pdf

from batch 104: A279551_A279556_Logarithmic_Deficit_and_Inverse_Source.zip (bundle Report 126) + A279551_A279556_Sharp_Deficit_Constants_and_Inversion_Source.zip (bundle Report 127) + A279551_A279556_Loglog_Correction_and_Refined_Inversion_Source.zip (bundle Report 129)

For Britt and Beaton's inversion-sequence classes 247 and 759 (OEIS A279551 and A279556, growth constants $\mu = 8$ and 9), with deficit $\mathcal{D}(n) = n \log \mu - \log a(n)$, $S = n^{1/3}(\log n)^{2/3}$, $N = \log n$ and $J = \log\log n$, three manuscripts on one commitment-tree spine prove: $\mathcal{D}(n) = \Theta(S)$, strict monotonicity, an inverse threshold, smooth boundary values of sharp Gevrey order 3, a genuine singularity at $1/\mu$ and that neither generating function is $D$-finite (Part I); the sharp constant, $\mathcal{D}(n) = \sigma_D S + o(S)$ with $\sigma_D = (3\pi^2/(2D))^{1/3}$, $D = 1/2$ and 1 (Part II); and the next term, $\mathcal{D}(n) = \sigma_D S + (7\sigma_D/3)SJ/N + O(S/N)$, with refined inverse and derivative-growth consequences (Part III, the base). The write proves that no form $Cn^g\mu^n \exp(-bn^\beta)$ is an asymptotic equivalent, which refutes as equivalents Britt and Beaton's numerical $n^{3/8}$ forms (their (4.12)–(4.13), which they call non-rigorous); the independent check extended this to a factor $(\log n)^\gamma$, restricted one sentence to $b > 0$ and corrected two twice-rounded entries of a residual table. No coefficient equivalent or amplitude, no coefficient of $S/\log n$, no ratio theorem, no effective constants and no unique dominant singularity are claimed. The method parallels `a202061-ascent-120-deficit` (the same scale, critical local root, potential, 7/3 term and excluded $n^{3/8}$; different sequences and normalizations, no cross-citation), whose Parts IV–V model the first open question. Nothing in the manuscripts was found wrong. *Merged from three manuscripts printed in dependency order (Reports 126, 127, 129); Report 127's repetition of Report 126's Sections 2–4 is printed once.* Unrefereed; not formalized.

### 192. Diagonally symmetric alternating sign matrices (OEIS A005163)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a005163-diagonally-symmetric-asms/article.pdf

from batch 104: A005163_Full_Leading_Equivalent_and_Individual_Amplitude_Source.zip (bundle Report 128) + A005163_Exact_Logarithmic_Scale_and_Inverse_Thresholds_Source.zip (bundle Report 124) + A005163_Linear_Growth_Fugacity_Pressure_and_Limit_Laws_Source.zip (bundle Report 121) + A005163_Diagonal_Roots_Growth_Bounds_and_Inverse_Brackets_Source.zip (bundle Report 120)

Let $a_n$ count the diagonally symmetric alternating sign matrices of order $n$ (OEIS A005163) and let $Z_n(s)$ weight each by $s$ to the number of its nonzero diagonal entries; $\alpha = \frac{1}{2}\log(3\sqrt{3}/4)$, $\beta = \frac{1}{4}\log 3$, $\kappa = 5/72$. Part I (the base) proves $a_n \sim C_* e^{\alpha n^2 + \beta n} n^{-5/72}$ along all $n$, and

$Z_n(s) \sim C(s)e^{\alpha n^2+nB(s)}n^{-\kappa}$ locally uniformly on $\Re s > 0$, the amplitude characterized by Fredholm determinants and a Fourier series but not evaluated; this is the leading term of Behrend, Fischer and Koutschan's Conjecture 11.1 (arXiv:2309.08446v3), whose $n^{-2}$ and parity-dependent $n^{-5/2}$ corrections are not proved. Part II, on which Part I rests, proves stability of the fugacity polynomials, weak cross-size interlacing, an exact calibration at $s = \sqrt{3}$, $\log a_n = \alpha n^2+\beta n-\kappa \log n+O(1)$, a holomorphic pressure with bounded error and a central limit theorem; Part III a second route to the pressure, the limiting root measure (half of its mass escapes to infinity), Hoeffding concentration, a local limit theorem and a large-deviation principle; Part IV elementary envelopes with certified $p < \beta < q$; each with an inverse. No digits or closed form of $C_*$ (an uncertified $C_* \approx 0.7235287$ from exact terms to $n = 177$ agrees with Behrend–Fischer–Koutschan's estimate), no rate, no effective constants, no root simplicity or strict interlacing are claimed; Part I's operator-theoretic sections were read but not re-derived line by line at intake. Nothing in the manuscripts was found wrong; the independent check made the interlacing of two notes weak and corrected three statements of uncertified evidence (among them the centre 0.7235287, first printed 0.7235288). *Merged from four manuscripts; Report 124 is printed in full because Part I depends on it, and the sections that Reports 120, 121 and 124 share almost word for word are printed once.* Unrefereed; not formalized.

### 193. Permutations with exactly one occurrence of 1234, 1243 or 12345 (OEIS A217057, A224179, A224248)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a217057-unique-pattern-occurrences/article.pdf

from batch 105: A217057_Unique_1234_Asymptotics_and_Conjecture_Refutation_Source.zip (bundle Report 134) + A217057_Rational_Structure_Logarithmic_Asymptotics_and_Inversion_Source.zip (bundle Report 135) + A217057_All_Orders_Logarithmic_Asymptotics_and_Inversion_Source.zip (bundle Report 137) + A224179_Leading_Amplitude_Diffusive_Cancellation_and_Inversion_Source.zip (bundle Report 136) + A224248_Exact_Formulas_Leading_Asymptotics_and_Inversion_Source.zip (bundle Report 138) + A224248_All_Orders_Asymptotics_and_Inverse_Thresholds_Source.zip (bundle Report 140)

Counts the permutations with exactly one occurrence of 1234 ($u_n$, OEIS A217057), of 1243 (A224179) and of 12345 (A224248), normalized by $A_n = |\mathrm{Av}_n(1234)| \sim C_0 9^n n^{-4}$ and $|\mathrm{Av}_n(12345)| \sim C_A 16^n n^{-15/2}$, through exact half decompositions glued by binomials. Part I proves $u_n/A_n \to R$, a series of positive terms, so $u_n \sim C_0 R\, 9^n n^{-4}$, and a rational certificate $R > 0.50009$ that refutes Conway and Guttmann's conjecture $R = 1/2$ (EJC 32(1) P1.3, Section 5.1); the write sharpens the bound to $R > 0.50153$ by an exact partial sum and explains why their fits pointed to 1/2. Part II shows that the generating function is the diagonal of a rational series, hence $D$-finite (through Bostan, Lairez and Salvy, whose hypotheses the write checked), with first logarithmic term $(43020\sqrt{3}/\pi)\log n/n^3$; Part III (the base) gives every fixed order, with $\kappa_4 = -5041845\sqrt{3}/\pi$, and inverses. Part IV proves $b_n/A_n \to R_{1243}$, a rational diagonal, non-algebraicity and a square-root cancellation; Conway and Guttmann's suggested 149/160 stays open. Parts V–VI give exact formulas for one 12345, $U_n \sim C_A R_5\, 16^n n^{-15/2}$ with $R_5 \geq 15/32768$, and every fixed order with no logarithms: the same method with the opposite outcome. No certified decimals of $R$, $R_{1243}$, $R_5$ or the corrections, no effective constants, no explicit annihilating equation and no algebraicity statement for 12345 are claimed. Nothing in the six manuscripts was found wrong; Report 138's uncited literature claims are pinned to Bóna–Burstein and Waite, and the independent check reran the refutation independently and corrected two wordings. The 1432 sibling is `a224182-unique-1432-order`. *Merged from six manuscripts in dependency order; Report 136's word-for-word repetition (up to one letter) of Report 134's Proposition 4 and Lemma 5 is printed once.* Unrefereed; not formalized.

### 194. Polynomial order for permutations containing exactly one 1432

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a224182-unique-1432-order/article.pdf

from batch 105: A224182_Polynomial_Growth_Order_and_Inverse_Bounds_Source.zip (bundle Report 139)

For the number $b_n$ of permutations of $[n]$ with exactly one occurrence of 1432 (OEIS A224182,

by reversal) and $A_n = |\mathrm{Av}_n(1432)| = |\mathrm{Av}_n(1234)|$, proves without a computer $M_{n-3} \le b_n \le (n+2)A_{n+2}$, hence $b_n = \Theta(9^n/n^3)$ with $1/2187 \le \liminf b_n/(nA_n) \le \limsup b_n/(nA_n) \le 81$, by minima skeletons read as Ferrers-board matchings, a shape-Wilf step, an entropy estimate, an inflation injection and a Burstein-type marked split; and an inverse bracket whose unrounded endpoints differ by 11/2. The exact image of the split is computer-assisted: a reduction to at most eight points proved by hand and an exhaustive certificate of 2,074 cores. The write adds that the generating function is not algebraic (Lemma 6.3 of Bóna and Burstein) and that the inverse is an instance of the transseries volume, and finds that the one-step bound $f_m(n) \le n \sum_{r<m} f_r(n-1)$ displayed in the proof of Mansour, Rastegar and Roitershtein is justified by their argument only with the factor $n^2$ (an avoider in $S_7$ has nine preimages), which gives one power less; so, among the sources read, the marked split is the only complete proof of the upper order, which the source and the placement record had treated as known. No limit of $b_n/(nA_n)$, no proof of Conway and Guttmann's 3/80 (the exact ratio rises to 0.03829 at $n = 17$ and first falls at $n = 18$), no leading equivalent and no $D$-finiteness are claimed; Regev's three-row asymptotic and Krattenthaler's growth diagrams are cited, not re-derived. Nothing in the source was found wrong; the independent check corrected three wordings. The sibling `a217057-unique-pattern-occurrences` uses the same normalizer by unrelated methods. Unrefereed; not formalized.

### 195. The weighted Dyck Newton triangle (OEIS A258219, A258220 and A292692)

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a292692-weighted-dyck-newton-diagonal/article.pdf`

from batch 105: `A292692_Weighted_Dyck_Diagonal_Asymptotics_and_Inversion_Source.zip` (bundle Report 141) + `A292692_All_Orders_Interior_Asymptotics_and_Inversion_Source.zip` (bundle Report 142) + `Weighted_Dyck_Upper_Edge_Transition_All_Orders_and_Inverse_Source.zip` (bundle Report 143)

Weight a peak at $(a, b)$ of a Dyck path by $k + a/b$ and let $p_N(k)$ be the total weight of the paths of semilength $N$ (OEIS A258219); its Newton coefficients $P_{N,m}$ form A258220, whose diagonal $a_n = P_{2n,n}$ is A292692. Part I proves Peter Bala's conjecture on A258219, the Riccati identity $F - 1 = kxF + xF^2 + 2x^2F_x$ and its Stieltjes continued fraction, through an exact identity for all weighted prefixes, and $a_n \sim Cd^n n!/\sqrt{n}$ with $d = (51\sqrt{17} - 107)/4$ and $C = \sqrt{2 + 2/\sqrt{17}}/\pi^{3/2}$, which establishes Kotěšovec's numerical asymptotic form (2017) and growth-rate conjecture (2024) on A292692 with the amplitude in closed form; also the first relative correction and threshold brackets. Part II (the base) proves every fixed order of the normalized ratio $R_{N,m}$ uniformly on compact subsets of $0 < m/N < 1$, with every diagonal coefficient in $\mathbb{Q}(\sqrt{17})$, and an inverse at every fixed order. Part III treats the upper edge $m = N - q$: exact two-sided bounds for $q \le N/2$, the profile $\sqrt{N}R_{N,N-q} \sim F_q$ for $q = o(\sqrt{N})$, built from the four-regular-map sequence A292186 (whose Riccati equation and leading asymptotic are credited as prior art), its expansion to every fixed order, its strict monotonicity and an inverse by polynomial reversion. The leading inverses reduce to the transseries volume's factorial core. The range $\sqrt{N} \lesssim q = o(N)$, the lower edge $m = o(N)$, effective constants and convergence of the expansions stay open. Nothing in the manuscripts was found wrong; the independent check corrected a false note of the write (positivity of $R_{N,m}$ at $m = 0$ comes from $p_N(0) > 0$, not from a Stirling number, which vanishes there) and made the transseries citations precise. *Merged from three manuscripts in dependency order (Reports 141, 142, 143); embedded copies and restatements are printed once.* Unrefereed; not formalized.

### 196. The proportional placement game (OEIS A398540)

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a398540-proportional-placement-game/article.pdf`

from batch 105: `A398540_Placement_Game_Leading_Asymptotics_and_Inverse_Source.zip` (bundle Report 144) + `A398540_Placement_Game_Logarithmic_Corrections_and_Inverse_Source.zip` (bundle Report 145) + `A398540_All_Orders_Logarithmic_Asymptotics_and_Inverses_Source.zip` (bundle Report 146)

In A. Serra's continuous placement game, $n$ uniform values arrive one at a time and must be placed at once into $n$ ordered slots; the winning probabilities $W_n$ of the proportional policy satisfy an exact

recurrence with beta-binomial weights, and OEIS A398540 records digits of $r_* = \lim W_{n+1}/W_n \approx 0.5754724$. Part I proves that $r_*$ exists and is computable, $527/1000 < r_* < 79/125$ by exact rational arithmetic, and $W_n \sim r_*\sqrt{2\pi n}\, r_*^n$, Serra's conjecture with a corrected constant; Part II the pointwise expansion $W_n/(r_*\sqrt{2\pi n}\, r_*^n) = 1 + 1/(4n) - (7/540)\log n/n^2 + B_2/n^2 + o(n^{-2})$, eventual strict decrease of the ratios, and that no pure inverse-power ratio expansion holds through $n^{-3}$; Part III (the base) every fixed order, with polynomials in $\log n$ of degree at most $j-1$, the explicit third order, inverses at every fixed order and nested rational rate enclosures with a proved contraction (not run). With proofs, the write shows that Serra's prefactor $\sqrt{2\pi}$ and its approach from below are wrong for the game (his Proposition 6.17 is a correct computation for an integral model that does not model the recurrence); that his integral equation of Open Problem 3, read through an ansatz of amplitude $C$, has relative error tending to $1 - C^2/(2\pi r_*)$, which is $1 - r_*$ at the true amplitude, not zero (as restated by the independent check); that his pure-power ratio expansion fails at every order $p \geq 2$; and that his inequality (198) expresses log-concavity, not log-convexity. Effective constants, ratio monotonicity at every index, exact logarithm degrees, certified digits, other policies and whether Report 146's constant $S_A$ equals $1/12$ (uncertified fits agree to $10^{-23}$) stay open. Report 146 was corrected in place before delivery and its earlier version was never received. Nothing in the manuscripts was found wrong. *Merged from three manuscripts in dependency order (Reports 144, 145, 146).* Unrefereed; not formalized.

### 197. Alternating Baxter involutions (OEIS A347546): the corrected count and its fixed-point refinement

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a347546-alternating-baxter-involutions/article.pdf

from batch 106: Corrected_Baxter_Involutions_Asymptotics_and_Inverses_Source.zip (bundle Report 149, base) + Baxter_Fixed_Point_Crossover_Profiles_and_Inverses_Source.zip (bundle Report 151)

OEIS A347546 counts the involutions among the doubly alternating Baxter permutations of length $n$. Part I shows that the recurrence of Min (J. Chungcheong Math. Soc. 34 (2021), Theorem 2.7), from which the OEIS data were computed, miscounts this class: in an even-length involution the two outer blocks are mutual inverses, not involutions, so the outer factor is the Catalan number $C_j$, not the involution count $o_j$, and $C_4 = 14 > 12 = o_4$. The class has 2168 members of length 20, not 2166; two omitted involutions are printed. From three published necessity lemmas of Min and Park (2006), accepted rather than re-proved, Part I proves the corrected recurrences, algebraic generating functions with dominant singularity $\rho = (\sqrt{5}-1)/4$, equivalents $\mathcal{A}_s(1+\sqrt{5})^m m^{-3/2}$ for the even and odd subsequences with every fixed order, and Lambert-$W_{-1}$ length inverses with ceiling brackets leaving at most two adjacent candidates. Its Catalan count of doubly alternating Baxter permutations is Guibert and Linusson's theorem (2000), re-derived and credited at the write. Part II refines the count by pairs of fixed points ($q = 1$ gives Part I back): every fixed order uniformly on compact complex windows of the sparse weight $q = w/m$ with entire profiles, an exact criterion for inverting in the weight, interior inverses, and Poisson-type laws with total-variation rate $O(m^{-1/2})$. The write proves that Min's odd formula is right and her even formula fails exactly from half-index 10 on, so the recurrence-defined sequence lies strictly below the class at $n = 20$ and every $n \geq 22$: Min's printed values at $n = 20, 22, \ldots, 26$ and 21 of the 42 terms of A347546 ($n = 20$ and $22 \leq n \leq 41$) are wrong, while its definition is right; nothing was sent to the OEIS or the journal. The independent check confirmed the six corrected values of Min's list as direct counts (literal brute force through $n = 26$). The leading Lambert centres are instances of the transseries volume. No effective constants, all-index monotonicity or fixed-weight central limit theorem. Unrefereed; not formalized.

### 198. Unlabeled circle graphs (OEIS A156808 and A156809)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a156808-circle-graphs/article.pdf

from batch 106: Circle_Graph_Enumeration_Error_Bounds_and_Inverses_Source.zip (bundle Report 150, base) + Circle_Graph_First_Corrections_and_Refined_Inverses_Source.zip (bundle Report 152)

Let $c_n$ and $g_n$ count connected and all unlabeled circle graphs on $n$ vertices (OEIS A156808, A156809) and $h_n = (2n-1)!!/(4n)$. Part I proves $c_n \sim g_n \sim e^{-3/2}h_n$ from chord matchings under the dihedral group of order $4n$ and asymmetric split-prime cores with recoverable decorations; relative errors $O(1/n)$ through a self-contained Poisson approximation of local patterns, the connected fraction $1 - 1/(2n) + O(n^{-2})$, a labeled equivalent, a Lambert-$W$ inverse with ceiling-safe brackets, and A156808(13) = 21,593,488,017. Its prime probability $e^{-3} + o(1)$ improves the bound $e^{-4}$ stated by Bassino, Bouvel, Féray, Gerin and Pierrot (arXiv, 2024) but follows from their own lemmas, and is not claimed as new; the write credits their Proposition 5.14 for the method of Part I's symmetric-matching bound, cited by neither manuscript. Part II, a sequel, proves $c_n = e^{-3/2}h_n(1 - \frac{15}{4n} + O(n^{-2}))$, $g_n = e^{-3/2}h_n(1 - \frac{13}{4n} + O(n^{-2}))$ and $c_n/g_n = 1 - \frac{1}{2n} - \frac{1}{n^2} + O(n^{-3})$, from a prime probability $e^{-3}(1 - 5/n + O(n^{-2}))$ and a canonical split-tree transfer, with first corrections for automorphism-weighted and labeled counts and a refined inverse. Its transfer to every fixed order is conditional on the matching prime expansion, and the second coefficients $\beta - 277/32$ and $\beta - 297/32$ depend on an unknown prime constant $\beta$. Open: $\beta$, an unconditional expansion beyond the first correction, effective constants and certified thresholds, exact counts beyond order 13. The 2026 journal version of Bassino et al. was not read, so priority is bounded. The independent check found no error and corrected one side condition ($\log(2u) > 1$ holds for every $L > 0$). The leading inverses are instances of the transseries volume's factorial core; the brackets are not instances of its staircase theorem. Siblings sharing a pipeline: `a123448-permutation-graphs`, `a005975-interval-graphs`. Unrefereed; not formalized.

### 199. All orders refinement for unlabeled permutation graphs: factorial expansion, inverse thresholds, and sampling in OEIS A123448

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a123448-permutation-graphs/article.pdf

from batch 106: Permutation_Graph_All_Orders_Expansion_Inverses_and_Sampling_Source.zip (bundle Report 153)

Let $a_n$ count unlabeled permutation graphs on $n$ vertices (OEIS A123448). For every fixed $R$ the report proves $a_n = \frac{1}{4}\sum_{k<R} h_k(n-k)! + O_R((n-R)!)$, with a coefficient generator that uses only finite unrooted and vertex-rooted counts, $k!h_k \in \mathbb{Z}$ and $h_0, \ldots, h_7 = 1, -4, -1, -94/3, \ldots, -40114096/315$. The main step is a finite-polynomial approximation (almost every permutation realizer has a recoverable large prime quotient with bounded decorations), after which Borinsky's factorial composition theorem transfers every fixed order. It adds strict monotonicity, threshold windows for every fixed order that keep the rounding, an explicit second Lambert-$W$ shift, and the all-orders total-variation distance $4/n - 15/n^2 + \cdots$ between the graph of a uniform permutation and a uniform graph class. The write adds $h_8 = -3687969659/2520$ from the OEIS term $a_{10} = 524572$, which the independent check confirmed by its own exhaustive enumeration, and classifies the inverses against the transseries volume (the centre an instance of its factorial core, the two shifts of its core reversion; the windows proved directly). Two statements of the placement record are corrected: the phrase "prepared for private review" occurs in no delivered file, only packaging negations; and Johnston's 2020 question whether $a_n = o(n!)$ was already answered in the negative by Bassino, Bouvel, Féray, Gerin and Pierrot (arXiv, 2024), whose proof gives $a_n \geq n!/30$ for large $n$, so the report adds the constant $1/4$ and every fixed order, not the dichotomy. Priority for the leading term is not claimed: El-Zahar and Sauer's $n!/2$ for two-dimensional posets (1988) is a close antecedent, and whether it implies $1/4$ is open. Not claimed: effective constants, certified thresholds, uniformity in $R$; the 1990 table values for $n = 15, \ldots, 20$ are quoted, not verified (the independent check corrected a note on their floating-point format). Siblings: `a156808-circle-graphs`, `a005975-interval-graphs`.

Unrefereed; not formalized.

### 200. Asymptotics and inversion for unlabeled interval graphs: leading equivalent and second order corrections

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a005975-interval-graphs/article.pdf

from batch 106: A005975_A005976_Interval_Graph_Asymptotics_and_Inversion_Source.zip (bundle Report 133)

Let $I_n$ and $C_n$ count all and connected unlabeled interval graphs on $n$ vertices (OEIS A005975, A005976) and $F_n$ the Fishburn numbers (A022493). From Hwang and Jin's Fishburn asymptotics (their (1.1), Theorem 25(ii) and Corollary 29, located at the write) and Gallai's orientation theorems, the report proves $C_n = F_n/2 - F_{n-1} - F_{n-2}\log n + O(F_{n-2})$ and $I_n = F_n/2 - F_{n-1}/2 - F_{n-2}\log n + O(F_{n-2})$, then the constant: $-(\log n + b_0)F_{n-2} + o(F_{n-2})$ with $b_0 = \log(6/\pi^2) + \gamma + 1 \approx 1.0795$. It adds the factorial form $K\, n!\sqrt{n}\,(6/\pi^2)^n(1+\alpha/n-\beta\log n/n^2+O(n^{-2}))$ and an inverse threshold kept between two ceilings, with constants and onset not effective. The proof locates the fibres behind the logarithm: simplicial vertices are the diagonal cells of the Fishburn matrix, and the dominant fibres are exactly two-to-one. The write read Hanlon (1982), who lists the asymptotic number of interval graphs as an open problem, Acan, and Bukh and Jeffs: none states $I_n \sim F_n/2$. This settles the placement record's caveat that the leading term might be classical, without establishing priority. Bukh and Jeffs's $\log I_n = n\log n - (1+\log(\pi^2/6))n + O(\log n)$ is sharpened. With the data extended to $n = 30$ from Hanlon's tables, the normalized remainder for $C_n$ crosses $b_0$ between $n = 26$ and 27, so thirty terms neither confirm nor refute $b_0$; the independent check corrected the quoted range of extrapolated limits. The Lambert start is an instance of the transseries volume's factorial core; the Newton steps and the bracket are analogues. Open: the remainder beyond $b_0$, a full power-log expansion, effective constants, the $n^{-2}$ coefficient of the factorial form. Shares the Hwang–Jin input with `a336070-weak-ascents`; siblings `a156808-circle-graphs`, `a123448-permutation-graphs`. Unrefereed; not formalized.

### 201. Diagonal Euler transforms (OEIS A252782 and A270917)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a252782-diagonal-euler-transforms/article.pdf

from batch 106: Diagonal_Euler_Transforms_Corrected_Growth_Crossover_and_Inverses_Source.zip (bundle Report 147, base) + Growing_Euler_Crossover_and_Attenuation_Inverse_Source.zip (bundle Report 148)

Let $A_\varepsilon(n) = [x^n]\prod_{j\ge1}(1-\varepsilon x^j)^{-\varepsilon j^n}$: the Euler transform of the $n$th powers, OEIS A252782 ($\varepsilon = 1$), and its distinct-colour analogue A270917 ($\varepsilon = -1$). Both entries state Kotěšovec's conjecture (2016) that $A_\varepsilon(n)^{1/n^2} \to e^{1/e}$. Part I proves that the limit is $3^{1/3}$: $\log x/x$ is maximal at $x = e$ over the reals but at $j = 3$ over the integers. The write adds $3^{mn}/m^m \le A_\varepsilon(n) \le 2^{n-1}3^{n^2/3}$ ($n = 3m+r$) with an elementary proof, so $A_\varepsilon(n)^{1/n^2} < e^{1/e}$ for every $n \ge 414$, and by exact computation for every $n \ge 8$: the conjecture is refuted, not merely sharpened, while the listed terms are right; nothing was sent to the OEIS. Part I then gives an exact expansion over logarithmic atoms, the three residue amplitudes with every exponentially small sector and a uniform tail bound, the first unrestricted-minus-distinct difference, power-log inverses to every order, convergent inverses of finite sector models, asymptotic residue-aware integer brackets, strict increase for every $n \ge 1$, and a crossover for a separately varying exponent $\tau$ with bounded $z = m^2(8/9)^\tau$, including a term of non-integer order $m^{-\delta}$, $\delta = 2.0959\ldots$, from parts of size five. Part II answers in part Part I's question on growing $z$: for $z \le Sm^{3/4}$ the coefficient acquires the attenuation $\exp(-8s^{4/3}/9)$, $s = z/m^{3/4}$, with quarter-power corrections and a term of order $m^{-0.6849\cdots}$, and the attenuation of a smooth model is inverted on compact windows; exact crossings are localized, uniqueness is not claimed. The write corrected one printed decimal of Part II's exponent; the independent check reclassified Part I's power-log inverse as an instance, not an analogue, of the transseries volume's core reversion. No Lambert $W$ occurs. Open: effective onsets and certified thresholds, all orders in the crossover regimes, $s \to \infty$, integer exponents, priority for $3^{1/3}$. Unrefereed; not formalized.

**202. Iterated Euler diagonals (OEIS A290353 and A290354)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a290354-iterated-euler-diagonals/article.pdf
from batch 106: Report225.zip (bundle Report 225, base) + Report226.zip (bundle Report 226) + Report228.zip (bundle Report 228); batch 138: ProveIt_Research_2026-10-08 (1).zip, its parabolic-amplitudes manuscript (Part IV)

Let $F_0(z) = z$ and $1 + F_{m+1}(z) = \prod_{d\geq 1}(1 - z^d)^{-[z^d]F_m(z)}$, so that $A(n, m) = [z^n]F_m(z)$ counts unlabeled rooted trees with $n$ leaves, all at level $m$ (OEIS A290353), and $a_n = A(n, n)$ is A290354. Part I proves $a_n \sim c\,(2n/\pi)^n n^{-4/3}$ with $c = T^{1/3}e^{-B}h(1) > 0$, $T = \pi/2$, where $h$ is the smooth, strictly positive density of a positive measure whose Laplace transform is the entire parametrization of $w \mapsto e^w - 1$ by its normalized Fatou coordinate: the form Kotěšovec conjectured on A290354 (2017), the digits $c \approx 4.4923$ remaining uncertified. It adds proportional-height asymptotics for $n/m$ in compact subsets of $(0, \infty)$, fixed and square-root height shifts with a floor phase, and a Lambert-$W$ inverse at exact values. Part II proves the first correction $1 + (-\frac{1}{18}\log^2 n + A\log n + C)/n + O(n^{-7/5})$, with $A$ and $C$ exact in $h(1), h'(1), h''(1)$ and a clock constant $\kappa$ in closed form in $\pi$, $\log 2$ and Catalan's constant. Part III proves $A(n, m)/(2T^{-n}m^{n-1}) \to e^{-1/(3\lambda)}$ when $m/(n\log n) \to \lambda \in (0, \infty)$, to every fixed inverse-logarithm order with rational-plus-zeta coefficients, through $h(0) = 2$, a Hankel representation of $h$ near zero and a smooth germ $t^{t/3}h(t)$. Part III conjectures $I(\beta) = \beta 2^{\beta/3}h(\beta)$ for the amplitude $I$ of the labelled analogue `a139383-iterated-bell-diagonals` (the sides agree to eight digits at $\beta = 1$, both uncertified); the write proves a consequence for the labelled array conditional on it (reciprocal note there). Part IV (batch 138) proves that identity for every $\beta > 0$ by limits of positive coefficient measures, with $h$ holomorphic on $\Re t > 0$, and from a geometric aggregate over all depths the leading amplitude $C = L^{L/3-1}I(L)/2$ ($L = \log 2$) of `a005121-strict-partition-chains`, Lengyel's constant; every statement is an implication from four unrefereed inputs of Part I and of the A139383 report, and reciprocal notes went to both. Its independent check found nothing wrong. Open: validated $c$ and $h$, higher corrections, effective constants, $n/m \to \infty$, the germ of $h$ at 0. Kaneiwa, Prellberg and Dudko–Sauzin are credited; no priority for the parabolic method. The independent check found no mathematical error, corrected two sentences overstating the novelty of Part II's restated interface, and recomputed $c = 4.4923298971$ (uncertified). The leading inverses are instances of the transseries volume's factorial core. Unrefereed; not formalized.

**203. Strict partition chains and their asymptotic expansions**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a005121-strict-partition-chains/article.pdf
from batch 106: Report223.zip (bundle Report 223)

OEIS A005121 (Lengyel's numbers) counts the strict chains from the least to the greatest element of the partition lattice $\Pi_n$. With $L = \log 2$ and $f_n = (n!)^2(2L)^{-n}n^{-1-L/3}$ the report proves $z_n = Cf_n\big(\sum_{j\leq M} c_j(L)n^{-j} + O(n^{-M-1})\big)$ for every fixed $M$, with $c_j \in \mathbb{Q}[L]$ ($c_1 = L/6$; $c_2$, $c_3$, $c_4$ explicit) and $C \approx 1.0986858055$ (uncertified), directly from the Stirling recurrence: a factorial majorant, a block-defect kernel expansion and a weighted first-difference operator of limiting norm $2\log 2 - 1 < 1$. The same argument on the disk $|q - 1| \leq 1/16$ of the length marker gives every fixed cumulant of the chain length. It adds a Lambert-$W_0$ inverse with a triangular recursion, eventual recovery of $n$ from $\log z_n$, and two-ceiling threshold enclosures. Against the transseries volume (write): the core is an instance of its factorial core, not of its Lambert core, the recursion of its core reversion, the recovery of its staircase rounding; the enclosure is proved directly, with a second route through the staircase theorem. The leading equivalent and $C > 0$ (Babai–Lengyel, 1992), the leading mean and the central limit theorem (Van Cutsem–Ycart, 1994) are prior and re-proved; formal higher corrections were announced by Prellberg (2002), and no priority is claimed for the coefficients. The write records that OEIS A005121's formula line "E.g.f. satisfies $Z(z) = \frac{1}{2}(Z(e^z - 1) - z)$" (in the entry since 2003) has the wrong sign: the correct equation is $2Z(z) = z + Z(e^z - 1)$; nothing was sent to the OEIS. A formal Laplace link with the amplitude $I$ of `a139383-iterated-bell-diagonals`, $C = L^{L/3-1}I(L)/2$, is an open

question; the independent check traced its $1.9 \cdot 10^{-4}$ numerical gap to the $(\log n)^2/n$ term of that report's first correction, after which the agreement is about $10^{-5}$ (not a proof). Part IV of `a290354-iterated-euler-diagonals` (batch 138) has since proved the link under four unrefereed inputs, and the marked amplitude $C(q)$ positive and holomorphic on $\Re q > 0$; a reciprocal note records both and re-scopes the two questions. Open: certified constants, dependence on the order, larger marking regions, distributional refinements, Van Cutsem and Ycart's variance conjecture. Unrefereed; not formalized.

### 204. Subdiagonal and superdiagonal partitions (OEIS A238875 and A238873)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a238873-diagonal-partitions/article.pdf

from batch 107: Two_Partition_Conjectures_Proofs_and_Quantitative_Extensions_Source.zip (bundle Report 156, base) + Subdiagonal_Partitions_All_Orders_Expansion_and_Integer_Inverses_Source.zip (Report 159) + Superdiagonal_Partitions_Exact_Exponential_Growth_and_Inverse_Source.zip (Report 158) + Superdiagonal_Partitions_Airy_Correction_and_Limit_Shape_Source.zip (Report 160) + Superdiagonal_Partitions_Full_Prefactor_and_Refined_Inverse_Source.zip (Report 162)

For partitions written $\lambda_1 \le \cdots \le \lambda_k$, let $s_r(n)$ count those with $\lambda_i \le i + r$ ($s_0$ is OEIS A238875, subdiagonal) and $A(n)$ those with $\lambda_i \ge i$ (A238873, superdiagonal). Part I proves both conjectures of Archibald, Blecher, Elizalde and Knopfmacher (Afr. Mat. 2025; journal conjectures, not OEIS ones): $0 \le P(R \le r) - s_r(n)/p(n) = O(1/\log n)$ uniformly in $r \ge 0$ for Dyson's rank $R$, so $s_0(n)/p(n) \to 1/2$, and $\liminf \log A(n)/\sqrt{n} \ge B_* = \sqrt{\pi^2/3 + 2\log^2 2} > \pi/\sqrt{3}$, so $q(n)/A(n) \to 0$; also a logistic crossover in $r/\sqrt{n}$ and a Lambert-$W$ inverse. Part II expands $s_0(n)/p(n)$ to every fixed order in $\beta = \pi/\sqrt{6n}$ ($c_0 = 1/2$, $c_1 = -1/8$, $c_2$, $c_3$ explicit), a second proof at $r = 0$ only. Part III proves the exact rate $B\sqrt{n} - O(n^{1/3}) \le \log A(n) \le B\sqrt{n} + O(1)$, $B = \sqrt{\pi^2/3 + 4\log^2 2} = 2.2829\ldots$, replacing Part I's $B_*$; Part IV the Airy term $\log A(n) = B\sqrt{n} - Dn^{1/6} + o(n^{1/6})$, $D = 1.5507\ldots$ (via Mallein's published killed-walk estimates), and the limit shape; Part V the full equivalent $A(n) \sim Kn^{-11/12}e^{B\sqrt{n}-Dn^{1/6}}$, $K = 1.2705\ldots$, *conditional* on the unrefereed preprint arXiv:2607.27504v1 (Yu-Tian Li) and labelled so throughout; Parts III–IV are the unconditional statements. Merged from five manuscripts in dependency order; six restated proofs are printed once, with pointers, and the local limits of Parts IV and V as two routes. The write proves that a partition has choosable initial intervals iff it is superdiagonal, settling Gus Wiseman's conjecture in A238873, which no manuscript addresses (and those in A387118, A388711, A387112); the argument is not new: Clément Garrot posted it in A387112 on 31 August 2026 (credited). It also derives the $r = 0$ defect $\beta/4 + (7/16 - 3/(2\pi^2))\beta^2 + O(\beta^3)$, answering Part I's question. Part V repairs two published displays (the conjecture paper's Theorem 2.1, a stray factor $q$ in Štampach–Šťovíček). The independent check corrected one wording ("decays like" for an upper bound). No explicit constants or onsets, nothing at fixed $r \ge 1$ beyond $O(1/\log n)$, no all-orders expansion of $A(n)$; nothing refuted. Unrefereed; not formalized.

### 205. Factorial-weighted Stirling products (OEIS A192563 and A192561)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a192563-factorial-stirling-products/article.pdf

from batch 107: Diagonal_Poly_Cauchy_All_Orders_and_Inverse_Thresholds_Source.zip (bundle Report 154, base) + Matched_Cycles_Exact_Certificates_All_Orders_and_Inverses_Source.zip (Report 155)

Two members of Munarini's 2011 OEIS family of sums of $k!$ times two Stirling numbers. Part I treats A192563, $a_n = \sum_m c(n,m)(m+1)^n$ with $c$ the unsigned first-kind numbers, the diagonal of the poly-Cauchy array A344639 of Bényi and Ramírez: with the exact saddle $r = \log n - \log\log n - \log\log\log n + o(1)$ it proves an expansion of $a_n$ to every fixed order in $r/n$ with explicit Gaussian contraction coefficients, the same for $A_{n,k}$ uniformly for $\alpha \le k/n \le \beta$ (nothing for fixed rows or columns, nor for any regime with $k/n \to 0$ or $k/n \to \infty$, a non-claim the independent check made explicit), threshold brackets from a smooth model and a nested-logarithm inverse; its re-proof of Bényi–Ramírez's Stirling product shows that $a_n$ equals Munarini's defining form for every $n$. Part II treats A192561, $b_n = \sum_k k!\, c(n+1, k+1)^2$, as a Gaussian Fock norm: an

exact positive expansion with an explicit tail at every $n \geq 1$, a rational-centre certificate that isolates $b_{10} = 1273270076895432$, the same certificate for finite products with negative real zeros, an expansion to every fixed order in $1/w$, $w = W_0(en)$, with rational coefficients through $A_{10}$, and a nested-Lambert inverse with a two-ceiling bracket. The two manuscripts use different methods and do not cite each other; they are merged as one family, read as Fock pairings, with a notation dictionary. The write adds, for the third member A192562, which neither treats, the identity $c_n = a_n - nA_{n-1,n} = \sum_m c(n-1,m)((m+2)^n - (m+1)^n)$ for its terms $c_n$, with proof; its asymptotics stay open. Part II's leading inverse is an instance of the transseries volume's factorial core; no bracket is an instance of its staircase theorem. Fixed orders only; no effective constants or onsets, so neither asymptotic inverse is a rounding rule; the $1/n$-scale correction for A192561 is not computed; Part I's novelty is provisional on Yakymiv's 2019 paper, and Part II claims no priority for its general certificate. Nothing refuted or corrected, and the independent check found no error. The two classical identities behind Part I's re-proof are formal in the Lean `Analysis/FabiusFunction` tree; the report's own statements are not. Unrefereed; not formalized.

**206. Stirling transforms of the Catalan numbers (OEIS A064856 and A086662)**
`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a064856-stirling-catalan-transforms/article.pdf`
from batch 107: `Report205.zip (bundle Report 205, corrected revision 2, base) + Report207.zip (Report 207)`

For the Catalan numbers $C_k$, both Stirling transforms, $\sum_k S(n,k)C_k$ (OEIS A064856) and $\sum_k c(n,k)C_k$ with unsigned first-kind numbers (A086662), studied from one Catalan density integral. Part I (second kind) proves an expansion to every fixed order in $W(n/4)/n$ with $n!$ kept exact and rational corrections in the Lambert saddle, the Bell comparison $\log(a_n/B_n) \sim (\log 4)n/\log n$, an all-fixed-order inverse with a two-ceiling enclosure, and that the single ceiling $\lceil I_0(a_n) \rceil$ is eventually $n+1$; it credits Spiridonov's 2004 Conjecture 3.3, the moment bound $B_k \leq M_{2k} \leq a_k$, to Bauer and Golinelli (2001), with no novelty claim. Part II (first kind) proves a resummed expansion with relative error $O_J(n^{-J-1})$, the expansion in $1/\log n$ with factorially divergent coefficients and an exact incomplete-gamma completion, explicit and resummed inverses with two ceilings, a local cycle law with a central limit theorem, the total-variation distance $\sim 3\sqrt{2/\pi}/(8\sqrt{\log n})$ to Ewens(4), and a boundary Gamma(3/2) law. Merged from two manuscripts, Part II filling the gap Part I leaves; the shared beta integral is printed in both, with a pointer. Revision 1 of Report 205 was never received. The write proves that Spiridonov's numerical model $a_k \approx (1.47^k + 1)A_k$ is not an asymptotic (the ratio tends to 0 for every fixed base $\beta > 1$), checks $B_k \leq M_{2k} \leq a_k$ on the thirteen listed terms of A094149, classifies the inverses against the transseries volume, and corrects, with a dated note, Part II's sentence that Part I "expressly deferred" the first-kind transform. The independent check separated the volume's quadratic part $h_{\mathrm{vol}} = 0$ from Part I's $h(\rho)$ (notation only). Fixed truncation orders only, existential constants and onsets, no single-ceiling rule, no relative asymptotics for $M_{2k}$ (later manuscripts of the same bundle are named as leads; batch 112 placed them as `a094149-sparse-graph-moments`, which proves $M_{2k} \sim 2B_{k+1}$ to every fixed collision order, and a reciprocal note here records that the question is answered and that $M_{2k}/a_k \to 0$ is now unconditional). Unrefereed; not formalized.

**207. Growth bounds for maximal families of disjoint partitions**
`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a068598-disjoint-partition-families/article.pdf`
from batch 107: `Maximal_Disjoint_Partition_Families_Growth_and_Inverse_Bounds_Source.zip (bundle Report 163)`

For $a(n)$, the number of inclusion-maximal families of pairwise disjoint strict partitions of $n$ (OEIS A068598), proves $1/5 \leq \liminf \log a(n)/(n \log n) \leq \limsup \leq 1/4$. The upper bound $a(n) \leq 3^{n-1} n^{\lfloor (n-2)/4 \rfloor}$ $(n \geq 2)$ is elementary and explicit, from an anchor and free-label encoding and the distinct-parts mass inequality (Spencer, via Nathanson); the lower constant $1/5$ injects queens configurations and rests on Luria and Simkin's theorem, while an elementary construction gives $1/7$. Also first-passage inverses with constants between 4 and 5 and finite Lambert brackets,

without monotonicity. Two statements of the OEIS entry are corrected (nothing submitted): Meeussen's graph comment counts *maximal* cliques (at $n = 8$: six vertices, 21 nonempty complete subgraphs, $a(8) = 3$), and Beregovsky's fits $A\exp(Bn^c)$ with $c > 1$ (2022, 2026) are not asymptotic: each exceeds the proved upper bound for $n \geq 1215$ resp. $n \geq 8579$, signs certified in ball arithmetic at the independent check. The write adds Simkin's 2023 improvement of the lower bound's linear term (read at abstract level), the two Lambert roots as instances of the transseries volume's factorial core and the brackets as analogues only, and an exact free-label identity that locates the families any proof of the constant 1/4 would have to count. The independent check corrected the write's Simkin interval to $[0.7096, 0.7109]$ and one sentence on the block construction. No limit, equivalent or expansion; the linear term belongs to a lower bound only; the monotonicity of $a(n)$ and the count of quadruple decompositions are open. Unrefereed; not formalized.

### 208. Odious and evil integer partitions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a067590-odious-evil-partitions/article.pdf

from batch 107: Odious_and_Evil_Partitions_All_Orders_and_Inverses_Source.zip (bundle Report 165)

For partitions into odious or evil parts (positive integers with odd or even binary digit sum; OEIS A067590, A067591) and into distinct such parts (A116492, A116491), proves Kotěšovec's July 2025 conjectures in A067590 and A116492, $p_-(n) \sim K_- n^{-5/8} e^{\pi\sqrt{n/3}}$ and $p_{-,1}(n) \sim n^{-3/4} e^{\pi\sqrt{n/6}}/(2 \cdot 12^{1/4})$, with the evil counterparts, expansions to every fixed order and a Bessel comparison with error smaller than every power of $n$; for each fixed multiplicity cap $m$ the odious-to-evil ratio is $m + 1 + O(n^{-R})$ for every $R$. The methods are sectorial Mellin flatness of $\prod_j (1 - e^{-2^j w})$, the eta transformation, uniform arc bounds and a completed Hankel contour; also a Lambert-$W$ inverse to every fixed order and two-ceiling threshold enclosures. The continuation of the Thue–Morse Dirichlet series (Allouche–Cohen) and the evaluation of $d = D'(0)$ (Allouche) are credited; the nine printed digits of $K_- = 0.22186\ldots$, which the write supported by two concordant floating evaluations, were enclosed at the independent check by an interval computation of $\log Q$ (given $d = -\log Q$; a machine computation, not a formal proof). The write relates the report to the Thue–Morse atlas in `Analysis/FabiusFunction`, whose real-axis flatness and shifted Dirichlet series are formal in Lean there, proving a bridge $D(s, 1/2) = (1 - 2^s)D(s)$; classifies the inverses as instances of the transseries volume; and proves that the ratio law is not uniform in $1 \leq m \leq n$; the independent check made three of its Lean citations precise (one statement holds only for $\Re s > 0$, one needs a range hypothesis, one step is formal only). No growing-cap uniformity, no term beyond all orders, no other certified constants, no onsets, no priority claim; nothing refuted. Unrefereed; not formalized.

### 209. Exact generating function for pop-stacked permutations: poles, differential algebraicity and faster enumeration

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a307030-pop-stacked-permutations/article.pdf

from batch 108: Pop_Stacked_Permutations_Exact_Generating_Function_and_Poles_Source.zip (bundle Report 164, base) + Pop_Stacked_Permutations_Asymptotics_and_Inverses_Source.zip (bundle Report 161) + Pop_Stacked_Permutations_Run_Distribution_Source.zip (bundle Report 166)

For pop-stacked permutations (OEIS A307030; by runs A309993), the base, bundle Report 164, derives from the run-overlap characterization an explicit exponential generating function, a quotient $P = N/D$ of explicit entire functions (near 0, a ratio of $\sin(zr/4)$, $\cos(zr/4)$ terms with $r = \sqrt{4e^z - 1}$). Every genuine pole is simple, with residue $-2e^{-2\zeta}(4e^\zeta - 1)/(\zeta + 2)$; there are infinitely many positive poles, so $P$ is not D-finite and $(p_n)$ is not P-recursive, as Claesson, Guðmundsson and Pantone (CGP, arXiv:1908.08910) predicted numerically; yet $P$ is differentially algebraic (an annihilator of order at most two, minimality not claimed). A Riccati equation computes $p_0, \ldots, p_N$ in $O(N^2)$ arithmetic operations against $O(N^4)$ in the cited papers (not a bit-complexity claim). The dominant pole $\rho = 1.1134390417\ldots$ and amplitude $C = 2e^{-2\rho}(4e^\rho - 1)/(\rho(\rho + 2)) = 0.6956885490\ldots$ are rationally certified and confirm CGP's decimals. Report 161 is a second route to $p_n = Cn!\rho^{-n}(1 + O(q^n))$ by a positive-operator argument without the formula,

with the elementary bound $(24/23)^{1/4} \le \rho \le 2$ and all fixed inverse orders. Report 166 adds the bivariate generating function, run mean $\mu n + \beta$ ($\mu = 0.44214954918886\ldots$) and variance $vn + \gamma$, a central and an $O(1/n)$ lattice local limit theorem, and an alternate proof of CGP's Conjecture 2 (fixed-run denominators); it credits a prior proof posted by Shivam Patel on MathDB (20 August 2026), which the write read and checked: Patel's proof is first, Report 166's a different one. The write proves (Remark 5.1, interval arithmetic) that CGP's printed list of positive poles (Section 3.3, p. 10) skips $\rho_{12} = 5.33887671271979\ldots$, every printed digit of the other fifteen being right (their repository file lists $\rho_{12}$), and finds their three printed complex pairs correct to 62–66 decimals, after which each drops two digits; it credits Part II's interleaving bound to Asinowski–Banderier–Hackl's Theorem 12 and moves twelve open items (effective gap and onsets, nonreal poles, a minimal ODE, bit complexity, certified $\beta, \gamma$, among others) to its questions. The independent check re-proved Remark 5.1 by its own phase argument, confirmed Patel's proof step by step and the recurrence against all 450 b-file terms, and corrected the write's description of ABH's Theorem 11 (its statement does include the costs Part I cites) and of Patel's MathDB status. No effective gap, error constants or onsets, no pole-sum expansion, no certified $\beta, \gamma$; OEIS terms are CC BY-SA 4.0. Merged from three manuscripts in batch 108. Unrefereed; not formalized.

### 210. Airy logarithmic asymptotics for bounded-indegree labelled DAGs: exact parking reduction and a Lambert-$W$ inverse

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a397711-bounded-indegree-dags/article.pdf

from batch 108: Bounded_Indegree_DAGs_Airy_Asymptotics_and_Inverses_Source.zip (bundle Report 157)

For labelled simple DAGs on $[n]$ with every indegree at most two (OEIS A397711), proves $\log a_n = 2n \log n - n - \kappa n^{1/3} + o(n^{1/3})$ with $\kappa = 3\zeta/2^{1/3} = 5.567271244467717\ldots$, $-\zeta$ the largest zero of the Airy function, and the threshold inverse $b + \kappa b^{1/3}/(2\log b + 1) + o(b^{1/3}/\log b)$ with $b = y/[2W_0(y/(2\sqrt{e}))]$, $y = \log x$. The same argument gives the $n^{1/3}$ Airy correction, with factor $(c-1)^{2/3}$, for every fixed indegree bound $c \ge 2$. The route: sink inclusion–exclusion, then Kung and Yan's generalized parking functions write the count as a Poisson excursion partition function with potential $q_i = \log(i/(i-1)) \sim 1/i$, and geometric blocks transfer Mallein's homogeneous area-tilt bounds to this singular potential (the same Mallein lemmas as Part IV of `a238873-diagonal-partitions`; the Airy zero is that of `a082161-airy-amplitudes`). Exact enumeration, generalized parking and the homogeneous Airy estimates are prior inputs. The theorems hold modulo Mallein's Lemmas 2.6–2.7 (arXiv:1307.4496v3), whose statements the write and the independent check read and whose proofs nobody here verified: the stated trust boundary. Exact counts to $n = 100$ are consistent with $o(n^{1/3})$ but cannot confirm $\kappa$ (windowed fits give $-5.50$ to $-5.54$ against $-\kappa = -5.567$, recorded as no confirmation), so the finite table is not evidence for the constant; the independent check withdrew the write's reading that the fits suggest no power prefactor. The write also shows that the OEIS entry's bound $g(n-1)^n$ is off by an exponential factor (nothing submitted). A logarithmic asymptotic only: no power prefactor, amplitude, $O(\log n)$ remainder or all-orders expansion, no uniformity in growing $c$, no certified inverse bracket. Steinsky's 2003 paper on coding bounded-indegree DAGs was not inspected in full, a priority limitation the report states. OEIS terms are CC BY-SA 4.0. Reciprocal notes went to `a082161-airy-amplitudes` and `a238873-diagonal-partitions`. Unrefereed; not formalized.

### 211. Asymptotics of assemblies of nested cycles: exact counts and fixed-order marked saddle expansions

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a308338-nested-cycle-assemblies/article.pdf

from batch 108: Nested_Cycle_Assemblies_Asymptotics_and_Inverses_Source.zip (bundle Report 167); batch 138: sharp_gumbel_assemblies.zip (Part II)

For $P(z) = \prod_{m \ge 1}(1 + z^m/m)$ and $F(z,u) = \exp\{u(P(z) - 1)\}$, the labelled counts $a_n = n![z^n]F(z,1)$ are OEIS A308338, and the coefficient of $u^k$ counts assemblies with $k$ nested-cycle components (A392471; components A007838). With $c = e^{-\gamma}$, proves $a_n \sim Cn!\, n^{c/2-3/4} e^{2\sqrt{cn}}$,

$C = e^{c(1/2-\log 2)-1}c^{1/4-c/2}/(2\sqrt{\pi}) = 0.0947779253\ldots$, and a complex-uniform expansion near $u = 1$ to every fixed order in $n^{-1/2}$ with coefficients polynomial in $\log n$, the first two explicit; a whole-circle estimate controls the secondary arcs. For the component count $K_n$, the mean is $\sqrt{cn} + (c/2)\log n + m_0 + \cdots$ and the variance $\sqrt{cn}/2 + (c/2)\log n + v_0 + \cdots$, so $\mathbf{E}K_n/\sqrt{n} \to e^{-\gamma/2} = 0.7493\ldots$; also a rounding-safe Lambert-$W$ inverse. This proves the conjecture of Riedel's January 2026 note: its boxed statement is $\mathbf{E}[X] \sim N\sqrt{n}$ with $N$ to be determined, and Theorem 7.1 gives $N = e^{-\gamma/2}$. The note's alternative, $n^M$ with $M$ "close to but not equal to" 1/2 (numerically $M \geq 0.4704567259$), is refuted, and its $N \approx 0.7719351399$ is not the constant (it equals $\mathbf{E}K_{4500}/\sqrt{4500}$ to $2.5\cdot 10^{-10}$); the write proves (Remark 7.3) that finite-$n$ ratios overshoot and two-point exponents undershoot 1/2 by order $\log n/\sqrt{n}$, the $(c/2)\log n$ term read as a power. Report 167 called Riedel's value a finite-data estimate without saying the question is settled; the placement record (and an earlier version of this entry) misquoted the note as conjecturing $\mathbf{E}K_n \sim Nn^M$ with fitted $M \approx 0.4705$. The write's Remark 10.1 records that Erlihson and Granovsky's covariance display (preprint eq. (4.54) = published (4.18), both read) gives, at $p = 1$, $C = 3$, $u = 1$, the negative variance $3/e - 18/e^2$ and contradicts Corollary 7.2 in this model, gives the corrected denominator $C\,\Gamma(p+2)$ and locates the dropped $C$; the independent check confirmed the correction against exact finite-$n$ variances at $n = 20000$ and found that their (5.72) as printed holds only for $C = 1$. The report does not use the display. The Gaussian regime with variance $\sim \sqrt{cn}/2$ and the component asymptotics $B_m/m! \to e^{-\gamma}$ are prior (Erlihson–Granovsky; Greene–Knuth). Fixed orders only: no beyond-all-orders terms, effective constants, certified $C$ or local limit; the source's five questions stand, with a sixth from the write (whether Erlihson and Granovsky's proof yields the corrected displays). Part II (batch 138) studies the largest outer component: its Gumbel Kolmogorov error is asymptotic to $\frac{(2+\sqrt{5})e^{-(3+\sqrt{5})/2}}{16\sqrt{c}}\log^2 n/\sqrt{n}$, no affine repair removes it, and the optimal continuous calibration has constant $\tau/(2e)$; also lattice and $E_1$ refinements and a two-term saddle. Its only analytic input is Part I's Greene–Knuth estimate, and it answers none of Part I's questions. Its independent check proved the affine minimax constant $E_* = W(1/e)/e$, answering Part II's Question 23.3, and corrected one rounding. OEIS terms are CC BY-SA 4.0. Unrefereed; not formalized.

**212. All-orders asymptotics for square lonesum decomposable matrices**

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a299907-lonesum-decomposable-matrices/article.pdf`

from `batch 108: A299907_All_Orders_Matrix_Asymptotics_and_Inversion_Source.zip (bundle Report 132)`

For square binary matrices that split, after independent row and column permutations, into lonesum blocks and zero lines (OEIS A299907; equivalently induced-$P_5$-free bipartite graphs on two fixed labelled shores of size $n$), proves from Kamano's exact bivariate generating function (credited) $a_n = (n!)^2 C\ell^{-2n}n^{-5/4}e^{\beta\sqrt{n}}(1 + \sum_j c_j n^{-j/2})$ to every fixed order, $\ell = \log 2$, $\beta = \sqrt{2/\ell}$, $C = 0.3394739586\ldots$ explicit, $c_1 = -0.5317063378\ldots$, $c_2 = -0.1635701301\ldots$, with a Gaussian recipe for every further coefficient; the proof treats the anisotropic saddle and all complementary arcs. Also all-orders inverse threshold brackets that keep the rounding uncertainty inside the ceiling, the exact sum $a_n = \sum_k k!B_k S(n+1,k+1)^2$ ($B_k$ from A000262; a second route to the OEIS double sum) and $a_{n+1} \geq 2a_n$. Ordinary lonesum matrices are Khera, Lundberg and Melczer's (2021), a different saddle; Khera's 2021 dissertation was not inspected, a stated priority limitation. Not claimed: effective remainders or onsets, certified finite-$n$ bounds, rectangular aspect ratios, a block-number limit law, beyond-all-orders terms. The write proves the two structural facts the source states without proof (each nontrivial component is a connected chain graph; the forbidden configurations are exactly the induced $P_5$), evaluates the source's recipe at order three, $c_3 = -0.26674650591\ldots$ in closed form (the independent check added $c_4 = 1.00342017659\ldots$), classifies the inverses against the transseries volume, and records that A299906's example prints $T(5,4)$ as 90446 for 90946 (nothing submitted); the check, which classified every binary matrix up to $5 \times 5$, found no error. Khera's dissertation and the shape of the $c_j$ are open questions. OEIS terms are CC BY-SA 4.0. Reciprocal notes went to `a122399-surjection-diagonal` and

a321941-asymptotic-coefficient-integrality (whose "A000262 occurs nowhere else" became stale). Unrefereed; not formalized.

### 213. Sparse binary matrices with line sums at most two: exact counts and all fixed-order asymptotics

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a197458-line-sum-two-matrices/article.pdf
from batch 108: Sparse_Binary_Matrices_All_Orders_and_Inverses_Source.zip (bundle Report 168)

For $n \times n$ binary matrices with at most two ones in every row and column (OEIS A197458), proves $a_n \sim (n!)^2 e^{2\sqrt{2n}}/(2^{5/4}\pi n^{3/4})$ with an expansion to every fixed order in $n^{-1/2}$: $c_1 = -17\sqrt{2}/96$, $c_2 = 6409/9216$, $c_3$ and $c_4$ explicit, and a finite exact Gaussian-moment rule for every later coefficient, plus a logarithmic expansion. The route is the labelled-component decomposition (isolated vertices, paths, even cycles) and a modified-Bessel diagonal $G(t) = P(t)I_0(\sqrt{t}(2-t)/(1-t))$, which satisfies an order-two ODE and an integer recurrence; a sectorial Bessel remainder, a whole-circle bound and off-circle connector estimates give the sharp fixed-order error, and the ODE recovers $c_1, \dots, c_4$ as a second route. Also an explicit Lambert-$W$ real inverse through its bounded correction and a two-ceiling enclosure of the integer threshold. The component generating function is classical (the Goulden–Jackson exercise cited in A062154 and A062156, not inspected), as are general bounded-degree enumeration (Caizergues and de Panafieu; Greenhill, McKay and Wang); this is a specialization and refinement, with no priority claim. Fixed orders only: no convergence, no exponentially small terms, existential remainder constants and thresholds; rectangular shapes and marked components are left open. The write derives the source's generating function in one line from the A062154 formula, proves the cycles-only ratio to A001499, and extends the proof to $A_{n+d,n}$ for fixed $d$ (all orders) and to the zero rows and columns, asymptotically independent Poisson(1) counts; it shows which inverse steps are instances of the transseries volume. The independent check corrected one sentence (the $e_n$ of Remark 7.3(b) are polynomials in $1/q$, not Laurent polynomials in $q$). OEIS terms are CC BY-SA 4.0. Unrefereed; not formalized.

### 214. Fixed positive permanent binary matrices (OEIS A089479): asymptotics, canonical block laws and reproducible certificates

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a089479-fixed-permanent-matrices/article.pdf
from batch 108: Fixed_Permanent_Matrices_Asymptotics_and_Inverses_Source.zip (bundle Report 184)

For each fixed $k \geq 1$, $C_{n,k}$ counts labelled binary $n \times n$ matrices with permanent $k$ (the triangle A089479). Proves $C_{n,k} = (n!)^2 k^{-1} 2^{\binom{n}{2}}\{\rho^{-n}P_k(n) + O_k(2^{-n})\}$, with $\rho = 1.4880785455\dots$ the classical root for acyclic digraphs and $P_k$ an effectively computable polynomial of exact degree $\Omega(k)$ (prime factors with multiplicity) whose positive leading coefficient is an explicit product of prime amplitudes. The route: marked perfect matchings ($kC_{n,k} = n!\, b_{n,k}$ for digraphs with $\mathrm{per}(I + A) = k$), finite suppressed strongly connected kernels (circulation bound $m - v \leq k - 2$), Robinson's component transform (read in de Panafieu and Dovgal, credited) and a self-contained complex root isolation. Also the complete $P_2$, $P_3$, $P_4$ with rationally certified coefficient intervals, explicit all-$n$ remainder constants $M_2 = 1000$, $M_3 = 3900$, $M_4 = 170000$, every fixed Stirling and finite spectral order, smooth and integer threshold inverse bounds, and a limiting independent prime-block size law with expansions in exponentially weighted total variation. The case $k = 1$ (the permanent-one/acyclic-digraph correspondence) is classical. Not claimed: growing $k$, explicit correction laws for composite $k$, a certified inverse solver, worldwide priority; decimal diagnostics are not certificates, the optional C++ audits are finite, and Robinson's 1995 paper was not retrieved. The report says that a "second independent rational implementation" reproduces the certified intervals; it is not in the package, as the write records (a dated note and Question 7). The write recomputed the rows $n \leq 6$, $k \leq 4$ and every certified constant, identified the case $k = 1$ with A003024's classical asymptotic, and found Jovovic's "up to permutation of rows/columns" right for either action alone, wrong jointly; the independent check counted all $2^{36}$ binary $6 \times 6$ matrices, found no error and named the joint orbits (A003087). Reciprocal

notes went to `a182220-source-boundary` and `a116379-bounded-identity-trees`. Unrefereed; not formalized.

### 215. Asymptotics for binary matrices with zero row and column slopes

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a222959-zero-slope-matrices/article.pdf

from batch 108: A222959_Zero_Slope_Matrices_All_Orders_and_Inverses_Source.zip (bundle Report 194)

For the square diagonal $C_n$ of OEIS A222959, the $n \times n$ binary matrices whose least-squares regression slope vanishes in every row and every column, proves, with $S_n = n(n^2-1)/12$ for odd $n$ and $n(n^2-1)/3$ for even $n$, $C_n = 2^{n^2} e^{-7/5} \sqrt{2/\pi}\,(2/(\pi S_n))^{n-1}(1 + 171/(350n) + 483051/(245000n^2) + O(n^{-3}))$, and an expansion to every fixed order with rational, parity-free coefficients given by a finite Wick-pairing and power-sum prescription. The main issue is global localization of a growing-dimensional Fourier integral, settled by a relative exponential bound outside one central tube through a dense arithmetic-progression inverse lemma and a defect-core decomposition. Consequences: the parity ratio $C_{2m}/C_{2m-1} \sim 6e^{-3}/(\pi m^3)$, parity-aware smooth inverses with odd-integer threshold envelopes, a meet-in-the-middle formula and the palindromic bound $C_n \geq 2^{\lceil n/2 \rceil^2}$. It credits the one-row fair-word counts (Prodinger; Richomme) and the Barvinok–Hartigan framework. The localization constants give only an existential threshold, the known small counts are strongly preasymptotic, and $c_3$ onwards are not computed; no practical onset, convergence or absolute priority. The package's data notes say the nine square values were recomputed by a C++ meet-in-the-middle count that is not shipped (the Python replay covers $n \leq 8$, $n = 9$ optionally), as the write records. The write reproduces $C_{10} = 615667968$ from the b-file the source could not reach, proves Wiseman's 2023 "Conjecture" in A222955 to be Richomme's theorem, and adds $T(n,k) \geq b_n^{\lceil k/2 \rceil}$ with equality iff $k \leq 6$, so $C_n \geq b_n^{\lceil n/2 \rceil}$, sharper than the palindromic bound; the independent check recounted 119 cells and corrected one classification (the smooth inverse is also a formal instance of the transseries volume's Lambert–$W$ template). OEIS terms are CC BY-SA 4.0. A reciprocal note went to `a380592-tied-football-seasons`. Unrefereed; not formalized.

### 216. All fixed-order asymptotics for square contingency tables: the density-one model and its controlled inverse

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a110058-square-contingency-tables/article.pdf

from batch 108: Report236.zip (bundle Report 236)

For labelled $n \times n$ nonnegative integer matrices with every row and column sum $n$ (OEIS A110058), with Canfield and McKay's leading equivalent $L_n = e^{1/4} 4^{n^2}/((4\pi)^{n-1/2} n^{n-1})$ (credited), proves that $\log(a_n/L_n)$ has a rational inverse-power expansion to every fixed order, with first coefficients $-3/2$ and $223/32$ (relative form $1 - 3/(2n) + 259/(32n^2)$) and every coefficient given by a finite connected-Wick algorithm. The proof combines Canfield–McKay's absolute localization theorem with Isaev's complex cumulant theorem; an independent formal Gaussian factorization checks all eighteen cumulants behind the two corrections. A compact-positive-density extension has coefficient polynomials in $1/[\lambda(1+\lambda)]$ and gives the correction factor $\Delta = (67/6 + 5/(3u))/n + O(n^{-2})$, $u = \lambda(1+\lambda)$. The report calls this "an eventual square case" of the Canfield–McKay correction-factor conjecture $0 < \Delta < 2$; it settles one regime (square tables, density in a fixed compact interval, $n$ large), not the conjecture (finite sizes, nonsquare tables and vanishing or diverging densities stay open, as the report says; Greenhill and McKay's sparse asymptotic for $\Delta$ is prior); the write narrowed the abstract, the Section 9 title and one sentence, keeping the first wordings, and tabulates $\Delta(n, n; n, n) \in (0, 2)$ for $n \leq 13$ (consistent, not proof). Its uniformity in the density rested on the report's own reading of the Canfield–McKay localization proof; the write's Lemma 8.2 derives the uniform superpolynomial bound the proofs use from the statement of their Theorem 3 alone, and the uniform rate stays the report's reading (a further question). Also a two-ceiling enclosure of the inverse counting function. Remainder constants are not explicit, there is no explicit starting index, and the exact computations check identities and small counts, not

finite-size errors; its comparison with Barvinok and Hartigan's exponent is not a correction. The independent check re-read the cited proofs and corrected one classification (the four-term reversion is also a formal instance of the volume's Lambert–$W$ template). OEIS terms are CC BY-SA 4.0. A reciprocal note went to `a380592-tied-football-seasons`. Unrefereed; not formalized.

### 217. All fixed-order asymptotics for symmetric packed matrices: dimension, trace, collision laws and controlled inverses

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a138178-symmetric-packed-matrices/article.pdf
from batch 108: Report238.zip (bundle Report 238)

For symmetric nonnegative integer matrices of total entry sum $n$ with no zero line and ordered, distinguished positions (OEIS A138178), with $L = \log 2$ and $I_n$ the involution numbers, proves $A_n = e^{L^2/4} I_n/(2L^{n+1})\,(1 - L^2/(6\sqrt{n}) + (-L^2/4 + L^4/18)/n + O(n^{-3/2}))$ and every fixed further order, uniformly for complex dimension and trace markers near one, through Jovovic's geometric transform (OEIS, credited; the marked extension is the report's). Dimension and trace have independent Gaussian limits on the scales $n^{1/2}$ and $n^{1/4}$; uniformly on compact sets of collision markers, including zero, the repeated diagonal and off-diagonal cells are asymptotically independent Poisson($L$) and Poisson($L^2/2$) counts, independent of the Gaussian pair, with the sharp leading total-variation distance $e^{-L-L^2/2}L^2(2+L)/\sqrt{n}$ (the intake re-derived the constant) and an $O(n^{-1})$ corrected Poisson approximation. Also an all-fixed-order Lambert-$W$ inverse with two-ceiling bounds. The leading equivalent of the binary companion A135588 and the collision mechanism are Cameron, Prellberg and Stark's (credited). It is the symmetric analogue of the matrix compositions of `a261781-matrix-compositions` and uses the devices of its Part IV (bundle Report 169), but answers none of that report's questions, and its own source notes say the models differ: a separate report, with a reciprocal note in `a260700-parabolic-double-cosets` (the same Fubini pole at $\log 2$; `a261781-matrix-compositions`' Part IV already notes it). The write checked all 501 b-file terms, which the source could not fetch, against its own exact computation and computes $C_3 = -L^2(50L^4 + 324L^2 - 1755)/6480 = 0.11772518\ldots$ (a further question: one implementation); the independent check, by a different exact route to $n = 2000$, found no error and confirmed $C_3$ numerically at three values of $L$. No effective onset, lattice local limit, large deviations or exponentially small sectors; the trace marker does not cross $v = 0$; no global priority. A reciprocal note of batch 113 records that one of its decreasing-margin variants, A104779, is treated in `a104779-total-kostka-sums`, with $a_n/A_n \sim 2Ce^{-L^2/4}L^{n+1}$; the classification question stands. Unrefereed; not formalized.

### 218. Symmetric sign matrices and free degree constraints: leading asymptotics, a critical window and controlled inverses

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a027832-symmetric-sign-matrices/article.pdf
from batch 108: Report239.zip (bundle Report 239)

For symmetric $n \times n$ matrices with entries $\pm 1$ (diagonal included) and nonnegative row sums (OEIS A027832), an exact encoding as graphs with one extra vertex of unrestricted degree (a structural fact Greenhill and McKay state, credited) reduces the count to degree-capped graphs. The report proves that freeing a fixed number $f$ of degree caps near half the order multiplies the all-capped count by $\Phi(\ell)^{-f}(1 + o(1))$, where $\phi(\ell) = \ell\Phi(\ell)$ ($\ell = 0.5060544689\ldots$), and, uniformly for $f = O(\sqrt{M})$, a finite-saddle formula and an explicit equivalent with a collective quadratic and a bounded-offset linear correction, by uniform dense degree-sequence enumeration (McKay–Wormald), relative concentration, an exponential tilt and balancing of the even degree-sum lattice. With McKay, Wanless and Wormald's all-capped asymptotic this gives $A_n \sim 2^{n(n+1)/2} q^n Z_0 e^{-\ell^2/2}$ times $\exp(\ell\sqrt{n} + J/4)$ for even $n$ and 1 for odd $n$, explicit constants, consistent with all 17 OEIS terms (ratios 0.9996 and 0.9982 at $n = 16, 17$, an intake check), plus the precision of a continuous inverse and eventual two-ceiling threshold bounds. A leading $1 + o(1)$ theorem only, conditional on the cited dense-enumeration inputs: no all-orders expansion, effective onset, caps beyond $\sqrt{M}$ or historical priority (Anderson's chapter and the unpublished Sillke–Flammenkamp work

were not inspected; the write matched Flammenkamp's table with $A_1, \ldots, A_{16}$). In one delivered section two carriage-return bytes replaced the backslash and "r" of a roman-font command, so the delivered PDF prints "meven" in eq. (7.10) and one more display; the write restored the command with a dated note and removed the file's `-text` attribute. The independent check found no error and corrected two rounded constants of the README. OEIS terms are CC BY-SA 4.0. Unrefereed; not formalized.

### 219. Maximum-density kings on cylinders and planes (OEIS A137432, A061593, A061594)

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a137432-maximum-density-kings/article.pdf`

from `batch 109: Rectangular_Kings_Aspect_Ratio_Transition_Source.zip (bundle Report 174, base, Part I) + Cylindrical_Kings_All_Order_Asymptotics_and_Inverses_Source.zip (Report 173, Part II) + Planar_Kings_Pole_Bounds_and_Inverses_Source.zip (Report 175, Part III)`

Count placements of $hw$ nonattacking kings on a $2h \times 2w$ board, one in each $2 \times 2$ block: $A(h, w)$ with cyclic columns, $a_n = A(n, n)$ (OEIS A137432), and $P_h(w)$ on the ordinary board (A061593, A061594 at $h = 2, 3$). Part I, for $w/h$ in a compact positive interval, gives a marked-star sum with relative error $O(1/h)$, a free energy with a transition at $w/h = \log 2$, every fixed order above it, $A = h^w C(\lambda_h) \sum_j c_j(\lambda_h) h^{-j}$, a critical window, three limit laws, Lambert-$W$ first-crossing brackets and, at fixed height, squarefree denominators whose irreducible factors have degree at most $\lfloor h/2 \rfloor + 1$, with equality at the heights $h = p - 3$. Part II proves $a_n = Cn^n(\sum_{j \le M} c_j n^{-j} + O(n^{-M-1}))$ with Kotěšovec's $C = 2e(e-1)^2/(e-2)^2$, exact $c_1, c_2, c_3 \in \mathbb{Q}(e)$ and the ratio expansion $a_{n+1}/(na_n) = e(1+1/(2n)+O(n^{-2}))$, which proves the A137432 conjecture (Kotěšovec, 2011) that the ratio tends to $e$. Part III bounds the degree and pole order of every factor of the planar denominators, gives Wilf's dominant term $(\alpha_h w + \beta_h)(h+1)^w$ with $\alpha_h \ge 1$ and a Lambert-$W$ bracket. Merged in dependency order: Part II's main theorem is Part I's at $\lambda = 1$ (the coefficients agree identically). Of Kotěšovec's book conjectures (the book itself not read), Conjectures 1 and 4 are proved and the upper half of Conjecture 2 on both boards; the write proves $2(h+1)^w \le A(h, w) \le 2^h(h+1)^w$ with its equality cases (so the book's strict bound for $a_n$ fails exactly at $n = 1$) and, reading Part III's Theorem 27.2, that the dominant planar factor has exponent exactly two at every height (Conjecture 3's printed $\lfloor (h+1)/2 \rfloor$ contradicts the exponents on its own page; not settled). Report 173's fixture names the OEIS b-file (to $n = 384$) but holds $a_{400}$ and $a_{800}$; the independent check identified its recorded SHA-256 as that of Matsuo's table and corrected three rounded digits and one range. Effective constants, growing windows and low-degree multiplicities are open questions. A reciprocal note went to `a201513-sparse-chess-placements` (a different, sparse regime). Unrefereed; not formalized.

### 220. Nonattacking bishops: the amplitude and the diagonal corrections (OEIS A002465, A238260)

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a002465-nonattacking-bishops/article.pdf`

from `batch 109: Report219-reproducibility.zip (bundle Report 219)`

For the number $B_n$ of placements of $n$ nonattacking bishops on the $n \times n$ board (OEIS A002465), proves $B_n = b\,\Gamma(n) q^n (\sum_{j \le J} c_j^{(n \bmod 2)} n^{-j} + O(n^{-J-1}))$ for every fixed $J$, with $r \in (1, 2)$ the root of $r/(1 - e^{-r}) = 2$, $q = e^r/r$ and the closed form $b = e^r/(2\pi\sqrt{r^2 - 1}) = 0.63126\ldots$ of the OEIS constant A238260 (given there only by digits and a limit definition); $c_1, c_2$ are parity-free and the parity split first appears at $n^{-3}$, $c_3^{(1)} - c_3^{(0)} = r^3/(4(r+1))$. The route is the two colour Ferrers boards, a Stirling convolution and a two-variable saddle; also a parity-preserving two-ceiling inverse with a Lambert centre. The write checked $b$ against all 105 printed digits of A238260, gives a Lambert form $b = -1/(\pi w_0 \sqrt{(1+w_0)(3+w_0)})$, $w_0 = W_0(-2/e^2)$, matches all 376 b-file terms, proves $B_{n+1} \ge 2B_n$ ($n \ge 1$, so the inverse's onset is explicit) and moves the angular-only order-4 receipt and the unproved parity-branch enclosures to its questions; the independent check found no mathematical error and corrected its description of Santos's Theorem 6. The exact enumeration and the leading equivalent with numerical $b$ are prior (Perott, Arshon, Kotěšovec,

Santos). Not a Part of `a201513-sparse-chess-placements`, whose general theorem assumes a finite move set; a reciprocal note went there. Unrefereed; not formalized.

**221. Maximal matchings of complete tripartite graphs (OEIS A297487)**
`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a297487-tripartite-maximal-matchings/article.pdf`
from `batch 109: Tripartite_Maximal_Matchings_Asymptotics_and_Inverses_Source.zip (bundle Report 176, base, Part I) + Unequal_Tripartite_Matchings_Transitions_and_Inverses_Source.zip (Report 178, Part II)`

A matching of a complete multipartite graph is maximal exactly when its unmatched vertices lie in one part, so every count is a sum of factorial ratios. Part I treats $A_n$, the maximal matchings of $K_{n,n,n}$ (OEIS A297487): with $\lambda = \sqrt{n/2}$, an expansion to every fixed order, uniform in a real marker of unmatched vertices, which gives the OEIS equivalent $A_n \sim 3\sqrt{2}(2n/e)^{3n/2}e^{\sqrt{n/2}-3/8}$ with four corrections; the unmatched count is within $O(n^{-1/4})$ in total variation of a parity-conditioned Poisson($\lambda$) law; the probability of a maximum matching to every fixed order; and a two-ceiling threshold bracket. Part II, an explicit sequel, treats unequal parts: the branches for strictly triangular proportions, and $K_{2n+d,n,n}$ across the face $a = b + c$ of the triangle inequality with a uniform moving window at scale $n^{1/3}$, Gaussian laws, four corrections of the boundary counts $M(2n, n, n)$ (no OEIS entry) and an exterior Poisson law. It settles one instance of Part I's question on unequal parts; more parts and higher orders stay open. The write compared all 100 b-file terms, proved Howroyd's OEIS program equal to the exact formula, and classified the inverses (the leading terms are factorial-core instances, the brackets not staircase instances); the independent check, with brute-force counts and exact $\log M(2n, n, n)$ to $n = 4000$, found no mathematical error and corrected two attributions of quoted wording. Unrefereed; not formalized.

**222. Labelled outerplanar graphs to all fixed orders (OEIS A097998, A098000)**
`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a097998-outerplanar-graphs/article.pdf`
from `batch 109: Labeled_Outerplanar_Graphs_Expansions_and_Inverses_Source.zip (bundle Report 170)`

For connected and all labelled outerplanar graphs (OEIS A097998, A098000), proves $c_n = An!\rho^{-n}n^{-5/2}(\sum_{j\le J} d_j(0)n^{-j} + O(n^{-J-1}))$ and the same for $g_n$ with amplitude $e^{\nu}A$, uniformly in a complex component marker, with four explicit corrections, a refined shifted Poisson law $1+\text{Poisson}(\nu+5\tau/(2n))$ for the number of components with total-variation error $O(n^{-2})$ (the limit $1 + \text{Poisson}(\nu)$ is prior), and a two-ceiling inverse with a Lambert start. It corrects a published constant: the amplitude $h \approx 0.018216$ of Bodirsky, Giménez, Kang and Noy (2007), Theorem 5.1, is $e^{\nu}A = 0.0080960474\ldots$, the printed value being $(9/4)e^{\nu}A$ (exponent and radius right); the write proves this with a hand bound and an interval enclosure and refutes the exponent and decimals of their introduction, finds the preprint's 0.017657 wrong too and Kang's thesis value 0.008095 inaccurate, and observes the same 9/4 in their Theorem 7.1 (graphs with no $K_{2,3}$ minor). The independent check confirmed the correction by Richardson extrapolation of exact counts to sixteen digits, traced the 9/4 to a misprinted constant in their Theorem 3.6 ($-(3/2)F_1$ for the $-(2/3)F_1$ of its proof), and found numerically that their series-parallel constants appear to be $3\sqrt{\pi}$ too large (an open question, with the cause of the 9/4 and a second analytic route). Unrefereed; not formalized.

**223. Planar Eulerian orientations: exact Lambert sectors, logarithmic corrections and inverses (OEIS A324312, A277493, A324314)**
`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a324312-planar-eulerian-orientations/article.pdf`
from `batch 109: Planar_Eulerian_Orientations_Lambert_Expansions_and_Inverses_Source.zip (bundle Report 171)`

For rooted planar Eulerian orientations by edges (OEIS A324312, and A277493 = 2·A324312) and quartic ones by vertices (A324314), with $m = n + 2$, $L = \log m$ and $T = \log L + C + \gamma$, proves $a_n = \kappa_0\mu^m/(m^2L^2)\{\sum_{j\le J} Q_j(T)/L^j + O_J((\log L)^{J+1}/L^{J+1})\}$ with six universal polynomials $Q_0, \ldots, Q_5$, the sharper $a_n = \kappa_0\mu^m\{h_m - k_m/2 + O(m^{-4}L^{-3})\}$ through the coefficients of exact

analytic Lambert models and any fixed number of such sectors, a relative correction $2/(mL)$ that no logarithmic truncation sees, and smooth inverse charts with a two-ceiling threshold bracket. Every count statement is conditional on Bousquet-Mélou and Elvey Price's $\Delta$-continuation (JCTA 2020, Propositions 8.2–8.3), whose proof for the general family they only indicate; the independent check found the same authors' later statement (refined preprint arXiv:2503.15046v3, Section 6.4) that the logarithmic behaviour has been proven for both weights, so the condition is a refereed proposition with an indicated proof, not a conjecture. The write corrected the source's citation of the continuation ODEs (Section 8.1, not 7, in every version), reproduced all printed terms by its own exact reversion and classified the inverses; a written proof of that proposition is its open question. The 3.0 MB `exact.json` is not staged (it regenerates byte-identically; README recipe). Unrefereed; not formalized.

### 224. Distinct sums of square roots: quantitative asymptotics and inverse indices (OEIS A397045)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a397045-square-root-sums/article.pdf

from batch 109: Distinct_Square_Root_Sums_Asymptotics_and_Inverses_Source.zip (bundle Report 172)

$a(n)$, the number of distinct values in $[n, n+1)$ of finite sums of square roots of positive integers (OEIS A397045), equals the number of vectors $(m_d)$ over the squarefree $d \geq 2$ with $\sum m_d\sqrt{d} < n+1$. Proves $\log a(n) = K(n+1)^{2/3} + O(n^{1/3})$, $K = (81\zeta(3)/\pi^2)^{1/3}$, which is the entry's conjecture including the constant (Kohlbecker's 1958 Tauberian theorem already implies the leading law, as the report says and the write works out), an exact-product saddle formula with relative error $O(n^{-1/3})$, inverse indices, and, unconditionally, that no equivalent $Cn^{\beta}\exp(Kn^{2/3})$ exists, because of a surviving pole at a zeta zero of least height. The write refutes, with proof, the entry's comment that integer radicands are equivalent to squarefree ones (the integer-radicand count is $3, 8, 21, 46, 104$ against $a(n) = 3, 7, 17, 35, 76$), proves that $a(n)$ increases strictly, and extends the table to $n = 75$; the independent check confirmed both and corrected one inference (the non-equivalence is visible from $n = 2$, not $n = 1$) and the letters of a cited theorem. Unrefereed; not formalized.

### 225. Diagonal multiple alignments: all fixed orders, column counts and inverse thresholds (OEIS A262810, A262809, A316677, A316674)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a262810-diagonal-alignments/article.pdf

from batch 109: Diagonal_Sequence_Alignments_Asymptotics_and_Inverses_Source.zip (bundle Report 177)

For the number $a_n$ of binary matrices with $n$ labelled rows, every row sum $n$ and ordered nonzero columns (OEIS A262810, the diagonal of A262809), proves $\log(a_n/K_n) = \sum_{j\leq J} C_j n^{-2j} + O(n^{-2J-2})$ for every $J$, with Kotěšovec's equivalent $K_n$ and explicit $C_1, C_2, C_3$, uniformly in a column mark; the column count has mean $n^2/(2\log 2) + n/4 + \cdots$ and a Gaussian limit on scale $n$; substituting $k = n$ in the classical fixed-$k$ equivalent misses a factor $e^{-1/12}$; a Lambert-$W$ inverse and a two-ceiling threshold enclosure; and the nonnegative analogue A316677 by a binomial coupling. The write quotes the four entries, checks both b-files by Howroyd's formula and records that the unmarked relation $2^{n-1}a_n$ is also in Eger (2015), credited there to Duchi and Sulanke (not read); the independent check confirmed every number and qualified the write's claim that Howroyd's formula is a different route (it is the same double sum in the other order). The report's README promises a source file not in the archive (disclosed). Unrefereed; not formalized.

### 226. Lazy closed walks in growing dimension (OEIS A328716, A328718)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a328716-lazy-closed-walks/article.pdf

from batch 109: Lazy_Closed_Walks_Asymptotics_and_Inverses_Source.zip (bundle Report 179); arrival fe7165a3c: uniform-lattice-bridges-research.zip (Part II)

For the number $a_n$ of $n$-step closed walks in $\mathbb{Z}^n$ with steps 0 and $\pm e_i$ (OEIS A328716, the diagonal of A328718), proves $a_n = C_\varepsilon (dn)^n \{\sum_{\ell\leq M} c_{\varepsilon,\ell} n^{-\ell} + O(n^{-M-1})\}$, $\varepsilon = (-1)^n$, for every

$M$, with Bessel definitions of Kotěšovec's constants $d$ and $C_\pm$ (so his equivalent) and explicit $c_{\varepsilon,1}, c_{\varepsilon,2}, c_{\varepsilon,3}$; an absolute complex-uniform marked expansion; the zero-step count to every order about a parity-conditioned Poisson law; occupied axes Gaussian on scale $\sqrt{n}$, asymptotically independent of the zero steps; parity inverse brackets of eventual width at most one; and a compact proportional extension. The write proves Kotěšovec's 2019 conjecture in A328718 that row $r$ is asymptotic to $(2r+1)^{n+r/2}/(2^r(\pi n)^{r/2})$ (a local limit theorem, classical, no novelty claimed); the independent check re-derived that proof and made its $1/N$ term, $-3D/16$, explicit (corroboration, not a theorem). Pin `83befe707` exists. Part II (a non-bundle manuscript of 7 October, written as Sections 13–21) gives a scalar saddle expansion for weighted closed walks with steps 0, $\pm e_i$, uniform in the dimension and in all nonnegative activities, to every fixed order with leading relative error at most $24/m$; in the dense isotropic regime the Gaussian estimate is relatively correct exactly when $D^3/N^2 \to 0$; a joint Poisson law in the sparse regime; and an exact counterexample at $D = 16$, $N = 26$ to the universal Gaussian return bound of Felker–Lyons and Ball–Sterbenz. It answers the host's question on sparse and dense ratios and advances two others. Its independent check confirmed the counterexample by two further exact routes, relabelled one comparison and added two precisions of scope. Unrefereed; not formalized.

### 227. Decorated logarithmic eta products (OEIS A330499, A330498)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a330499-decorated-eta-products/article.pdf

from batch 110: Decorated_Eta_Oscillatory_Asymptotics_and_Inverses_Source.zip (bundle Report 185, base, Part I) + Report211-repro-v1.zip (Report 211, Part II)

Two decorations of an eta-type product with $L = -\log(1-z)$. Part I proves $a_n\rho^n/n! = \pi^2/(12(e-1)) + \sum_j n^{-3/4-j/2}H_j(\sqrt{n})$ at every fixed order, which identifies the decimal $0.478656\ldots$ of A330499, with cusps, signed subsequences and two-ceiling inverses. Part II, with the harmonic kernel that Part I excluded, proves $n\rho^n a(n)/n! = \log 2 + \sum_j n^{-1/4-j/2}H_j(\sqrt{n})$, and so Kotěšovec's conjecture $c = \log 2$ in A330498 exactly as stated. The kernels (an eta identity, a signed Voronoi sum), main terms and scales differ, so both Parts are printed in full. The write proves that the two leading oscillations share frequencies, damping and phase (their amplitudes differ) and classifies the inverses against the transseries volume (Theorems 9.1 and 21.2 are not staircase instances); Report 211's release receipt was not delivered. The independent check confirmed the proof of the conjecture and the shared oscillation, and corrected one notation attribution. Fixed orders only; eight open questions. Unrefereed; not formalized.

### 228. Spiral Fibonacci sequences (OEIS A094926, A094925, A078510)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a094925-spiral-fibonacci/article.pdf

from batch 110: Hexagonal_Spiral_Fibonacci_Asymptotics_and_Inverses_Source.zip (bundle Report 190, base, Part I) + Square_Spiro_Fibonacci_Growth_and_Inversion_Source.zip (Report 191, Part II)

Two Spiro–Fibonacci constructions with square-root delays and different growth classes. Part I (the hexagonal spiral) proves $a_n \sim C\varphi^n$ for both seeds, which proves Scheucher's 2015 conjectured equivalents in A094926 and A094925 (the certified digits agree with all 106 printed digits of A258639 and all 77 of A094925), with every fixed order, the sharp $\sqrt{n}$ phase and inverse envelopes. Part II (the square spiral, A078510) proves $a_n = Ce^{H(n)}(1 + O(w^3/\sqrt{n}))$ with $C > 0$ and an inverse with an inverse-logarithmic hierarchy; it settles no OEIS conjecture. The Parts share a construction family and a method but no theorem, as the report says. The write proves that Scheucher's heuristic form $a(n) = (a(n-1) + a(n-2))/(1 - cd^{-\sqrt{n}})$ in A094926 cannot hold with constants (corner rows drop the excess ratio by $\varphi^{-2}$); the independent check confirmed this to $n = 40000$ and clarified one provenance date. Growing orders, closed forms, effective onsets and other seeds are open. Unrefereed; not formalized.

### 229. Exponential towers (OEIS A096537, A096542)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a096537-exponential-towers/article.pdf

from batch 110: A096537_Corrected_Asymptotic_and_Airy_Limits_Source.zip (bundle Report 192, base, Part I) + A096537_Fractional_Corrections_and_Gamma_Laws_Source.zip (Report 193, Part II)

Part I proves $b_n = a_n/(n!)^2 \sim \sqrt{2/(\pi\rho)}\,\rho^{-n}n^{-1/2}$ through critical towers $H_h(1/(eh)) \sim Kh^{1/6}e^{-\alpha h}$, with an Airy window and size law, a uniform lattice local limit, Gaussian tree heights and a Lambert inverse. Part II, a declared sequel, answers two of Part I's questions: $b_n\rho^n\sqrt{n} = C + c_1 n^{-1/3} + o(n^{-1/3})$ with $-0.61 < c_1 < -0.25$ (an Airy integral, its sign interval-certified), a Gamma prefactor for every fixed positive shift, and a Borel boundary. The write refutes with proof Kotěšovec's A096537 conjecture, whose amplitude $2\pi$ should be $2\sqrt{2\pi/\rho}$ (ratio $\sqrt{2/(\pi\rho)} = 1.00355251532641107954\ldots$), and A096542's formula $T(n,1) = n \cdot \text{A096537}(n)$, false for every $n \geq 2$ (it is $n \cdot \text{A096537}(n-1)$). The independent check confirmed both refutations by its own recurrence and qualified one sentence on short-range fits. Report 193's three byte-copy certificate files are not shipped. Unrefereed; not formalized.

### 230. Modified Entringer connection constants and spectral limits (OEIS A386381, A386363)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a386381-modified-entringer/article.pdf

from batch 110: Modified_Entringer_Asymptotics_and_Spectral_Limits_Source.zip (bundle Report 180)

Identifies Kotěšovec's OEIS amplitude of A386381 as $c = \pi C(1) = \pi A(\pi/2)$, where $(zA')' = (\sec + \tan)A$, and certifies it by an enclosure; proves every fixed order, uniformly for a complex boundary mark on compact sets, the Fredholm form $C(\lambda) = \det(I + \lambda K)$, an infinite-Bernoulli limit law, a large-mark law with its far tail, simple negative limiting zeros and two-ceiling inverses. The OEIS equivalent is Corollary 5.2, its decimal a truncation of the certified $c$; the entry's divisibility conjectures are out of scope. The write recomputed all 265 b-file terms and classified the inverses against the transseries volume; the independent check recomputed every number by its own routes and corrected one provenance sentence. Six open directions. Unrefereed; not formalized.

### 231. Certified asymptotics for tableau defect clusters (OEIS A394326)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a394326-tableau-defect-clusters/article.pdf

from batch 110: Tableau_Defect_Clusters_Certified_Asymptotics_Source.zip (bundle Report 183)

The hard-rod extraction behind A394326 is an exact first-return quotient $C = 1 - D/N$ of signed trace-class Fredholm determinants; a computer-certified unique simple pole gives $|a_n - c\rho^{-n}| \leq 2500(100/63)^n$ for all $n$, with $1/\rho < 1.6180189139 < \varphi$, every fixed spectral order, growth from $n = 600$ and two-ceiling inverses. The write refutes with proof the entry's formulas $a(n) \sim A\varphi^n$ (false for every $A$) and $a(n) \sim a(n-1) + a(n-2)$; its residual comment ($< 0.0002$) is compatible. The independent check replayed the certificate chain, found the pole independently at $\rho = 0.61803976\ldots$, and proved the residual comment for every $n \geq 30$ and the strict decrease of $a_n/\varphi^n$ from $n = 30$ (the tails resting on the certificate), which settles the write's two questions. The 1.1 MB `inverse_proposals.json` is shipped, since the mandatory certificate reads it. Unrefereed; not formalized.

### 232. Rounded running means (OEIS A065094, A065095)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/a065094-rounded-running-means/article.pdf

from batch 110: Rounded_Running_Means_Bessel_Asymptotics_and_Inverses_Source.zip (bundle Report 186)

Every bounded perturbation of the running-mean recurrence is $AL_{n-1}(-1) + O(\sqrt{n})$ with an exact positive-weight amplitude, a pointwise optimal envelope and nested rational brackets; the floor and ceiling amplitudes $A_-$, $A_+$ are certified to 69 truncated digits. Also every fixed order, eventual log-concavity, the range inverse to every order and two-ceiling thresholds. This proves the unsigned

conjecture of both entries that $a(n)$ is asymptotic to $C \cdot \mathrm{BesselI}(0, 2\sqrt{n})$, and Kotěšovec's numerical equivalents, all four of whose decimals are truncations of the certified digits. The independent check recomputed $A_{\pm}$ to 80 digits by a different route and corrected one Lean declaration name. Six open questions. Unrefereed; not formalized.

**233. Hidden oscillations in square-root factorial sampling (OEIS A326805)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a326805-square-root-factorial-sampling/article.pdf
from batch 110: Square_Root_Factorial_Sampling_Oscillations_and_Inverses_Source.zip (bundle Report 187)

For $S(x) = \sum_k x^{\sqrt{k}}/\Gamma(1+\sqrt{k})$ and $a(n) = \lfloor S(n) \rfloor$: every fixed order of the discrepancy as a Lambert-$W$ chirp of envelope $n^{1/8}(\log n)^{3/8}$, the exact limit set $[-C_*, C_*]$ with $C_* = 4(4\pi)^{-7/8}$, an exact finite-harmonic formula, a sector bound, a convergent Lagrange inverse, and the failure of $\lceil W(Y/2) \rceil$ in both directions. It proves Kotěšovec's conjecture $a(n) \sim 2ne^n$ exactly as stated, with relative error $o(n^{-r})$ for every $r$. The transseries exposition of A336293 concerns a different model. The independent check confirmed the proof and every number and corrected two Lean declaration names. Four open questions. Unrefereed; not formalized.

**234. Compositions with a length-dependent minimum (OEIS A098131, A098132, A098133)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a098131-minimum-length-compositions/article.pdf
from batch 110: Minimum_Length_Compositions_Asymptotics_and_Inverses_Source.zip (bundle Report 189)

For compositions of $n$ into $k$ parts each at least $k+s$, $s$ fixed, proves every fixed order with $4n = e^{2v}v(v+2)$ and $t = e^{-v}$, $C_{s,1}$ in closed form and $C_{s,2}$ by a finite formula, interlaced half powers for exact minima, fixed rare-excess laws and integer brackets, and shows that the leading smooth inverse misses $v/48 + O(1)$. The three entries state no asymptotic formula or conjecture; the write recomputed every b-file term and tested the second order numerically. The independent check confirmed every coefficient by a second route and corrected one transseries statement (the exact equation for $v_0$ has an admissible dominant core). Five questions and five restrictions stand. Unrefereed; not formalized.

**235. Partitions on maximal Schreier supports (OEIS A239950)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a239950-maximal-schreier-supports/article.pdf
from batch 111: Report195_TeX_and_Reproducible_Code.zip (bundle Report 195)

For partitions whose least part equals the number of distinct part sizes, proves $p_n = Kn^{-3/4}e^{2\sqrt{Sn}}(\sum_j c_j n^{-j/2} + \cdots)$ to every fixed order, $S = \mathrm{Li}_2(2/3) - \mathrm{Li}_2(1/3) + \log(3/2)\log 2$, with $c_1 < 0$ and $c_2$ explicit, a Gaussian least-part law, a Lambert-$W_{-1}$ inverse with two-ceiling brackets, and a rational certificate that $F(q)/q$ has a real zero in $(-4/5, -3/4)$, so the generating function is no finite eta quotient. The write recomputed all 1001 b-file terms by an independent route and classified the inverses against the transseries volume; no claim of the source was found wrong or unproved. The independent check confirmed every number by a third route and corrected one OEIS attribution. Strict increase is observed, not proved. Unrefereed; not formalized.

**236. Partitions whose number of distinct sizes equals the maximum multiplicity (OEIS A239964)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a239964-sizes-equal-max-multiplicity/article.pdf
from batch 111: Report196_TeX_and_Reproducible_Code.zip (bundle Report 196)

Proves $a(n) \sim (\pi H'(1)/(12\sqrt{2}))n^{-3/2}e^{\pi\sqrt{2n/3}}$ and $P(D = M) \sim \pi H'(1)/\sqrt{6n}$, with $H(z) = \prod_j (1 - e^{-jz})$, through an exact inclusion–exclusion diagonal transform (Manyama's OEIS product formula credited), with all fixed orders in Bessel and terminating-polynomial form, $c_1, \ldots, c_4$, and a Lambert-$W_{-1}$ inverse with two-ceiling brackets. The write found three of the six printed

decimals rounded in the last place; the independent check confirmed every constant by two routes and added two precisions. No claim of the source was found wrong; strict increase is observed, not proved. Unrefereed; not formalized.

**237. Distinct multiplicity values in integer partitions (OEIS A373271, A373273)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a373271-distinct-multiplicity-values/article.pdf
from batch 111: Report197_TeX_and_Reproducible_Code.zip (bundle Report 197, base, Part I) + Report198_TeX_and_Reproducible_Code.zip (Report 198, Part II); arrival 7b8ed30e7: ProveIt_Research_2026-10-07.zip, its partitions manuscript (Part III)

One exact identity, $P(q)\sum_m w(m)(1-Q_m(q))$, binds two statistics of the multiplicities of a random partition. Part I ($w=1$, A373271) proves $a(n)/p(n) = (6n)^{1/4} - 23/16 + Kn^{-1/4} + O(n^{-1/2})$, the $-23/16$ needing a telescoping tail, with totals, eventual increase and a shrinking two-ceiling inverse. Part II ($w(m)=m$, A373273), whose package is adapted from Part I's and which answers its weighted-value question for that weight, proves a five-scale mean with leading term $\sqrt{6n}\log n/(4\pi)$, an absolute expansion and an inverse; its repetitions of Part I's estimates are kept with notes. The independent check of Parts I–II confirmed every table, constant and integral and added one precision. Part III (a non-bundle manuscript of 7 October, written as Sections 25–34) answers Part I's question on fluctuations, $\mathrm{Var}_n D_0 \sim v_0 n^{1/4}$ with $v_0 = 0.3080002444\ldots$ and a central limit theorem, Part II's question on weighted fluctuations, a Gumbel law for $D_1$ with $\mathrm{Var}_n D_1 \sim n$, and, for the fluctuation laws, the question on the weights $m^\alpha$ (Gaussian below $1/2$, a $\log n$ variance factor at $1/2$, non-Gaussian above); its mean of $D_1$ is a second route to Part II's first two terms. Its independent check confirmed every constant, table and label and restricted one re-scoping: a second term is proved only for $0 < \alpha \le 1$. Unrefereed; not formalized.

**238. Extreme-part conditions in integer partitions (OEIS A350879, A237753, A237751, A118096, A117086)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a350879-extreme-part-conditions/article.pdf
from batch 111: Report216-reproducibility.zip (bundle Report 216, base, Part I) + Report217-source-v2.zip (Report 217, Part II) + Report218-reproducibility.zip (Report 218, Part III)

Conditions that couple the largest part $M$ and the smallest part $L$. Part I: $M = kL + b$ and $M \ge kL + b$ for fixed $k \ge 2$, $b \ge 0$ (A350879 columns, A237753, A237751) to every fixed order, with finite sieves and a $W_{-1}$ inverse. Part II: $\max = 2\min$ (A118096) to every fixed order on the scale $e^{\pi\sqrt{2n/15}}$, with inverses, increase and log-concavity. Part III: the deficit $D = p - a$ of A117086 ($\min \mid \max$), $D \sim (\pi/(2\sqrt{6n}))p$ to every fixed order, with residues, a certificate at $n = 1000$ and inverses. Part II's generating function is the $L = 2$ summand of Part III's, and Parts I and III share the transfer machinery (Part III's repetitions kept with notes). The leading equivalents of A350879, A237753 and A118096 are Kotěšovec's, re-proved. The write refutes with proof Kotěšovec's first form in A350879, $k!\,\pi^k \exp(\sqrt{2\pi n/3})/\cdots$, for every $k \ge 2$ (a factor $\pi$ is missing under the root, and the quotient tends to 0), and proves that A117086 increases strictly. The comparison with Pittel (2007) stays open. The independent check confirmed the refutation, the monotonicity proof and every count and coefficient, and corrected one provenance statement. Unrefereed; not formalized.

**239. Power weighted Dyck moments (OEIS A290569, A216966, A218221, A227887, A338634, A291333)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/a290569-power-weighted-dyck-moments/article.pdf
from batch 111: Report201.zip (bundle Report 201, base, Part I) + Report199.zip (Report 199, Part II) + Report202.zip (Report 202, Part III) + Report204.zip (Report 204, Part IV)

Dyck-path moments with descent weights $h^p$ (the array A290569). Part I, for every fixed $p > 0$: $Z_n(h^p) \sim C_p d_p^n (n!)^p n^{-1/2}$ with relative error $O((\log n)^{-2})$, which re-proves Kotěšovec's posted general formula, harmonic and parity-harmonic transfer, absolute amplitudes for positive

polynomial weights and two-ceiling inverses. Part II ($p = 4$): the first correction $-(1 + \pi)/(16n)$ for A227887 and the cumulants A338634, shifted moments to every order, and the parity theorem, which proves A338634's conjecture that $a(n)$ is odd exactly when $n$ is a power of 2 (modulo 2 the S-fraction gives $A^2 + A = x$). Part III: relative first corrections in three regimes, a second route to Part II's. Part IV: the diagonal $p = \tau n + \sigma$ (A291333) to every fixed order uniformly, with a Gevrey remainder and an integer-safe inverse. Part I's main theorem and the absolute amplitudes rest on Kriecherbauer and McLaughlin's Theorem 1.5, read only as restated by Claeys, Krasovsky and Minakov (a disclosed trust boundary); Part III's general first-correction coefficient is conditional on an unverified Freud envelope and is an open question (numerical evidence to six digits at $p = 3$). The independent check confirmed the parity proof and every constant, and corrected a b-file claim (A216966 and A227887 have no b-file) and one attribution. Unrefereed; not formalized.

### 240. Sparse graph moments: closed tree walks (OEIS A094149)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a094149-sparse-graph-moments/article.pdf

from batch 112: Report214-reproducibility.zip (bundle Report 214, base, Part I) + Report213-reproducibility.zip (Report 213, Part II) + Report212-reproducibility.zip (Report 212, Part III) + Report210-reproducibility.zip (Report 210, Part IV) + Report209-reproducibility.zip (Report 209, Part V); batch 138: sparse_graph_moments_research.zip (Part VI)

The moments $M_{2k}$ of the spectral measure of $G(N, 1/N)$, which count closed tree walks (A094149), in one chain printed newest first. Part V: $M_{2k} \geq 2(B_{k+1} - B_k) - F_{2k}$, with equality exactly for $k \leq 3$. Part IV: the one-hub and secondary-edge families $G_k$, $J_k$ exactly. Part III: the dominant-vertex class has size $(2 + o(1))B_{k+1}$, with a reduction of the full equivalent to an unproved tail estimate, the scalar correction $W^3/(2k)$ and two counterexamples to a Poisson-averaging heuristic. Part II proves that estimate, so $M_{2k} \sim 2B_{k+1}$. Part I: every fixed collision order, $a_k/(2B_{k+1}) = 1 + \sum_{j \leq p} A_j(k) + O(\log^{10p+6} k/k^{p+1})$ with $A_j \sim W^{3j}/(2^j j!\, k^j)$, and a two-ceiling inverse. The essential external input is Bauer and Golinelli's coefficientwise counting theorem. Repeated proof blocks are printed once, the omitted copies keeping their numbers through notes. The bounds on the OEIS formula line hold and are not sharp; no OEIS conjecture exists. Report 212's spectral-transfer heuristic is an open question. The independent check recomputed every number through $k = 256$ and completed one OEIS quotation. A reciprocal note in `a064856-stirling-catalan-transforms` records the answer to its question on $M_{2k}$. Part VI (batch 138, Sections 57–64) extends Part I's all-orders theorem uniformly to compact mean degree $c$ in $G(N, c/N)$ ($c = 1$ is Part I), evaluates the logarithmic correction, gives sharp and optimal Poisson laws for the deficit and a critical growing-degree factor $\exp(c^2 w^2/k)$; it answers the host's mean-degree question and its suborder question in part. The write derived from it that Part V's weighted lower bound is asymptotic to the moment for each fixed $c$; the independent check found nothing wrong. Unrefereed; not formalized.

### 241. Labeled fat trees (OEIS A055779, A295623)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a055779-labeled-fat-trees/article.pdf

from batch 112: Report203.zip (bundle Report 203)

For the $M$-weighted labeled fat trees $A_n(M)$ ($A_n(1)$ is A055779), proves $A_n(M) = Cq^n n^{n-2}$ times an expansion in $1/n$ to every fixed order with rational $c_j(r)$, uniformly over real $M \geq 1$ (Zaslavsky's exact formula re-proved), with $c_1$, $c_2$, $d_2$ explicit, a fixed-$M$ inverse with centre $t = Y/W(qY)$ and two ceilings, and an elementary Poisson block-deficit appendix. The leading amplitude is implicit in Kotěšovec's A162695 formula; no OEIS conjecture exists. The write corrected two rounded decimals of the source and recorded two false readings of OEIS letters; the independent check confirmed every number by brute force and extrapolation and corrected the write's own rounding of Kotěšovec's $p$. The block-count crossover is motivation without proof, an open question. Unrefereed; not formalized.

### 242. Uniform rooted trees and binary partitions (OEIS A003238, A003318)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a003238-uniform-trees-binary-partitions/article.pdf
from batch 112: Report215-reproducibility.zip (bundle Report 215)

A computer-assisted proof ($2^{24}$ exact coefficients, a cut positive operator, 256-bit directed intervals and a positive Abelian transfer) that neither A003238$(n+1)$/A018819$(n)$ nor A003318$(n+1)$/A000123$(n)$ converges, the profile oscillation exceeding 0.00001839189788: the second ratio is exactly the one whose convergence Erdős and Loxton (1979, p. 328) left open, so their question has a negative answer. Also a holomorphic dyadic profile, expansions to every fixed inverse-logarithmic order and two-ceiling inverses of relative width. The write refutes with proof Cloitre's A003238 conjecture $\log a(n) \sim c \log^2 n$ with $0.4 < c < 0.5$: $c = 1/(2\log 2) = 0.7213\ldots$. The radial sieve, which neither the intake nor the write reran, was checked by other means at the independent check, which found nothing to correct. Certified profile values, onsets and smaller certificates are open. Unrefereed; not formalized.

### 243. Many-color rooted trees and their symmetries (OEIS A242249, A255517, A242375, A255523)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a242375-many-color-rooted-trees/article.pdf
from batch 112: Report222.zip (bundle Report 222)

Rooted trees with $q$ colours on the non-root vertices (all classes A242249, identity classes A255517): a common complex domain and every fixed order uniformly for integer $q \geq 40$, palette jets, the asymmetry crossover $\exp(-1/(e^2\alpha))$, a Poisson leaf-pair law with signed corrections and the sharp total-variation constant, Aut $\approx (C_2)^J$, diagonal corrections for A242375 and A255523, and palette sizing. The write proves Kotěšovec's A242249 conjecture $d(k) \sim ek$ (2014) and its A255517 analogue (2015), which the source does not name: the growth base is $ek \pm 1/(2e) + O(1/k)$. The independent check confirmed both proofs and every table entry and qualified one transseries instance claim. Labelle (1992) was not read. Unrefereed; not formalized.

### 244. High maximum outdegree in rooted unlabeled trees (OEIS A244407, A244410, A244372)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a244407-high-outdegree-rooted-trees/article.pdf
from batch 112: Report227.zip (bundle Report 227, base, Part I) + Report229.zip (Report 229, Part II)

For $T(N,k)$, the rooted unlabeled trees with $N$ nodes and maximum outdegree exactly $k$ (A244372), Part I proves the plateau $T(k+m+1,k) = f_m$ $(m < k)$ and $T(2k+1,k) = f_k - 1$, so A244407$(n) = f_{n-1}$ and A244410$(n) = f_n - 1$, a uniform relative error $O((N+1)\rho^k)$ with sharp base, the two-hub sector, every fixed order of both sectors, the Rayleigh hub depth and Lambert-$W$ inverses. Part II, its continuation, gives the exact third sector $T(3k+r+1,k) = f_{2k+r} - b_{k+r} + U_r + kV_r$ and the next boundary $-7$ ($-6$ at $k=1$), answering Part I's question. The whole A244372 b-file satisfies every exact identity; Goh and Schmutz (1994) were not compared, so no priority is claimed; the positivity of $U$ is tested only (to $r = 1200$ at the independent check, which found nothing to correct). Unrefereed; not formalized.

### 245. Beyond the leading count of commutative magmas (OEIS A001425)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a001425-commutative-magmas/article.pdf
from batch 112: Report200.zip (bundle Report 200)

For the isomorphism classes $U_n$ of commutative binary operations on $n$ elements: exact fixed-moved-type sectors with convergent rational amplitudes, an explicit bound on everything omitted for every fixed defect $D$ and $n \geq \max(8, 4(D+1))$, the first three sectors explicitly, labeled against unlabeled rare symmetry ($P_n \sim T_2(n)$, $Q_n \sim 2T_2(n)$, the typical group one transposition), and an exact-Gamma two-ceiling threshold enclosure with a Lambert-$W$ initializer. No OEIS conjecture

exists. The independent check recomputed every count by a literal orbit walk and made one OEIS citation precise. The papers of Harrison and of Tamura were not read. Unrefereed; not formalized.

**246. Graphs with a distinguished maximal independent set (OEIS A340021)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a340021-distinguished-maximal-independent-sets/article.pdf
from batch 112: Report206.zip (bundle Report 206)

Proves $U_n = I_n(1 + O(n^2 2^{-3n/8}))$ with $I_n = 2^N b_n/n!$, $b_n$ the expected number of maximal cliques in $G(n, 1/2)$, by a uniform Burnside reduction; every fixed algebraic order through a discrete Poisson carrier; a finite-sector expansion in powers of $1/\log n$; the black-size law with exact switches $t_k = 2^{k+1}W((k+1)/2)$ and logistic windows; and finite Newton inversion with two-ceiling envelopes. The source read the OEIS entry only as a cached snapshot; the write checked all 41 b-file terms. The independent check confirmed every count and table entry and recorded a second preamble package of the write. Unrefereed; not formalized.

**247. Self-powered binomial sums (OEIS A360592, A360479, A360747)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a360592-self-powered-binomial-sums/article.pdf
from batch 113: Report230.zip (bundle Report 230, base, Part I) + Report235.zip (Report 235, Part II)

For $a_p(n) = \sum_k (n - pk)^{pk} \binom{n-pk}{k}$ ($p = 1, 2, 3$: A360592, A360479, A360747), Part I proves every fixed order for every $p$ through an exact residue-conditioned Poisson representation (a critical resummation at $p = 1$, with no odd quarter powers), sharp total-variation constants for the inactive count when $p \geq 2$, Newton and Lagrange inverses and a two-ceiling envelope. Part II, its $p = 1$ continuation, answers Part I's question on the critical marked law within the binomial family: the globally optimal parity-conditioned binomial, at distance $\sim (5/8)\log 2\,\sqrt{2/\pi}\,\mu^{3/2}/n^2$. The write refutes with proof the first corrections of the three OEIS refinements (an extra $1/(8c)$ at $p = 1, 2$; a nonzero $n^{-1/4}$ term at $p = 3$, whose true value is 0); their leading equivalents are right. The independent check confirmed the refutation to $n = 10^{24}$ from the live formulas and fixed one duplicate PDF destination. Unrefereed; not formalized.

**248. First maximum frequencies in ordered set partitions (OEIS A386374, A386375)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a386374-first-block-maximum/article.pdf
from batch 113: Report208-reproducibility.zip (bundle Report 208)

For ordered set partitions whose first block is tied for largest ($a_n$, A386374) or strictly largest ($u_n$, A386375): $|a_n/n! - \sum_m c_m r_m^{-n}| \leq 11$ by subtracting each positive pole, every fixed order as a moving-window root-free finite sum, the two envelope scales $\limsup nP_n \log\log n/\log n = 2/e$ and $\liminf nP_n/\log\log n = 2\log 2$ with the phase minimum, $u_n/a_n \sim \log 2\,\log\log n/\log n$ phase-uniformly, and a finite-step Newton inverse with two-ceiling enclosures. Gourdon's 1996 thesis is the credited prior for the moving poles and the scale; Bender and Gao's paper was not read in full. The independent check confirmed every number and made two finite ranges precise. Unrefereed; not formalized.

**249. Exact pole sectors and integer recovery for ordered tuple relations (OEIS A173217, A301466, A301468)**
generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a173217-ordered-tuple-relations/article.pdf
from batch 113: Report220-reproducibility.zip (bundle Report 220)

For $H_d(n)$, the sets of $n$ distinct ordered $d$-tuples on an active ordered vertex set ($d = 2, 3, 4$: A173217, A301466, A301468): an exact-amplitude pole decomposition, a global tail bound after the signed-Stirling transform of the Fubini poles, integer recovery from $O(n^{1-1/d})$ exact poles with error below $13/(18n^{d-1})$ and a certified rational-interval algorithm, all fixed orders of the collision kernel on compact complex sets, and two-ceiling inverses. Its Appendix A found that the proof of the transseries volume's `q2:thm:fubini` drops the constant 1/4 (the theorem survives for

$n \geq 1$) and that `q2:rem:weighted` needs the maximum, not the minimum, of $\rho_x$ and reverses the large-weight direction; the volume was repaired separately (`d92db8d06`). The independent check found the same missing constant in `q2:thm:weighted` (repaired in `08bc7a5c8`), confirmed every number, and corrected a statement on A000670's conjectures and a line count. It does not resolve the two-transform question of `a260700-parabolic-double-cosets`, which gained a reciprocal note. Unrefereed; not formalized.

### 250. Fixed-couple ménage rows (OEIS A332709)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a332709-fixed-couple-menage/article.pdf
from batch 113: Report221.zip (bundle Report 221)

For the ménage permutations with $\pi(1) = k$, counted by $T(n,k)$: $T(n,k) - T(n,k-1) =$ A127548$(n-2k+4)$ on the natural domain, and row unimodality and discrete concavity with all equality cases, proving both conjectures displayed in A332709; an exact mean decomposition and a position-uniform all-orders row expansion through $n^{-10}$; the exact row total variation $2(1/(n-2) - A(n,1)/U_n)$; fixed-column inverse models and a two-candidate threshold enclosure. The write shows that the A258667 conjecture holds as a leading equivalence (what the linked Lean file proves) but fails as an asymptotic expansion, at order $n^{-4}$; the independent check confirmed this and settled the same conjecture in A258664–A258666 and A258673 the same way. The full texts of Kagey (2024) and of Gu and Zhao were not read. Unrefereed; not formalized.

### 251. All fixed order asymptotics for restricted two-covers and line graphs (OEIS A060053, A014500)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a060053-two-covers-line-graphs/article.pdf
from batch 113: Report224.zip (bundle Report 224, "reconstructed edition 2")

For $H_n = e^{-1}\sum_m (M(m))_n/m!$, the restricted two-covers $V_n$ (A060053) and $U_n$ (A014500) and the labelled line graphs $L_n$: an exact-saddle expansion to every fixed order with a pure power remainder, transfer through signed polynomial multipliers, Lambert coefficients to every fixed order with explicit first and second orders, $L_n/U_n = 1 - r^6/(48n^3) + \cdots$, and model inverses kept apart from certified thresholds. The line-graph multiplier rests on Cameron, Prellberg and Stark's imported exception classification. The write shows that A132219, "Number of line graphs on $n$ labeled nodes", cannot count them: its $a(4) = 66 > 2^6$, the true value being 60 (it follows an obsolete arXiv multiplier); the independent check confirmed the corrected counts to $n = 7$ without that classification and added two precisions. Unrefereed; not formalized.

### 252. Record sum diagonal expansions and integer inverses (OEIS A368246)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a368246-cycle-minima-sums/article.pdf
from batch 113: Report234.zip (bundle Report 234)

For the permutations whose cycle minima sum to $n$ ($a_n$, A368246) and $b_n = a_n/(n-1)!$: for every fixed order an expansion with constant $e^{-\gamma}$, no $n^{-1}$ term and every algebraic correction periodic with mean zero, via $T(se^{-s}) = s$; explicit terms through $n^{-3}$ with a Gamma amplitude; delayed singularities at $\pm i$ and the natural boundary; and a two-ceiling inverse with finite Newton centres. This proves Kotěšovec's posted conjecture $c = e^{-\gamma}$ as stated, as Giuliano, Szewczak and Weber's Theorem 2.1 (2013) already gives. The independent check confirmed the proof both ways, corrected the archive's file count, and qualified one "proved" (the constant-shift observation is derived, not proved, beyond $d = 1$). Unrefereed; not formalized.

### 253. All fixed order asymptotics for total Kostka sums (OEIS A104779, A321652, A068313)

generating-functions-and-asymptotics/oeis-sequence-asymptotics/
a104779-total-kostka-sums/article.pdf
from batch 113: Report237.zip (bundle Report 237)

For the total $a_n$ of the Kostka matrix (A104779, of which A178718 is a duplicate): $a_n = C\sum_{s\leq S} H_s I_{n-s} + O(I_n n^{-(S+1)/2})$ with $C = \prod_j (1 - 1/j!)^{-1}$ and $I_n$ the involution numbers, 40-

digit interval certificates of the constants, the involution saddle at every fixed order, a Lambert-$W$ and Newton inverse with two-ceiling thresholds, the nonunit margin profile in total variation, A321652 and A068313 on the scale $C^2 n!$, and an exact dominant/alternating sector split. Schwob's identities are imported (arXiv v1). Not a Part of `a138178-symmetric-packed-matrices`, whose transform and scale differ; reciprocal notes went there and to `a260700-parabolic-double-cosets`. The independent check confirmed every constant and count and traced the residual alternation of the sector note to a third exponentially small effect. Unrefereed; not formalized.

**254. All order asymptotics for hypergraphs without singleton intersections (OEIS A323297, A323296)**

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a323297-singleton-free-hypergraphs/article.pdf`
from `batch 113: Report240.zip (bundle Report 240)`

For the labelled 3-uniform hypergraphs without singleton edge intersections ($A_n$, A323297; $B_n$, A323296), from Howroyd's component decomposition: an explicit finite rule for every fixed saddle correction with relative remainder $O((r/n)^{J+1})$, a reversed logarithmic inverse through three corrections that is an exact formal instance of the transseries volume's Lambert template, a total-variation distance $\sim (10/3)\phi(1)r^3/n$ for the exceptional components, deletion by saddle shift, two independent fluctuation scales for the component count and the edge defect, and transfer to $B_n$. The independent check confirmed every count, expansion, reversion and table entry and made one OEIS comparison precise. Unrefereed; not formalized.

**255. Distinct distances in a cubic lattice (OEIS A275672)**

`generating-functions-and-asymptotics/oeis-sequence-asymptotics/a275672-cubic-distances/article.tex`
from `batch 138: A275672_Research.zip (arrival 2c7f4fd68)`

OEIS A275672 (Vladimir Reshetnikov's entry of 2016) is the largest number of points of $\{0,\dots,n-1\}^3$ whose pairwise Euclidean distances are all different. Exhaustive diameter-case C++ searches with verified witnesses give the new terms $a(7) = 10$ and $a(8) = 12$ (computation records, not formal unsatisfiability certificates); verified constructions for every $n \le 30$ give $a(9) \ge 13$ and $a(10) \ge 14$. A rational Gaussian certificate proves $\limsup a(n)/n \le (6\kappa)^{-1/2} = 1.84889\ldots$; also an elementary lower bound, the Lefmann–Thiele lower bound and structural properties. The growth exponent and strict monotonicity are not determined. Placed without a write (a single manuscript, by Vladimir's direction of 9 October 2026): the delivered PDF was not staged, and the delivery README, which serves as the report README, still names `a275672.tex` and the PDF. Nothing was submitted to the OEIS. Unrefereed; not formalized.

## 12.3 Asymptotics, records and discrepancy

**256. The Exponential Scale of Gregory Sign Thresholds**

`generating-functions-and-asymptotics/gregory-coefficient-sign-thresholds/gregory_sign_threshold.pdf`
from `gregory_sign_threshold_research.zip`

For $(x/\log(1+x))^m = \sum_n G_n^{(m)} x^n$ and $\mathrm{Gr}(m)$ the least $N$ beyond which $(-1)^{n-1}G_n^{(m)} > 0$, proves $\log \mathrm{Gr}(m) = m + 1 - \gamma - (\zeta(2)+\zeta(3))/m + C_2/m^2 + O(m^{-3})$ with $C_2$ explicit, plus a finite algorithm for every coefficient of the complete expansion; in particular $\mathrm{Gr}(m)/e^m \to e^{1-\gamma}$ and $\mathrm{Gr}(m)/3^m \to 0$. This replaces Conjecture 6.1 of Xu and Zhao (arXiv:2609.11072v1), whose printed lower bound already conflicts with its own reported $\mathrm{Gr}(5) = 113$; that value is independently certified here.

**257. Exact Records in the Square-Root-of-Three Walk**

`generating-functions-and-asymptotics/sqrt-three-walk-records/article.pdf`
from `sqrt3_record_recurrence.zip`

Proves the four-phase record recurrence for $S(N) = \sum_{n\le N}(-1)^{\lfloor n\sqrt{3}\rfloor}$ in the corrected form on page 7 of Bruin and Fokkink, *On the records and zeros of a deterministic random walk*

(arXiv:2503.11734v2), where it is experimental and attributed to Arias de Reyna and van de Lune. Includes minimality of every claimed first passage, a rational generating function, closed forms, sharp logarithmic extrema, and an extension to every even integer trace $D \geq 4$.

**258. Sharp discrepancy for two families of binary substitutions**
generating-functions-and-asymptotics/binary-substitution-discrepancy/article.pdf
from oeis_discrepancy_counterexamples.zip; batch 73:
OEIS_Quadratic_Morphic_Words_Research_Package.zip; batch 114: binary_morphic_fluctuations.zip

For the fixed word of $0 \mapsto 1$, $1 \mapsto 101010$, the position sequences A284365 and A284366 both carry a conjectured upper position error below 2, and both conjectures are false: the first counterexamples are at ranks 2,977,771 and 8,933,313, with common error $(2977771\sqrt{21} - 13645857)/2 \approx 2.00487$ and sharp common upper bound $(6 + \sqrt{21})/5 \approx 2.11652$. Proves complete discrepancy limit intervals for every substitution $0 \mapsto 1$, $1 \mapsto (10)^m$, with infinite counterexample families and exact recurrences for their indices. Part II, added in batch 73, treats the disjoint family $0 \mapsto 1$, $1 \mapsto 10^a1^b$: the prefix discrepancy is bounded exactly when $a \leq b + 1$, with sharp, unattained position-error envelopes from a four-state Bellman system, and sharp bounds for A284368–A284371. The density question of Parts I and II stays open. Part III (batch 114) answers Part II's question on the unbounded regimes, with $X(N) = \#0 - q\,\#1$ over the first $N$ letters: at $a = b + 2$, $\limsup X(N)/\log N = -\liminf X(N)/\log N = (a-1)/(2\log a)$, every value in between is a limit point, and Part II's cycle $P_k$ is the first prefix of weight $k(a-1)$; for $a \geq b + 3$ the cluster set of $X(N)/N^\alpha$, $\alpha = \log q/\log\lambda$, is a closed-form interval $[-\mathcal{C}_{a,b}, \mathcal{C}_{a,b}]$ with a continuous anti-periodic graph-directed phase profile, and at $(a, b) = (8, 1)$ the endpoints are $\pm\sqrt{10}$ and $X(N)^2 + 5X(N) \leq 10N$ for every prefix. For every word with $a$ zeros and $a - 1$ ones beginning with 1 it gives an exact, coefficientwise positive convolution formula for complete-block prefix histograms, exact moments, quantitative Gaussian and local limit laws, a large-deviation principle with explicit pressure, the sharp variance bounds $(a-1)/a^2 \leq v \leq (a^2-1)/12$, and the zero-height profiles, counted by $(a-1)2^{a-2}$, A001787$(a-1)$. The existence of the Gaussian limit (Paquette–Son) and the invariant-section identity (Solomyak, Rajabzadeh–Safaee) are credited; the first local (Edgeworth) correction, only sketched, is an open question. The independent check of the Part III write found eleven places of Part II where a batch-73 rename had left Part I's constant $C$ in place of $\widehat{C}$, so that the bound (13.9) for A284369 read $E_0 < 2.1547$ although $E_0$ reaches 3.6437; all eleven now read $\widehat{C}$, giving the sharp $2 + \sqrt{3}$ of Theorem 12.1, with a dated note. Unrefereed; not formalized.

**259. Exact eventual maximizers of Rudin–Shapiro autocorrelations**
generating-functions-and-asymptotics/
rudin-shapiro-autocorrelation-maximizers/article.pdf
from rudin_shapiro_maximizers.zip

For the aperiodic autocorrelation $C_m(k)$ of the level-$m$ Rudin–Shapiro sequence, classifies the maximizing shift exactly for all large $m$: $k_m^* = (2^{m+1} + (-1)^m)/3$, with $A_m = b_{m-2}$, settling the discrete uniqueness-and-location conjecture of Tarnu (J. Approx. Theory 287 (2023) 105866), repeated as Conjecture 1.5 of Choi–Tarnu (2025), including the limit $k_m^*/2^m \to 2/3$. The certificate is a rational polytope argument with exact arithmetic.

**260. Apéry arrays and conjectural zeta accelerations**
generating-functions-and-asymptotics/apery-array-zeta-accelerations/article.pdf
from Apery_OEIS_Conjectures_Proofs.zip, apery_oeis_conjectures.zip

Proves the entire shifted-diagonal $\zeta(3)$ family recorded as conjectural in OEIS A143007, and both the super- and subdiagonal $\zeta(2)$ families of A108625, at every nonnegative offset: for all $k \geq 0$, $\zeta(3) = H_k^{(3)} + \sum_{n\geq 1}(2n+k)(3n^2+3nk+k^2)/(n^3(n+k)^3 T_{n-1,n+k-1}T_{n,n+k})$. The main diagonals are the classical Apéry sequences A005259 and A005258. Proofs are finite telescoping identities with rational lattice potentials and explicit limit estimates; no computer test or unproved conjecture enters. Adds rational error bounds, numerator and denominator recurrences, a three-term recurrence along every shifted diagonal, and sharp fixed-offset asymptotic constants. Merged from two reports whose boxed identities were character-identical up to macro names.

**261. Sharp logarithmic remainders for Ford's renewal recurrence**
generating-functions-and-asymptotics/ford-renewal-logarithmic-remainder/article.tex
from batch 138: ford_renewal_refinement.zip (arrival b28d0850b)

For $a_j = (j+1)\log(j+1) - j\log j - 1$, $g_0 = 1$ and $g_n = \sum_{j=1}^{n} a_j g_{n-j}$, with $\sum_j a_j \rho^j = 1$ and $\gamma = (\sum_j j a_j \rho^j)^{-1}$, proves $E_n = \gamma\rho^{-n} - g_n \sim 1/(n^2 \log^2 n)$ with every fixed inverse-logarithmic order, coefficients $c_k = (k+1)!\,[u^k](1+u)e^{\gamma_E u}/\Gamma(1-u)$, and an exact positive measure representing $E_n$ for $n \geq 1$ (complete monotonicity, positive definite shifted Hankel matrices); explicit bounds from $n = 4096$, two sum rules, a $1/n$-sector expansion, product and inverse laws, and a theorem for a broader class of moment-increment renewal recurrences. It refines the bounded error of Ford's Lemma 3.7, which is credited; $\rho$ is OEIS A246746. Placed without a write (a single manuscript, by Vladimir's direction of 9 October 2026): the delivery README serves as the report README, and the delivered PDF was not staged. Unrefereed; not formalized.

## 13 The Jacobian conjecture and Keller maps

Four reports continuing the formal project `Algebra/JacobianConjecture`, which verifies in Lean and Rocq Alpöge's three-variable counterexample and a degree-six stable representative in five variables (its stabilization to all higher dimensions in Lean only); its further degree reductions are SymPy certificates. None of the four reports' own theorems is formalized; each README names the project declarations it builds on. Three study the project's three-variable map, one of them its iteration; the fourth studies a different map, Gao's five-dimensional $F_6$.

**262. The fiber geometry of Gao's Keller map $F_6$**
jacobian-conjecture/gao-f6-fiber-geometry/article.pdf
from ProveIt_Six_Sheets_Research.zip, ProveIt_Keller_Fibers_Research.zip

For Gao's five-dimensional six-sheeted Keller map $F_6$ (arXiv:2608.00222v1, §4.5.1), classifies every geometric fiber: the cardinalities are exactly $0, 1, 2, 3, 4, 6$, all seven root-multiplicity patterns occur, and off the hyperplane $c = 0$ the inverse points are the simple roots of an explicit sextic. The omitted set is a smooth closed surface isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$, of codimension three, with the single effective omitted parameter $(21/32, 5/32, 25/128)$. Gives an explicit finite normalization of rank six, the two-component nonproperness locus, the discriminant's normalization and local analytic types, global monodromy $S_6$ with two arithmetic certificates, no proper intermediate field, real fiber cardinalities $\{0, 1, 2, 3, 4\}$, and an obstruction to equivalence with a stabilization of a map in at most three variables. The map is Gao's; the fiber theory is the reports'. Merged from two manuscripts proving the same core in identical normalizations; the second one's different proofs are kept as marked second routes. Not to be confused with the project's formally verified dimension-five stable map.

**263. Arithmetic local–global dichotomies for the three-variable Keller map**
jacobian-conjecture/arithmetic-local-global-fibers/article.pdf
from ProveIt_Arithmetic_Local_Global.zip; batch 58: ProveIt_Dense_Integral_Hasse.zip; batch 64: ProveIt_Beyond_Split_Fibers.zip

For the project's three-variable Keller map, gives an explicit infinite family of integral targets with a preimage in every $\mathbb{Z}_p^3$ but none in $\mathbb{Z}^3$ (each fiber is three rational points, so these are not rational Hasse failures); classifies the split integer-root plane $C = 2$ by a gcd condition, with locally soluble parameter density $27/(4\pi^2)$ surviving the removal of $S$-integral fibers; proves the exact integral 2-adic fiber law $(21, 7, 3, 1)/32$ for fiber sizes $0, 1, 2, 3$, determined modulo 8; and contrasts it with a rational Hasse principle over every number field. The complex stratification is credited to Gao and the finite-field histograms to an external note. An intake remark records an integral triple collision over the project's own collision value. Not a new disproof of the Jacobian conjecture. Part II, added in batch 58, answers Part I's density question: such failures are Zariski dense in every plane $C = c \neq 0$, hence in $\mathbb{A}^3$, and none lies on $C = 0$. It gives a complete local criterion for completely split fibers, finds finitely many integral split fibers on each plane, and proves $\#\mathcal{H}_c^{\mathrm{sp}}(T) \sim \kappa(c) T^{2/3}$

for the completely split failures in square boxes $|A|, |B| \le T$, with $\kappa(c)$ an explicit Gamma factor times an Euler product. Part III, added in batch 64, gives a complete local-integral certificate for every target, nonsplit fibers included, and on $C = 2$ counts the integral Hasse failures in the same boxes: $\frac{27}{2\pi^2} T \log T + O(T)$, almost all of them nonsplit.

**264. All-degree rigidity of a weighted Keller class**
`jacobian-conjecture/weighted-keller-rigidity/article.pdf`
from `ProveIt_Weighted_Keller_Rigidity.zip; batch 44: ProveIt_Affine_Weighted_Keller_Classification.zip`

For polynomial maps of weights $(-1, 1, 2)$ written through $t = xy$ and $v = x^2 z$, classifies the Keller maps in every degree over any field of characteristic zero: those with $p, q$ affine in $v$ and $r$ affine in $(t, v)$ are an explicit tame family and a two-parameter diagonal orbit of the known map, so every noninvertible member has coordinate degrees $(7, 6, 4)$ and sixteen monomials, with no input degree bound. Extends to $r = R(t) + \gamma v$, where the possible maximum degrees of the noninvertible members are exactly 7 and every even integer at least 8, and shows that with $r = 1 + t + v$ the fixed degree-seven coefficient scheme is the double point $\mathbb{Q}[\varepsilon]/(\varepsilon^2)$, with an explicitly obstructed tangent (a torus times it when the two coefficients of $r$ are freed). The classification does not cover unrestricted weighted maps. Sharpens to all degrees the project's degree-seven sparsity search; it does not show degree seven minimal in dimension three. Part II, added in batch 44, removes the restriction on the coefficient of $v$ in $r$: whenever $p, q, r$ are affine in $v$ that coefficient is constant, so the classification covers the whole affine-in-$z$ class; it adds dual-number slopes of every degree and a universal second-order obstruction.

**265. Dynamical degrees of the three-variable Keller map**
`jacobian-conjecture/keller-map-dynamical-degrees/article.pdf`
from `ProveIt_Keller_Dynamics.zip (batch 70)`

Studies the project's three-variable Keller map $F$ under iteration, which neither the project nor the other reports touch. Over any field of characteristic other than three, $\deg F^n$ satisfies $d_{n+2} = 6d_{n+1} + d_n$ with $d_0 = 1$, $d_1 = 7$ $(1, 7, 43, 265, \dots)$, so the first dynamical degree is $3 + \sqrt{10}$; every positive weighted degree of every iterate is explicit, and every coordinate's downward-completed Newton polyhedron has exactly two vertices. Outside characteristic three, composing with a shear $z \mapsto z + y^2 h(xy)$, $h \neq 0$ of degree $m$, gives first dynamical degree $(2m + 7 + \sqrt{4m^2 + 32m + 61})/2$; in characteristic zero these maps are all Keller, noninjective, of generic degree three and in one tame left–right equivalence class, yet their dynamical degrees are unbounded. In characteristic three the second coordinate is $y$ exactly and the degrees follow separate closed formulas (first dynamical degree 4 or $m + 3$); in characteristic two the map is purely inseparable of degree two. A monomial-dominance escape theorem gives a forward-invariant complex escape region, and integral points there have maximal arithmetic degree and are Zariski dense. No Lean or Rocq source is supplied; the formalization route is a plan.

## 14 Galois theory and radicals

Two reports continuing the formal project `Algebra/PolynomialFormulas` (Lean and Rocq root formulas through degree four, Abel–Ruffini, and Lazard's quintic solver); none of its theorems is formalized.

**266. Specialization-safe radical solvers**
`galois-theory-and-radicals/specialization-safe-radical-solvers/article.pdf`
from `ProveIt_Radical_Solvers_Research.zip; batch 52: ProveIt_Optimal_Kummer_Atlases.zip`

Gives an exact cyclotomic-resultant criterion for the characteristics in which equal-cardinality subset sums of the roots of an irreducible prime-degree polynomial can collide, and evaluates it: the bad characteristics are $\{5\}$ for quintic pairs, $\{2, 7\}$ for septic pairs and $\{2, 7, 29\}$ for septic triples. Classifies the collisions completely in characteristic 29 (exactly the cyclic polynomials $(X - \eta)^7 - b$) and in characteristic 2 (translated $Z^7 + aZ^3 + bZ + d$, realizing the nonsolvable

$\mathrm{GL}_3(\mathbb{F}_2)$). Constructs separating subset invariants in every characteristic, reconstructs all roots from one coherent $p$-th radical on each Fourier chart once the intermediate-field invariant data are given (generalizing Lazard's pivot-one scheme for quintics; a coefficient-only solver is left as further work), realizes every Fourier support by rational polynomials with full affine Galois group, and shows the Picard group of the cyclic configuration quotient is $\mathbb{Z}/p$, so no single global Kummer coordinate exists. The Picard result has a written proof only. Part II, added in batch 52, answers Part I's question on the least number of charts: $\mathrm{CH}^*(\mathrm{Conf}_p(\mathbb{A}^d)/C_p) = \mathbb{Z}[t]/(pt, t^{d(p-1)})$, so every cover by regular Kummer charts, even with nonlinear character sections and different characters on different charts, needs exactly $d(p-1)$ charts ($p-1$ for polynomials: four for quintics, six for septics), and the Fourier charts attain it. This too is a written proof, by integral intersection theory; its checker verifies only finite identities.

**267. Finite separating ranges for sextic block resolvents**
`galois-theory-and-radicals/sextic-block-resolvent-separators/article.pdf`
from `ProveIt_Finite_Sextic_Separators.zip` (batch 41); batch 57: `ProveIt_Three_Plus_One_Resolvents.zip`; batch 70: `ProveIt_Zero_Parameter_Sextics.zip`

The project finds parameters separating the sextic pair and triple block-resolvent descriptors by an unbounded `Nat.find` search. This report proves in writing that on the curve $(t, 2t^2)$ one of $t = 1, \ldots, 196$ always separates both families, sharply for arbitrary complex test sets, by orbit counting. The bound is not formalized; formalizing it would make that search bounded, and two further unbounded searches of the project remain. Part II, added in batch 57, shows that deciding radical solvability of an irreducible sextic needs no separating parameter: the sextic is solvable exactly when its triple resolvent has a base-field root at one parameter chosen in advance, or its matching resolvent has one at each of three distinct parameters, because a base-field matching root of a nonsolvable sextic comes from a Galois-invariant pentad, of which there is at most one, and occurs at no more than two parameters. Part III, added in batch 70, sharpens this: one compressed matching test at $t = 0$ together with one triple test decides, by an integer certificate that the five values of a pentad are never all equal and by reciprocal transport; and since a pentad's points span an affine space of dimension at least two, the polynomial of false-positive parameters has degree at most one and does not vanish at zero, so any two distinct matching parameters suffice. It adds a denominator-free Pfaffian of order twenty for the constant triple resolvent and a cubic $15 \times 15$ matrix polynomial whose determinant is the cube of the constant matching resolvent.

## 15 Hilbert's tenth problem

Twelve reports on Diophantine representations of computation that control their witnesses, beside the formal project `Computability/HilbertTenthProblem`, which proves MRDP in both directions in Lean, the universal pair $(58, 4)$ and the single-fold unary exponential representation of Jones and Matiyasevich. The first three (canonical certificates; probabilistic, quantum and continuous computation; liveness beyond halting) continue that project. The other six, added in batches 76, 78 and 79 (groups as substrates, signal machines, stochastic and thermal exactness, quadratic orthant certificates, polynomial witness histories, smooth finalizers), treat further substrates or witness domains and use MRDP at most as an imported theorem. Batch 79 also extended five of the older reports. Batch 80 added Parts V and VI to the signal-machine report and replaced the programs of three batch-79 Parts (canonical certificates Part XIX, quadratic orthant certificates Part VI, signal machines Part IV) by their authors' corrected code editions, whose manuscripts are unchanged. Batch 82 opened `five-particle-binary-automata` and `fixed-universal-polynomials`, the latter on the programme's own fixed universal polynomial, and extended the signal-machine report (Parts VI–VIII) and the group-substrate report (Part V); batch 83 opened `periodic-turmite-first-revisits` and added Part XX to the canonical certificates and Part V to the fixed universal polynomials; batch 88 added Parts II–IV to the turmite report (a literal periodic Langton ant and a Diophantine certificate for the fixed machine

it simulates) and Parts VI–VIII to the fixed universal polynomials (the candidates below the 84-operation record). Batch 91 brought the pipeline's Reports 49–71 (Report 68 was never delivered): Part XXI of the canonical certificates (fixed-arity sandpile polynomials), Part VI of the group-substrate report (a chronological matrix certificate), Parts IX–XIII of the signal-machine report and Part VII of the five-particle report (smaller recognition radii); its two OEIS manuscripts became `a196460-clipping-tables`. The last three entries below are batch 82's and 83's new reports. None of the reports' own theorems is formalized; each README names the project declarations it builds on or relates to.

**268. Canonical Diophantine certificates**

`hilbert-tenth-problem/canonical-diophantine-certificates/article.pdf`

from `Diophantine_Causal_Computation.zip, canonical_diophantine_certificates.zip, ProveIt_Linear_Size_Diophantine_Certificates.zip, Witness_Faithful_Diophantine_Compilation.zip, canonical_trace_polytopes.zip, diophantine_counter_programs.zip (batch 60); batch 61: Causal_Diophantine_Queue_Compilation.zip; batch 62: Names_to_Numbers_Diophantine_Heaps.zip, order_free_diophantine.zip, ProveIt_Priority_Pumping_Diophantine.zip, one_shared_fallback_research.zip; batch 63: Diophantine_Certificates_Without_Histories.zip; batch 78: Exact_Wiring_Diophantine_Report.zip, Topology_Is_Not_Free_Interaction_Nets.zip, no_ghost_wires.zip, sandpile_diophantine_research.zip; batch 79: Positive_Spectrum_Diophantine.zip, Spectral_Guards_Without_Time_Expansion.zip, no_borrowed_firings.zip, Compressed_Queue_Diophantine_Research.zip, Eager_Tree_Calculus_Research_Package.zip; batch 80 (corrected code edition of the last): Eager_Tree_Calculus_Research_Package_corrected.zip; batch 83: Literal_Periodic_Sandpiles_and_Diophantine_Certificates_Package.zip, Real_Exactness_of_Binary_Sandpile_Certificates_Package.zip; batch 91: A_Fixed_Arity_Integer_Certificate_for_Binary_Sandpile_Stabilization_Package.zip, Finite_Legal_Binary_Target_Firing_in_Periodic_Sandpiles_Package.zip, Repeated_Legal_Target_Firing_in_Periodic_Sandpiles_Package.zip, Unrestricted_Finite_Global_Sandpile_Stabilization_Package.zip (Part XXI)`

Diophantine representations that control their witnesses, not only their existence. For Petri nets and interval-guarded translations, a convex quadratic integer polynomial whose natural zeros are in bijection with the commutation classes of enabled executions of length $T$ (its real zero set is a rational polytope with exactly those lattice points), with quartic second routes, a Cartier–Foata variant for bounded causal height, exact commutation tests, #P-completeness of counting and an exponential lower bound for normal-form automata. Single-fold quartics for multisets and nested repetitions of counter actions independent of the repetition counts, for "$p(i) \geq 0$ for all $i \leq T$" with $O((d+1)^3)$ witnesses independent of $T$, for memory logs with $24L$ witnesses, and for FRACTRAN and scheduled SKI runs. For cyclic-tag and deletion-tag systems and fixed-read FIFO networks, certificates with one coordinate per consumed symbol and one guard slack, with no intermediate queue quantified; faithful word memory with polynomial insertion at both ends needs two natural coordinates. Proves that a single-fold (finite-fold) universal halting polynomial exists exactly when every enumerable set has such a representation, an open problem not solved here. Parts X–XIII, added in batch 62 from four manuscripts: quartic certificates for dynamic heaps and local graph programs, with birth-order normalization of fresh names, two-stack universality and a proof that perpetual memory safety has no existential representation; for polynomials nonnegative on the real orthant, degrees two and three represent exactly the semilinear sets and degree four every enumerable set, with single-fold quadratics for semilinear sets and order-free self-assembly compilers; for first-enabled-rule multiset and FRACTRAN programs the sharp pumping threshold $2H + 1$, a uniform single-fold quartic and c.e.-completeness of eventual periodicity; and conservative priority reaction networks with one shared fallback, with single-fold trace and minimum-fuel quartics and a least-reservoir relation whose single-fold (finite-fold) representability is again equivalent to that of every c.e. set. Part XIV (batch 63): one quartic per run-length routing topology with exactly $4n + 4m + s$ witnesses and a unique root iff the run terminates, and a fixed polynomial-time rank function whose single-fold (finite-fold) graph is equivalent to the corresponding principle for all c.e. sets. Part XV, added in batch 78 from three manuscripts: exact port wiring for Lafont's interaction combinators. If $M \neq N$ are perfect matchings of $2n$

ports and $Q$ is a uniformly random one, closing by $Q$ gives different parities of cyclic wires with probability between $2/5$ and $2/3$, both bounds sharp and the equality cases classified; bounded six-rule compilers with $O(C_0 + F + T)$ natural coordinates, degree four and one witness per legal schedule (scheduled histories, not commutation classes), built on a minimal ten-state boundary calculus, a ten-pattern annihilation kernel and memory certificates that are ledger variants of Part V's $24L$ (whose product lemma two of the manuscripts re-prove; no novelty is claimed for it); and a canonical permutation-orbit certificate with an implemented fixed-schedule compiler. This answers Part X's question on an interaction-combinator frontend for bounded scheduled histories; quotienting by independent schedules stays open. Part XVI (batch 78): for abelian sandpiles on a finite undirected multigraph with a sink, an exact support criterion for the odometer and cubic integer polynomials with exactly one natural zero for every input, carrying the odometer, the stable endpoint and canonical burning ranks instead of a toppling history, with $10n + 6E$ coordinates in the compact form (a directed counterexample fixes the scope); on $\mathbb{Z}^d$ with a stable periodic background, an exterior collar that certifies global halting exactly, with a unique canonical radius that by Cairns's imported universality theorem has no computable bound; and the exact minimal eventual period of stabilization along an input ray. It answers Part XIV's question on infinite-background abelian computation for undirected graphs, and its single-fold equivalence for Cairns's relation is an instance of Part IX's. Batch 79 added a second route inside Part XVI, written independently: $12n + 16E$ quadratic residuals in $11n + 12E$ natural witnesses with exactly one natural zero, carrying the odometer, whose sum of squares is a quartic with an integral rejection gap, and on $\mathbb{Z}^3$ one fixed radius-one normal form with 47 finite-deviation fields whose field solution exists, uniquely, exactly when the input stabilizes (neither ledger improves the other). Part XVII (batch 79, from two manuscripts proving one theorem by two routes): complete discrete sign charts of positive-base exponential polynomials along a loop, verified at run endpoints only (a row of order $m$ has at most $2m - 1$ sign runs, sharp), with one witness per input in arithmetic augmented by power atoms $y = b^n$ ($O(D^3)$ witnesses for fixed bases, $O(D^4)$ with the bases as inputs) or, for an externally fixed bit bound, in an ordinary quartic; the complete sign-chart graph of $2^n - y$ has a single-fold (finite-fold) representation exactly when every c.e. set has one, which sharpens a proposition of Part IX; and $\Re((3 + 4i)^n)$ changes sign with positive density on every residue class. Part XVIII (batch 79): for reliable FIFO actions along a straight-line trace grammar, a quartic with exactly $6g + 3c + 1$ natural witnesses and one zero exactly when the trace connects the supplied words, quantifying no intermediate queue; exact macro repetition counts, infinite repeatability as a word equation, a Fine–Wilf pumping threshold and unique-witness quartics for supplied infinite loops; its re-proofs of Parts I and VI in one coordinate are printed as pointers. Part XIX (batch 79): for Barry Jay's Tree Calculus under eager evaluation and every externally fixed bound $N$ on distinct calls, a quartic with $3N^2 + 19N$ natural witnesses whose zeros project to the terminating applications (memoized proof DAGs, made well-founded by additive call costs), a canonical refinement with exactly one witness, a strict bracket-abstraction compiler from weak call-by-value $\lambda$-calculus, and a literal universal tree of 175 constructor-DAG nodes with an r.e.-complete halting domain, whose code of about $4.5 \cdot 10^{10}$ bits is charged by 175 quadratic equations and never materialized. Since batch 80 Part XIX ships its authors' corrected code edition, whose evaluator rejects non-exact or negative codes before any state change, the repair proposed by the Hilbert's-tenth research tree's review; its manuscript is unchanged. Part XX (batch 83, from two manuscripts, the pipeline's Reports 35 and 36) delivers the universal periodic input loader that Part XVI asks for: one explicit stable periodic background on $\mathbb{Z}^3$ and a finite loader for every finite two-sided binary tape of the Neary–Woods machine $U_{15,2}$, with finite total activity exactly when the machine halts, and finite binary prism certificates whose natural zeros are the canonical odometers (a re-derivation of Part XVI's certificate, marked as such); replacing a product $fg$ by $f(g + k + c)$ makes every nonnegative real zero natural, with the same summand count, degree, support and coefficient height, by an argument different from the real gates of `quadratic-orthant-certificates`, which it credits. The research tree's review is printed with its findings, among them a real-algebraic limit on fixed-arity compression.

Dated notes of batch 88 record that Parts II–IV of `periodic-turmite-first-revisits` build the turmite analogue of Part XX for the same machine (no shared theorem, and no ordinary-input loader), and that the candidates below the 84-operation record are tabulated with their status in `fixed-universal-polynomials`, whose Parts VII–VIII refute the first-index deletions and the square/product 82 candidate (one 83-operation candidate there is still unresolved). Part XXI (batch 91, the pipeline's Reports 50 and 52–54) answers Part XX's question on an integer-only packed interface: fixed-arity positive-integer polynomials of exact degree 18 on one raw tile-and-patch input, with the input encoding paid and no real-witness claim, for binary global stabilization (2,566 witnesses), finite legal binary target firing (3,308), repeated target firing (3,865) and unrestricted global stabilization (3,262). The zero sets of the last two are r.e.-complete conditionally, through Cairns's universality theorem and through Part XX's loader respectively, by outside computable reductions, not a paid compiler. Their witness sets are infinite, so the fixed-arity compression questions are touched, not answered, and Part XX's sentence that nothing there is a fixed-arity universal polynomial is scoped to Part XX. Six printed passages of Cairns's paper that two manuscripts call misprints are checked, and each printed form is shown to fail; whether the corrected constructions prove Cairns's theorems stays open. The power macro is a lemma of `periodic-turmite-first-revisits` and the containment step the Jones–Matiyasevich masking formalized in the Hilbert's-tenth project, both credited; Part VI of `group-theoretic-substrates` shares both, and an earlier sentence here that no other theorem is shared is retained, with its refutation in that report's Remark 1.1. Gate counts treat integer constants as free and are not comparable with Part XX's; the 84-operation record is unchanged. Merged from twenty-seven manuscripts; not a Lean formalization.

**269. Diophantine laws of probabilistic, quantum and continuous computation**
`hilbert-tenth-problem/probabilistic-quantum-and-continuous-computation/article.pdf`
from `ProveIt_Diophantine_Quantum_Normalization.zip, quantum_diophantine_research.zip, ProveIt_Diophantine_Barriers.zip (batch 60); batch 62: Quartic_Probability_Landscapes.zip, ProveIt_Reversible_Equilibria.zip, Diophantine_Certificates_for_Coherent_Computation.zip, proveit_rational_rotations_research.zip, dense_rotations_research.zip, strict_contractions_research.zip, Diophantine_Contracting_2Adic.zip, Precision_Is_Memory_Diophantine_Neural_Computation.zip; batch 64: Rational_Tubes_Integer_Zeros.zip; batch 76: Infinite_Quantum_Runs_Diophantine_Certificates.zip`

One fixed quartic represents the output law of every effective probabilistic or quantum program as masses of projected roots; exact normalization budgets with a unique-witness quartic certificate; normalized two-outcome probabilities are exactly the weakly computable reals, and conditional comparisons are $\Sigma_2^0$-complete. Deciding whether a seven-outcome rational quantum instrument in dimension four has an impossible outcome word is undecidable (via Neary's matrix-mortality theorem), with bijective quartics for words of fixed length. One rational piecewise-affine map of the cube realizes every disjoint pair of enumerable sets as exact acceptance and robust rejection, with quartic safety certificates for a computable critical-noise radius. Parts V–VII, added in batch 62 from eight manuscripts: Boolean circuits compile to coercive integer quartics of height at most 4 whose real zeros are Boolean, with uniform Hessian and rounding constants, so that halting probabilities of fair-coin programs are root counts, component counts or Euler characteristics; birth–death chains with defect environments have the sharp spectral gap $(3-2\sqrt{2})/4$ and a root mass that is rational exactly for eventually periodic environments; unique-witness quartics for coherent contraction trees over $\mathbb{Z}[e^{i\pi/4}]$, and dense rational SO(4) gate groups from the rotations $(3 \pm 4i)/5$, $(3 \pm 4j)/5$ with c.e.-complete state transfer or c.e.-complete membership; a fixed 1/2-Lipschitz rational piecewise-linear map of $\mathbb{R}^4$ and a 1/2-contraction of $\mathbb{Z}_2$ with $\Sigma_1^0$-complete exact half-space reachability and exact extinction; and unique-witness quadratics for rational clipped neural networks, whose tied four-layer synthesis is equivalent to Hilbert's tenth problem over $\mathbb{Q}$. Part VIII (batch 64): complete integer tube certificates, sums of squares of quadratics with at most one witness, for reaching open rational targets under rational polynomial flows, and global mass-action normal forms that carry them to chemical reachability. Part IX, added in batch 76 from one manuscript: for time-homogeneous rational quantum loops (completely positive,

trace-nonincreasing), the group inverse and Cesàro projection as the unique solution of four linear equations with no rank choice, all stopping moments by one finite recurrence certified by a Faddeev–LeVerrier circuit, unique seven-coordinate natural wires extending Part VI's quartic compilation to rational inputs, an exact weak-clock resolvent with an exponential limit law, a geometric clock on the Eisert–Müller–Gogolin instruments giving every device the law $(16/25)(9/25)^n$ while forbidden-record existence stays undecidable (also in dimension four, conditional on Part II), a two-state counterexample to the unrestricted reading of Lardizabal's rationality conjecture, and no uniform existential Diophantine graph for computably scheduled one-dimensional loops. Its three arguments repeated from Parts I, II and V are pointers. The report now points to the sibling reports `stochastic-and-thermal-exactness`, which answers Part VII's structural-subclass question negatively in part (dated notes), and `signal-machine-collision-certificates`, which shares no theorem. A dated note (batch 79) applies the finalizer of `smooth-diophantine-finalizers` to Part V's Boolean-circuit quartics, whose zero sets are singular at every real zero. Merged from thirteen manuscripts; not a Lean formalization.

### 270. Beyond halting: recurrence, progress deadlines and Diophantine liveness

`hilbert-tenth-problem/liveness-beyond-halting/article.pdf`

from `Beyond_Halting_Diophantine_Liveness.zip, Quadratic_Diophantine_Dynamics.zip (batch 62); batch 63: quadratic_infinite_computation.zip; batch 79: clock_spectra_research.zip`

For computable finitely branching systems, existence of a recurrent run with a computable progress deadline is $\Sigma^0_3$-complete and equivalent to a nonempty effectively closed class of accepting schedules, and to an accepting schedule of hyperimmune-free degree; fixed deadlines are $\Pi^0_1$, bounded gaps $\Sigma^0_2$, unrestricted recurrence $\Sigma^1_1$-complete, with strict separations and every c.e. degree as an exact deadline spectrum. An auxiliary-free degree-two polynomial with coefficients at most eight gives the transition relation of counter networks (degree two is minimal); a second, incoming-edge encoding has height six, a sum-of-squares quartic companion and an exact planar realization; and a two-stack quartic certifies finite runs meeting prescribed deadlines. Part VI (batch 79) answers in part the report's question on deadline spectra beyond c.e. cones: clock problems are, up to uniform computable transformations, the upward effectively closed mass problems of Baire space; every $\Delta^0_2$ degree, and every degree of a unique path through a recursive tree (among them $\mathbf{0}^{(r)}$ and $\mathbf{0}^{(\omega)}$), is a least clock degree of an input of a fixed two-stack system; unions of incomparable cones have no least degree, and a least clock degree is hyperarithmetic and contains a self-modulus. Its degree-two two-stack certificate is a second route, at lower degree, to the report's quartic compiler. Merged from four manuscripts; not a Lean formalization.

### 271. Groups as Diophantine substrates

`hilbert-tenth-problem/group-theoretic-substrates/article.pdf`

from `batch 76: arithmetic_van_kampen.zip + three_commutative_phases_research.zip; batch 78: Order_Is_Not_a_Moment.zip + ProveIt_Affine_Matrix_Diophantine_Research.zip; batch 82: Universal_Matrix_Semigroup_and_Diophantine_Certificates_Package.zip; batch 91: Chronological_Diophantine_Matrix_Certificates_Package.zip (Part VI)`

Low-degree certificates for computation in groups, all through the integral Heisenberg group $\mathsf{H}$. Part I: with Sanov's chart (the free group of rank two as the integral points of $x+t+4xt-yz=0$), quartics whose zeros are in bijection with padded factorizations and that accept exactly the words of van Kampen area at most $m$, with $(2s+13)m-4$ witnesses, so that minimal arity tracks the Dehn function; proof-DAG compression, and exactly $k+\ell$ product gates for $[a^{2^k}, b^{2^\ell}]$, optimal in the stated calculus; a height bound rests on an imported Cornulier–Tessera lemma. Part II: three injective commuting phase homomorphisms into $\mathsf{H}^N \times \mathbb{Z}^m$ whose ordered product realizes every integral quadratic system fibre-bijectively, hence, with MRDP, three fixed free abelian subgroups of a Heisenberg power with c.e.-complete product membership, a proposed, unrefereed negative answer to the three-subgroup question of König–Lohrey–Zetzsche and Roman'kov; two-phase decidability, known for integer exponents and credited, is new only for natural and mixed exponents. Part III (batch 78): for three letters the realizable pair counts $K_{AB}$ at fixed $K_{AC}, K_{BC}$ form a gapless interval, with a 24-witness quartic single-fold over complete profiles for $c=2$, holes for four

letters, and decidable three-generator submonoid membership in $\mathsf{H}^r$ given one multiplicity, a partial answer to a question of the Hilbert's-tenth programme. Part IV (batch 78): the hit set of an affine unimodular curve in any subgroup of $\mathrm{SL}_d(\mathbb{Z})$ is finite (at most $d-1$ points) or exactly periodic, with sharp periods and no uniform algorithm; input degree two, via the repository's quadratic loader (reconstructed and attributed), is the least that loads a noncomputable c.e. set; and an affine abelian compiler into $\mathrm{SL}_{6g+2}(\mathbb{Z})$. Part V (batch 82, the pipeline's Report 32): one fixed positive semigroup of 229 literal $\mathrm{SL}_4(\mathbb{Z})$ matrices whose membership simulates the Neary–Woods machine $U_{15,2}$ on finite tapes, hence r.e.-complete conditionally on that simulation chain, with a family of bounded-length sum-of-squares certificates, a 94-tile certificate and an exact fibre generating function; it answers Part IV's question on a complete universal matrix certificate in part (positive semigroup, bounded length), re-derives the research tree's $U_{15}$ construction (credited) and prints its machine table a third time in the collection. Part VI (batch 91, the pipeline's Report 55): one fixed-arity polynomial of exact degree 12 with 41,309 positive witnesses and 184,016 gates recognizing histories of every length of the programme's five-register matrix countdown with fixed contexts, which answers in part Part V's question on a uniform unbounded polynomial and Part IV's on a fixed-arity unbounded checker; the arbitrary-duration packing it calls missing was supplied the same night by the research tree by another route (1,393 operations, at degree at least 35,587), recorded in a dated note, not a retraction. Its matrix-interface equivalence and its reading as a family of programs (a Neary–Woods premise, an unimplemented compiler) are kept as open items, not refuted. Its power module and containment lemma are shared with Part XXI of `canonical-diophantine-certificates`; Remark 1.1 retains and refutes earlier sentences of both reports that said otherwise. Built from six manuscripts; the first five share no theorem but a classical Sanov lemma, the Heisenberg setup printed once; one Mihailova step is a second route to the probabilistic-quantum report's. Not a Lean formalization.

### 272. Collision geometry is linear: Diophantine certificates for rational signal machines and conservative particle dynamics

`hilbert-tenth-problem/signal-machine-collision-certificates/article.pdf`
from batch 78: Collision_Geometry_Diophantine_Signal_Machines.zip + Signal_Machine_Diophantine_Certificates.zip; batch 79: Conservative_Signal_Frontend_Corrected.zip, Sparse_Lattice_Diophantine_Certificates.zip; batch 80: Sparse_Lattice_Diophantine_Certificates_corrected.zip (corrected code edition), Three_Mass_Reversible_Computation.zip + Exact_Targets_Three_Mass_Units.zip, Single_Unit_Three_Mass_Decidability (1).zip + Four_Mass_Decidability_Package.zip; batch 82: Timed_Four_Mass_Quartic_Certificates.zip (Part VI), Dimension_Independent_Particle_Thresholds_Package.zip + Sparse_Orbit_Geometry_and_Exact_Counting_Package.zip + Binary_Planar_Four_Particle_Shuttle_Package.zip (Part VII), Unbounded_Compact_Clean_Clocks_Package.zip + Canonical_Histories_and_Infinite_Fibers_Package.zip (Part VIII); batch 91: A_Sharp_Reversible_Four_Particle_Timing_Threshold_Package.zip + Uniform_Timed_Charts_for_Four_Mass_Units_Package.zip (Part IX), Periodic_Signal_Machine_Collision_Certificates_Package.zip + Five_Signal_Nonsemialgebraic_Periodic_Validity_Package.zip + Five_Signal_Diophantine_Certificate_Source.zip + Five_Signal_Rational_Rotation_Family_Source.zip (Part X), Planar_Signal_Realization_Source.zip + Invertible_Signal_Returns_Source.zip + Projective_signal_returns_source_and_evidence.zip + Five_signal_homogeneous_realization_source_and_evidence.zip (Part XI), Five_signal_branching_source_and_evidence.zip + Clocked_signals_source_and_evidence.zip (Part XII), Bounded_certificates_and_encoded_gap_counts_source.zip + Positive_POWER_reductions_and_exact_compiler_costs_source.zip + Low_arity_Diophantine_compilers_sources.zip (Part XIII); superseded, not printed: Conservative_Signal_Diophantine_Frontend.zip (batch 79), Single_Unit_Three_Mass_Decidability.zip (batch 80)

A rational signal machine computes by collisions of finitely many kinds of particles moving at fixed rational speeds on a line. Once a complete finite collision history is fixed, its realizations are exactly the natural zeros of a sum of squares of integer affine residuals, of degree two, with one witness tuple each; typed endpoint conditions on maximal adjacency intervals exclude unrecorded

and omitted simultaneous collisions, and causal depth controls the denominators. Part I eliminates every event coordinate: the realizing initial gaps form a relatively open rational polyhedral cone cut out by at most $3(n-1)+(3m+1)E$ conditions, which gives small integral realizations, an NP upper bound for bounded synthesis, a canonical first-hit quartic, exact quasi-polynomial seed counts, an exactly solved first-collision tournament and a four-speed accumulating clock with denominators $2 \cdot 3^k$. Part II keeps event times and positions as witnesses on a rational grid, adds halting, proves full column rank (strict convexity in the witnesses) and bounds the canonical witnesses by $60^K \max(1, M_0)$ for the four-speed interface of Durand-Lose's universal machine, which is not transcribed. Part I's convex-quadratic barrier is a special case of Part XI of `canonical-diophantine-certificates`, printed as a second route. No fixed-arity polynomial for unbounded halting is claimed. Part III (batch 79) writes out a fixed reversible, number-conserving universal signal machine with 114 meta-signals, 445 two-to-two rules and exactly 18 live signals on every encoded input, compiled from Morita's 15-state, 6-symbol reversible universal Turing machine through Durand-Lose's conservative stacks; its event dynamics is an 18-dimensional partial piecewise homogeneous integer-linear map, and every fixed horizon has an explicit canonical quadratic certificate (4,196,998 auxiliaries per step for the literal machine). This answers Part I's question on an audited universal frontend, except for an ordinary-integer loader; the manuscript is the corrected edition of an earlier delivery, which is superseded. Part IV (batch 79) compiles a fixed horizon of one-dimensional partitioned number-conserving lattice dynamics into one quartic sum of squares with a unique natural witness, corrects a sign in Morita and Imai's construction, and proves decidability and effective semilinearity at total mass at most two; since batch 80 it ships its authors' corrected code edition, whose polynomial terms must be exact integers (a floating-point false zero is now rejected), the repair proposed by the Hilbert's-tenth research tree's review. Parts V and VI (batch 80) find the least conserved mass of an undecidable observation in Part IV's weighted model. Part V: when many symbols have weight one, every separated reversible two-counter machine compiles to a globally reversible, mass-conserving cellular automaton of radius at most four (radius one by blocking) whose canonical simulation has total mass exactly three; a universal source machine gives one fixed rule, so three is the least mass with undecidable pattern occurrence there, and each fixed horizon has a degree-two certificate with a unique natural witness. Its declared extension runs the source forward, then its inverse, then relabels, so that exact-pair reachability is undecidable at mass three as well, with an exact first-target clock and a compact certificate. Part VI: with at most one weight-one symbol the timed relation at mass at most three is effectively Presburger, and exact reachability stays decidable at mass four, where a binary radius-six shuttle with hit times $k^2+(2d-11)k$ is not Presburger; so integer-state number-conserving automata need at least five particles, and since batch 82 the binary five-particle upper bound of `five-particle-binary-automata` makes five exact for binary rules. Batch 82 added a third source to Part VI: for a fixed input of mass at most four, disjoint half-open charts and a quartic with exactly one natural witness tuple for each complete timed configuration (four witnesses and seven squares for the binary shuttle, exact even over the nonnegative reals). Part VII (batch 82, three manuscripts): with at most one weight-one symbol, exact and translated reachability and pattern occurrence are decidable from every input of mass at most four on $\mathbb{Z}^d$, for every $d$ and without reversibility (at $d=1$ a second proof of Part VI's theorem); semilinear visited sets, eventually quasipolynomial box counts of sharp degree, quadratic clocks, the effect of drift and canonical first-visit quartics; and a binary radius-six planar shuttle with exact wedge and drift counts and a one-step quartic. Its sharp four-versus-five boundary in every dimension is conditional on two Reports of the five-particle report. Part VIII (batch 82, two manuscripts): for four fixed nonuniversal three-mass sources, emitted horizon-free polynomials (about 480 to 610 operations, exact degrees 2344 and 1192) with a positive zero exactly when the cleaned automaton first reaches its exact target; every accepted fibre is infinite, and after the outer height is made canonical the outer history is unique and projects bijectively onto an infinite native fibre, a theorem implied by Part III of `fixed-universal-polynomials`; both rest on inherited research-tree results. Batch 91 added Parts IX–XIII from fifteen manuscripts, the pipeline's

Reports 49, 51, 56–67 and 69. Part IX: a binary number-conserving rule, globally reversible with radius at most 90, whose four-particle orbit hits an anchored word exactly at the times $k^2 + (2d - 23)k$, so four particles are the least for a non-eventually-periodic occurrence set under reversibility too; and, conditional on Part VI's sources, one finite family of timed charts for every ordered input of mass at most four, with uniform stationary-frame Presburger reachability and a single-fold quartic, while original-frame untimed reachability need not be Presburger. Part X: for a supplied ultimately periodic complete macro with at most four live signals, infinite validity is a finite formula of quadratic sign tests, compiled into a quartic with one natural witness tuple; Zeno behaviour and rational deadlines are decided, and every valid such macro on rational data has a rational accumulation time (Part I's accumulation question in part); with five live signals a half-contracted irrational rotation has a valid set that no finite sign formula describes, yet explicit degree-twelve positive-integer certificates exist (81 witnesses and 44 equations for one machine). Part XI realizes every invertible rational planar return, every fixed-centre projective lift and, locally, every compatible invertible $3 \times 3$ homogeneous return with five live signals, and classifies and certifies the strict iteration kernels. Part XII compiles every fixed two-counter program into one number-preserving table with exactly five live signals and an exact nondestructive branch test (Durand-Lose's five-signal counters credited; no universality claim), pads every instruction to exactly $30D$ and pays for native gap initialization inside two degree-twelve power modules. Part XIII compresses the certificate of each fixed horizon to three, two or one positive witnesses, shrinks the power module from 26 to 12 leaves, classifies when no witness is needed (restated with proof in `a196460-clipping-tables`) and counts the encoded gap triples exactly; Report 68, which it cites, was never delivered. Report 64's strict bound $25D/18 < T < 25D/9$ is attained at $a = 1$ and is corrected in a remark; the paid compiler for unbounded histories is absent, so no universal operation, degree or witness bound changes. The research programme reviewed the archives before placement and the written Parts after it (`5b9101184`, which corrected five summaries that could be read as one scalar witness). The three-mass manuscript re-proves Part IV's mass-two theorem by the same argument (printed as a pointer), and the single-unit manuscript reruns Part IV's semilinearity architecture one mass higher (marked step by step). Four large packet and fixture files of Parts III and IV, five rule and certificate exports of Part V (20.8 MB), the 70 emitted circuit files of Part VIII and nine heavy evidence files of Part XIII (about 35 MB) are regenerated rather than stored. *Parts I and II are two independent manuscripts proving one theorem by two routes, both printed in full; Parts III and IV come from two further manuscripts, and Parts V and VI from four more, each Part a manuscript and a companion built on it (a fifth archive is superseded by its attribution revision and not printed); batch 82 added six more, a third source of Part VI, three for Part VII and two for Part VIII, and batch 91 fifteen more, two to four for each of Parts IX–XIII, whose restatements of one another are numbered remarks;* not a Lean formalization.

**273. Stochastic and thermal exactness: exact events in uniformly convergent systems**
`hilbert-tenth-problem/stochastic-and-thermal-exactness/article.pdf`
from `batch 78: ProveIt_Exact_Erasure_Research.zip + ProveIt_Exact_Convergence_Research.zip + Thermal_Arithmetic_Diophantine_Hamiltonians.zip; batch 79: Mixing_Does_Not_Remove_Arithmetic.zip, Coercive_Green_Diophantine.zip + Well_Conditioned_Diophantine_Computation.zip`

One pattern in five settings: an approximation converges uniformly at a known rate, while an exact event of the same system is undecidable. Part I lifts integer matrix mortality to strictly positive rational column-stochastic matrices with no common invariant complex line, entries in any band around $1/n$ and total-variation contraction $\eta^T$; a product has rank one exactly when the source product vanishes, so exact erasure is undecidable for at most eight positive $7 \times 7$ matrices (from an imported mortality bound) and many-one complete for a nine-state family with a variable alphabet, with horizon-$T$ quartics having one witness per erasing word. Part II builds a fixed turn-based stochastic game with discount $1/2$, reward $1/4$ and full-support dyadic rows: value iteration converges at rate $2^{-n}$, yet exact attainment of the known fixed point $\frac{1}{2}\mathbf{1}$ from a dyadic terminal payoff is c.e.-complete; the one-player case stays open. Part III turns natural arithmetic

circuits into diagonal Hamiltonians $H = (1 + \sum_i n_i)Q$ of degree at most five whose zero states biject with the witnesses; compact resolvent and finite thermal trace are equivalent to finite witness multiplicity and undecidable on a padded universal family, and a uniformly computable critical fugacity satisfies $t_* < 1$ exactly on halting inputs. The quadratization, no-cutoff and finite-fold steps are those of `canonical-diophantine-certificates`, printed as pointers; defects that the research tree found in three shipped programs are disclosed, not patched, and Part II's two largest certificates are regenerated by its code rather than stored. Part IV (batch 79): every nonzero rational polynomial $P$ of translation rank $r(P)$ is realized by $s = r(P) + 1$ states (optimal) with commuting, invertible, strictly positive doubly stochastic rational matrices in any band around $1/s$, acceptance $1/s + \varepsilon q^{|n|} P(n)$ and total-variation contraction $\theta^L$; on a fixed alphabet, with the input on a rational quadratic Bézier curve or a line segment, the exact hit of the equilibrium value is c.e.-complete, with no computable horizon. It reuses Part I's lift, printed as a pointer. Part V (batch 79, two manuscripts proving one theorem by two routes): uniformly coercive operators $\alpha I - A$ on computable path forests, and on a connected graph of maximum degree three, with polynomial-time approximable inverses, whose unique normalized integer Green certificates exist exactly on halting inputs; the rational Green values are classified by a Pell equation, finite support and attainment of a strongly convex energy are undecidable, and membership over $\mathbb{Z}[S^{-1}]$ is decidable for every finite prime set $S$. *Merged from six manuscripts; Parts I–IV share no theorem;* a sibling of `probabilistic-quantum-and-continuous-computation`, not a Lean formalization.

**274. Quadratic orthant certificates: maximal parallelism, timed irreversible races, a fixed Waterfall universal machine, active membranes and reset Petri nets**

`hilbert-tenth-problem/quadratic-orthant-certificates/article.pdf`

from `batch 78: Maximal_Parallel_Diophantine.zip + Total_Quadratic_Diophantine_Semantics.zip + Waterfall_Diophantine_Certificates.zip; batch 79: Membrane_Motif_Research_Package.zip, Universal_Membrane_Research_Package.zip, Reset_Petri_Net_Certificates.zip; batch 80 (corrected code edition of the last): Reset_Petri_Net_Certificates_corrected.zip`

Parts I–III, V and VI compile fixed finite systems into integer polynomials of degree two, nonnegative on the real orthant, whose natural zeros biject with the semantic objects and whose unsquared products express a condition that balance equations cannot; Part IV uses degree four. Parts I–III sit at the level $D_2^+ = \mathrm{SL}$ of the degree classification in Part XI of `canonical-diophantine-certificates`. Part I certifies one maximal parallel round of a multiset-rewriting system with $d + 2m + 4H$ natural auxiliaries, and also histories and exact first halting; the resource-cover rank $\rho(A)$ gives the sharp counting exponent $m - \rho(A)$, fixed-horizon counts are eventually quasipolynomial, counting is #P-complete, the deadlock of $A + B \to C$ has no convex quadratic, and a program-uniform certificate needs degree four. Part II gives irreversible exclusive-site systems with positive integer delays a total semantics: one quadratic per catalogue has exactly one natural zero for every delay vector, recording the earliest-completion outcome and certified permanent absence, which has no computable grounding bound. Part III proves the exact macrostep behaviour of Iijil's fixed 46-clock Waterfall program for the Neary–Woods universal machine $U_{15,2}$ and compiles $k$ steps into one quadratic with $35k$ witnesses and an empty or singleton fibre; its arity grows with $k$. Re-proofs and generalizations of Part XI's results are printed as second routes; the research tree's reduction to 107 witnesses at $k = 7$ and its independent finding of the same Neary–Woods erratum are credited. Iijil's two machine data files are third-party and not under MIT-0. Parts IV–VI were added in batch 79. Part IV certifies one maximally parallel step of polarizationless noncooperative active membranes with division and dissolution by a sum of squares of quadratic residuals (degree four, one polynomial per finite schema), with exact inclusion maximality, an exact variable and residual ledger, the sharp next-population bound $2^n - 1$ and a four-rule system that defeats identical-subtree sharing. Part V gives two literal universal active-membrane frontends for the Neary–Woods machine (affine: 528 three-counter instructions and 2,544 rules; prime: 8,408 two-counter instructions and 34,605 rules), both c.e.-complete, the transcendental density of the prime language, of Turing degree $\mathbf{0}'$, and outcome quadratics whose products are strong selector gates, so that every zero over the real

orthant is natural and every fibre is empty or a singleton. Part VI builds a unit-weight reset Petri net (539 places, 771 transitions, 233 reset arcs from three places) with c.e.-complete exact-target reachability, exactly one accepting word at each length of an arithmetic progression and a canonical certificate with $2813h$ witnesses, natural over the nonnegative reals; it re-derives Part V's program layer without citing it, which is printed once. Seventeen heavy schemas and traces of Parts V and VI are regenerated rather than stored. Since batch 80 Part VI ships its authors' corrected code edition, whose net builder rejects unknown places and non-exact or negative counts and whose peak compiler requires Boolean flags, the repair proposed by the Hilbert's-tenth research tree's review; nets, schemas, witnesses and counts are unchanged. *Merged from six manuscripts: the first three share no theorem, and Parts IV and V are complementary;* not a Lean formalization.

### 275. Polynomial witness histories: unique linear certificates with finite polynomial unknowns

`hilbert-tenth-problem/polynomial-witness-histories/article.pdf`
from batch 79: `unique_polynomial_histories.zip` + `one_coordinate_certificates.zip` + `Linear_Boundary_Transport_Research.zip`

Certificates whose unknowns are a fixed number of finite polynomials rather than integers: identities linear in them (in Part II of degree two, every product through one shared stride), a clock that makes the whole witness unique with 0/1 coefficients, and no computable input-only bound on its degree or support. Part I compiles cellular computation over $\mathbb{N}[X, Y]$ with $s^3 + s + 5$ polynomial unknowns and $3s + 3$ affine-linear rows (or $3\lceil \log_2 s \rceil + 6$ unit-coefficient rows, or nine weighted rows), with a unique Boolean-coefficient witness exactly at a first singleton acceptance, exact degrees, a $\Sigma_1^0$-complete fixed matrix and a single quadratic. Part II answers Part I's question on one coordinate negatively: coefficient-bounded univariate linear systems over $\mathbb{N}[X]$ are decided by a finite-state procedure, so no universal univariate linear compiler has computably bounded witness heights; eleven degree-two identities with the self-calibrating stride $S = X^{m+2h+2}$ keep the full unique Boolean history, in bijection with Part I's witnesses. Part III, independent of both: linear boundary transport uniform over every nonzero commutative ring, a compiler with $q + d$ linear identities in $q + 2d$ sorted polynomial unknowns, a row-tagged single equation, an exact integer/rational energy gap and a Hilbert-space range theorem. A fixed number of polynomial unknowns is not a fixed number of integer unknowns: no ordinary fixed-arity representation is claimed, the no-bound results are the bounded-search argument of Part IX of `canonical-diophantine-certificates`, and undecidability of linear equations over polynomial semirings is older work (Narendran). Three expanded-polynomial exports of 1.0–1.6 MB are regenerated by the shipped code rather than stored. *Merged from three manuscripts, one answering another;* not a Lean formalization.

### 276. Smooth everywhere, universal on the integers: smooth Diophantine finalizers

`hilbert-tenth-problem/smooth-diophantine-finalizers/article.pdf`
from batch 79: `Smooth_Quartic_Diophantine_Certificates.zip`

For integer polynomials $f_i$ of degree at most two with homogeneous quadratic lifts $q_i(x, t)$, the finalizer $F = \sum_i q_i(x,t)^2 + (tu - 1)^2 + g(z_1) + g(z_2) + 2g(z_3)$, $g(z) = 2z^2 - z$, is a quartic in five more variables whose natural zeros are in bijection with those of the system and whose integer zeros are exactly two sign-related copies of its integer zeros, while its affine scheme is smooth over $\mathbb{Z}$ with geometrically integral fibres, certified by an explicit integral Bézout identity $AF + \sum_v B_v F_v = 1$ of product degree at most seven. It has a nonnegative point over $\mathbb{Z}[1/2]$, points over every $\mathbb{Z}_p$ and modulo every integer, rational points dense in its real locus and exactly two real path components, so no local obstruction remains. With first weight one, universal smoothness holds exactly for total guard weight 4 or 16, and three guards (five auxiliaries) are optimal within positive weight-four guard families; also a bounded counter-program frontend with a unique natural witness, c.e.-completeness on the compiled class with no computable bound on the least integer witness, and power-of-two padding to higher degree. General smooth-variety undecidability is Poonen's; the counter adapter and the quartic normalization are those of `canonical-diophantine-certificates`, credited; no finite-fold or single-fold MRDP theorem is

claimed. A dated note in `probabilistic-quantum-and-continuous-computation` applies the finalizer to that report's Boolean-circuit quartics. From one manuscript; not a Lean formalization.

**277. Five-particle binary automata: reversible universality, a literal universal source, clocks, lazy evaluation and quartic certificates**

`hilbert-tenth-problem/five-particle-binary-automata/article.pdf`

from `batch 82: Reversible_Binary_Five_Particle_Package.zip (base), five-particle-binary-portable.zip, Literal_Universal_Reversible_Source_Package.zip, Reversible_Startup_Optimization_Package.zip, Cellular_Clock_Domination_Package.zip, Exact_Lazy_Reversible_CA_Evaluator_Package.zip, Event_Budgeted_Quartic_Certificates_Package.zip, Two_Parallel_Conservative_Involutions_Package.zip, Sparse_Parallel_Particle_Evaluation_Package.zip, Canonical_Parallel_Quartic_Certificates_Package.zip; batch 91: Two_scale_recognition_radius_sources.zip, Shorter_exactness_windows_sources.zip (Part VII)`

Twelve "Research Reports" (14–20, 26–28, 70, 71) of one pipeline, on one object, that continue the signal-machine report above. Part I: every finite partial-injective guarded two-counter machine compiles to a globally reversible, number-conserving binary cellular automaton of explicit radius with five-particle encodings and an anchored halt word, and every deterministic two-counter program to a binary conservative one; with that report's mass-four decidability, five is the exact particle threshold for binary and for globally reversible binary number-conserving automata (answering its threshold question), with fixed-horizon certificates of degree at most four and a literal instance from the 8,408-row source of `quadratic-orthant-certificates`, whose source lemmas are re-derived (second presentation). Part II prints the universal reversible two-counter source that Part I left existential (122,622 controls, 141,561 rows), with positive return and periodicity r.e.-complete on its clean family and quartic certificates. Part III: history-tag swaps that give empty startup in 138 source steps and first-encounter optimality among static orientations, lifted to the cellular clock. Part IV: a lazy evaluator equal on every finite support to the frozen ordered product of about $2.7 \cdot 10^{11}$ involutions. Part V: event-budgeted quartics with exactly one natural witness for a fixed source, mass, horizon and budget. Part VI: two full-shift number-conserving involutions that agree with the ordered rule on admissible states, a sparse evaluator and fixed-horizon quartics for the new rule; their arithmetic layer re-proves lemmas of `canonical-diophantine-certificates` and of the signal-machine report, credited. Part VII (batch 91, Reports 70 and 71) answers Report 26's first question with two further full-shift extensions of the same admissible dynamics: a two-scale isolation lemma gives the sufficient radius $108D + 149 + 3J$ (55,027,013 for the universal ledger), with a four-particle input separating the new rule from Report 26's and the injective number-conserving binary rules of radius two shown to be exactly the five shifts; triple exactness lowers the radius to $76D + 105 + 3J$ ($76D + 104 + 3J$ for the same rule; 38,722,713 and 38,722,712), so the earlier tiny-source comparison, 978 against 540, becomes 581 and 409. The earlier evaluator and certificates do not transfer to the new rules, and the true or minimal radii, Report 70's reliance on the undelivered Report 68 and an independent audit of the main proofs are open items. No intrinsic universality, minimal radius, unbounded or fixed-arity representation, executed universal startup or novelty is claimed; the threshold corollaries rest on the signal-machine report's lower bound. Heavy regenerable tables, sources and traces (over 200 MB) are rebuilt by the shipped code rather than stored. The Hilbert's-tenth research tree reviewed seven of the first ten Reports, and Reports 70 and 71 before and after the write (without re-auditing their proofs), without finding a defect; not a Lean formalization.

**278. Fixed universal polynomials: a literal Grill instance, exact degree laws, native witness fibers and index restoration**

hilbert-tenth-problem/fixed-universal-polynomials/article.pdf

from batch 82: Fixed_Universal_Grill_Polynomial_Package (1).zip (base), Native_Grill_Exact_Degree_Laws_Package.zip, Entire_Native_Witness_Fiber_Package.zip, Failure_of_Positive_Index_Restoration_Package.zip, Counterexample_Height_Expansions_and_Rotation_Discrepancy_Package.zip; batch 83: Exact_Negative_Index_Obstruction_for_Positive_Diophantine_Interfaces_Package.zip; batch 88: Nonredundant_Main_Comparisons_in_Positive_Diophantine_Certificates_Package.zip (Part VI), Failure_of_First_Index_Deletion_Package.zip (Part VII), Structural_Obstructions_for_the_Free_Coefficient_83_Candidate_Package.zip, Full_Positive_Collapse_of_the_Square_Product_82_Candidate_Package.zip, Exact_Certificate_Height_and_an_Even_Rank_Obstruction_Package.zip (Part VIII); superseded, not printed: Fixed_Universal_Grill_Polynomial_Package.zip (batch 82)

Eleven "Research Reports" (23 revision 1, 24, 25, 33, 34, 37, 39, 41, 43, 45, 46) on the Hilbert's-tenth programme's own fixed universal polynomial and its native arithmetic block, built on its research artifacts at pinned commits. Part I: one literal 397,488-phase Grill program and a complete circuit of 3,600,546 operations with 797,135 positive witnesses, whose language theorem is conditional on pinned imports and whose exact total degree 69,339,973 is unconditional algebra. Part II: exact degrees $63N - 4$, $12N + 10$ and $87N + 16$ for every run table of the native Grill template. Part III: the entire positive fiber of the native component at fixed ports and a two-term height law with a $\zeta(1/2)$ second term, conditional on pinned native soundness and completeness. Part IV: the programme's proposed positive index restoration fails for one compiler child, which has infinitely many positive zeros with negative restored index at every input; one canonical family of them is counted, $N(B) \sim \kappa \log\log B$, with a rotation discrepancy and a Kesten obstruction. Part V (batch 83): for the two interfaces left open there, an exact necessary-and-sufficient arithmetic predicate for a full positive zero with negative restored index, with positive reconstruction of every witness; whether the predicate ever passes stays open. Part VI (batch 88): on explicit thinned scale progressions every condition of that predicate but one congruence holds infinitely often while the congruence fails, so the main projection and the main norm are nonredundant on positive tuples, and every odd-index hit of the congruence needs a Jacobi symbol $-1$ (its analytic existence proof was not recertified by the programme). Part VII: deleting the first-index quotient from the tree's 85- and 86-operation polynomials gives 81- and 82-operation candidates with full positive zeros at every input and every genuine compiler, so with the empty-set compiler they accept a rejected input. Part VIII, three sources on the one- and two-operation reductions of the 84-operation polynomial: structural constraints on every full positive zero of the free-coefficient 83 candidate (which the research tree has since refuted), infinitely many positive zeros of the square/product 82 candidate at every input (a second construction of a research-tree theorem), and the exact height of the displayed certificates with an even-rank obstruction. "82" and "83" name several polynomials; a front-matter table lists every candidate below the record with its status. A dated note of batch 91 records that the r.e.-complete fixed sandpile polynomials of Part XXI of `canonical-diophantine-certificates` are not ordinary-input universal polynomials and do not bear on the record. Results that re-derive research-tree constructions are credited there; Report 23's first edition is superseded by its revision; a 61 MB circuit and a 1.6 MB program are rebuilt by the shipped code; two third-party papers are not redistributed. No new MRDP theorem, optimality or operation record is claimed, and no Part lowers the 84-operation record; the programme reviewed every source but certifies only parts of some; not a Lean formalization.

**279. Periodic turmites: first revisits, a literal Langton ant and its Diophantine certificate**

`hilbert-tenth-problem/periodic-turmite-first-revisits/article.pdf`

from batch 83: `Polynomial_First_Revisit_and_Exact_Pattern_Queries_for_Turmites_Package.zip` (Part I); batch 88: `Literal_Periodic_Langton_Ant_and_Visit_Boundary_Package.zip` (Part II), `Arithmetic_Initialization_for_a_Periodic_Langton_Ant_Package.zip`, `Complete_Positive_Certificate_for_the_Literal_Periodic_Ant_Package.zip` (Part III), `Exact_Structured_Construction_of_the_Fixed_Ant_Coefficients_Package.zip`, `Exact_Fusion_of_the_Ant_Background_Polynomials_Package.zip` (Part IV)

For an explicit cyclic L/R turmite on a periodic colour tile with finitely many binary-encoded defects and a binary start, the first repeated lattice position is decidable in polynomial bit time: the algorithm returns the exact first repeated arrival with a witness, or a no-revisit certificate with a periodic tail, through an affine lane calculus with an exact intersection primitive. It also compiles finite unions of head-relative observation clauses into an exact first hit (no polynomial bound claimed for the Boolean expansion; a Presburger route through Cooper's algorithm, which is formalized in `Logic/PresburgerArithmetic`, is a second, unexecuted proof), and proves that no undecidable language reduces to globally one-visit runs read by such observations. This is the negative side of two research-tree notes on ant and toggle-router universality; turmite universality itself is third-party. Part I claims nothing after the first revisit, no loader or universal atlas, no novelty. Parts II–IV (batch 88, Research Reports 40, 42, 44, 47 and 48) answer its open question on a literal two-visit interface: one fixed doubly periodic binary background, a fixed start and a finite loader make the binary RL ant (Langton's ant) simulate the fixed fifteen-state Neary–Woods machine $U_{15}$, visiting every cell at most twice, with halting read off a residue-and-heading port; with Part I this gives a sharp one-versus-two-visit boundary. A fixed polynomial system initializes the board from two positive sentinel integers, and one fixed polynomial $F_2$ with 465 positive witnesses and exact degree 2,304,000 recognizes halting of $U_{15}$ on the two sentinel-decoded side words (Report 44, a recovered edition). Its source uses 2,307,457 operations with prescribed coefficients. When all coefficients are built from the literals 1 and 3, Reports 47 and 48 reduce the dense-prefix construction to 14,658,934 operations; later programme packets reduce that count further. $F_2$ recognizes a fixed r.e.-complete language only through the published $U_{15}$ encoding: it is not an ordinary-input loader for the programme's input convention and does not bear on its 84-operation record. Seven primitive cell maps are numerical data taken from the figure sources of Maldonado, Gajardo, Hellouin de Menibus and Moreira, credited and not covered by MIT-0; 27 heavy regenerable files were excluded with recipes. A dated note of batch 91 records that Part III's fifteen-equation power system recurs in Part XXI of `canonical-diophantine-certificates`, Part VI of `group-theoretic-substrates` and Parts X–XIII of `signal-machine-collision-certificates`, whose Report 67 shrinks it. From six manuscripts, each reviewed by the programme without a defect (it certified only what it read); not a Lean formalization.

## 16 Convex geometry

One report, opened by batch 138 and placed without a write (a single manuscript on a new topic, by Vladimir's direction of 9 October 2026).

**280. Optimal simplex products for projection-body volume**

`convex-geometry/optimal-simplex-products/main.tex`

from batch 138: `ProveIt_Research_2026-10-08.zip`, its simplex-products manuscript (arrival `b28d0850b`)

Over affine images $K$ of products of simplices in dimension $n$, proves $\max |\Pi K|/|K|^{n-1} = P_n = \max \prod_i s_{d_i}$, $s_d = (d+1)d^d/d!$, for the projection body $\Pi K$: block size 13 is the unique optimal rate ($\rho = 2.80996473256\ldots$), two balanced candidates suffice in every dimension, and from $n = 100$ (sharp) the optimal multiset is unique and follows an exact rule modulo 13, with the sharp

uniform gap $G = s_{14}^8/(s_{13}^4 s_{12}^5) = 1.0010398138\ldots$; also $P_{n+13} = s_{13}P_n$, the periodic asymptotics $(\rho/e)^n n^{-1/2}$, and the optimum at a prescribed number of factors or facets. Within the product class only: Brannen's conjecture over all convex bodies was refuted earlier by Feng, Hu, Liu and Xu, credited. The delivered PDF was not staged; `main.tex` inputs `../common/preamble.tex`, whose copy is staged as `common/preamble.tex` inside the report, so the source does not build where it is placed. Unrefereed; not formalized.

## 17 Probability

Nine reports on limit laws, concentration, sampling algorithms and a probabilistic model of mathematical physics, which opened this category in batch 138. Each is a single manuscript that continues nothing in the tree, placed without a write by Vladimir's direction of 9 October 2026: the delivered layout, file names and PDF are kept with the wrapper removed, the delivery README serves as the report README, and beyond light checks at intake nothing was re-derived. Their claim ledgers and non-claims stay in the delivered files. The last two study the same Gaussian quadratic form under different normalizations and are cross-referenced.

**281. A sharp analytic renormalization threshold for binary broadcasts on trees**
`probability/tree-broadcast-renormalization/article.pdf`
from batch 138: `ProveIt_Research_2026-10-08 (1).zip, its tree-broadcast manuscript (arrival b28d0850b)`

For the level sum $S_n$ of a binary broadcast on the $b$-ary tree with edge correlation $\theta$ and $X_n = (b\theta)^{-n}S_n$, proves that $e^{-\mathrm{Var}(X_n)t^2/2}\,\mathbf{E}e^{tX_n}$ converges locally holomorphically if and only if $\theta > b^{-3/4}$, a fourth-cumulant threshold distinct from the reconstruction threshold $b^{-1/2}$; a uniform analytic residue in the window $\theta_n = b^{-1/2}e^{\gamma/(2n)}$; lattice local expansions to all orders; and, at $b = 2$ and criticality, majority success $p_n = \frac{1}{2} + 1/\sqrt{\pi n} - \frac{5}{4\sqrt{\pi}}n^{-5/2} + O(n^{-7/2})$. The package-level files of its archive are staged in `bundle/`; the README cites `../VALIDATION.json`, which is `bundle/VALIDATION.json` here. Unrefereed; not formalized.

**282. Sharp Poisson asymptotics for the longest arithmetic progression in a random set**
`probability/random-progressions-sharp-poisson/paper.pdf`
from batch 138: `sharp_poisson_random_progressions.zip (arrival b28d0850b)`

For the longest nonconstant arithmetic progression $U_n$ in a Bernoulli-$p$ random subset of $\{1,\ldots,n\}$, proves $P(U_n < r) = e^{-\lambda}(1+c_p\lambda^2 r^2/n)+O((\log n)^{3/2}/n)$ uniformly in $r$, with $c_p = (q/p)(5/3-\pi^2/9)$ (twice the integral of the squared binary entropy); the uncorrected error is exactly of order $(\log n)^2/n$, with a lattice-periodic coefficient and explicit limsup and liminf. No improvement of a deterministic Szemerédi or van der Waerden bound is claimed. Placed here rather than at the delivery's path under `Combinatorics/Ramsey/Research`, whose directories refine deterministic theorems formalized there; the delivered `INTEGRATION.md`, README and `notes/` still name that path. Unrefereed; not formalized.

**283. Exact marginal crossover for a quartic Bose gas**
`probability/marginal-bose-crossover/article/marginal_bose_crossover.pdf`
from batch 138: `marginal_bose_research.zip (arrival 36f1dd8ef)`

The complete convergent expansion of $G(s) = \sum_n \Psi(s\sqrt{n})/n$, $\Psi(u) = \int e^{-x^4-ux^2}\,dx$, with radius $\sqrt{8\pi}$; the constant left unspecified by Łebek and Jakubczyk (Phys. Rev. A 102 (2020) 013324, Sec. IV.2) is $\pi/2$ in their variable $\theta = s^2/8$. Also a uniform $O(\log(1/\varepsilon))$ evaluation count, a Lambert-$W$ inverse expansion for the critical temperature and an all-orders lattice expansion. Unrefereed; not formalized.

**284. An exact small-step criterion for random reshuffling**
`probability/random-reshuffling-criterion/article.pdf`
from batch 138: `reshuffling_research.zip (arrival fcfe6ef6a)`

For quadratic components with a common minimizer, one block of sampling without replacement

eventually beats sampling with replacement for every initial state if and only if $H \ker V \subseteq \ker V$; otherwise $\lambda_{\min} = -\frac{m(m-1)(4m+1)^2}{72(n-1)}\lambda_{\max}(G_K)\,\eta^4 + O(\eta^5)$, with an exact rational classification and certificates. Unrefereed; not formalized.

**285. Uniform exponential-tilting guarantees for coefficients and constrained sampling**
`probability/uniform-exponential-tilting/paper.pdf`
from `batch 138: uniform_tilting_research.zip (arrival fcfe6ef6a)`

Coefficients of products of positive linear factors (Poisson-binomial laws) to relative accuracy $\varepsilon$ from $O(\log(1/\varepsilon))$ Fourier evaluations after a coarse exponential tilt, with certified intervals; and a tilted probabilistic-divide-and-conquer sampler for two-row contingency tables needing fewer than $5\sqrt{n}$ expected proposals for every margin vector. Unrefereed; not formalized.

**286. Sharp and computable corrections for nonuniform coupon collection**
`probability/nonuniform-coupon-collection/article/coupon_collector.pdf`
from `batch 138: coupon_collector_research.zip (arrival 2c7f4fd68)`

An explicit remainder $R_s$ for the order-$s$ Poisson–Charlier approximation of the coupon-collection probability for every probability vector, pointwise smaller than that of Hwang, Li and Zacharovas (arXiv:2605.29633, Lemma 5.3), and the optimal scalar constant $K_{m,s} = \frac{m^s}{2^s s!}(1 + \frac{s(-2s^2+5s+1)}{9m} + O(m^{-2}))$. The nearest collection reports, `a261781-matrix-compositions` (a coverage-window expansion for a different array) and `a122399-surjection-diagonal` (Poisson defects), share no theorem with it. Not re-derived at intake. Unrefereed; not formalized.

**287. Deterministic relative approximation of product-distribution total variation**
`probability/product-total-variation/article/fourier_total_variation.pdf`
from `batch 138: fourier_tv_research.zip (arrival 2c7f4fd68)`

A deterministic approximation, to relative accuracy $\varepsilon$, of the total variation distance between two product distributions in $O((N/\varepsilon)\log^2(2/(\varepsilon d)))$ operations, with extensions to Markov paths and to the Bayes-risk curve. Two delivered run logs are staged as records. Not re-derived at intake. Unrefereed; not formalized.

**288. Sharp Gaussian quadratic concentration from signed spectral moments**
`probability/gaussian-quadratic-concentration/spectral_concentration.pdf`
from `batch 138: spectral_concentration_research.zip (arrival 2c7f4fd68)`

For $Q = g^{\mathsf{T}}Ag - \operatorname{tr} A$ with $g$ standard Gaussian, operator norm $L$ and $v = \operatorname{tr} A^2$, proves the optimal coefficient $c(s)$ in $P(Q \geq t) \leq \exp[-c(s)\min(t^2/v, t/L)]$ at cubic spectral balance $s = \operatorname{tr} A^3/(Lv)$, with $c(0) = (\sqrt{5} - 1 - \log\varphi)/4 = 0.18871\ldots$ against the unrestricted $(1 - \log 2)/2$; also a $\operatorname{tr} A^4$ Gauss–Radau refinement and an exact binomial example. Companion of the next entry: its two-mode extremizer $\sqrt{2}\,UV$ ($A = \operatorname{diag}(2^{-1/2}, -2^{-1/2})$, $s = 0$) attains this $c(0)$ at $t = v/L$ (checked at intake). The notations differ ($s$, $v$, $L$ here; $\delta = \sum c_i^3$ at $\sum c_i^2 = 1$ there), and the option `--k` of `exact_certificates.py` is a replication count, not the spectral $k$. Unrefereed; not formalized.

**289. Sharp two-mode extremality for Gaussian quadratic fluctuations**
`probability/gaussian-quadratic-two-mode/article.pdf`
from `batch 138: two_mode_gaussian_extremality.zip (arrival 2c7f4fd68)`

For $Q = \sum_i c_i(g_i^2 - 1)$ with $\sum c_i^2 = 1$ and fixed $\delta = \sum c_i^3$, proves that the two-mode form $a(g^2 - 1) - b(h^2 - 1)$ maximizes $\mathbf{E}f(Q)$ for every $f$ with convex $f''$; hence all absolute moments of order $p \geq 3$, the skewness–kurtosis frontier, spectral rigidity and a Laplace envelope; at zero skewness the extremizer is $\sqrt{2}\,UV$ and $\mathbf{E}Q^4 \leq 36$ (not 60). Companion of the previous entry, with the same nearest source (Pang, arXiv:2609.05914): the two share no theorem, neither cites the other, and they meet at zero skewness, where this extremizer attains that entry's $c(0)$. Unrefereed; not formalized.

# 18 Set theory

### 290. ZFC closure kernels and small inconsistency-search machines

set-theory/zfc-closure-kernels/article.tex

from batch 143: zfc_closure_research.zip + zfc_closure_research (1).zip + zfc_exact_fiber_hulls.zip (9e5b610259); batch 144: zfc_machine_research.zip + zfc_subset_kernel_research.zip + zfc_subset_only_machine_research.zip + zfc_tm_pure_subset_research.zip (17d0f2eaa), zfc_one_schema_research.zip (39409c8c9), zfc_machine_research (2).zip (83dec60c2); batch 145: zfc_small_machine_research.zip (9c98495586), zfc_small_machines_20261016.zip (96836f9eea); batch 146: zfc_consistency_machine_research.zip + zfc_consistency_proposal.zip (47f3a36029), zfc_empty_seed_research.zip + zfc_small_machine_proposal.zip (af518abcf8); batch 147: zfc_small_machine_research (2).zip (9439d775aa); batch 148: zfc_consistency_machine_report_20261016.zip (5e5603303d); batch 149: zfc_consistency_research.zip (de047657e) + zfc_substitution_engine.zip (be9e617d2); batch 150: zfc_consistency_completion_proposal.zip + zfc_epsc2_proposal.zip + zfc_minimal_rule_research.zip + zfc_qnand_research.zip (47e887108); batch 151: zfc_consistency_proposal.zip + zfc_exact_hull_research.zip (121a0de1bb); batch 152: zfc_consistency_machine_report_20261016.zip + zfc_consistency_machine_research_2026-10-16.zip + zfc_contraction_research.zip + zfc_epsilon_research.zip + zfc_minimization_20261016.zip + zfc_small_machine_report_2026-10-16.zip + zfc_small_machine_research.zip + zfc_small_machine_research_2026-10-16.zip + zfc_small_machine_research_20261016.zip (da6dabcd89); batch 153: zfc_consistency_research_2026-10-16.zip + zfc_small_machine_research_2026-10-16 (2).zip (4e720ccbbf); batch 154: zfc_consistency_research_2026-10-16 (2).zip + zfc_consistency_research_20261016.zip + zfc_unary_closure_research.zip (61e7783656), zfc_finite_kernel_research.zip (3d731d8cf4), zfc_cyclic_algebra_research.zip (2b10c9072e); batch 155: proveit_zfc252_research_2026-10-17.zip + zfc_consistency_research_2026-10-17.zip + zfc_finite_closure_2026-10-17.zip + zfc_fused_transport.zip + zfc_research_2026-10-17.zip (fab89c5d26); batch 156: zfc_consistency_252_research_20261017.zip + zfc_consistency_research_20261017.zip + zfc_reset_hod_research_2026-10-17.zip (2447a98214); batch 157: zfc_sparse_certificates.zip (85633fd2f5); batch 158: zfc_geometric_counter_research_2026-10-17.zip (e92db213bc), zfc_sub252_research_20261017.zip (a2bf288dea), permutation252-reduction-study.zip (073543e11b), cachemp252_research_2026-10-17.zip and zfc252-reduction-research-2026-10-17.zip (80e26b89e9); batch 159: recovered part of zfc-state-reduction-research.zip (same arrival)

Fifty-six manuscripts reconciled without rewriting the seeded base article. Bounded-image, seeded and exact-fiber kernels are compared mathematically with Foundation/Choice-free ZF, then ZFC. Later work gives subset-only bounds, a pure/mixed equivalence without Extensionality and the hereditary-saturation interpretation of equality-free PSC. An incoming-profile monochromatic decoder supplies another bootstrap; empty-input EPSC is equivalent to seeded PSC. Ordinary CP/GIC equality erasure, observational and bisimulation interpretations, Foundation-free Infinity, total codecs and complete reference proof checkers retain their separate scopes. These are ordinary arguments, not delivered proof-code or binary compiler certificates.

A finite-sort countermodel satisfies the entire reduced emitted C/G calculus, with either quantifier row, while making its target false. The same obstruction covers the conjunction/grammar rewrites and the offset PSC wrapper. It retires the unchanged abstract calculus as a consistency-search target. Consequences for literal tables remain conditional on their missing simulation. The full ordinary equality-free theories retain their mathematical calibrations. A later argument formalizes the finite-table obstruction in PA at the ordinary mathematical level and isolates a guard-independent fresh-hull criterion. Complete Hilbert and ground-clause reference searches restore cross-variable instantiation; neither is compiled into the reduced tables. A tagged repair corrects one Capture-to-Collection extraction in Friedman's class-theory manuscript, without refuting its global equivalence theorem. New binding-witness and shared-template tableau frontends supply complete ordinary reference algorithms, with no compiled state bounds. Dynamic schema parameters permit late witnesses and variable identification. A diagonal counterinterpretation separates composite substitution from fresh and signed-atomic instantiation. Two independently rebuilt standalone Lean files prove binding-syntax identities and a specified PSC/modal derivation invariant; they do not verify the Python frontends.

Batch 150 adds captured EPSC with genuine-Bottom low codes and a closed relative falsity alternative; EPSC2 with two matrix parameter slots and arbitrary internal binders; a protected-witness closure rule preserving the EPSC theorem set, even with purely logical bound proofs; and a complete quantified-NAND EPSC tableau. Their ordinary proofs and executable references remain separate from the 271-state EPSC, 255-state decoder and 341/343-state clean-initializer controls. The decoder's 606,702-step blank prefix and 57,150 folds are Python evidence. A separate original-word grammar control has 269 states; full dispatcher pricing of all 6,720 distinct sampled recipes confirms that sample's minimum of 269. Neither new closure-rule nor quantified-NAND frontend has a compiled table.

Batch 151 derives outgoing observational equality by a four-node formula, a reserved-zero falsity translation and explicit SK carrier terms. The strict EPSC proxy control has 288 states; all 264 recipes reproduce. Its generic parser-accepted-input compiler claim fails for a negative start label; the start-zero public-253 payload is unaffected. The full term has 192,272 BCL bits, not a measured binary evaluator cost. A finite certificate excludes strict step quotients of literal253, not smaller blank-equivalent machines. EFPC replaces EPSC's hull by an exact fixed point using the full schema's Separation resources. One universally closed instance cannot be replaced in isolation, as a three-object counterexample shows. Two standalone Lean files pass 13 empty-axiom audits at their local resource/Boolean scope. All 20,480 compiler recipes and 148,368 symbol-wise fusion records reproduce; the best EFPC control is 271 versus EPSC270. Fuel/abstraction-limit failures remain unresolved. A narrow modal-list correction leaves the cited main interpretation intact. These are measured comparisons and local proofs, not a new state record.

Batch 152 adds raw deductive equivalence of one-parameter EPSC plus fixed adjunction with full EPSC, and represented exact hulls at each positive parameter arity. EPSC1 alone remains open. A complete signed-name calculus uses explicit contraction; local stream transducers are not a full TM. Ground epsilon critical formulas and two-slot matrices give an ordinary finite Boolean-refutation characterization. A regular-ordinal countermodel refutes the unsupported closure deletion. Default-last four-slot EPSC2 and dense scoped quantified-NAND codes retain their equality-free scopes. The raw inhabited $\mathrm{SR}_2^+$ schema uses a shared world constant and an arbitrary-transitive-cover bisimulation interpretation. Its finite-pool tableau uses one common existential witness. FAK12 gives a finite assignment/class presentation with 28 Skolem clauses and 62 literals, two constants, 15 proper functions and two binary relations. These ordinary calibrations and reference algorithms have no compiled new state bound.

A four-row 253-state variant has exactly the same bootstrap tape after 312,935 steps, 40 earlier than the baseline. Its ordinary same-halting argument retains the register boxes, all 8,391 affine cases and an independent 2,981-pair register certificate. The 283-state fixed-emission control's 1,272 compiles reproduce; 453,939 row mutations leave 707 bounded runtime survivors, not new blank-start machines. A separate clean initializer gives 315 states. The 218,872 initializer-reuse candidates give no survivor. Eighteen regenerated clean controls and 36 whole-tape boot checks give shared quotients 334/336, 321/323 and 309/311. The original-word comparison 341-to-334 combines four raw initializer states and three quotient states; comparing quotients gives 338-to-334. Public 253 still has no smaller exact quotient. The final broader protected-basis replay reaches 268, improving its saved 269 control while matching the existing 268 benchmark. No lower bound follows from bounded search failures.

Code 0's delivered description as closed is corrected: it has free variable 0 although semantically true; the emitter's freshness offset remains valid. Archived Lean components compile unchanged on copies, with their limited scope and standard axiom audits recorded. No old patch is installed and no Coq build is run. At the batch 152 cutoff, Lean derived actual CP/GIC quotient ZF0, ordinary Infinity and a one-way abstract-enumerator consistency implication. The exact binary tape is related to RM331 after 313,429 steps; these local results leave effective Foundation/Choice translations, full enumerator and binary runtime correspondence, and the internal ZFC endpoint open.

Batch 153 adds checked arbitrary-cutoff enumerator starts and exact absorption of dirty

selector/scratch registers into a virtual clean root. The six nonformula fields may vary, with safe formula data, the genuine root and the zero logical register tail retained. The archived 19-source Lean closure freshly builds, with only standard axiom dependencies. A standalone 12-window certificate justifies a same-count 253 direction rewrite by an ordinary positive-progress whole-tape argument. The 597,080 bounded transition-alternative checks give no deletion; every regenerated ledger row agrees. The runtime252 diagnostic has a literal 999,186-step prefix, a reconstructed-frame 2,794-step instruction, and one million accelerated register instructions. It proves neither a safe root nor divergence.

A three-component normalized class interpretation gives full EPSC1, Extensionality, IR, Separation, Union, Power Set, Infinity, the stated local graph Choice and parameter-free Replacement, while Pairing, a particular EPSC2 instance and one-parameter Replacement fail. The old hereditary quotient is the entire interpretation. This is an ordinary relative deductive separation, not strictly lower consistency strength; other EPSC1 interpretations remain open. Its finite 7/20/760-object diagnostics explicitly fail full EPSC1 and are not offered as models of that schema.

The scanner obstruction is reached after 1,011,713 literal blank steps. Original and edited continuations take 476 and 556 steps to different PCs; this refutes the advertised instruction simulation, not ultimate halting equivalence. Radius6 checks 63,756 state pairs against 335,369 windows and accepts none. All 432 outgoing-equality recipes actually compile, best290; arbitrary-payload translation remains open. The 12,002 permutation grammars have minimum270, with the prior268 outside their family. Three million substring proposals retain268. Bounded failures supply no optimality claim. Both 23-page scratch PDFs rebuild and are visually reviewed. Dossiers retain all scopes, pins, stale-command corrections and the exact replay adapter.

Current Lean has since completed the numerical encoded/decoded model comparison, actual CP/GIC/IR sentence-theory refutation equivalence, complete structured-root enumeration, and the 219-node path/structured correspondence. At that earlier checkpoint, the distinct literal program and complete binary transports remained separate; later progress is recorded below. Foundation/Choice and internally arithmetized ZFC conclusions retain their exact theorem scope.

Batch 154 adds finite fusion of GIC/EPSC instances into one matrix-dependent instance, with exact selector/bootstrap costs and a complete single-instance refutation-search specification. It is not one fixed finite membership theory. An ordinary one-parameter EPSC1 interpretation needs only exact Binary Union; its support countermodel keeps all finite coarse bounds but still lacks Pairing. This is not a consistency-strength separation or a refutation of a different internal-well-founded interpretation. The finite assignment/class kernel has 15 clauses, 35 literals, 174 term occurrences, nine proper functions and genuine equality, improving source 34's 28/62/244 payload. Its ideal equality-aware ground search is complete; no whole-kernel search or binary compiler was executed.

The E7/E9/E8 cyclic algebra routes reflect empty-premise identities, not arbitrary additional premises. Their 7/9/8 equation counts have 81/75/90 both-side term nodes; E7's alternative 88-symbol convention includes seven equality signs. Ground generator invariance is not a substitution schema. Restricted phase grammars have exact optimizers. The 14-sentence finite class source and 10,182-node membership wrapper preserve the tagged Capture repair, while the expanded modal query and binary backend remain absent. Current 299/333-state E7 controls use a placeholder query and establish no consistency-machine bound.

Source 38's original native-model bridge freshly builds all 26 archived Lean sources, with 14 standard-axiom probes; newer current modules already close much of its pinned gap. A 3,530-window/7,044-obligation certificate gives a guarded three-to-one public253 path replacement. Its ordinary halting equivalence and full-tape bootstrap at 312,895 steps keep 253 states. All advertised finite suites, bounded searches, context inventories and exact compiler/certificate artifacts reproduce. Two rotation sweeps score 118,977 layouts and retain 268 states. The billion-step negative diagnostic concerns the old identity-layout runtime252, not the new register-permutation candidate. Five scratch PDFs rebuild cleanly, with all fonts embedded and selected theorem pages visually reviewed.

Since the source pins, actual native-model/refutation and literal-program transports have advanced, as have the new candidate's exact 82,552-step clean bootstrap and box/executor/framing proofs. Closed decoder coverage, global binary composition and full ZFC/Choice/internalization must retain their own verified boundaries; these arrivals alone establish none of them.

Batch 155 adds a global dirty-stack barrier for the actual candidate order: replacing a marked pop by a push strictly lowers the candidate code and supplies an earlier unmatched theorem push. With PL=NP=D=0, arbitrary old formula, stack and cache values are safe. Exact accelerated blank reachability and whole-tape runtime arguments give an ordinary consistency calibration for the older unpermuted runtime252 table. Four narrow Lean modules pass; the actual enumerator invariant and enormous new bootstrap are not kernel-serialized. This is distinct from the current register-permutation252 table, whose closed Lean theorem already gives literal blank nonhalting iff Con(ZF), including Foundation and Power. Con(ZFC) implies its nonhalting; the reverse Choice transport and internal PA proof remain open.

The finite represented-EPSC and nine-function closure kernels reach 901 bits with native equality or 202 bytes with explicit equality under distinct signatures. Guarded witness sharing, member selection and genuine equality congruence are retained. Payload size alone gives no machine bound. Bilateral ray/support models strengthen the unary obstruction, while raw EPSC1 plus exact BinaryUnion is deductively equivalent to EPSC1 plus profile adjunction and full EPSC. Uniform finite support forces trivial action for any fixed- dimensional Extensionality/Pairing interpretation, including finite parameters; invariant interpretations and bare unary equiconsistency remain open.

Two shifts with fixed GEN/UI cost 22 variable-output controls; a two-map finite-support universal transport costs 24. These exclude the remaining logic, storage, search and initializer. Complete compiler controls have 265 states; borrowed-register initializers yield 299/300 and fused congruence 271. The new 4,549-pair compression certificate compares two versions of the fused program, not the original emitter. A composed public253 shortcut saves 120 moves with the same complete tape at 312,855 moves and no state reduction.

All advertised executable finite suites and bounded searches reproduce, including 20,736 fusion recipes and 117,474 affine initializer checks. The one supplementary subtree/commutation diagnostic without a driver is not independently replayed. Five scratch PDFs have 30/28/27/34/29 pages, embedded fonts and visual review. Dossiers preserve the pinned comparisons, corrected manifest adapters, ordinary/formal boundaries and all unsuccessful proposals. No 251 result, global minimality, Coq rebuild or internal PA/ZFC endpoint is claimed.

Batch 156 supplies an ordinary set-coded ZFC reconstruction for the existing permutation252 table. It distinguishes the closed external Con(ZF) Lean theorem from Choice and internal-arithmetic formalization. Cofinal padding uses the upstream closed-zero decoder; hidden-coordinate logical controls refute the weakened fragment without becoming CP/GIC models or consistency separations. The exact eight-family gap generator has no complete binary implementation. Whole-state compiler controls have279 and corrected286 states, with retained calibration, startup and missing-row obligations; these savings cannot be subtracted from252.

The unchanged base12 numerical checker has an exact sparse history. Canonical row count is at most $\lfloor\sqrt{3b+1}\rfloor$; the square-root lower example concerns unfolded occurrences, not distinct rows. Explicit fixed circuits and bounded rejected-MP intermediates support conservative bit complexity and an ordinary arithmetic presentation, not a delivered PA representation proof or binary-state bound. Productive continuations combine positive action lengths with decreasing pure-jump rank. The separate55,471,132,048 and166,413 deletion families find no sufficient certificate; bounded diagnostic stops are not semantic impossibility proofs. Involutive ground tours give the exact orbit with restoring stages, but still need a calibrated ZFC query, complete writer, compiler and state price.

Six counters permit arbitrary formula, stack, additive parameter and cache. A marked-occurrence injection controls all candidate commands, while an initial AX guarantees eight good entries above its exact dirty tail. The sharper finite credit is exact only for its conservative syntactic

weights. Subsequent Lean checks the actual late-credit scheduler and all six arbitrary-data starts; no new binary initializer or sharper-credit theorem is implied. The HOD interpretation supplies ordinary selector Choice and an internal formula-tree decoder route; its complete all-code formal implementation remains separate. The arbitrary-row fusion domains cover31,626 pairs and over32 billion row recipes through finite sufficient-window pruning, with no certificate. This is negative evidence for the stated criterion, not a lower bound.

Source47/48 finite suites and full searches reproduce, including64 compiler alternatives,166,413 incoming-action recipes and both continuation ledgers. Source49's two complete fusion ledgers, differential/closure controls, all238 rotations and fixed20,000/46,413-proposal stochastic runs reproduce as well. Three scratch PDFs have38/37/26 pages and visual review; two source49 underfull paragraphs are acceptable. All delivered files and the base article are unchanged. No new state record, global minimality, Coq rebuild or finished internal PA/ZFC endpoint is claimed.

Batch 157 supplies weighted quotation/height bounds and exact shorter parsing fuel on every code. Normal constructor trees biject with all naturals, collapsing only $B_0(Z, Z)$. Canonical local certificates retain all twelve C254 rules, native equality, both gates and fixed falsity. The quoted interface uses local forward equations; the constructor-only checker removes numeral arithmetic. Its small-certificate bounds describe existence, not every redundant accepted input. No binary codec, complete enumerator/compiler or new blank initializer is delivered. Current IOpen Cantor graphs close the elementary pairing step; the full coded Choice and PA/ZFC machine endpoints remain open.

Two mathematical reviews and ten substantive replay jobs pass, with one explicit provenance-only skip. The complete132,048-recipe degree-one journal reproduces but repeats source48. The saved degree-two search has10,845,350 pruned and35,893,594 remaining assignments; a fresh600-second run reproduces 723,940 prefix nodes and validates its own partial partition. Neither finds a survivor. The34-move window identity and residual fiber obstruction exclude only a specified finite-label representation, not spare storage or alternative simulations. They establish no blank reachability or state lower bound. Both25-page source PDFs rebuild cleanly and receive visual review. All62 ancillary files remain unchanged; five main/delivery files are archive-only. The offline helper recovers all67 files and replays the substantive checks.

Batches 158–159 add five complete sources and one explicitly salvaged incomplete source. Parity-complete scheduling, no-filler MP and cache-result MP change the proof-block interface without dropping any C254 rule. The geometric253-state runtime's inherited startup provably escapes; the different cache-result252-state machine has a reproduced707393-step safe blank startup, 2998-entry source correspondence and8361 affine cases, but no new Lean endpoint.

HF acyclic-pattern seeds and exact role-allocation/progress certificates broaden research interfaces. Erasing598 transfers nevertheless increases a complete backend from314 to888 states. The5432 affine-code census has minimum281 in its own backend; its additive proxy bound22 is not a TM-state bound. Completed incoming-action families extend older partial records; shared target2 and fold families are not counted twice. Larger searches remain partial, and sufficient checker failure is not semantic inequivalence.

The new countdown method verifies181 local guarded contracts, with exact source matrices, positive and zero cases and full relative tape interfaces. All138 modified tables still fail later cases. No smaller consistency machine is established. The user's target remains provability in PA; the maintained external252/ZFC equivalence is already closed, but internal PA remains open.

At the user's request,327 intact local entries of the damaged negation archive are salvaged. Its ZIP directory, independent theory checkers and four journals are absent. All32 available partition journals reproduce32514048 recipe assignments; this is partial coverage and not exhaustive replay of rejected traces. Available255-state table/frame controls reproduce; the full verifier fails on a missing import, and complete search/invariant claims remain unverified.1133 ancillary files are preserved unchanged,40 article/delivery files stay recoverable from Git archives, and standalone recovery tools restore original paths. Full source/reconciliation reviews distinguish ordinary proofs, fresh finite checks, historical evidence and missing material. No delivered article/PDF or Lean/Coq

build was rewritten or rerun by intake.

Machine engineering credits Mira-Cult, Jeff, Wade and mira-emmy's prior 253-state reduction. Prepared/runtime-only 252 cores, blank-start 256 and Wade 269 experiments remain distinct. New 250/249/248 raw tables are compiler benchmarks; ternary primitive-Bottom 251/252 variants retain the same obstruction. Complete-equality EPSC and full-word controls have 270 states, with the prior full-word control at 269. A complete-original-logic conjunction layout reaches 268, above the attributed 253 benchmark. Its certificate compares the compressed program with the revised source word; the logical rewrite and binary simulation are separate bridges. All 1,024 costs in source 18's PSC histogram reproduce exactly, including five 283-state minimizers and the fixed winner; an initial review discrepancy was an accounting error. Unbundled historical sweeps remain limited. Independent register certificates and exact regenerated tables do not certify binary execution. Current Lean modules check raw-model PSC resources, generic register bisimulation, primitive-Bottom ternary coding/semantics, the full 312,975-step literal 253-state bootstrap, and unconditional named-variable Tarski completeness for primitive Bottom. Later Lean proves a named-to-original-numerical proof adapter, a whole-tape relation to the exact RM330 endpoint, and outgoing equality with reserved-zero semantics. The numerical model bridge compares decoded and encodable bundles under IR and explicit valuations; its original vacuous empty-domain premise is refuted on record. At that earlier checkpoint the binary and Foundation bridges were open; the later closed252/ZF theorem and external Choice reduction are recorded above. The internally arithmetized PA/ZFC endpoint remains open. No new internally certified sub-254 consistency machine is claimed. Placements `b07d1a27b`, `ac4287e5b`, `3cc6ba5b1`, `f31ba244b`, `520bcb2a1`, `30e6d1fd1`, `a9a313b79`, `a6823de93`, `65c776f4be`, `b123527ff4`, `6135ffef4b`, `d1636c9af8`, `799bcd094d`, `7755fdc9a8`, `e2ceaa6f8c`, `78b528440a`, `901dd50056`; complete articles recoverable from committed arrivals, ancillary files preserved. Unrefereed.